The Dynamics of Political Participation in the Lives of Ordinary Americans

by

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Abstract

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This dissertation is an attempt to understand the dynamics of political participation within the lives of ordinary Americans. The analyses show that although few people may get involved in any single year, many get involved in some political activity at some point over their lives.

After describing the shape and characteristics of political participation as it evolves over time, the dissertation examines three factors that distinguish those who are apt to participate from those who are not in previous cross-sectional analyses: education, residential mobility, and parenthood. The chapter on education shows that education does not dramatically change the shape of participation trajectories within people's lives: the more highly educated do appear to have a higher probability of participation at any moment, but this "education gap" does not change shape as people age. The chapter on residential mobility suggests that moving to a new place does not disrupt the non-voting participation of individuals as strongly as might have been expected given the literature on vote turnout. The chapter on parenthood shows that men and women react differently to the transition to parenthood and to the aging of their children.

This dissertation contributes to current knowledge in four ways. First, by carefully conceptualizing the dynamics of political participation, this work provides a nuanced and complex view of what the over-time questions really are. Second, by articulating and testing the different dynamic implications of the cross-sectional literature, this project adds new evidence and perspectives. Third, this dissertation shows that the shift from analysis of cross-sections to analysis of dynamic processes implies a shift in the objects of analysis and

the possible causal relations among them, thereby helping social scientists in general study dynamic processes more effectively. Finally, this work provides a detailed account of the development of political participation over the life-spans of individual Americans, thereby adding to accounts of political socialization in general.

Professor Laura Stoker and Professor Henry Brady
Dissertation Committee Chairs

To Cara and Laura

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Chapter 1 Introduction

This dissertation is an attempt to understand the dynamics of political participation within the lives of ordinary Americans. By *political participation*, I mean actions taken by individuals to influence the selection of officials and the choice and/or implementation of policies.[1] By *dynamics*, I mean:

- The *changing rates* at which people participate over the course of their lives.
- The *timing* of participation episodes in the life-course of the individual.
- The *persistence* of spells of activity and inactivity.
- The *sequences* of activities and issues pursued by individuals.

Whether one opposes or favors democracy, no one disagrees that participation is the essence of democracy. In addition, many people believe that within democracies participation can bring great benefits — aiding current advanced democracies in their task of "perfecting" democratic government — or great dangers — allowing uninformed citizens to choose cruelty and irrationality that could threaten the stability and quality of democratic governments. Whatever one's scientific or normative commitments, one ought to care to learn more about political participation.

Both those who fear the mob lurking within democracy and those who hope for the self-actualizing individual would agree with Schlozman (2002) when she writes,

> Citizen participation goes to the heart of democracy. In fact, it is difficult to imagine democracy on a national scale without the right of citizens to take part freely in politics (page 433).

[1]The main indicators of such activity utilized here will include acts such as writing letters, email, or making phone calls to elected officials; going to meetings, rallies, demonstrations or protests; donating money; or otherwise getting involved in community and/or campaign activities.

The concept of democracy is, after all, at its core "government by the people." Since governing is action, so must a democracy involve governing action by citizens.[2] Thus, concern for (or against) democracy must imply some concern about citizen participation.

In her review article in *Political Science: The State of the Discipline (2002)*, Schlozman aptly summarizes the major debates about the benefits of political participation under three main themes. First, places where many people participate are supposed to be nicer places to live — places where public and private goods and services are provided more efficiently due to cooperation and trust. Second, individuals who participate are supposed to become better humans; their activity allegedly helps them learn and grow in certain positive ways, including being more empathetic and appreciative of "the needs and interests of others and of society as a whole" (page 437). Third, a society where all people can freely participate avoids tyranny and allows individuals to protect their interests from encroachment by others.

While Schlozman presents these arguments that participation is a good without taking a stand herself, no one to my knowledge has as elegantly summarized the opposing point of view — perhaps because there is no large current body of literature extolling the dangers of citizen participation. However, the relatively rare opponents also make arguments that might convince us to care about this phenomenon. For example, even though Nie, Junn and Stehlik-Barry (1996) are not opponents of participation, they do remind us that the number of "seats in the political theater" may remain fixed, and that increasing participation may be futile if the outlets for citizen action are finite. Their observation forces one to imagine what might occur if all the populace were stimulated and enabled to participate, but if the ability of the political system to absorb such activity were unable to expand in turn. A lack of legitimacy and inadequate channels for problem solving via non-violent political participation have been suggested to exist at the root of many of the "democratic breakdowns" in which democratic polities have been supplanted by authoritarian or totalitarian regimes in this century (See, for example, Linz, 1978). Thus, one might care about political participation from a concern that too much participation threatens democratic stability.

[2] Of course, the definition of who counts as "the people" continues to be contentious, and at various times in history extremely small subsets of the human subjects of purportedly democratic governments were the only ones whose participation mattered (Smith, 1997).

More recently, Fiorina (1999) has suggested yet another concern about political participation. He argues that those who currently participate in the United States tend to hold political views that are much more extreme than those held by most of the populace. He writes that, "...when engagement is largely the domain of minority viewpoints, obvious problems of unrepresentativeness arise. When they do, civic engagement has a dark side that is not sufficiently recognized by its proponents" (page 403). According to this view, the fact that zealot minorities dominate the spaces of political participation drives, dissuades, and disillusions more moderate individuals from engaging in such activity. This creates a cycle in which moderate, representative, individuals cede more and more of the political arena to extremists who produce grid-lock (by proposing unrealistic policies) and outcomes that do not maximize the public good (since the activists represent only small minorities of the population). He suggests that rather than denigrate those who do not participate, we should be more concerned about those who do get involved:

> It is time to abandon the notion of political participation as part of human nature. It is not; it is an unnatural act ... Contrary to the suggestions of pundits and philosophers, there is nothing wrong with those who do not participate; rather, there is something unusual about those who do. All too often they are the people "nobody sent." (citing Rakove (1979, page 318) page 415).

This dissertation does not engage directly with any of these normative arguments about democracy and the good life. In fact, many of the answers to these debates hinge on understandings about the state of the world — and, in particular, understandings about the nature, causes, and consequences of political participation. I hope that my work here helps provide just such an empirical foundation for the theories that currently stimulate such arguments and that the results presented in the chapters that follow provoke further interest about democracy and its citizen components.

1.1 Why should we care about dynamics?

Political participation seen as a dynamic process occurring within the lives of individuals tends to look different from political participation seen as a line dividing those who engage in politics from those who do not. Examined as a dynamic process, it encourages scholars to ask new questions; it suggests new understandings of past findings; it requires new ways of thinking about causal effect on participation; and, in the end, reveals new information about the political life-cycle.

Asking new questions Most previous research has not focused on political participation as a dynamic process within individuals, but rather has pursued an understanding of the variations in political participation across individuals studied at a single moment in time. Because of this static focus, it has been enough to say, for example, that "Education facilitates participation." Although it has certainly been important to know *who* participates, this kind of general statement about the relationship between education and political participation leaves open several questions, the answers to which are essential for understanding what it is that education does to encourage participation. Considering participation as a dynamic process forces one to ask new questions like: "Education *when*, predicts participation *when*? In what way, if at all, does education enable people to continue participating despite interruptions such as moving to a new home or having a baby?" These questions show that education may relate to an evolving process of political participation in several ways. For example, education may act as an initial condition determining nearly the entire future trajectory of participation (i.e., early education may prepare a person to leap into action when necessary, even if the opportunity/necessity doesn't occur for years into the future). Or perhaps education may function as a time-varying influence such that when education changes, so does participation. This simple example using education reveals that even well-established cross-sectional findings leave open many important questions. In fact, a single static theory may have many different implications in the context of a dynamic process. As I study the changing rates, timing, persistence, and sequence of political participation within the lives of individuals, I shall be forced to grapple with questions about how to specify and articulate the relationship between political participation and time — questions about which the past literature does not offer much help. I hope that this struggle to ask the unasked questions (and the attempt to answer them) will expand how we think about the participation process.

Understanding past findings In addition to conceptualizing the dynamics of political participation, and in order to build on past literature, this dissertation articulates (and tests) the many different dynamic processes that grow from cross-sectional findings. For example, if education in the form of "civic skills" (such as writing letters, attending meetings, speaking in public, or organizing groups) is the main driver of participation (Verba, Schlozman and Brady, 1995), then dynamic implications would include: (1) that past participation should increase the likelihood of participation in the future (since past participation should provide experience and skills) and (2) that gaining any education (degree oriented or not) should increase the likelihood of participation (again, via acquisition of civic skills). If education mainly acts to confer status in a social network (Nie, Junn and Stehlik-Barry, 1996), then the dynamic implications would include: (1) that early degree-granting education should matter much more than late; (2) that vocational or non-degree granting education should not matter; (3) that different education degrees should function differently in different generations (since position relative to others in a cohort is what determines network position); and (4) that past participation should increase the likelihood of future participation (since past participation should create relationships between politicians and participators), but only in the kind of participation or issue-area experienced in the past. The move to study political participation over time thus provides many opportunities to assess, extend, and more deeply understand past findings.

Understanding causal processes Studying dynamic processes requires understanding causality in ways that are different from those employed and operationalized in cross-sectional analyses. For example, as I described above, a single causal agent can influence the overall trajectory of a dynamic process by occurring early, late, or throughout the process. Furthermore, the trajectory may be altered by changes in rate of occurrence (e.g. increasing or decreasing the number of moments in which people participate), changes in intensity of occurrence at each moment in time (e.g. changes in the number of participatory acts in a single moment), or changes in the patterns of occurrence (e.g. increasing or decreasing the length of spells of participation in a given activity). Although this dissertation is about political participation in the lives of individuals, the questions I ask are in fact the major questions that one must ask about dynamic processes in general.

Some political scientists have begun thinking carefully about dynamic processes. For example, Carpenter has used the probability theory of stochastic processes (i.e. dynamic processes with a probabilistic or random component) in deriving theoretical understandings about the behavior of bureaucrats and government agencies (See, for example, Carpenter, N.d.). The most comprehensive attempt to clarify the concepts that confront students of empirical dynamic processes has been Pierson (2000). Pierson's article, however, focuses only on one characteristic of dynamic processes, namely, path dependence. Although path dependence is one important way that processes may evolve over time, my preceding comments show that there are several other aspects of dynamic processes which are important to recognize and understand. In this dissertation I articulate and illuminate major conceptual questions about evolution and causality in the context of dynamic processes in a way that I hope will help scholars interested in the many different ways that politics evolves over time within many different types of units.

Understanding the political life-cycle Although examining the rate, timing, persistence and sequence of political participation as a dynamic process is a new perspective, the study of political socialization has long been concerned about the dynamics of political behavior over the life-span. By providing a detailed explanation of the development of political participation over the lives of individuals, I hope to help scholars of political socialization develop a more complete account of life-span development in general.

In summary, this dissertation contributes to current knowledge in four ways. First, by carefully conceptualizing the dynamics of political participation, this work provides a nuanced and complex view of what the over-time questions really are — questions left open by the cross-sectional literature. Second, by articulating and testing the many different dynamic implications of the cross-sectional literature, this project adds new evidence and perspectives to the current literature on political participation. Third, I show that the shift from analysis of cross-sections to analysis of dynamic processes implies a shift in the objects of analysis and the possible causal relations among them, thereby helping social scientists in general study dynamic processes more effectively. Fourth, I provide a detailed account of the development of political participation over the life-spans of individual Americans — thereby adding a significant piece to accounts of political socialization in general.

1.2 Developing Hypotheses about Dynamic Processes

When writing down hypotheses in cross-sectional analyses, it is common to think of causality in terms of an "effect" being detectable or not. If $Y_t(u)$ can be thought of as the outcome that one desires to explain, say the response of some individual, u, to a treatment t, the most common way to think of a causal effect can be written:

$$Y_t(u) - Y_c(u), \tag{1.1}$$

where $Y_t(u)$ represents the value that Y would have for individual u if that person received the treatment t, and $Y_c(u)$ is the value that Y would have for that same individual if she did not receive the treatment.[3] In general, cross-sectional analysts only observe Y_t or Y_c

[3] I am following the notation of Holland (1986) here. There are many other ways to think about causality as summarized and critiqued by Brady (2002). However, what is known as the Neyman-Rubin-Holland

for a given person, not both. But, the general idea of causality as representable by a scalar difference is maintained in those analyses, such that the actual difference computed is between individuals who are as similar as possible with the only difference being that one received the treatment and the other did not.

In some ways this conception of causality still holds during the analyses of dynamic processes — in fact, in the chapters in this dissertation I do observe both Y_t and Y_c for the same person. I have a whole series of observations about each person — and a series of observations on the "treatment" as well — so that the questions are not about simple differences, but about how one series changes in response to another series. In this case, the "effect" is no longer easily representable as a scalar difference, but is instead a function, and inference is about the difference between the shape of this estimated function and the shape of the function that is hypothesized to relate the two series.

Suppose we have one variable that is changing over time, Y, that we want to model as a function of another variable that is changing over time, X. There are 4 ways that X can come to influence Y, holding constant, for the moment, the ways that past Y affects present Y:

Life-cycle If X occurs early in the life of a person, is the effect on Y different from the situation where X occurs later in the life of a person? Does acquiring a college degree "in step" with one's birth cohort entail a different kind of boost to political participation than acquiring it later? That is, the effect of X on Y occurs conditional on age.

Type The type of X that occurs may matter for the level of Y. For example, an additional year of education may provide a college degree or vocational training. All types of additional education may not be equal when it comes to enhancing political participation. That is, even if X is a continuous variable, it may have a non-linear effect on Y — such that a unit change in X may influence Y differently depending on where on the X scale this unit change occurs.

Timing The effect of X on Y may be delayed or immediate. Perhaps acquiring a college degree does not have an immediate effect, but prepares a person to respond to calls for participation later. That is, there may be a lag in the effect of X on Y.

understanding of causality is assumed by most statistical analyses of cross-sectional data in political science.

History How often X has occurred in the life of a person may change the effect of an additional unit of X. Take the rare example of a person who has received 4 BAs — it is probable that each BA received does not have the same participatory impact as the previous one (and, in fact, might in this case have diminishing returns). That is, I'm using the word "history" here to refer to the number of past Xs (and the amount of time passed since the last occurrence of X, too).

As I noted above, all of these ways for X to affect Y ignore the fact that past Y may affect present and future Ys in a variety of ways as well (via the moment in the life of a person that Y occurred; via the particular type or sequence of types of Y that occur; via delayed or immediate response; or via the number of previous Ys or the time since the last Y occurred). What makes understanding this relationship even more complicated is that, in theory at least, each of the 4 causal dimensions can combine with the others. Of course, not all of these possible causal relationships are plausible given what we know about education and political participation. However, a surprisingly large subset remains.

It is common to think about causality in dynamic processes by naming them: "path dependent", "Brownian", "Weiner", "Poisson", and "Markov" are all names that are applied to processes to describe the particular combinations of life-cycle, type, timing, and history that drive particular processes forward.[4] In general, the process of political participation does not fall neatly into categories where, say, present Y is completely determined by Y one period in the past, or by a value of X at the point where the series of Y began. Thus, labeling in this case tends to obfuscate rather than illuminate. Instead, I will try to talk directly about life-cycle, type, timing, and history in terms of the particular variable and hypotheses being examined in each chapter. This will also help map the results of the dynamic analyses back to the cross-sectional findings.

[4] I have labeled the kinds of causal agents in such a process with reference to the actions of individual humans here. Most people are familiar with certain of these labels being applied to States, pieces of radioactive matter, or patterns of missile strikes. However, just as a person may behave according to a Poisson process, so too can a war of missile strikes have a life-cycle whose constant rate may depend crucially on initial conditions.

1.3 Relating Previous Time-Constant to Time-Varying Findings

Given the complex ways in which any political participation may be caused, it may appear difficult to generate dynamic hypotheses from previous static findings. In fact, it is not so difficult, mainly because previous scholars have talked about causal effects as the difference in outcome between treatment and control within the same person. Cross-sectional analyses have had to make do with comparisons between individuals due to the limitations of data, but the theories at stake all imply a within person effect. Thus, much of this dissertation will involve distilling the implied within person effect from the scholarly consensus, derived from between person analyses.

Of course, this tight link to previous work raises another question: How should I interpret within person findings that do not support the bulk of previous between person results? How should I make sense of differences between the two sets of information about the same phenomenon? This is the central challenge of this dissertation. To my knowledge, there is no well-defined method for bridging the two kinds of analyses. I will confront this problem in each chapter to come.

1.4 Data

Description The skeptical reader of the discussion so far might argue that while observing the dynamics of political participation within people's lives would be interesting and informative, it is impossible because all available data are from cross-sections or short (2 to 6 year) panels. This is largely correct with one (and only one) important exception. This dissertation relies almost entirely on data from the Political Socialization Study (Jennings and Stoker, 1997).[5] This panel study began in 1965 with in-person interviews with a national random sample of 1669 high school seniors and their parents.[6] The high school seniors and one of their parents were reinterviewed in 1973 and 1982. In 1997 the members of the Youth generation were interviewed again, but their parents were not. The final dataset contains data from four interviews, over 33 years, with 935 members of the high school class of 1965 (often labeled "G2" for "generation 2" in this dissertation); and data from three interviews with 898 members of the Parent generation (often labeled "G1" for "generation 1" here).[7]

[5] At various points in the following chapters I also use data from the National Election Studies and the United States Census as well.

[6] The respondents were selected based on a national equal probability sample of high schools, and students were randomly selected within high schools.

[7] While nearly everyone in the G2 cohort was born in 1947 or 1948, the mean (and median) age of members of G1 in 1965 was 46 (with the middle 50% of G1 between 42 and 50 years old). Thus, while the G2 cohort

During the interviews (in 1973, 1982 and 1997 for G2, and in 1965, 1973, and 1982 for G1), respondents were asked to recall, explain, and date past political participation. This dataset has the strength that it allows me to reconstruct annual participation histories over long periods for two generations of Americans (from roughly 1951-1982 for individuals in the older "Parent" generation and from 1965-1997 for individuals in the younger "Youth" generation).[8] The timing information allowed me to build mini-time-series for each person that stretch for a minimum of 18 years and a maximum of 32 years depending on the type of activity.[9] In fact, this is the *only* source for histories of participation that last beyond 6 years, and it is also the *only* source which allows the recovery of annual information and descriptions of the issues and candidates motivating involvement. It is only now, after nearly 30 years, that this aspect of the dataset can be exploited. Thus, in fact, I can study the dynamics of participation, but only because of this amazing and unique data source.

Measurement All empirical research involves measurement in which scholars strive to map the domain of content of a concept to observable phenomena. "Political participation" as a concept refers to any activity done in an effort to influence politics. From my point of view, an adequate yet pragmatic measure of "political participation" must therefore capture information about activities which can be easily recognized as aiming to influence "politics." The survey questions that I analyze in this dissertation seem to me pragmatic yet adequate representations of the idea of political participation. They include questions about activities during campaigns, such as attending campaign rallies or meetings; donating money for an issue, party or candidate; displaying a campaign button or sticker or sign; or volunteering to work for a candidate, issue or party. The survey questions I use here also include a battery that focus expressly on non-electoral activities such as contacting public officials; writing letters to the editor; attending a demonstration or protest; or working with others in the community. Most people would agree that these questions do ask about activities aimed to influence politics. These questions are also particularly useful given my interest in "ordinary" Americans since they represent what "political participation" means for the

ages from 18 to 57 over the course of the 4 waves of the study, the G1 group ages from roughly 46 to roughly 63 over the first 3 waves of the study.

[8]Members of the Parent generation were asked to recall their past participation in 1965, and so their data stretches back to 1951 (the earliest date any of them recalled participating). Members of the Youth generation were first asked this battery of questions in 1973, and were asked to recall back to 1965. Thus, their series begins in 1965.

[9]See Appendix A for wording and details about the political participation questions in the Political Socialization Study.

non-elite majority of the country better than what this concept might mean for other types of political actors, like senators, party leaders, or social movement activists.

Figure 1.1 shows the raw data on participation in non-electoral activities reported by a random sample of 20 members of the Youth generation from among all of those individuals who did any of these types of activities over the 32 year study period. Each panel of the figure shows the data for a particular person, and the height of each line represents the number of activities reported by that person in a particular year. For example, Person 354 reported attending a protest or demonstration in 1969. Then, in 1977, 1978, and 1979, he contacted an elected official. Also in 1978, he did some work with others in his community, and in 1982, he did some community work again. This shows up as "spikes" of height 1 for each of 1969, 1977, 1979, and 1982, and a spike of height 2 for 1978. In the rest of the dissertation I will analyze these acts and others separately since different types of participation tend to have different patterns of growth and change over the life-span of individuals. This figure shows, however, that even if we combine 4 types of action, moments of participation tend to occur at irregular intervals, some individuals display multi-year spells of activity, and many show sparse sequences of participation over the years. This kind of data provides the raw material for understanding the dynamics of political participation across the lives of ordinary Americans.

Caveats Of course, no dataset is perfect. We pay for the unique strengths of this dataset with 3 main weaknesses. First, ample psychological evidence suggests that (1) people are fairly competent at remembering the times at which significant life events occurred, and (2) people are not very competent in remembering the times at which life events occurred that were not somehow important to them (Tourangeau, Rips and Rasinski, 2000). Furthermore, such literature suggests that certain ways of asking questions about timing can increase accuracy of recall (Belli, Shay and Stafford, 2001) — and the Political Socialization study did not use those methods. Thus, to the extent that moments of participation are rare (which all evidence to date suggests) and important to the respondents, then the data I analyze here will be a reasonable representation of the participation histories of the two generations interviewed. To the extent that these events are not important, then I expect to have a mis-representation — either because people are over-reporting due to social desirability bias (as they do to questions about vote turnout in the NES) or, more likely, are under-reporting

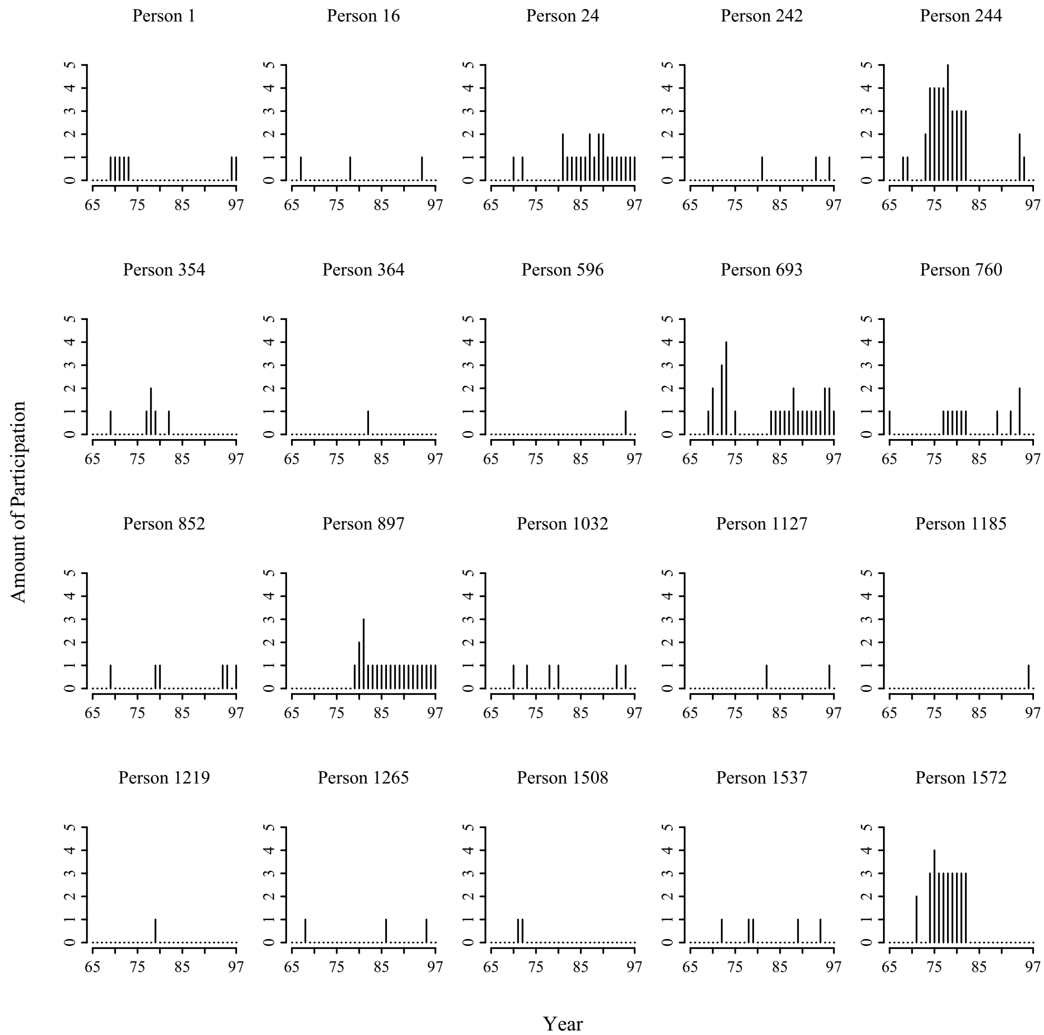


Figure 1.1: Profiles of Individual Non-Electoral Participation: Youth Generation

Note: This figure indicates the number of acts of political participation reported by each respondent for each year between 1965 and 1997, considering only the following four types of political acts: Working with others in the community, Contacting elected officials, Attending protests or rallies, and Writing letters to the editor.

due to forgetting.[10] This is a serious problem, and the analyses in this dissertation include some adjustments for this problem with memory.

Second, the dataset analyzed here does not contain several variables that have figured prominently in previous research, the most important being "mobilization" or the

[10]It is highly likely that forgetting will be the dominant problem. For example, Pierce and Lovrich (1982) found that 44% of known signers of a petition in Idaho did not recall having signed it after a year. Of course, as Hansen and Rosenstone (1983) note, petition signing may be like telling a passerby the time, and thus unremarkable compared to other activities in which more resources must be expended.

extent to which people are recruited to participate by others. To the extent that mobilization drives participation and is heavily correlated with, say, education or lack of residential mobility, then we would expect that coefficients estimated to represent the influences of education and residential mobility would be capturing the effects of mobilization. To the extent that different individuals have different propensities to be recruited which are fixed over the life-span, I can deal with this problem via different adjustments I've made for cross-sectional heterogeneity. However, to the extent that the characteristics of individuals that lead to mobilization are not available and mobilization itself is not available, then the coefficients estimated here will not be unbiased (even in the case of perfect memories).

Third, the Political Socialization data over-represent certain parts of the population and under-represent others. For example, the "youth" cohort consists only of those individuals who made it to their senior year of high school in 1965 — and about 24% of their cohort dropped out before that time (Jennings and Niemi, 1981). As time has progressed, individuals have dropped out of the panel study, such that by 1997 the sample was more female, wealthier, and more educated than the population of people in that age group at that time (Jennings, Stoker and Bowers, 2001). However, the amount of attrition has been far less than would be expected for a study conducted over 30 years and the effect of the attrition on political variables has been quite small. The retention rate for this study was 58%, once 68 known deaths were taken into account.[11] Thus, the histories collected and analyzed here will both probably under-represent the true amounts of participation done by the individuals due to forgetting and will over-represent the amount of participation in the nation due to the relatively higher SES of the cohort.

Finally, given theories about the motivations for political participation, the temporal resolution of the data is not ideal. For example, if participation tends to be self-motivated, then people would need to perceive some reason to get involved before participating. If participation tends to be mobilized by others, then participation must be requested before it can occur. In either case, I would expect that the time frame over which such perception and recruitment occurs would be measured in weeks or months, not years. The data that I generated from the Political Socialization dataset are, to my knowledge, the

[11] Comparisons between those individuals who dropped out of the study after the first wave and those individuals who stayed in until 1997 show that those who stayed were slightly more liberal and slightly more politically involved than those who did not. However, being a panel member after 1965 does not explain more than 2% of the variance in any of the political variables. See Jennings and Niemi (1981, Appendix A) and Jennings and Markus (1984) for more analyses showing that panel respondents continued to represent their cohort quite well over time.

most temporally fine-grained information about political participation that currently exist, and they are annual quantities. So, despite the fact that this dissertation presents the most detailed information currently available, with respect to the phenomena under study, it is crude.

Although I might say that it is worth writing this dissertation on the simple grounds that some information is better than none, others might argue that any misleading information is worse than none. The rejoinder, of course, is that one can never know anything without attempting to do the knowing. I argue that this project is worthwhile as an attempt to discover something about the dynamics of political participation within the lives of individuals: by keeping the limitations in the fore but also by using *the very best information on the dynamics of political participation ever available*, I hope to provide some knowledge and be able to discern when the results may be misleading — and to articulate this. The limitations articulated here do not make the study of the dynamics of political participation a lost cause or fruitless. Rather, they remind me that we are at the frontier of new knowledge where we can hope to discern perhaps messy shapes that will guide further, more targeted, fine-tuned, and accurate, exploration.

1.5 Organization

Chapters 3 through 5 of this dissertation are devoted to investigations of how certain key cross-sectional findings translate into the dynamic context. However, since the data used in this dissertation are unique, Chapter 2 is devoted to description. How much do individuals appear to participate over their lives? When they get involved, how long do they tend to remain active? If they act in one period, are they more or less likely to act again in the subsequent moment? These questions are not about the kind of causal relationships that dominate the literature, but it is important to have some understanding of them before asking questions about what causes the timing or duration of participation within people's lives.

After Chapter 2, the other chapters of the dissertation contain analyses of the consequences of education, residential mobility, and parenthood on the long-term dynamics of non-voting political participation. Education, residential mobility, and parenthood (as it intersects especially with gender) have all been identified in previous research as major aspects of people which influence who participates.[12] By choosing variables identified as

[12] Of course, previous literature also suggests a number of other important influences on participation, but

important in the cross-sectional research, I can assess and expand on previous findings. Since these variables are also seemingly well-understood, it will be all the more useful to the scholarly community for me to articulate new questions about their relationships to participation, and to show how the cross-sectional theories create many different implications for their functioning in a dynamic process. Another benefit of focusing on these three factors is that they manifest themselves in time in three different ways. After an initial burst of activity, formal education tends not to change much as individuals age. Residential mobility occurs as rapid interruptions: individuals begin and complete moves over the span of months, and many move often throughout their lives. Finally, although children usually spend around 20 years living with their parents, their influence on political participation probably occurs over a discrete range of time, roughly coinciding with the school-years.[13]

Chapter 3 focuses specifically on the relations between education and political participation. The fact that people with more education participate more than people with less has received some of the strongest support of nearly any social scientific relationship, with confirmation across data sources, historical periods, and geographies. However, it is not clear if education has a consistent effect on participation across the life-span, nor do we know whether it merely enhances the levels of participation expected of individuals at each moment in time or if it actually changes the shape of the trajectory of political participation over time.

Chapter 4 analyzes another relationship that has also received strong confirmation in the literature on political participation: that individuals who are recent arrivals to a place are less likely to get involved in politics than long-time residents. This finding has often been interpreted to mean that if a person moves to a new place, she ought to be less likely to participate during her first few years in the new location. The Political Socialization Study allows just such a test since the data contain observations of the participation of individuals both before and after moves.

Chapter 5 departs somewhat from the focus on the well established relationships

I cannot pursue them all in this dissertation. Also, as Chapter 2 demonstrates, this dissertation does not provide enough space to adequately analyze all of the different dynamic aspects of political participation in depth. However, since none of these aspects has been studied before, I will present at least some information about each of them, and will go into depth for one or two.

[13] Highton and Wolfinger (2001) have also shown that vote turnout differs between young adults who remain in their parents' houses during their 20s and young adults who leave home. Thus, both parents and children influence the political participation of each other while they live together (and probably also when they no longer live together as well).

of education and residential mobility of the previous two chapters. This chapter instead asks whether the transition to parenthood has participatory consequences. The idea that becoming a parent might decrease participation seems intuitively obvious, but recent work that has explored this idea has shown that parents are not that different from non-parents when it comes to participation. My analyses will investigate how participation changes for individuals as they make this transition and as they add more children to their families.

The aim of Chapter 6 is to summarize the findings of the dissertation and to put these results into perspective with the current consensus built on cross-sectional data. This chapter will also provide the opportunity to discuss the many questions these chapters raise — questions that I hope will provide fruitful avenues for future research on political participation.

Chapter 2 The Shape of Participation

Political participation understood as a dynamic process within the lives of individuals is different from political participation thought of as a dividing line in society. Understanding inequality in political participation is an important way to understand how power and influence are distributed in a society at any given moment — and this is the perspective that has been pursued most vigorously by political scientists so far. The point of this dissertation is to add another perspective to the current one — to think about what divides moments in individuals lives during which they participate from those moments during which they do not. Previous work clearly has much to offer research in this area — knowing *who* participates suggests *when* people might participate. I also hope that understanding when individuals are apt to participate will reflect back on the work which has concerned itself with inequalities across individuals — in essence to fill in a part of the picture that has been, until now, largely unexamined. As I laid out in the introduction, this work is a first step on the road to understanding the dynamics of political participation, and thus this dissertation proceeds by examining two of the most important findings about what divides those who participate from those who do not (i.e. education and residential mobility), and by adding a third that has recently been examined in hopes it would answer questions about why women tend to lag behind men in participation (namely, bearing and rearing children). Although the testing and elaboration of causal hypotheses distilled from these previous findings may seem to be the core of the dissertation, it is at least as important to understand what political participation looks like across the lives of individuals. In a sense, the new data on political participation provided to us from the Political Socialization study is about a new and different phenomenon even though the types of activities studied are the same. Because it is a process occurring over time, new questions arise: How much do people tend to get involved over their lives (or at least over large parts of their lives) compared with over the single years studied by cross-sectional surveys? Does participatory activity tend to occur in multi-year spells? Or as single, sporadic, moments of involvement? To what extent does the amount of participation in one year relate to the amount in another year? Does the history of participation as it cumulates over time have an impact on the probability of participation at a given moment in time? When people do get involved, for how many years do they tend to remain involved? How long do they tend to wait between moments of activity? Clearly, there are as many questions about what this process *is* that are just as interesting as traditional questions about how this process is *caused*.

The point of the this chapter is to describe the dependent variable of the dissertation and to provide some initial answers to these questions. Each question is a different window looking out onto the same phenomenon. At first thought it might appear a trivial exercise: take political participation and vary it over time within peoples' lives. What should become clear from this chapter is that adding a single, temporal dimension to a well known phenomenon adds much more than one dimension of complexity to the enterprise of studying it. In the three subsequent chapters I will only be able to focus on one manifestation of this dependent variable. So, this chapter serves not only describes the dependent variable as it will be used throughout the dissertation, but also presents the many dependent variables that could (and should) be analyzed as this phenomenon begins to be studied.

2.1 Cumulative Participation

In previous research, the question "How much do people participate?" tended to be answered with "At this moment, X% reported doing some form of political participation." That is definitely interesting and important information. However, that question might also be asking about the quantity of participation individuals can be expected to engage in over time — or over large portions of their lives. That is, there is a distinction between "How much are people participating now?" and "How much do people tend to participate over their lives?"

Figure 2.1 shows an answer to the first question for each year that the National Election Studies (NES)(Sapiro, Rosenstone and the National Election Studies, 2001) asked individuals about their non-voting campaign participation. To allow for some comparability with the Political Socialization data, I have restricted the analyses of the NES to individuals who roughly match the age cohorts of the Parent and Youth Samples. Since about 50% of the Parent Sample in the Political Socialization Study was born between 1915 and 1923, I used the birth years of 1913 to 1925 for the approximately equivalent NES Cohort. Since 96% of the Youth Sample was born in 1947 or 1948, I chose an NES Cohort born between 1945 and

1949. Also, since the Political Socialization Youth cohort is defined by having finished high school, the NES Cohort chosen to represent these people only included respondents who had received high school degrees or higher levels of education. The height of each black bar shows the proportion of NES respondents in the age cohort of the Political Socialization parents (labeled “G1” for “Generation 1”) in that particular year who reported engaging in the participation types displayed: Attending campaign rallies or meetings, displaying political buttons or signs, contributing “other work” to a campaign, or donating money. The height of each gray bar shows the same information, this time for the second generation (“G2”), of people born in 1945-1949.

These plots suggest that non-voting, campaign oriented participation is highly variable across years in the aggregate (each plot has a few years in which the proportion reporting participation in the past year is much higher than surrounding years); that the most common of these acts (among these two cohorts) is donating money (on average, across, all years, 11% of G1 and of G2 donated money); followed by displaying buttons and signs (which were very popular among the G1 cohort in the 1950s and early 1960s) (on average, over all years, 10% of G1 and 11% of G2 did this); attending campaign rallies and meetings (cross year averages: 7% and 10% for G1 and G2 respectively); and finally “other” campaign work (cross year averages: 4% and 5% for G1 and G2).

Figures 2.2 and 2.3 show the distribution of activity within people’s lives for each of 8 types of political participation over 33 years for the non-campaign activities and 18 years for the campaign oriented activities of the Political Socialization data.[1] In each case, the height of the bars shows the proportions in the two samples reporting 0,1,2... acts of each type over the years.[2] The gray bars show the proportions for the Parent sample, and the white bars show depict the Youth sample. The x-axes for electoral and non-electoral activities run from 0 to 20 acts.[3]

[1]The question wording for these questions is described in Appendix A. The NES didn’t collect information about non-electoral activities (such as those shown in Figure 2.3). Also, the Political Socialization study did not collect detailed timing information about electoral activities in 1997.

[2]The first generation, or Parent sample (born around 1920), is labeled “G1”, and the second generation, or Youth sample (born in 1947 and 1948), is labeled, “G2”.

[3]Individuals were allowed to report multiple acts of a given type in a year (up to three or four acts). For ease of presentation, I have limited the number of acts shown here to a maximum of 20. However, at least one person reported doing as many as 70 acts of *Community Work*, 49 acts of *Contacting Officials*, 47 acts of *Letter Writing*, 26 acts of *Demonstrations*, 28 acts of *Rallies/Meetings*, 29 acts of *Buttons/Signs*, 20 acts of *Other Work*, and 28 acts of *Donating Money*. So few people engaged in more than 20 acts (or even more than 1 or 2 acts as the graphs show), that allowing the axes to stretch to the limits of the data hindered comparison and interpretation.

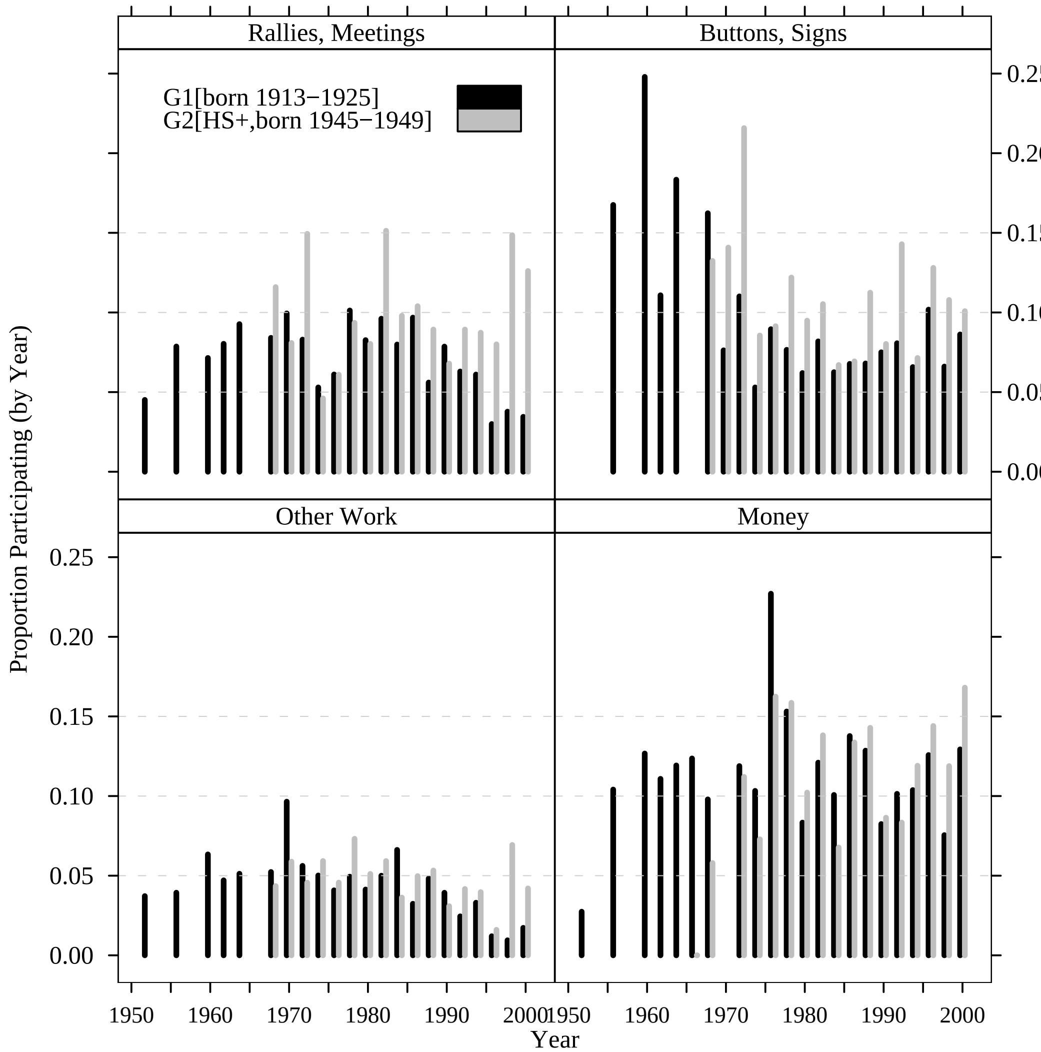


Figure 2.1: Proportion Participating in Each Cross-Section: Two Cohorts from the NES

Figure 2.2 shows that, for each act, over 60% of the respondents (in both groups) had not engaged in it. never reported participating in any of the four campaign oriented types of activities over the period of the study.[4] Like the NES respondents, the types of electoral activities chosen by the Political Socialization respondents for their political activity were, in order of frequency, donating money (28% of G1 and 30% of G2 reported doing this at least one time), wearing buttons/displaying signs (26% of G1 and 37% of G2

[4]Members of the Parent generation were allowed to report activities from before 1965. The earliest reported activity among that group occurred in 1951. Members of the Youth generation were asked these questions first in 1973, and the question referred to activities done since 1965.

reported doing this at least one time), attending rallies and meetings (25% of G1 and 33% of G2 reported attending at least one campaign related rally or meeting), and finally doing "other" campaign work (15% of G1 and 21% of G2 reported doing this at least once). It is sensible that many more Political Socialization respondents reported participating in these acts than did NES respondents — the Political Socialization respondents had many more opportunities (18 years) than did the NES respondents (1 year). However, the comparison of the information from the NES and the Political Socialization survey provides yet another hint that the phenomenon of political participation examined within the lives of ordinary Americans may be quite different from the phenomenon examined across Americans at a single point in time — more common over the life-span than within a single year.

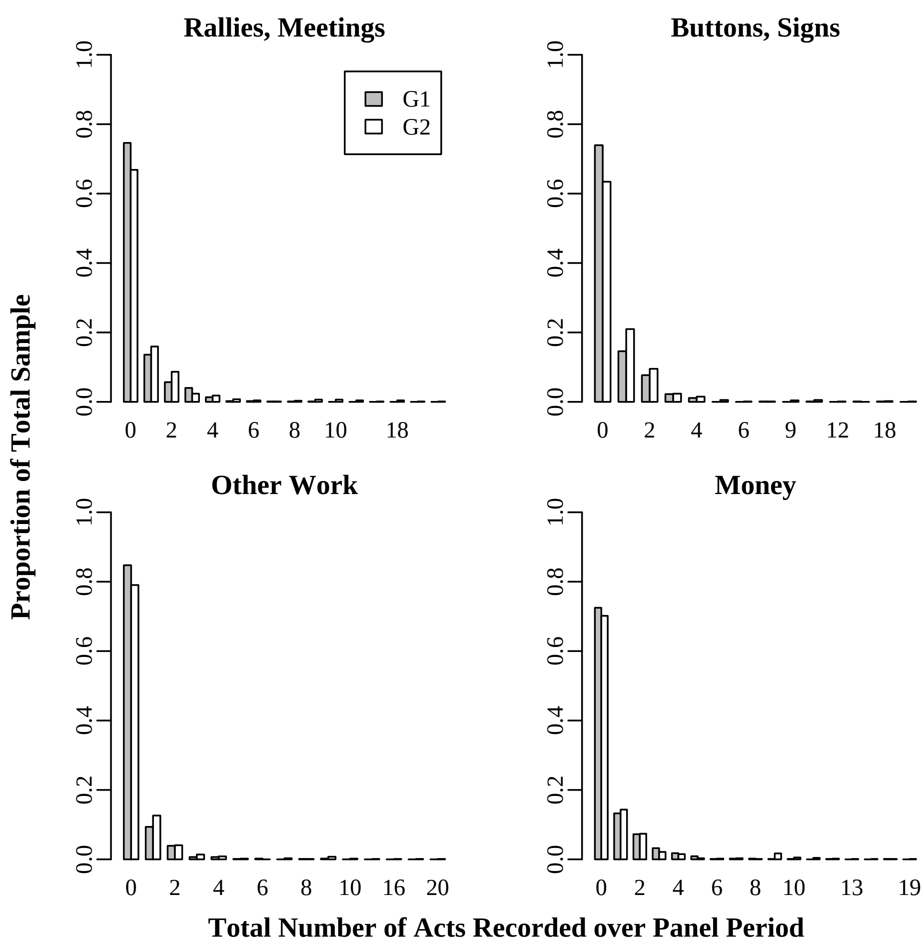


Figure 2.2: Cumulative Acts of Electoral Activity

Figure 2.3 provides similar information about the Political Socialization respondents, this time for non-electoral activities. The NES did not ask questions about these kinds of activities more than 3 times over the 1952 to 2000 period. So I do not know if the amount of participation shown here over 33 year periods within people compares to the amount that would be expected from questions asked of a single cross-section. This set of activities includes two that are very rare compared to the electoral acts (at least among G1): only 5% of G1 and 14% of G2 wrote any letters to the editor and only 3% of G1 and 24% of G2 engaged in demonstrations or protests over this period. This set also includes two that are very common compared to the electoral activities: 36% of G1 and 65% of G2 did some work with others in their community and 38% of G1 and 68% of G2 contacted a public official in some way over this period. This shows that the two generations have very different repertoires of political behavior when it comes to demonstrations and protests and letters to the editor (probably accounted for by some of G2 attending college in the late 1960s and by the fact the G2 are better educated than their parents). It also shows that members of G2 tend to be more participatory than their parents in all other kinds of activities — electoral and non-electoral. In fact, majorities of G2 contacted officials and did some community work, and the figure shows that at least 1% of G2 did such activities more than 2 times — 1.5% reported doing 6 acts of community work and 7 acts of contacting as they aged from 18 to 50.

The pattern of participation for voting is very different from that displayed in Figures 2.2 and 2.3. Most respondents reported voting in multiple presidential elections (as can be seen in Figure 2.4). The Parent Generation were only asked about 5 presidential elections (64,68,70,72,76,80) and G2 were asked about 8 presidential elections (68,72,76,80,84,88,92,96). About 68% of G1 reported voting in all 5 presidential elections, with only about 6% reporting that they had not voted in any. The analogous figure representing steady turnout for G2 is 38% voting in all 8 presidential elections that occurred between 1965 and 1997, but only 1.7% reported never voting. It makes sense that fewer of G2 should appear to be habitual voters than their parents in this figure. First, the members of G2 are younger than G1 by about 20 years, and younger people are supposed to be less likely to vote that older people for a number of reasons (See Highton and Wolfinger, 2001, for example). Second, the members of G2 have more opportunities to not-vote than the members of G1 — thus their pattern of voting is more apt to be sporadic across the 8 periods. It is an open question whether, by age 70, if the members of G2 will be more likely

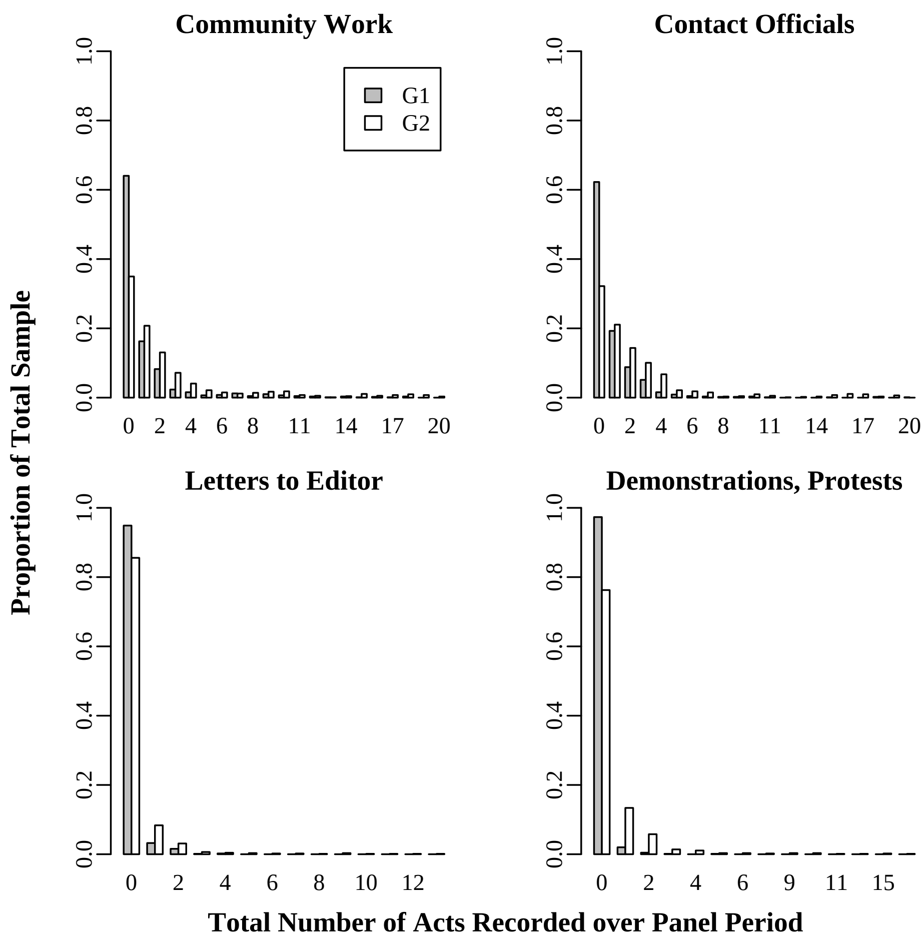


Figure 2.3: Cumulative Acts of Non-Electoral Activity

vote in every election, or will still vote in most, but not all of them.[5]

So we've seen two answers to the question "How much do people participate." In general, when allowed to look back over multiple years, it appears that people participate much more than when allowed to look back over a single year. This is sensible, since one can imagine that each moment in time provides some chance for a person to participate. And even if at any one moment few individuals take advantage of this opportunity, the group of participators does not consist of the same people moment to moment. Thus, more people get involved (at some point in their lives) in the political system than we had previously

[5]The fact that the members of G1 showed higher voting participation than G2, but that the members of G2 displayed higher non-voting participation that G1 is interesting, and given the current debates about the decline in "social capital" and the "greatest generation" probably deserves further study. Could the "greatest generation" only be "great" when it comes to voting but not other participation?

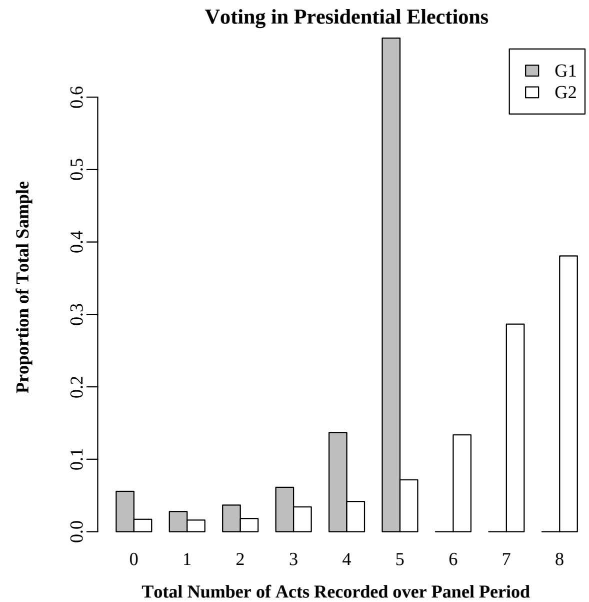


Figure 2.4: Cumulative Number of Acts of Voting

thought given the cross-sectional data.

However, these answers lead to another question: "Among people who only do one act, which act is it? Among people who do many acts, which acts do they choose to do?" Perhaps the repertoires of activities chosen by those who participate infrequently are different from the repertoires chosen by those who participate frequently. The previous results suggested that participation may not be distributed as unequally across the population as previously thought (at least, when seen from the perspective of a life-time rather than a single year). That finding did not mean that participation was a frequent occurrence, however. In fact, even looked at over time, very few individuals turn out to be habitual participators. If these few highly active people tend to do different kinds of activities than the many people who participate once or twice in their lives, then concerns about inequality might not be diminished but just refocused on what in particular the highly active minority are doing.

Figure 2.5 shows the answer to these questions for both G1 and G2. Each line shows a smoothed version of what proportion of the sample reported doing a given act among those who reported doing 0,1,. . . 20 acts over the study periods.[6] This tells us that,

[6]Without smoothing these graphs became unreadable in their complexity. The point of smoothing here was to look at the overall relationships without getting caught up in the details of each point. I choose 70% as the bandwidth after comparing plots of the raw data with a variety of other bandwidths (from 35% to

for G1, contacting officials was the most common type of activity among those who did few acts (of those who only did one act over the period, around 30% chose contacting officials as that act, about 25% chose community work, around 10% of this single-act group wore buttons, displayed signs, or donated money). Among G2, both contacting officials and community were common types of acts among those who did few acts — and they continued to be most common by far, even among the frequent participators in that cohort. The black lines on each panel refer to the non-electoral activities, and the gray lines show the electoral activities. If the lines on these plots stacked up and did not cross, this would suggest that certain acts are always easier (or at least more common) than others — regardless of whether a person has been an active participator (engaging in many acts over time), or has only done one thing ever. This is not the case here. Certain acts seem to be chosen by individuals who participate rarely (namely contacting officials and community work among G2 and G1) and other acts are found only among those individuals who participate more often (namely donating money, attending meetings and rallies, and doing "other" campaign work (among both G1 and G2)).

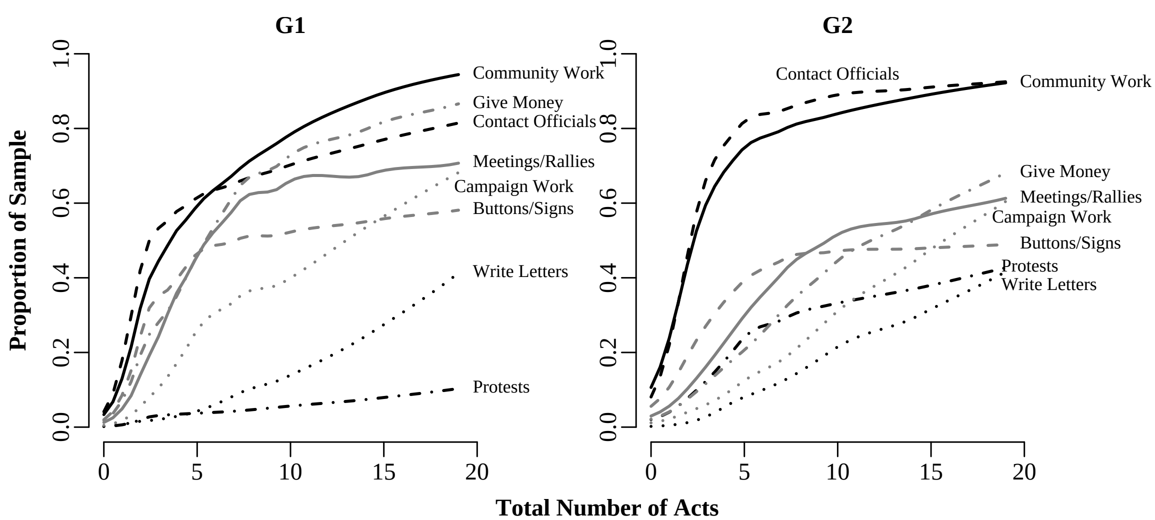


Figure 2.5: Frequency of "Lifetime" participation in certain actions

Note: The curves were generated with a local linear binomial scatterplot smoothing algorithm with a bandwidth of 70% of the nearest neighbors of each point included in the smoothed estimates.(see Loader, 1999, for more details about this smoothing method)

70%). No substantively important details are lost with the current choice of bandwidth.

Although the two generations are not very different in terms of the rank of activities among the infrequent and frequent participators, there are large differences in the proportions involved in different activities. Among members of G2, there appears to be a clear distinction between two non-electoral activities (contacting officials and community work) and all the rest of the activities. Working with others in one's community appears to be nearly a ubiquitous ingredient of the participation bundle of both generations especially as the frequency of activity increases: nearly 90% of people in both generations who engage in over 15 acts have community work as one of the acts that they do. However, among the older generation, donating money surpasses contacting officials in frequency among the most participatory members of that generation and over 60% of high participators in G1 attend campaign meetings and rallies. The fact that the older people are more apt to donate money shouldn't be that surprising given that they have had more time to earn money over their lives — and probably earn more in general — than their children. The frequent participators among the younger generation are more apt to do campaign activities than the infrequent participators (with donating money and campaign meetings and rallies as the most common types of electoral activities). Protests and demonstrations are nearly never a part of even the mostly highly active member of G1's repertoire of activity, but around 30-40% of the highly active G2 reported attending such activities (slightly more than writing letters to the editor) — and protesting is on par with giving money among those who did fewer than 5 acts in this cohort. This is yet another indication of the effect of coming of age in the late 1960s and early 1970s for this group of people.

In general, even the most participatory members of G2 do not appear to engage in electoral activities as much as the frequent participators of G1. This could merely be a function of age — and could suggest that campaign activities, at least over the period 1965 to 1982 were largely dominated by people older than 40 years old. It could also suggest that the repertoires of young people of this cohort are predominantly focused on direct contact with officials and local level activities outside of the established structures of elections, rather showing partisan allegiance that may come with intensive activity in campaigns. This too, could be seen as a period effect if one believed that the alienation expressed by a vocal minority during protests of the 1960s and the Watergate scandals of the early 1970s was felt by other members of that age cohort, effectively pushing them out of partisan politics. However, there is no evidence that this cohort was or became any less partisan than their parents, and thus, perhaps the fact that the activists among G2 eschewed electoral politics has more to do with structures of mobilization and recruitment by the parties focusing on older people. In the end, these interpretations are merely speculation and deserve further investigation in future studies of this dataset.

These descriptive analyses have shown that members of the Class of 1965 are, in some ways, more participatory than their parents: they are more likely to get involved at all, and are more likely to do more activities (from age 18 to 35 and from 18 to 50, for electoral and non-electoral activities respectively) than their parents (from age roughly 50 to roughly 70). However, their voting patterns appear more sporadic. And, the frequent participators among them appear to eschew electoral activities in favor of two activities not tied to campaign cycles — work with others in the community to solve local problems, and contacting elected officials — whereas the most participatory of their parents appear to have a more "balanced" bundle of activities, including community work and contacting (and donating money) as well as other activities tied to campaign cycles. What we'll see in the next section is that G2 also tends to be more participatory than G1 if we look at the proportion participating at any one year over the study period.

2.2 Average Trajectories

When describing changes over the lives of individuals scholars have often explained that such changes (in everything from attitudes, to income, to mental and physical health) can be understood as the product of three general kinds of causal factors that are often labeled age, period, and cohort effects. This impulse to disaggregate a temporal phenomenon into parts is not new and makes sense in many situations. This typology of age (or lifecycle), period, and cohort effects is useful for this project insofar as it provides an efficient vocabulary for talking about the types of causal agents at work influencing the trajectories of political participation engaged in by people over their lives.

Age effects are those influences on a given process seen to stem from the fact that individuals change in certain regularly patterned ways as they move through life. What "age" refers to is not that somehow passing through birthdays in and of itself matters for any particular substantively interesting outcome, but that, within any historical era, individuals tend to engage in certain activities and to have certain experiences at more or less similar moments within their lives — and some of these activities are seen as relatively "life-changing" such as initiation ceremonies and emancipation from parental-authority figures, child bearing and rearing, and physical decline, among may other kinds of milestones that determine the "lifecycle" of an individual. That is, age is best thought of as a general (rough, imperfect) measure of what kinds of experiences an individual has had and what kinds of experiences an individual has not had. It is these experiences that constitute an "age effect" or a "lifecycle" effect so termed. Of course, different individuals pass through their lives differently, and so although the relationship between age and lifecycle sometimes appears eerily close (when most of one's college friends have commitment ceremonies or weddings within a 2 year period), it is clear that broadly speaking age does not equal lifecycle, and even experiences associated with a lifecycle noted above may have different importance for different people.

Period effects refer to historical events that (in theory) influence all individuals equally. A war, a depression, a drought, or a fad, all place imprints upon the current and future behavior of individuals who have been exposed to such things. Thus, the political behavior of individuals could be explained in part both on the basis of their lifecycle experiences, and the period in history during which they lived.

Cohort effects are literally understood to be the interaction between the effects of historical period and lifecycle. The best example of cohort based explanation that comes easily to mind is that used by scholars to explain that the "greatest generation" were those who had "come of age" during the Great Depression and possibly also during World War II — that somehow, experiencing such historical events before having passed through many lifecycle experiences (or while particularly "impressionable" for experiential, psychological, or physiological reasons) have created a cohort of individuals moving through their lives with extraordinarily high civic dedication compared to other ages.[7]

This section shows the dependent variable as it will be used in future chapters. Since this dissertation is a first step toward understanding this phenomenon, it seemed sensible to choose a feature of this dynamic process which is akin to that studied in the cross-sections — the probability of participation at a given moment in time. Figures 2.6, 2.7, and 2.8 show the proportions of respondents in both generations reporting participation in each of the activities over the years.[8] Age, period, and cohort effects are clearly evident

[7] Age, period, and cohort have long been seen as being in a relationship such that knowing two out of the three tells you the other one — that is, that they are linearly dependent on one another. There has been an extensive debate in sociology about the estimability and utility of formal models aiming to disentangle one kind of effect from another. For examples of this debate and for technical details about age-period-cohort analysis see Palmore (1978); Fienberg and Mason (1979); Rodgers (1982, 1990).

[8] I omit confidence intervals from these figures to avoid clutter and because statistical comparison between the two generations is not my main intent here.

in these figures — with age effects mainly apparent from the upward trends in some of the graphs; period effects showing via large bumps and dips; and cohort effects possibly revealed in the overall levels of participation for the two generations. There are five main features of these plots that I want to note in particular:

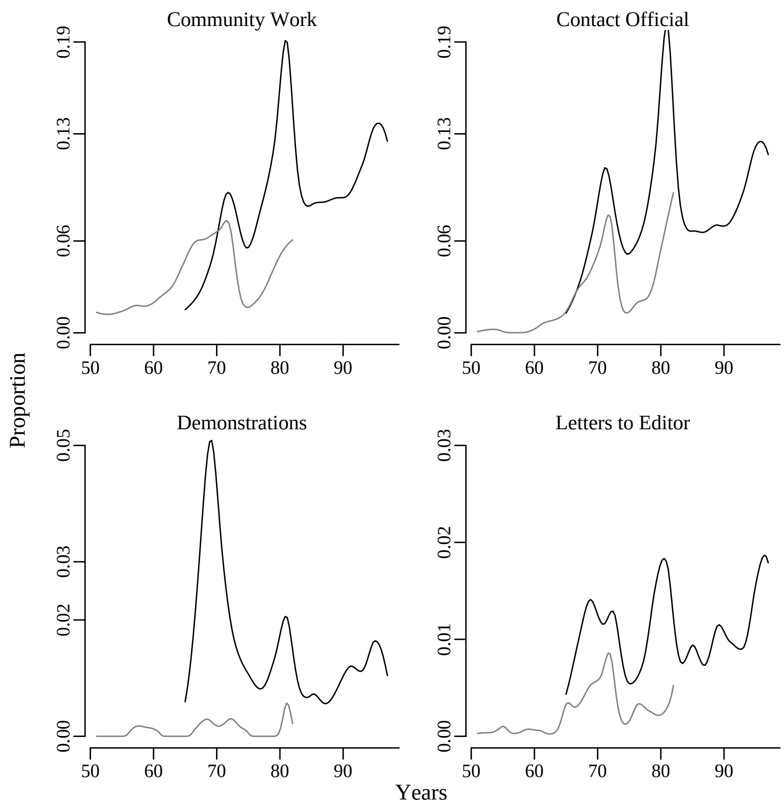


Figure 2.6: Overall Non-Electoral Participation by Generation

Note: Light gray lines represent the Parent generation (born around 1920) and the dark gray lines represent the Youth generation (born in 1947-48). Smooth lines are locally quadratic binomial fits using a neighborhood of 20% the nearest points.

Generational Differences 80% of the Parent generation (G1) were born between 1910 and 1926, with 50% born between 1915 and 1923. These people tended to enter adulthood during a period that spanned both the Great Depression and World War II. Previous literature would suggest that this cohort should be the most active in civic life (See, e.g. Putnam, 2000), however, it is clear that the Youth generation participates in political activities over this study's period at higher rates than their parents despite the fact that some of their parents are from the "Greatest Generation." Of course, these differences might be explained

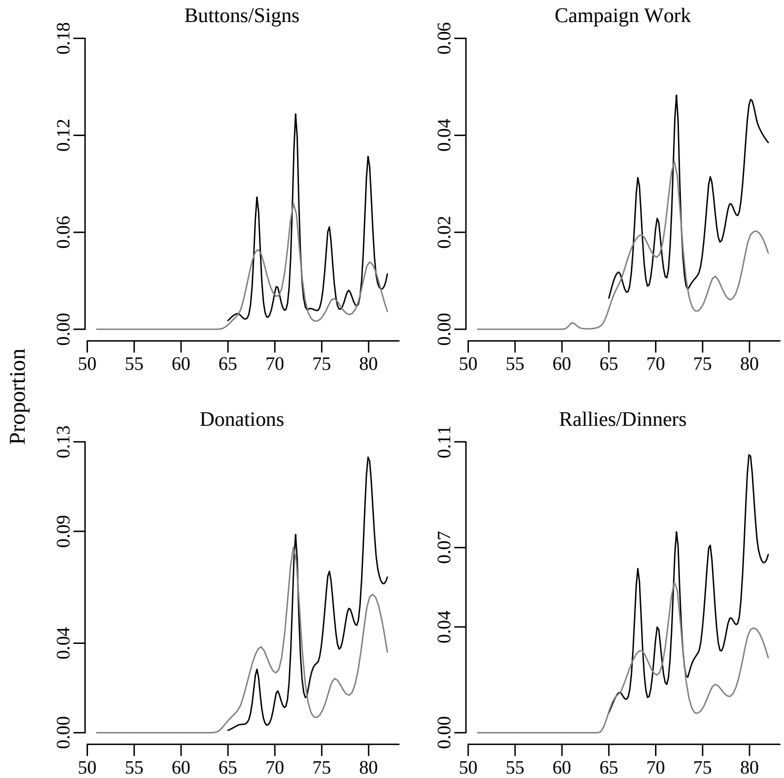


Figure 2.7: Overall Electoral Participation by Generation

Note: Light gray lines represent the Parent generation (born around 1920) and the dark gray lines represent the Youth generation (born in 1947-48). Smooth lines are locally quadratic binomial fits using a neighborhood of 20% the nearest points.

by the fact that G2 happen to be more educated than G1 — and that the particular sample of G2 here happens to be better educated than their birth cohort in the population (since they were a sample of high school seniors and some of their cohort didn't make it that far in high school). I take up this question about education and participation in Chapter 3.

Trends *Community Work*, *Contacting*, *Letters to the Editor*, *Campaign Rallies/Dinners*, *Other Campaign Work*, and *Campaign Donations* all display increasing proportions over time for G2. There appears to be much less trend for G1. This makes sense as a life-cycle effect since the life-stages that G2 pass through (age 18-50) tend to involve much more political and personal change, in general, than the life-stages passed through by their parents over this study (age 50-70).[9] Another life-cycle effect that is apparent here is that

[9]Although children leaving home, retirement, and aging also ought to be personally consequential for members of G1.

the participation levels of the G1 and G2 generation are often quite similar in the 1960s and 1970s (when G2 are young), but then the participation levels of G2 pull away from those of their parents both generations age.

Electoral Cycles Campaign oriented activities display clear links to a presidential year cycle — with much less activity during congressional election years and years without national elections. Election effects can be seen as period effects in that elections occur for all people at once — and we can see that both generations show similar rises and falls of activity with the electoral cycles. There are also weaker electoral cycles evident in the non-electoral activities, which suggests that even activities which are not explicitly about national campaigns may be affected by them. We also see that the effects of all elections are not equal — clearly certain elections must have emphasized the wearing of buttons (in 1972 and 1980) for example, while others did not as much (in 1976 for example).

Period Effects The most dramatic period effect that is not about elections that is visible in these displays is that for attendance at protests and demonstrations during among G2 in the late 60s/early 70s. We will see in later chapters that this effect is largely driven by those members of G2 who went to college during this period. That is, that members of the "Protest Generation" mainly tended to participate directly in protests if they happened to be in college during that period — thus making this apparent period effect into something more like a cohort effect (where a historical event mainly matters for the political behavior of one particular group of individuals due to their particular experiences (such as lifecycle) at that point.

Peaks over Interview Years The community work and contacting officials series show high peaks in the proportions of respondents reporting participation during those years, and the immediately preceding years. This evidence that people are more likely to report more activity the closer the year is to the interview date indicates that respondents are probably forgetting to report acts that they engaged in further away from the interview date.[10] I will attempt to account for this memory problem in two different ways in the multivariate analysis.[11]

[10] In studies of voting turnout, the main concern is about people reporting more turnout than they had really engaged in. In this study, it appears, that the main problem will be under-reporting — i.e. forgetting to report the occurrence of events that actually happened — and a possible correlation between time since the interview date and rate of forgetting.

[11] The first adjustment will be to allow the intercepts in the multivariate models to be different for those years that are interview years (and presidential election years, given the periodicity seen here). The second

Figure 2.8 shows the proportion of the members of G1 and G2 who reported voting in the presidential elections of this period. As before, the gray line represents G1 and the black line G2. This time, the data are not smoothed, and the simple proportions reporting participation are plotted here (with circles for G1 and triangles for G2) and joined by lines to create a visual trajectory. For voting, life-cycle seems to be the driving force. The percent voting among G1 is more or less constant across the 1964, 1968, 1972, 1976, and 1980 elections. But the percent of G2 voting increases from just over 50% in 1968 (at age roughly 21), to levels comparable to (and higher than) their parents as of 1976 (at age roughly 29) and remains that way, or trends slightly upward, until 1996 (roughly age 50).

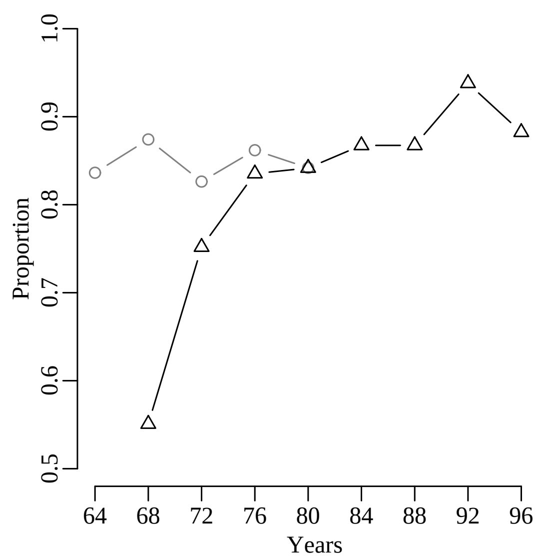


Figure 2.8: Voting in Presidential Elections by Generation

Note: Light gray lines and circles represent the Parent generation (born around 1920) and the dark gray lines represent the Youth generation (born in 1947-48).

The only apparent period effect in the data on presidential elections occurs in 1992 (after data collection for G1 had ceased). A particularly large proportion of G2 reported turning out to vote in the presidential election of 1992 (well above 90%). The increase from 1988 to 1992 is probably not an anomaly. In the National Election Studies, 1992 also

adjustment is to use a class of models (known as multilevel, hierarchical, mixed effects, or random coefficients models) that allows the analyst to take into account unmeasured attributes of individuals. This should help prevent individuals who are particularly good at remembering participation (and individuals who are particularly bad at remembering) from confounding the analysis.

stands out as the year with the highest turnout in about 30 years (75% of NES respondents reported voting in that election, and the next highest turnout between 1965 and 1996 was in 1968 with 76% reported voting.) This increase also receives confirmation from federal voting statistics, which report 55% turnout in 1992 (compared to 49% in 1996 and 50% in 1988, with 55% and 61% vote turnout estimated for 1972 and 1968 respectively).

Despite concerns that the Protest Generation could be seen as a particularly alienated and inactive cohort of citizens (at least compared to the Great/Civic Generation of their parents), the data shown here suggests that this distinction between generations may be drawn too strongly. Although activist members of G2 appear to avoid electoral participation (other than voting), the cohort as a whole are as engaged in politics as their parents, and becoming more so as they age. Of course, this comparison is rough, and does not account for all of the many other differences between these two generations (most notably education). However, this section has added more data to what has been shown before about the differences between these two generations. It has also shown that, at least when it comes to aggregate political participation, age, period, and cohort effects are all operating.

2.3 Transitions and History

To what extent does participation at one moment relate to previous activity? If by engaging in an activity in the past, people are more apt to get involved in the future, this would suggest some kind of learning is occurring. It is also possible, however, that once individuals get involved in politics they later evaluate the experience as negative and abstain from further participation. It also seems possible that certain kinds of participation are just so costly (in time or money) or the opportunities are so rare (such that action-spurring events and mobilization touch individuals infrequently throughout their lives), that any one act of participation will be followed in time by no acts.

To explore these possibilities, this section contains two sets of descriptive analyses. The first focuses on the transitions from one year to the next. If a person did an act last year, how likely is she to do that act again this year? The second examines the impact of the entire of history of activity on present acts. If a person has done, say, 1 or 3 or 6 acts across her adult years, how likely is she to do an act this year?

2.3.1 Transitions at Lag 1 year

Consider, for example, the cross-tabulation of community work one period in the past by community work in the "present" among members of the Youth generation:

Table 2.1: Transitions From One Period to the Next in Amount of Community Work Among G2

		Present			
		0	1	2	3
Past	3	24	5	8	185
	2	106	33	283	5
	1	890	1154	46	5
	0	27090	903	90	27

Out of all 30855 person-years, 27090 included 0 acts of community work followed by 0 acts of community work, 901 included 0 acts followed by 1 act, and 890 included 1 act followed by 0 acts. It is usually easier to look at this kind of table as a "transition matrix" which uses row percentages as an estimate of the probabilities of observing the different types of movements between states.

$$\mathbf{T} = \begin{pmatrix} .9637 & .0321 & .0032 & .0010 \\ .4248 & .5508 & .0220 & .0024 \\ .2482 & .0773 & .6628 & .0117 \\ .1081 & .0225 & .0360 & .8333 \end{pmatrix}$$

Of the people who did 0 acts of community work in the past year, 3.2% did one act in the current year. Of the people who did 1 act in the past, 42.5% of them did 0 acts in the present. Notice the large numbers on the main diagonal. These numbers imply that among the few people who manage to start participating at a certain rate (say doing 1,2, or 3 acts in a year), many are apt to continue — at least across adjacent periods.

Figure 2.9 summarizes the information in **T** and Table 2.1 graphically — using shaded squares to provide a quick sense of which kinds of transitions are most common. The shading of the squares is proportional to the number of person-years in that transition-category. The area above the diagonal in represents movements from less activity one period in the past to more activity in the present. The diagonal represents continuance of the same level of activity across adjacent periods. And the area below the diagonal represents transitions from more to less activity. The actual numbers of person-years in each square is printed on the plot. This figure shows that movements from less activity to more activity do happen — there were 903 moments of 1 action that followed a moment of no action. However, the transition matrix **T** shows that these 903 moments only represent 3% of the possible transitions from a moment of no action. Thus, this square is white. That is, the fact that a square has color (or not) only has to do with the proportion of the activity observed in the present conditioned on a past value. For example, of those years where people did 3 acts of community work, 185 were followed by years where people continued to do 3 acts. This is about 88% of the total number of years in which people did 3 acts, and so it is colored in nearly as dark as the square representing the 27,090 person-years where no activity followed no activity.

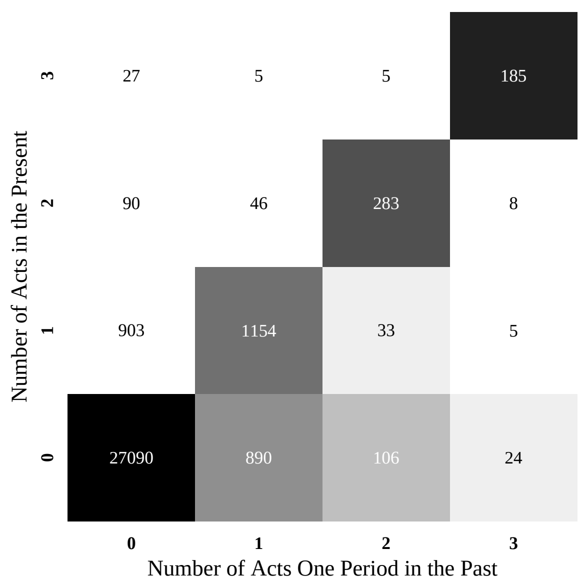


Figure 2.9: *Community Work* — Transitions from One Year to the Next: Youth

To avoid presenting 16 different matrices like **T**, Figures 2.10 and 2.11 present transition plots for each of the acts of participation, for each generation. You can see how the transition matrix **T** for community work among G2 maps onto the panel in the upper left corner of the Figure 2.10. The highest value (.96) is colored black and occurs at x=0 and y=0 (i.e. present participation is 0 following 0 past participation). As the legend shows, the darkness of color is proportional to the values in the squares, so the dark black squares contain values near 1 and the light gray (and white) squares contain values nearer to 0.

One general pattern that is evident for G2 is stability across adjacent periods — especially for 0 and 3 acts. Periods that contain zero acts are more apt to be followed by "empty" periods than by moments full of activity; persons engaging in 3 acts are more apt to follow 3 acts in the next year than otherwise. Doing 1 or 2 acts in the past year is also strongly related to continuing to do 1 or 2 acts in the present, but not quite as strongly as 0 and 3 acts — and larger proportion of 1 and 2 act years are followed by decreases than increases. In fact, for all types of activity except for *Community Work*, 1 act in the past is more likely to be followed by 0 acts in the present than by 1 or more acts.

The other general pattern concerns the paucity of shaded squares above the 45 degree line and the row of shaded squares at the bottom of each chart: people are much more likely to transition *to* 0 acts than *from* 0 acts. It seems as if people are likely to either continuing participation at the same level as they did in the previous period OR stop altogether (rather than "ramping up" and "tapering off" their level of activity over the years).

Figure 2.11 shows the same kind of information, this time for the Parent Generation (G1). Given 0, 1 or 2 acts in the past, these people are much more likely to do 0 acts in the present. That is, the predominant pattern is not of spells of continued activity like that shown for G2 in Figure 2.10, but of isolated, sporadic moments of action. It is only for *Community Work* that we see much evidence of years of activity following one another (and also for *Donating Money* for a very small number of years (7 out of 9!)). This figure adds to what we have already seen about the G1 cohort — they tend to participate less than the G2 cohort, and now we see that they tend to participate for only one year at a time (except for *Community Work* in any significance way). Thus, they are more infrequent participators as well as overall less participatory as a group.

Figures 2.10 and 2.11 tell a story where the Youth generation participation appears either habitual (years with 0,1,2,3 acts following each other) or sporadic (years with no activity following years with some activity and vice-versa); and where the Parent generation is not habitual, but instead sporadic. In both cases, most of the person-years in the dataset contain zeros followed by zeros — that is, non-voting political participation is rare. [12]

[12]It is possible that the appearance of habitual behavior among the Youth is an artifact of the survey

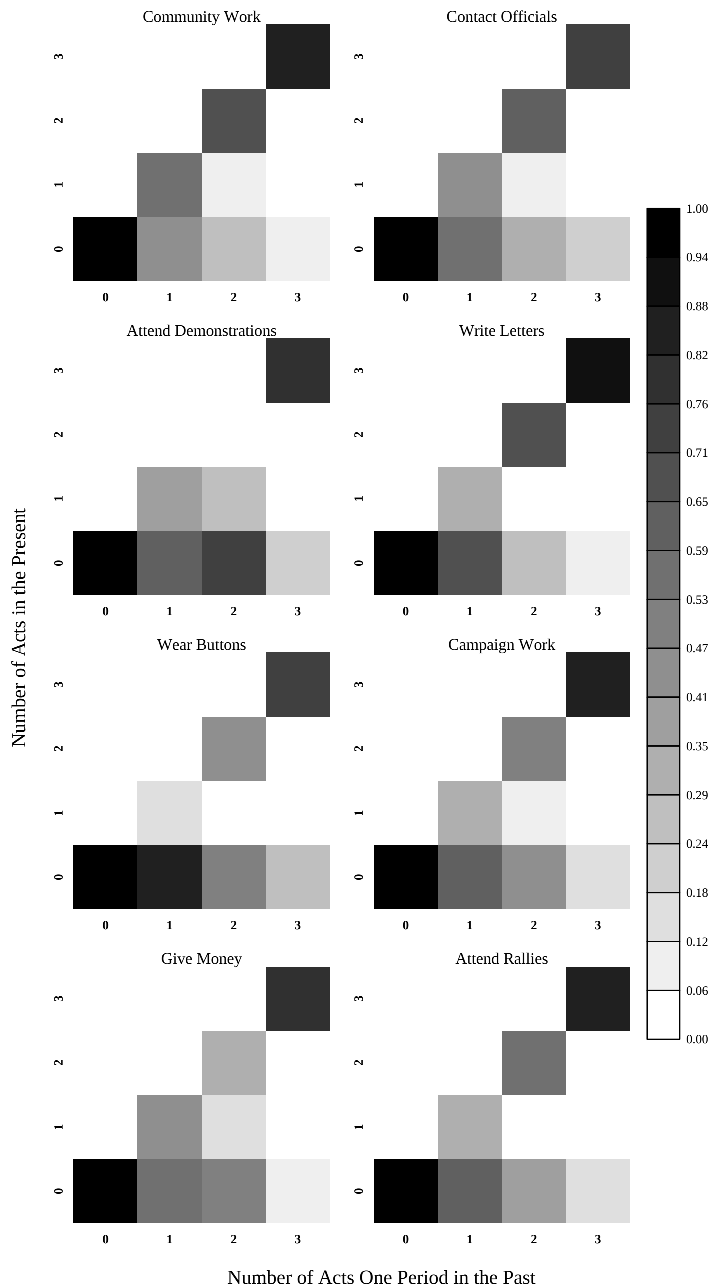


Figure 2.10: Participation Transitions from One Year to the Next: Youth

Note: The colors show the proportion of person-years where activity in the present (shown on the y-axes) followed activity one period in the past (shown on the x-axes). The key at right shows the proportions represented by the colors.

procedure — where respondents (of both G1 and G2) were allowed to name ranges of dates as they remembered their past activities, and where some respondents used ranges to mean "every year between X and Y dates" but other respondents probably used ranges to mean "some year in between X and Y dates, I don't remember exactly." Unfortunately, given the data, there is no way to distinguish between these two

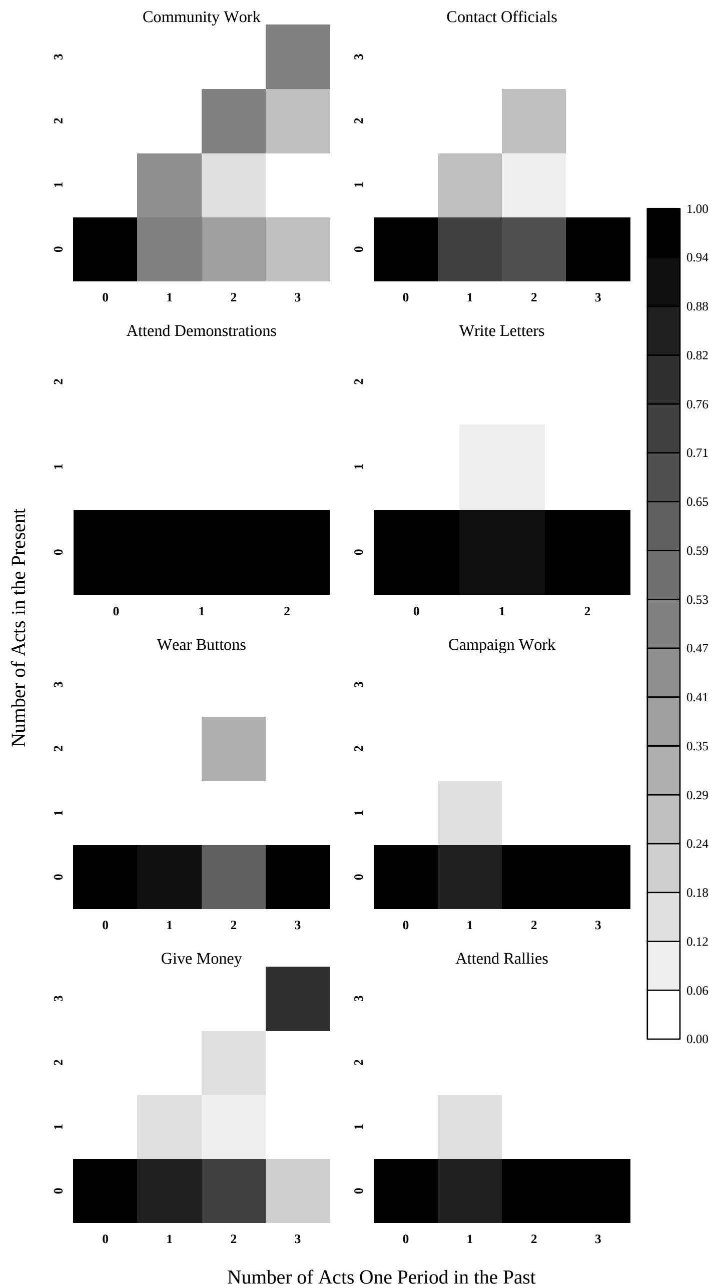


Figure 2.11: Participation Transitions from One Year to the Next: Parents

*Note:*The colors show the proportion of person-years where activity in the present (shown on the y-axes) followed activity one period in the past (shown on the x-axes). The key at right shows the proportions represented by the colors.

possibilities. However, the lack of such a pattern among the Parent generation (who responded to the same interviews, administered with the same techniques) suggests that the hypothesis that some subset of the youth (or all of the youth at some point in their lives) are involved in multi-year spells of activity while much fewer of the parents spend more than one year doing any particular type of political participation.

This pattern of sporadic participation from year to year is not merely an artifact of the particular cohorts in the Political Socialization study. The panel studies conducted by the National Election Studies show similar patterns over the short-term. These datasets have the strength that the respondents were only asked about their participation in the past 12 months, thus forgetting is probably a minor problem and dating the partipation to a particular years is easier. The weakness of these panel studies, however, is that they only cover 3 waves, usually 2 years apart and so ask about participation only every other year rather than yearly. Thus, they can only shed limited light on questions such as those raised above. That said, they will be useful for checking and corroborating the longer term longitudinal data from the political socialization study.

Figure 2.12 shows the kinds of information about participation available from two of these datasets. The information for the 1956-1960 NES Panel Study is in the left column of figures, and the information for the 1972-1976 NES Panel Study is in the right column. These figures show that most respondents in the 1956-1960 and 1972-1976 NES Panel Studies did not engage in electoral participation in either of the presidential election years in the studies (shown by the dark black boxes at 0,0 for each activity). However, among people who participate at all, a pattern of participating in only one of the two panels is more common than participating in both. The two exceptions to this "rule of rare activity" are *Wear Buttons* and *Give Money* in the 1956 to 1960 panel. Of the NES respondents who reported wearing buttons in the 1956 campaign, about 52% (n=94) did not wear them in the 1960 campaign, but about 48% (n=88) did it again. Of the NES respondents who reported donating money in the 1956 campaign, about 51% (n=63) did not give more money in the 1960 campaign, but about 49% (n=60) did it again.[13] Comparing within rows of this figure one sees differences between historical periods for button wearing and money giving, but not for campaign work or rally attendance. Overall, this figure provides a quick bit of corroboration against Figure 1.1, that non-voting participation in the USA seems both rare in any one cross-section of the public Verba, Schlozman and Brady (1995), but also is sporadic within people across time.

[13] I did not break down the NES samples by age or education for this example, nor did I present all of the different panel years, because this analysis is meant to convey the simple point that the sporadic participation patterns of the Political Socialization respondents are not merely idiosyncratic to that sample, but are broadly shared by all Americans, at least since the 1950s.

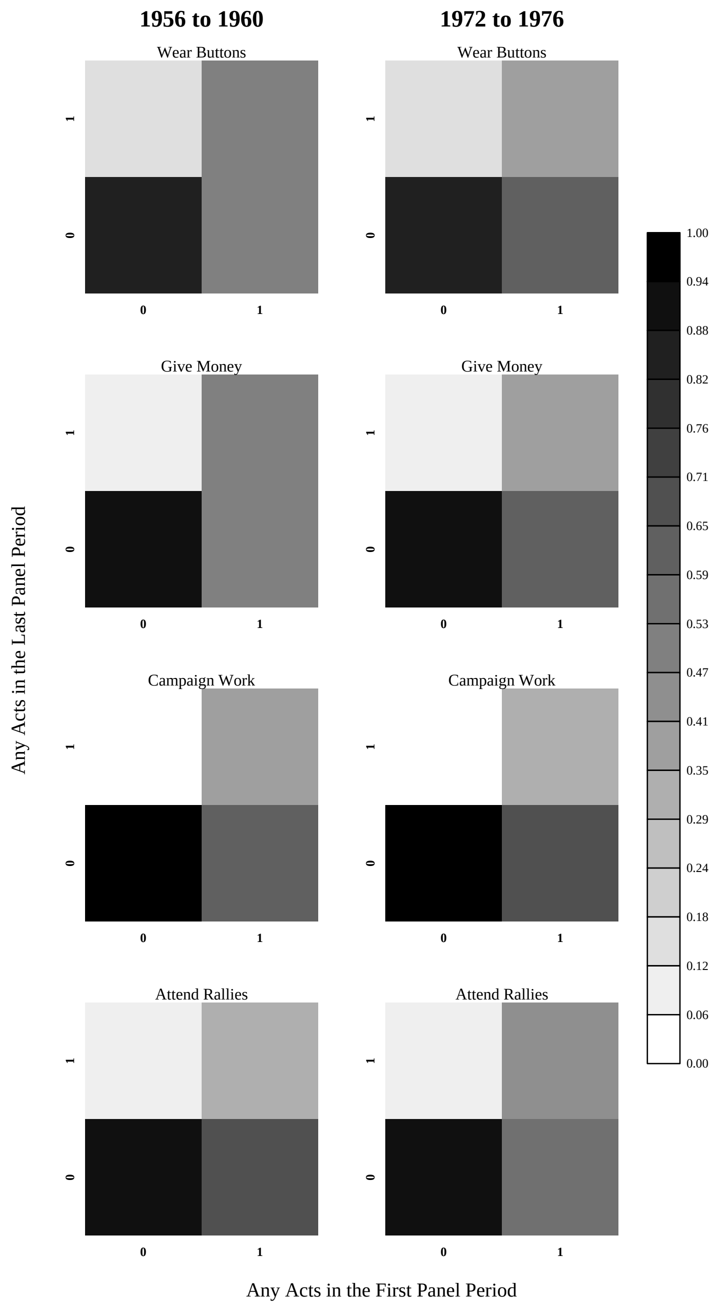


Figure 2.12: Participation Transitions across 4 years: National Election Study Panels

*Note:*The colors show the proportion of respondents where activity in the last panel period (1960 or 1976) (shown on the y-axes) followed activity in the first panel period (1956 or 1972) (shown on the x-axes). The key at right shows the proportions represented by the colors.

2.3.2 Transitions up to Lag 6

The figures presented in the last section provide a strong sense that the phenomenon of non-voting political participation is not well characterized by habitual or steady behavior. However, it may be characterized by learning. Merely looking at transitions from one year to the next ignores the possibility that present participation may be related to participation further in the past. It would be impossible to neatly visualize all of the possible relationships between the probability of participation in the present and the occurrence of participation at 1,2,. . . , 30 years in the past. For example, participation in the present could be related to participation 4 years in the past, but not 1, 2, 3 years or 5 years or more in the past — or it could be related to participation 3, 4, and 5 years in the past (with perhaps decreasing strength). That is, thinking only about the past 6 years, the present year can be thought of as being influenced by actions in any of those 6 years, or actions in any combination of those 6 years. A process in which the present is influenced by all of the previous 6 years (alone and in combination) is called a Markov process of order 6 — this is different from an Autoregressive process of order 6, where the influence of actions 6 years in the past only influence the present indirectly, via the interceding years. The figures presented in the previous section represented Markov processes of order 1 — where nothing further in the past than 1 year is allowed to influence the present. I only mention these names here in order to be clear about how I will test for how the past may affect the present beyond merely the previous year. One can test for the presence of a Markov process by using a simple generalized linear model where the probability of engaging in 0, 1, 2, or 3 acts is predicted using lagged versions of that variable (lagged up to 6 periods in the past), and all of the interactions between the different lagged variables.[14] For a Markov process of order 3, the structural part of the model looks like this:

$$\begin{aligned}\exp(y_{it_0}) = {} & \beta_0 + \beta_1 y_{it_{-1}} + \beta_2 y_{it_{-2}} + \beta_3 y_{it_{-3}} + \\ & + \beta_4 y_{it_{-1}} y_{it_{-2}} + \beta_5 y_{it_{-1}} y_{it_{-3}} + \beta_6 y_{it_{-2}} y_{it_{-3}} \\ & + \beta_7 y_{it_{-1}} y_{it_{-2}} y_{it_{-3}}.\end{aligned} \tag{2.1}$$

In (2.1), the number of acts of participation in the present y_{it_0} is modeled as a

[14] Also, since the dependent variable here is a count of the number of acts, I specified a Poisson probability distribution for it, and restricted the sum of the independent variables be greater than 0 using the natural log function (this is commonly known as a Poisson generalized linear model with log link or a Poisson regression with log link).

function of the number of acts of participation in the three previous periods, plus all of the interactions between the acts in the three previous periods. The fact that the left hand side is written $\exp(y_{it_0})$ is merely to indicate that the fitted values produced by calculating the sum on the right hand side must be greater than 0. This allows the amount of participation at each of the past lags to influence the amount of participation in the present, in all possible combinations. One can estimate such models for any number of lags, although as one adds more lags, this begins to produce prodigious numbers of interaction terms which makes the models unstable due to multicollinearity and loss of degrees of freedom. Having estimated a series of models each representing a Markov process of different order, one can use goodness of fit statistics to answer the question of,"How many years in the past does one have to go until the amount of past participation ceases to matter for the amount of present participation?" In particular, I used the Akaike Information Criteria (AIC) as the measure for comparing models rather than the log-likelihood. The AIC for a model is a trade-off between goodness of fit as measured by reduction in log-likelihood and the number of parameters needed to achieve this reduction (and it weights the number of parameters more so than the standard likelihood ratio test). I estimated 6 models per act, one for each order of Markov process up to 6. I stopped at six because of the complexity of the model and because I think autoregressive techniques are better for dealing with longer lags. Since this model would require 65 parameters to allow each of the preceding 6 years to affect one another as they predict the present period, given the sparsity of information in the dataset, I decided that AICs were the appropriately conservative measure of goodness of fit in order to assess the order of the best Markov process model for this data.[15]

I collected the AICs produced from each set of models and plotted them in Figures 2.13 and 2.14. These plots show how much the goodness of fit changes with the addition

[15]Given that my dataset contains multiple individuals "stacked" one above another, the simple method for estimating Markov process models illustrated by (2.1) could break down, since, for example, lag 1 of the dependent variable could have the 1965 value for one person affecting the 1997 value for the previous person. I "tricked" glm into distinguishing among the series on different people by using a weight function that applied 0 weight to the 1965 values for each person (thus preventing lags from crossing individuals). In a Markov process model, the present response depends on all available information about the previous states (for order 2, for example, it takes into account both the independent effects of the lagged 1 and 2 period states plus their interaction). In an autoregressive model, the present response only depends on previous states additively (so that, for any given order of past dependence, an autoregressive model involves fewer parameters). The AICS presented here are based on likelihood functions which are themselves based on the assumption that all of these observations are independent within person (an assumption that is surely false). All of the analyses in future chapters will take this kind of dependence into account using random coefficient models. For now, my goal is to provide a rough sense for one kind of temporal dependency in these data without delving into more complicated models.

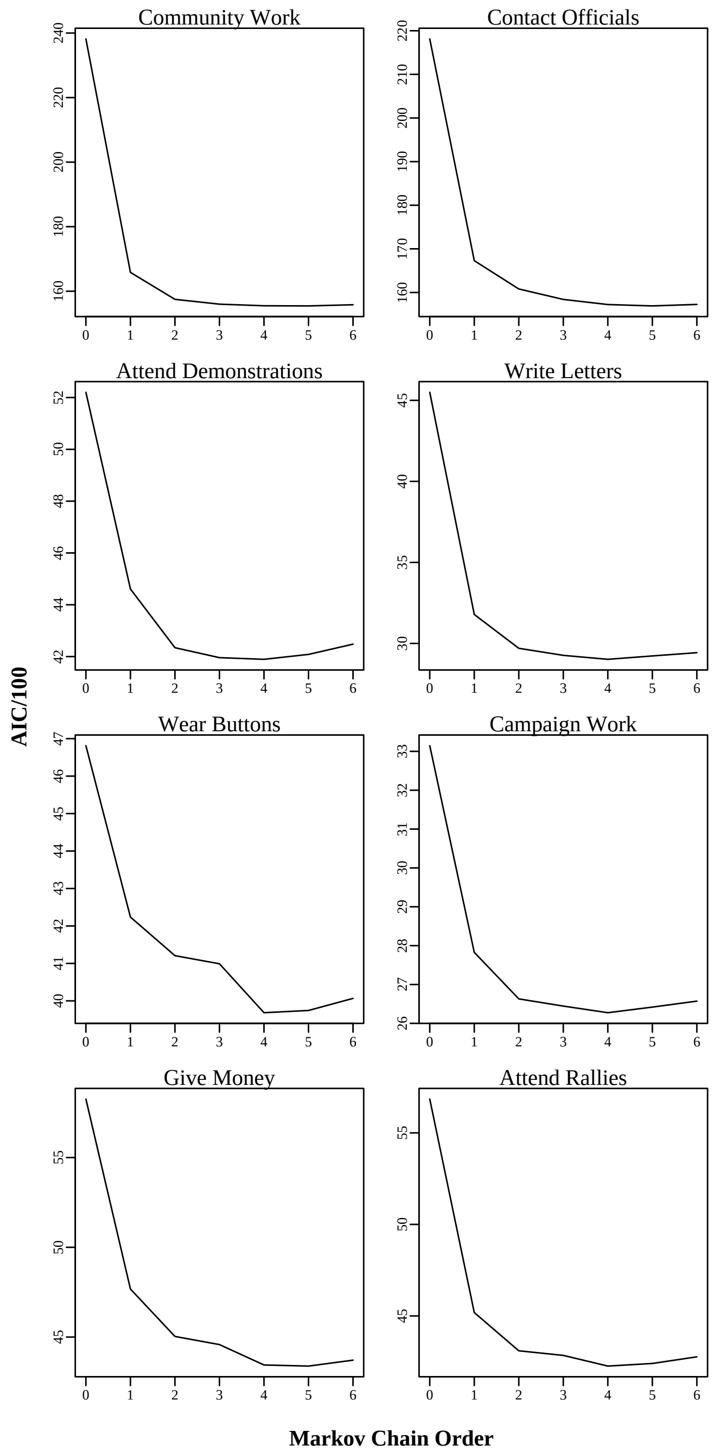


Figure 2.13: AICs for Markov Chains of Increasing Order: Youth

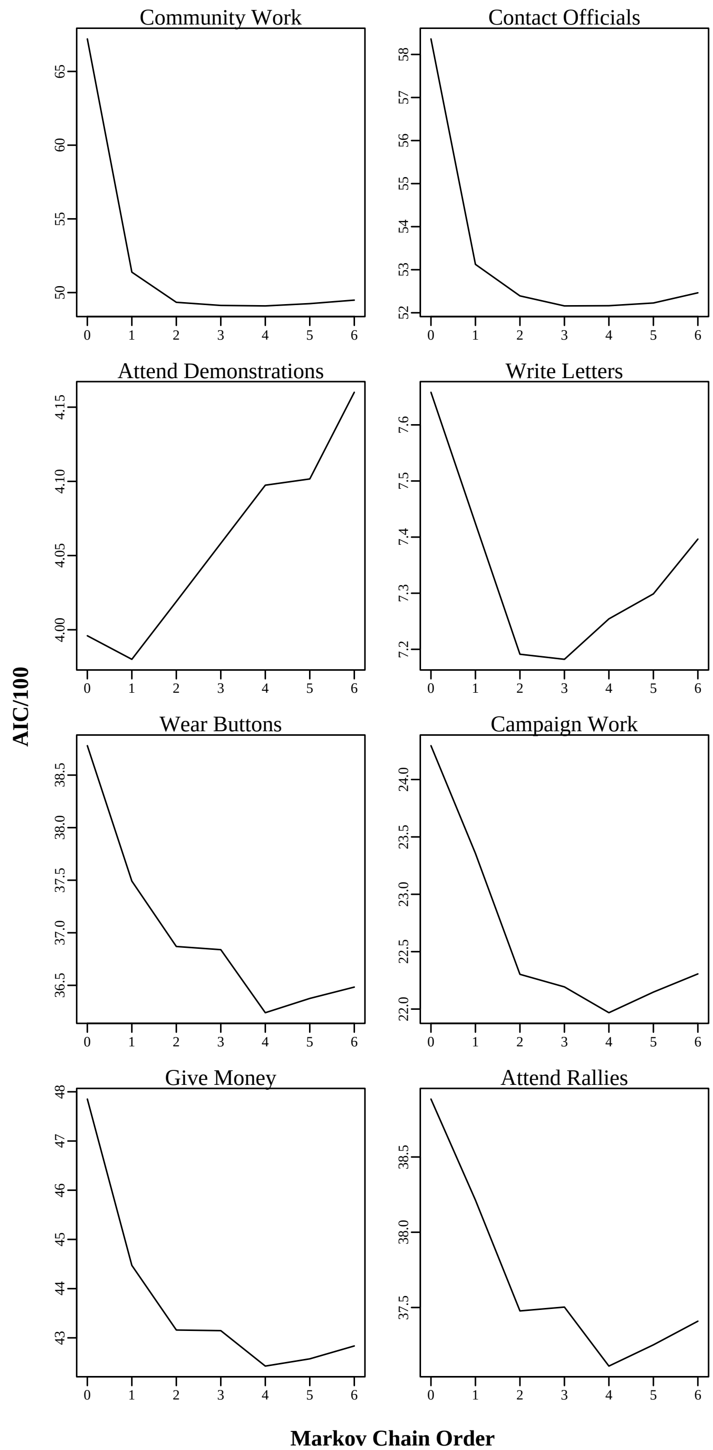


Figure 2.14: AICs for Markov Chains of Increasing Order: Parent

of more lags to the models.

These figures show that the biggest improvement in model fit is from the null model (with only a constant) to the model with 1 lag for all acts.[16] Roughly, after about 4 lags, the number of parameters that must be added to the model start to increase the AIC more than the lower in log-likelihood can decrease it, and so the AIC begins to increase slightly for most of the acts. Others of the acts, like letter writing and demonstration attendance among the Parent generation have so little temporal structure since they are so rare among this group, that the AICs begin to rise quite quickly, ceasing to justify the addition of lags to the model. So, the AICs together imply that these processes, in general, have at most a four period memory — but that the first period in the past is the most important component of this memory.

What does this tell us about the processes that produce participation? It suggests that the probability that a person will do a certain amount of a particular type of political participation at any one time is not strongly related (on average) to the amount of that activity he or she did in far in the past. Although, the amount of participation done in recent years (and recent elections) may influence the amount of participating in the present. This suggests that people may have relatively short memories when it comes to such behavior — and that perhaps civic education campaigns that aim for life-long effects may be misplacing their effort.

Taken together, the descriptive analyses presented here suggests that future analyses ought to take care to model the temporal dependence in these data paying careful attention to the dependence among adjacent years, and worrying less about direct long run relationships.[17] These analyses also remind us, again, that participation as a dynamic process raises new questions, many of which I will not answer in this dissertation: What predicts transitions between the number of acts, as depicted in Figures 2.10 and 2.11? In addition to electoral mobilization campaigns, what drives the longer term transitions (out to 4 years) explored in Figures 2.13 and 2.14? Are these patterns of dependence constant across the life-cycle? Or across genders and socio-economic status?

[16] Each plot is done on a different scale and the changes in AIC are not comparable between models. What matters here is the relative size of the changes for the different models of a given variable.

[17] This is just what I do in the other chapters in this dissertation, specifically using a Continuous Autoregressive structure at lag 1 (CAR1) for the within person errors which accounts for auto-correlation at lag 1 and then dies out exponentially over more distant lags.

2.3.3 Cumulative History

Another way the past may influence the present is via the cumulation of experience, networks, and other resources as people participate over their lives. For example, it is well known that older people tend to participate in politics more than younger people (See, e.g., Highton and Wolfinger, 2001). Of course, we know that "age" has no causal effect in and of itself. As a variable it is merely a placeholder for a host of other attributes of individuals and their environments that appear to change in regular ways over the life-cycle. So far we've seen that the amount of participation occurring in the immediate past has a strong relationship with the amount of participation occurring in the present. However, the behavior of human beings is not simply a Markov process — in which the present only depends on the single period in the past. It is possible that the entire history of political participation of a person matters for present action. Public (and private) policies such as civic education or Service Learning are premised on the idea that early experience with political participation has later consequences (either via the expectation of some cumulative positive effect of acts of participation on future participation; or via the expectation that early initiation of participation will have a larger effect than initiation later in life). I cannot fully explore these ideas about history and learning in this dissertation, however, I can provide one rough glance at the patterns as they exist in the Political Socialization data.

Figures 2.15 and 2.16 show one look at the relationship between the *total number of acts* that a person has done in the past, and her propensity to do that act again in the present. In this case, I present data showing what proportion of the sample report engaging in an act at every value of the total number of past acts. For example, the panel for *Community Work* among the Youth generation is shown in Figure 2.15. It shows that the proportion of people doing any acts of *Community Work* in a given year is nearly zero for those moments when nearly no acts have been done in the past (of those years for which individuals have done no past acts, 3% contain at least one act). However, this proportion rises quickly — of those years that occur after 1 act has occurred sometime in the past, 7% contain some acts of *Community Work*; after 10 acts have occurred in the past, the proportion of moments of participation is 29%. The total number of acts possible for the non-electoral types is 99 (maximum of 3 acts per year recorded over 33 years in the study period); and the total number possible for electoral types is 3 acts per year over 18 years = 54 acts. Only for *Community Work* do any respondents even approach these limits. And, once 30 acts have been done in the past, those few individuals are virtually guaranteed to continue participating (this suggests that those few highly participatory folks do so steadily at relatively lower rates over the years rather than packing all of their participation into a few short years at higher levels).[18]

The information presented here is different from that shown previously in Figure 2.3. In those figures the data displayed were the total number of acts that the respondents had reported across the entire study period (and the displays were truncated at 20 acts for ease of interpretation). These charts here show the influence of the past on the present, such that someone who eventually did 10 acts is part of the proportions for the acts between 0 and 10 (and the time difference between that person's 1st and 10th act is ignored).

The overall story here is clear. The proportion of individuals acting in a given year increases as the number of past acts increases. This increasing relationship is quite sharp for the first few acts (up to about 10), and then it appears to slow and flatten into a habitual-looking pattern.[19]

The story for the Parent generation shown in Figure 2.16 is similar, but with fewer individuals participating. Very few of G1 engaged in demonstrations or wrote letters, and so there appears to be little interpretable relationship here (only 6 members of that generation engaged in more than 1 demonstration or protest over the study period, only 13 wrote more than 1 letter to the editor). However, for those acts where some of G1 participated, having done more acts in the past increases the proportions of individuals who do any acts in the present.

Of course, these analyses are not meant to sustain a rigorous causal story. Instead, they are meant to provide just one final description of how participation at one point in time in a person's life may be related to participation at past points. And we've seen some hints that participation is related both to the amount done in immediately adjacent periods, and to the total quantity done in the past (whenever in the past it occurs).

[18]Only 15 members of G2 ever had more than 30 acts over the study period.

[19]I am not sure what to make of downward turns in the lines shown for many of the acts here or in Figure 2.16. It is clearly not merely an "end-effect" of the smoothing process, since it tracks closely to the actual proportions displayed. This is worth some further investigation.

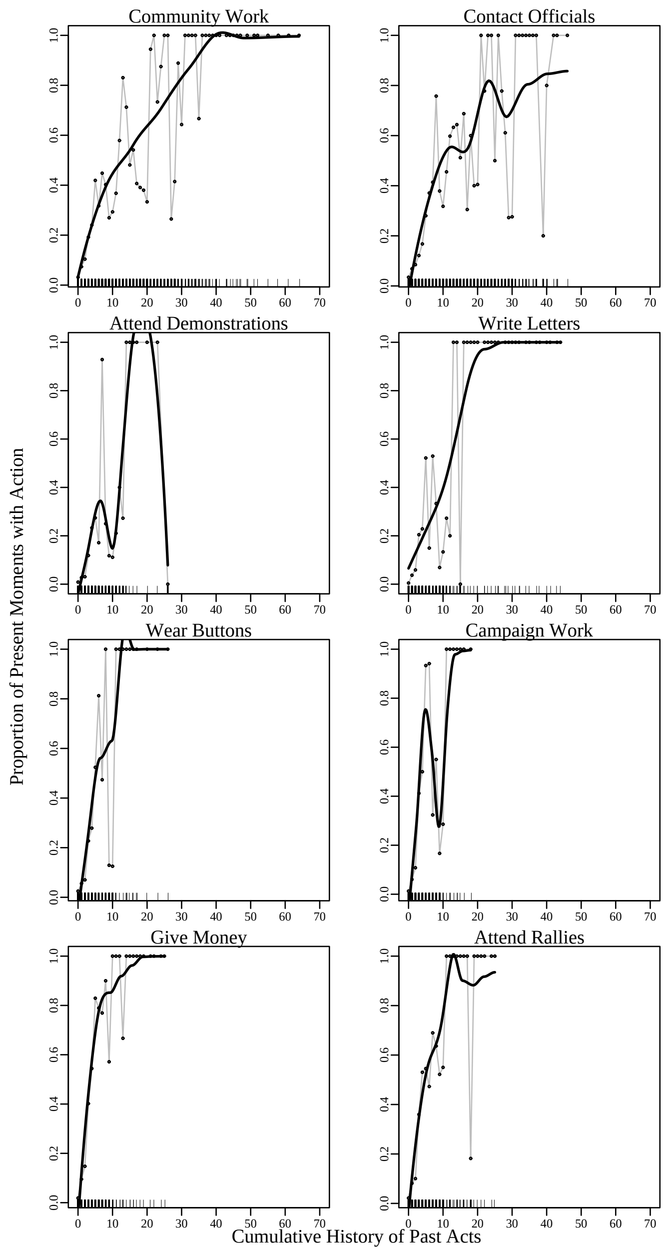


Figure 2.15: Influence of Past History of Present Participation: Youth

Note: The gray lines represent the proportion of the sample reporting present participation given the number of acts of that type that have occurred in the past. The thick black lines smooth over the jaggedness of the proportions (using a local regression of degree varying from 0 to 2 with a bandwidth of 50% of the nearest neighbors). Confidence bands for the smooth lines are omitted since the smoothing is not directly of the data but of means. The rug plot on the bottom of each graph shows the density of points on the x-axes.

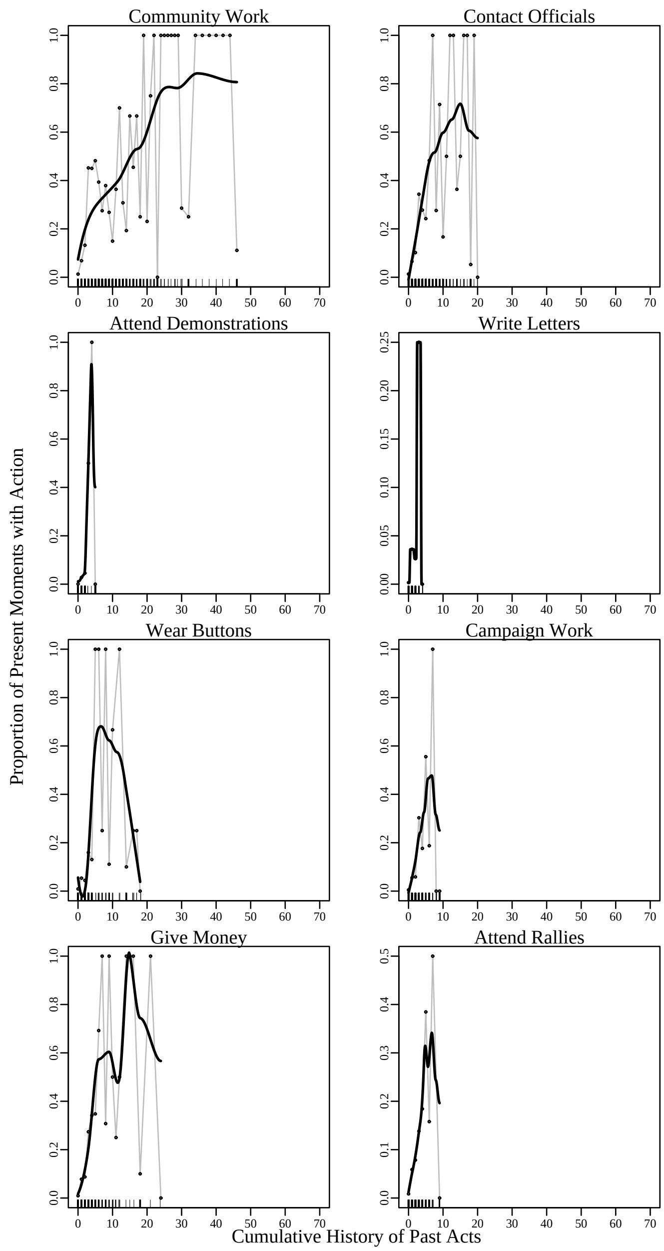


Figure 2.16: Influence of Past History of Participation on Present Participation: Parents

Note: The gray lines represent the proportion of the sample reporting present participation given the number of acts of that type that have occurred in the past. The thick black lines smooth over the jaggedness of the proportions (using a local regression of degree varying from 0 to 2 with a bandwidth of 50% of the nearest neighbors). Confidence bands for the smooth lines are omitted since the smoothing is not directly of the data but of means. The rug plot on the bottom of each graph shows the density of points on the x-axes.

2.4 Durations

Although little has been written on the persistence of spells of participation, it is clearly an important aspect of the phenomenon.[20] In the Political Socialization data, spells of political participation can have persistence to the extent that the same act is done repeatedly over one or more years. That is, a spell has duration of 1 year if it occurs only for a single year, and a duration of 2 years if it occurs over 2 years, etc.

Figure 2.17 shows the durations of spells of participation in the Political Socialization data — excluding the spells of duration 0 since the great majority of years do not include any action (i.e. a duration of 0). In general, the Parent generation not only engaged in fewer activities (as shown elsewhere in this chapter), but their engagements tended to last less time per spell as well. We also see here that most spells of participation lasted a single year. For example, 32% of the spells of button wearing and sign displaying done by G2 occurred for a single year, and less than 1% were spread across multiple years (meaning that the individual did this activity every year for more than one year). The only activities for which there appears to be appreciable persistence are community work and contacting officials among G2, where about 4% of the spells last 2 years for each of those types of activity. It is no surprise that campaign-oriented activities such as donating money, "other" campaign work, attending campaign meetings and rallies, and wearing buttons or displaying signs should occur largely as single-year spells since elections very rarely occur in adjacent years (and since this kind of activity does appear linked to elections, as it should, given what we saw in Figure 2.7. That most non-electoral activities should appear as single events, distributed sporadically across a person's life, however, is a different picture of participation than one in which either individuals are apathetic non-participators or they are habitual generators of social capital.

The other kind of durations that are relevant for understanding participation as a dynamic process are (1) the time it takes a person to engage in his first act of participation (especially interesting if time can start when they enter adulthood); and (2) the time people wait in between acts (for those people who do more than 1 act of a given kind). This data is the inverse of the information on persistence of acts presented above — which essentially showed the time between moments of inactivity. Now, I'll be interested in the time until

[20]See Berinsky, Burns and Traugott (2001) for the only other study that I know of to examine the durability, or fragility, or participation over multiple opportunities for action (in their case, vote turnout over 5 elections).

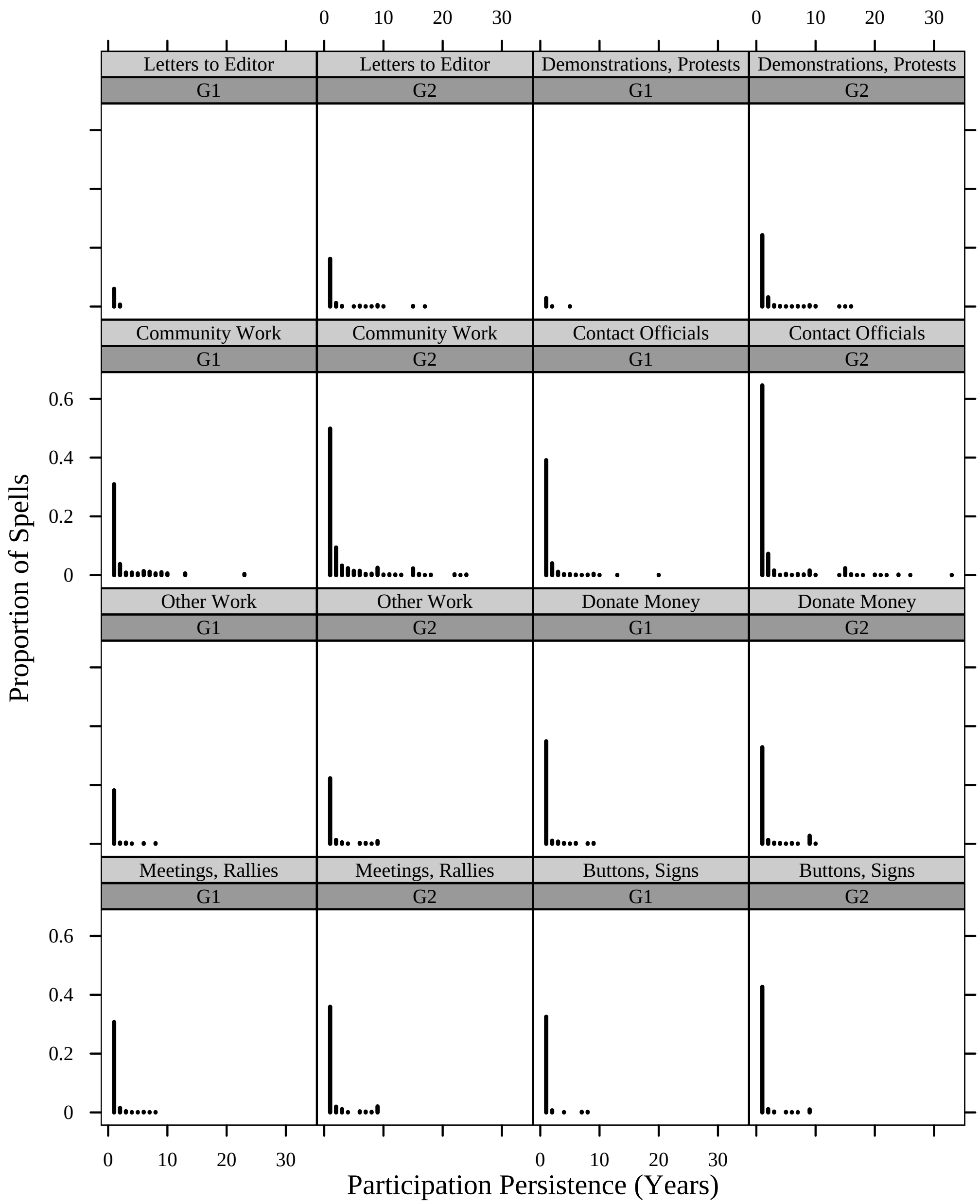


Figure 2.17: Persistence of Spells of Participation

Note: Spells of duration 0 were excluded from this figure.

and between moments of activity.

Figure 2.18 shows the distribution of the time individuals spend before engaging in their first act (among those who do any acts of a given type). For G1 time=0 is 1951 (which is the earliest that any of them remembered doing any acts when they were asked the questions in 1965). For G2 time=0 is 1965. The data for G2 are perhaps more interesting here, since nearly all of them were age 18 at time 0 — thus we can interpret their data as beginning roughly at the beginning of adulthood for them. The members of G1 were born at different times clustered around 1920, and so, the length of time since they did their first act does not map so cleanly onto the lifecycle.

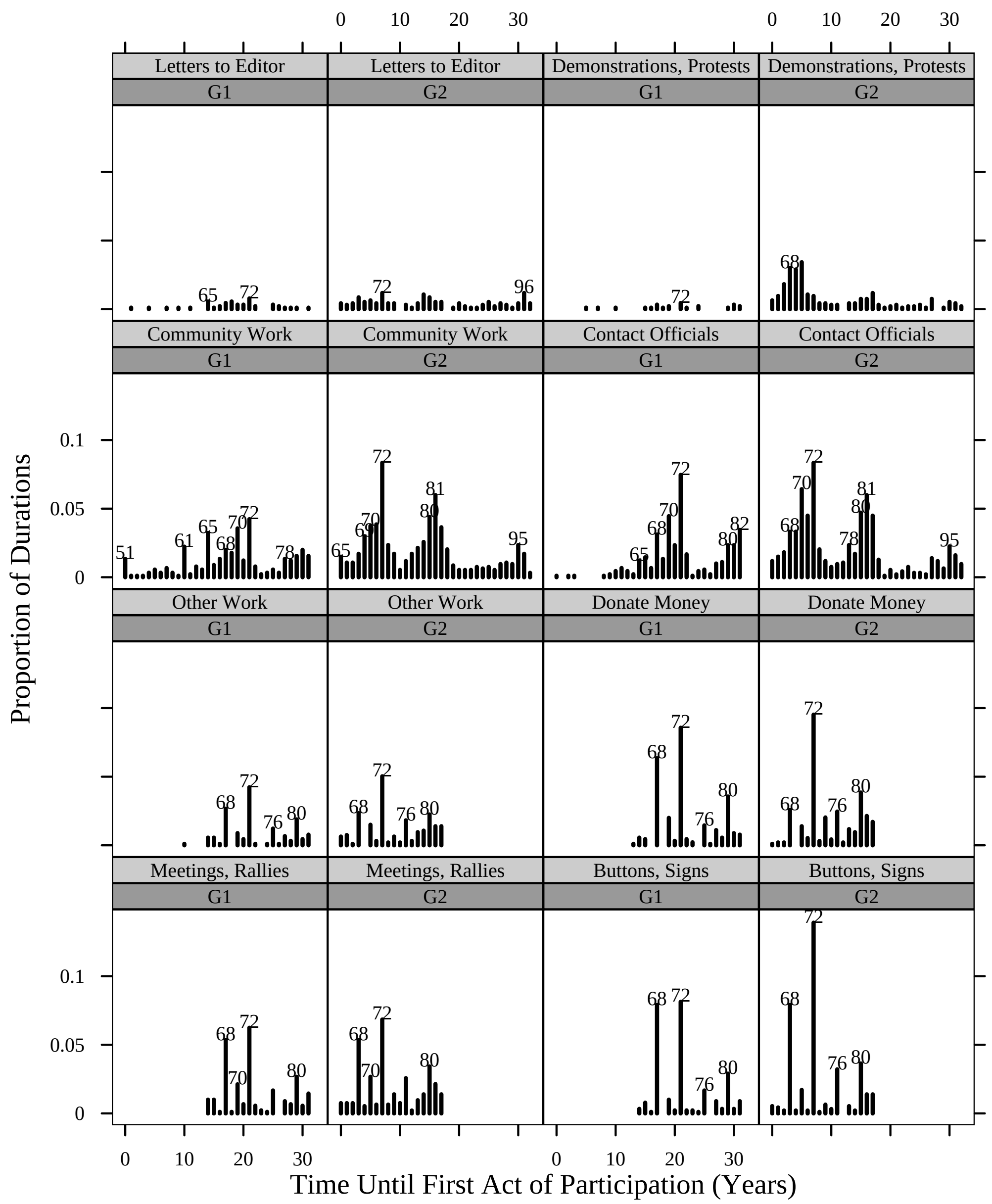


Figure 2.18: Time until First Act of Participation

Note: Durations that denote no participation are excluded from this graph. Peaks in this graph are labeled with the calendar year that they occurred. For example, nearly 10% of the members of G2 reported wearing a button or displaying a sign for the first time in 1972.

What is clear, here, however, is that people wait varying amounts of time before doing their first act. The electoral activities display more spikes — and these spikes are 4 years apart — signifying that people tend to engage in their first campaign-oriented act during presidential election years.[21] Age does not seem to be as strong a factor as campaigns when it comes to first entry into campaign oriented participation — both G1 and G2 had large, and similar, numbers wear buttons or display signs for the first time in the 1968 and 1972 presidential election campaigns, while fewer first wore buttons in 1976 and 1980. A similar pattern of first involvement tied to presidential elections occurs for campaign work and donations of money — with 1968 and 1972 being elections in which larger proportions did their first act than 1976 and 1980 for both generations (although, for the first generation this probably is not the first act in their adult life, but merely the first such act recorded in this study).[22]

The non-electoral activities (shown in the top two rows of the figure) do not show as clear a relationship with electoral cycles. They do display spikes around the interview dates of 1973 and 1982 (both sets of interviews occurred half way through those years, so it would be sensible for people to remember more activity in 1972 than 1973, and 1981 than 1982). What these rows also show is that the individuals in this dataset did different political acts for the first time all through their lives (within the study period). For example, there is no evidence here that most people did Community work for the first time while very young or waited until they were older to do it. Rather, with some variations, people did their first (and often only) act of community work anywhere from the age of 18 to the age of 50 (for G2) and from the age 35 to about age 70 (for G1). The one act that appeared to be more the province of the young than the old (and of one particular generation, G2) was demonstrations and protests, where the late 60s brought the most individuals to this type of activity for the first time among the Youth generation, and where very few of the Parent generation did this at all.

Figure 2.19 shows the durations that are missing from the previous plot — the amount of time people tend to wait between acts after their first act. Since most people who do more than 1 act only do 2 acts, most of the pattern in this figure is the amount of

[21] These spikes are labeled with the years in which they occurred, and we can see that they occur largely in presidential election years. Also, it is worth remembering that the survey did not ask detailed timing questions about campaign-oriented activities in the 1997 wave — thus the blank spaces after 1982 for these activities in this graph and others throughout this dissertation.

[22] I dropped G1's pre-1965 information here to enhance comparability with G2.

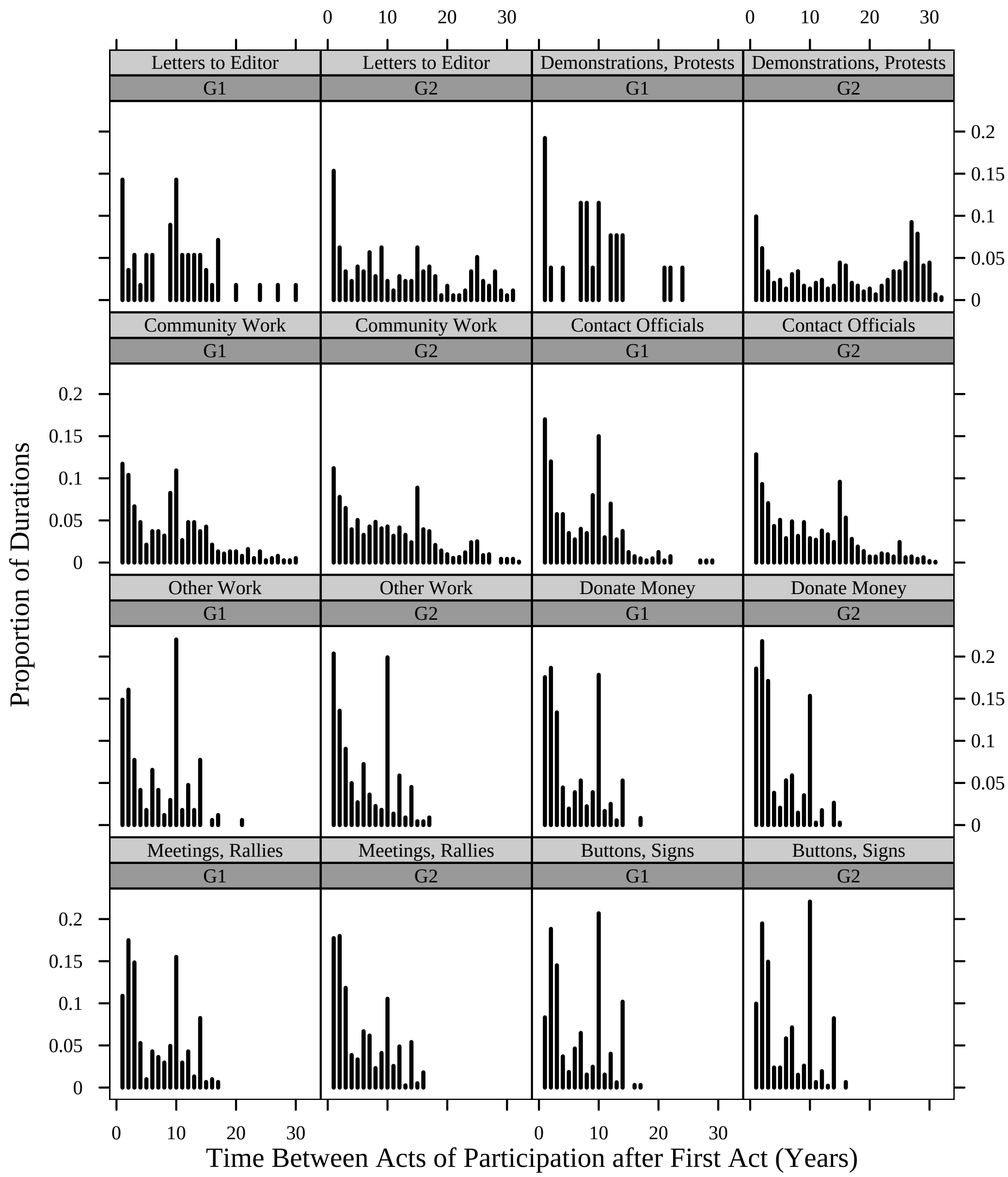


Figure 2.19: Time Between Acts (Starting with First Act)

time between the first and second acts of a given type (although there is no distinction here between the 1st-to-2nd act span and the durations between other acts). For this figure, there is no clear calendar date associated with the spikes — since different people did their first, or second, acts at different dates, the time until the next act can span different dates as well. However, most of these plots show a declining pattern from an inter-act duration of

1 year to about 10 years.[23] Then a spike or increase again around 10 years, and a more or less steady decline to the end of the series (except for *Demonstrations and Protests* which shows a concentration of around 25 year durations between acts among the G2). This figure suggests individuals tend not to wait very long between spells of participation — although there is diversity and appreciable proportions (of the small number of participators) who wait anywhere from 1 to 30 (or 18) years between spells.

2.5 Summary

In this dissertation, I will be analyzing data akin to that shown in Figures 2.6 and 2.7. They show the proportions participating in a given act, in a given year. Within a person, this kind of curve can be called a "trajectory" of political participation over their lives. Although it will be fascinating and important to understand the cumulative amounts, transitions, history, persistence, and durations of political participation, those particular dependent variables belong to future projects.

I should note here that I have not even completed the task of describing this phenomenon. For example, I have not done any analysis bearing on sequence: Do individuals appear to engage in certain types of activities in sequence — perhaps starting with certain campaign-oriented activities and then moving onto other types? Nor have I looked at the transitions, history, persistence, or duration *across* acts: How long after doing one type of act are individuals likely to do another of a different type (or the same type)? Nor have I asked questions about short-term versus long-term patterns (which would allow me to use additional short-term panel datasets that currently exist). And data on how individuals may react to their own evaluations and judgments of particular episodes of activity do not even exist.

Also, in this chapter, and in the others in this dissertation, I have given voting only cursory attention. Ample evidence points to the fact that voting behavior is not like other kinds of participatory behavior — and the very structure of the voting data that was only gathered about presidential elections in this dataset would require a different analytic setup that the one I use here.[24] So, since presidential voting is constrained to occur only every four years, since it has been so well studied, and since studying the other kinds of participation is a big enough job as it is, I have left it out of this dissertation nearly entirely. It will be

[23] There are no inter-act durations of 0 years since that would only occur within a spell.

[24] See both Wolfinger and Rosenstone (1980) and Verba, Schlozman and Brady (1995) for more detailed discussion about the differences between voting and other forms of political participation.

interesting and important to attempt to compare the findings here with information about voting behavior over time. However, again, this dissertation leaves that task for the future.

So, what have we learned from the different perspectives on the dynamics of political participation shown here? First, we've seen that the process among the older generation is different in many ways from the process within the lives of the younger generation. The older generation is less likely to engage in non-voting participation (although its members seem to have a more persistent pattern of voting (or of non-voting)). The younger generation appears more involved (more cumulative acts over the study period, higher proportions acting at any given moment, longer spells of activity) in nearly all of the activities measured here than the older generation. The only exception to this pattern is for voting, where the younger generation are less likely to vote in every available presidential election than the older generation (and they are less likely to be persistent non-voters than the older generation). Of course, I have not attempted to rigorously compare the two generations, controlling for the obvious differences in education and other resources which give the members of G2 a participatory advantage over the members of G1 (and presumably over the other less, well-educated members of their own cohort).

Second, I've shown that participation in the present depends, to some extent, on participation in the past. And that the past matters both as defined as the amount of participation occurring immediately preceding the present, and also as the total history of participation up to each point in time. The more individuals participate in the past, the more they seem to do so in the present — but from moment to moment, more individuals appear to stop participating than to start or increase. Of course, if individuals continuously moved out of the state of participation, then participation would decline over time — and I have shown that this is not the case here, especially with the upward trends in participation among the Youth generation as they age. But, as I explained in the Transitions section, it is true that, among people who have engaged in a single act in the past, most of those do not do so again immediately. However, these people may still participate again in the future, it is just that they wait more than a single year (or often more than 4-5 years) to do so again.

Third, I've suggested that participation tends to occur in one year spells, rather than multi-year bursts. But, the amount of time individuals tend to wait between one year spells tends to be short rather than long (although there is great diversity in the amount of time individuals wait between activities, and the amount of time individuals wait before doing any activity at all after entering adulthood). These findings suggests that "sporadic" or "irregular" are apt descriptions of these mini-time-series within people's lives. In other words, while political participation does not seem to be a steady hobby for many people, it is neither completely absent, or particularly restricted to any part of the life-cycle in particular.

It is worth considering this discovery a bit more. In Figure 1.1, I suggested that political participation might be sporadic in this way. And I worried that current accounts of participation based on relatively constant aspects of people (like education) might not map sensibly onto a phenomenon that appeared to turn on and off so rapidly. In this chapter, I have presented multiple kinds of evidence, all of which point to the fact that political participation is indeed sparse, and does turn on and off quickly within the lives of ordinary Americans. This leaves scholars of political participation with a puzzle: how can our cross-sectional findings be so robust and powerful for a phenomenon that clearly doesn't appear to behave in a constant or slowly changing manner? I pursue the answer to this question throughout this dissertation.

I've also shown in this chapter that more Americans (at least of these two cohorts) are engaging in participation than one would think given the cross-sectional findings. This difference is sensible: people have more opportunities to participate over many years than over a single year. However, the findings that more than 20% or so of both G1 and G2 did some of these kinds of non-voting participation over the periods of their lives studied here, suggests that an understanding of participation as a very rare activity, engaged in by a small subset of dedicated activists is incorrect. Rather, it looks like many individuals do engage in politics at one time or another in their lives, but that they don't face a threat or opportunity big enough to warrant the costs of action more than once or twice in their lives. The question that is then raised is what influences the occurrence of these participatory moments. That is the question I will pursue in the next three chapters.

Chapter 3 Education and The Life-Time Dynamics of Political Participation

This chapter engages with the predominant finding in the political participation literature to date: that individuals who have more formal education are more likely to get involved in politics than those who have less. The robustness of this relationship across countries and decades has spurred scholars question the mechanism by which it arises. That is, discovering that education is a strong predictor of political participation has raised the question about what it is, exactly, that education does to facilitate political participation. The most recent answers to this question provide two mechanisms: education influences political participation via provision of "civic skills" and "civic status". These theoretical mechanisms have received empirical support from findings that individuals who have money, time, or organizational abilities — that is, people who have the "civic skills" to participate in politics — are those more likely to do so. And, individuals who know a lot of other people, particularly politically active people — that is, people who have "civic status" — are also more likely to participate in politics than those who have less extensive and politically involved social networks.[1]

This chapter attempts to take the research on this relationship in a new, and different, direction. Specifically, the previous findings leave open the question of dynamics. Whether the effect of education has to do with civic skills or civic status, we still do not know *when* acquisition of education might be most important in driving future political participation. When might the influence of education take effect? Immediately? Or years after obtaining these civic skills or status?

In this chapter, I suggest that education has important effects on ways that political participation evolves over the lives of individuals, and that the timing of "doses" of

[1]The most recent and extensive articulation and defense of the "civic skills" and "civic status" points of view are provided by Verba, Schlozman and Brady (1995) and Nie, Junn and Stehlik-Barry (1996). Verba, Schlozman and Brady (1995) coined the term "civic skills." I use the term "civic status" to describe the findings and argument of Nie, Junn and Stehlik-Barry (1996).

education matters for how these participation trajectories play out. I sidestep the issue of whether, or to what extent, "education" really is a proxy for "civic skills" or "civic status", mainly due to a paucity of data available to distinguish between the two mechanisms. The Political Socialization data do not contain direct measures of civic skills or civic status, and so this chapter cannot directly address this particular debate.[2] In addition, the skills versus status dichotomy does not completely describe the ways that education can matter for participation, mainly because it fails to account for the indirect effects of education on other resources and on residential stability. However, understanding something about the dynamic effects of education is useful, whatever it is that education comes to represent. And, in fact, perhaps the findings here may suggest that one or the other of the current hypotheses is more plausible.

3.1 How does education facilitate political participation?

Formal education provides a variety of civic advantages to those who have attained it — and these advantages are not entirely describable in terms of civic skills and/or civic status.

Skills Formal education tends to teach civic skills — knowledge about the political process, abilities to communicate with politicians and to mobilize others via organizing. Verba, Schlozman and Brady (1995) also show that highly educated people are more likely to use civic skills in their daily jobs. Thus, education can directly teach skills, or can route people into jobs in which they learn, use and practice such skills daily.

Status The fact that education may allocate people into kinds of jobs in which they learn and practice civic skills alerts us to the fact that formal education is not only formative (i.e. increasing abilities) but also allocative (i.e. placing people into particular social, political, cultural, or economic locations). Nie, Junn and Stehlik-Barry (1996) emphasize that formal education places individuals into positions in social networks from which (1) they feel efficacious about influencing politics and (2) they are exposed to more mobilizing efforts by both neighbors and friends, as well as by activists and politicians.

Engagement and Interest Formal education may provide something beyond skills and status. Past literature suggests that education also seems to be associated with more

[2] A preliminary item response analysis of civic skills and civic status using the best available measurements (from the Citizen Participation Survey) did not provide clear enough scales to use with the Political Socialization data in this dissertation. For the results of this analysis, see Appendix C.

interest in politics and feelings of efficacy — both of which in turn facilitates participation. It is possible that interest in politics precedes education (at least college education), but it is also possible that education increases expressions of political interest as it allocates individuals to different social circles where articulations of political interest (and information) are prized.

Resources Formal education is the best predictor of life-time earnings. Since having money is necessary before a person can make a contribution to a candidate, party or campaign, formal education has an indirect effect on participation as it influences income (Verba, Schlozman and Brady, 1995). Education also provides individuals with resources of time — in so far as money can buy it.[3]

"Roots" One way in which formal education *inhibits* participation is that it increases the mobility of those that are highly educated. Thus, Verba, Schlozman and Brady (1995) note that the highly educated (who ought to be the highly participatory) have "shallower roots" in any given community, and therefore their observed participation is probably lower than it would be if they were to stay in one place.

The point here, is that education is a powerful predictor of political activity for a number of reasons and that it tends to operate via civic skills, civic status, interest, resources and roots. Rosenstone and Hansen (1993) summarize the distinction between the operation of skills and status succinctly:

> ...When political participation requires that knowledge and cognitive skills be brought to bear, people with more education are more likely to participate than people with less education. Participation, that is, requires resources that are appropriate to the task.
>
> On the other hand, education also indicates both the likelihood that people will be contacted by political leaders and the likelihood that they will respond. Educated people travel in social circles that make them targets of both direct and indirect mobilization. Politicians and interest groups try to activate people they know personally and professionally. (Rosenstone and Hansen, 1993, page 76)

In short, education both changes the individual (by providing abilities and motivations that can help the individual get over the costly thresholds of various types of

[3] Actually, one fascinating finding of both Verba, Schlozman and Brady (1995) and Burns, Schlozman and Verba (2001) is that one's amount of leisure time does not have a direct or powerful effect on one's political participation.

participation) and it changes the individual's place in some social, political, or economic context. Although the previous literature has covered a lot of ground to arrive at this rather nuanced understanding of a statement like "Education facilitates participation", we still don't understand much about *when* changing skills or status matters to political participation understood as a dynamic process.

3.1.1 New Questions and Dynamic Implications

How might we imagine that education drives political participation, seen as as dynamic process? Figure 1.1 in Chapter 1 and the findings in Chapter 2 illustrate that political participation is sporadic and episodic and thus doesn't map onto seemingly stable or slowly changing attributes of individuals like education in a simple way. At first glance these findings would seem to imply that explanations of political participation based on initial conditions can't explain the "shifting involvements" that we see. Individuals just don't gain (or lose) abilities and social network status quickly enough to drive the short moments of involvement apparent in Figure 1.1, and in the dataset more generally. This is the puzzle with which I ended the last chapter.

As I discussed in Chapter 1, any dynamic process, can be driven either by initial conditions or by other things that happen as it evolves. In the case of education and political participation, this distinction is between the initial attainment of education and the continuing development of education over the life-span.

What is an initial condition? In the case of political participation, what is an initial condition? In general, in the American political system, only adults have a say in politics — and thus, a simple answer to that question is that initial conditions, from the perspective of political participation, are those conditions that characterize the skills and status of a person when they attain adulthood. Adulthood, itself, is a slippery concept. In this chapter I consider age 18 as providing the break between childhood and adulthood. Individuals begin to be treated differently by the law and government at age 18.[4] My aim here is most emphatically *not* to arrive at a conceptually coherent understanding of "adult" versus "child," but rather to suggest that a reasonable understanding of "initial conditions", from the point of view of political participation, are those civic skills and civic status acquired by an individual in the moments just before he or she left high school/achieved 18 years of

[4]Of course, for most of the Youth cohort in the Political Socialization data, the voting age was still 21 years old until 1972. However, they could still be drafted at age 18, and many were involved in non-voting political participation even before they could vote.

age.[5]

So, although there are some reasons to divide the life of individuals into before and after some transition to "adulthood", one can also make a pragmatic argument that, given the periods of the lives of individuals that we've actually observed, two kinds of data on education seem relevant to understanding the process of political participation: the conditions which held up until age 18 and those which held afterwards. Should we, however, expect different causal influences from education attained at different times in a person's life?

3.1.2 Hypotheses about Education and Political Participation

Figure 3.1 depicts five different, stylized, trajectories for political participation to take across the life-course of an individual. In each graph, the x-axis starts at adulthood. Each trajectory is defined by how it relates to the attainment of education at different points in time in different ways. These graphs represent different hypotheses about how education may relate to the evolution of participation over the adult lives of individuals. The figure shows five out of the many possible ways that either an initial condition OR a life-time attainment of education may change the probability of participation.[6]

[5]One could argue that the distinction between "initial conditions" and the "varying conditions" across life is an artificial one — born perhaps out of metaphors with laboratory experiments in which stochastic processes are begun and stopped at the whim of researchers. I think, however, that such an argument is important — changes within the lives of individuals are probably much more continuous than often analyzed. However, a distinction between how a person was raised — what skills and status they attained as a function of largely their parents' and pre-adult schooling — and how a person decides to live after some (perhaps gradual) emancipation from parental rule — the skills and status obtained via life under one's own control (more or less) — is a reasonable one. One way around the problem of dividing life into "initial" versus "not initial" would be to claim merely that the *timing* of the acquisition of civic skills and civic status might matter (as described more in detail below), and that, given the current data sources, we observe individuals only between ages 18 and 50 (for the "youth generation"), and thus we take their responses at age 18 as indicative of civic skills and status acquired at that time or before, and we then measure the changes in civic skills and civic status more or less continuously from that point on. Most avenues for political participation are not open to individuals younger than age 18, but increasing avenues open to individuals after that age — as a function of laws and societal norms about when individuals ought to be trusted to act autonomously or at least when we, as a society, feel comfortable holding people responsible for their own behavior (and thus punishable more severely than if we considered the behavior the responsibility of some other agent, such as parents). Thus, it makes more sense to speak of "early" versus "later" in life.

[6]The purpose of this figure is to suggest how one might begin to think about causality in this kind of dynamic process — not to make an argument that, say, participation trajectories are characterized as portrayed (e.g. with each additional "dose" of education immediately increasing the probability of participation without any subsequent decline).

These graphs also only show one possible representation of a participation trajectory — as a counting process in which the influence of education is cumulative. Thus, they ignore issues of sequence or persistence, or non-monotonic relationships between the probability of participation and education. The slope of the step-functions can be thought of as the rate of change in participation. And, of course, the moments in which the steps occur represents the timing of participation.

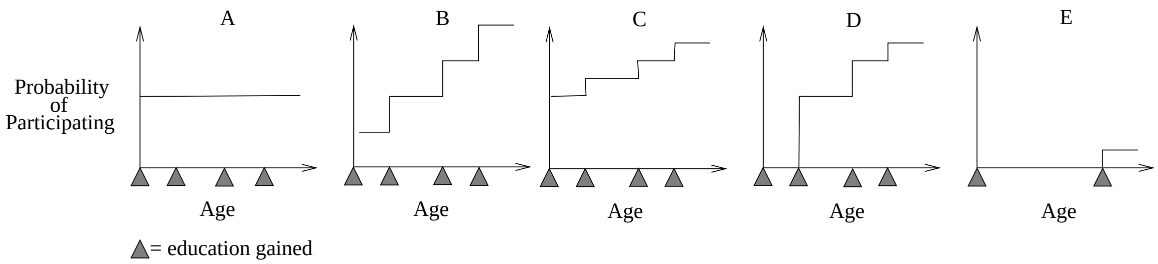


Figure 3.1: Hypotheses about Education and Probability of Participation

The first figure, labeled “A”, represents the situation in which the education attained at the beginning of the person’s political life determines the entire trajectory — this would be the story of “life-long persistence” from the perspective of political socialization theory. Each little triangle represents the occurrence of an event that one might think would stimulate participation, such as a moment in which the person gained additional education. In the story in which only pre-adult history matters, where you begin completely determines the whole story. Although, there is significant evidence of the importance of pre-adult political socialization for the life-time political development of individuals (See, e.g. Jennings, Stoker and Bowers, 2001) we also know that political orientations change in adulthood (See, e.g Jennings and Niemi, 1981).

The second figure, labeled “B”, represents a situation in which each acquisition of education matters the same amount. The amount that education attained early in life increases the probability of participation is the same as the amount of increase provided by acquisitions later in life. Thus, any person who gains education at any time in their life has their participation ratcheted up one unit. This figure represents a simple version of the “life-long openness” model that ignores the history of the person in favor of an emphasis on the readiness of a person to react to new situations — and the impact of such reactions is the same over time. If “B” is the correct story, then participation is malleable, and the timing of the attainment of education only affects final outcomes insofar that starting earlier leads to more opportunities for the acquisition of education. Similarly, the timing of the acquisition of education would not matter for aggregate outcomes such as the total amount of civic involvement by the American public, and we wouldn’t have to wait 20 years before we could see the effects of public policies aimed at changing the civic involvement of individuals.[7]

[7]This figure does not allow for any indirect effects of participation upon itself. That is, if early education

The third panel, labeled “C”, presents a situation in which initial conditions matter most, but in which the acquisition of later education does have some (small) impact — an impact that is the same no matter when the intervention occurs over the life-span. Thus, it represents a situation in which story “A” predominates over story “B”, where education gained over the life-span has some effect, but in which timing during adulthood doesn’t matter in terms of the amount of increase provided by an additional “dose” of education.

The fourth panel, labeled “D” depicts a trajectory in which education gained in pre-adulthood does not matter at all, but in which the first moment of education acquisition during adulthood has a major effect on the probability of participation — and the cumulation of subsequent education gained has a diminishing effect. This implies that the educational status achieved in early adulthood is most important in determining the life-time political participation trajectory of an individual — compared to other, later, educational attainments. It also implies that the first “dose” of education is the most important, and subsequent doses matter less — whether or not these doses occur at specified moments in the life-span. This story would be congruent with an understanding that each act of participation provides education itself.

The fifth panel, labeled “E”, shows a situation in which initial conditions do not matter, but in which only the timing of the acquisition of education in the life of the person matters. Receiving education late in life produces only a small increase in the probability of participation by virtue of the lateness of the intervention, not because of a kind of diminishing marginal returns process such as that occurring in “D”.

3.1.3 The Importance of Timing

In each of the hypotheses portrayed above, the probability of participation at a given moment for a given individual depends on when in their lives they received various infusions of education. Previous authors have also noticed that the timing of education is important. For example, Burns, Schlozman and Verba (2001) state that:

> Those who are born to parents of limited educational attainment begin life with a participatory disadvantage that is difficult to overcome; and any educational deficit upon entering adulthood will be magnified thereafter by the institutions of everyday life.(Burns, Schlozman and Verba, 2001, page 366)

leads to participation, then one would imagine that early participation (stemming from early education) would lead to more participation in the future — by increasing the size of the “steps” in figure B. Although this is plausible, I do not devote a figure to it just to keep things simple.

This kind of language about a "participatory disadvantage" suggests that these authors believe that experiences of participation build upon themselves — that participating in the past may lead to more participation in the future. The evidence presented in Chapter 2 supports this idea. Also, this quote alerts us to the distinction between the influence of a person's parents versus the influence of the experiences of the person as an adult. I think that it is plausible that past participation spurs future participation because one could think of past activities as providing skills relevant toward engaging in that activity again in the future. Engaging in political activity tends to change a person's civic status as well — one becomes known by activists, politicians, and others, as a person who may be called upon to act in the future. However, it is also important to note that past participation, in addition to providing education, also provides an opportunity for an individual to *evaluate* the experience of participation in that particular act. Such evaluation could be negative or positive (See Hirschman, 1982, for one conceptual analysis of disillusionment and cynicism born of public action). It is also possible that this evaluative component interacts with the education that a person brings to the given activity — some base of education may be necessary in order for an activity to be seen as worthwhile or enjoyable.

What does this mean for the importance of timing? There are several of different ways this could work: First, if the evaluatory component is small, and the civic skills/status gained in doing each act is large, then each act makes subsequent acts easier to do and/or increases the probability of exposure to other opportunities for participation via requests from activists or politicians. That is, if some kind of "increasing returns" situation exists (and a subsequent "power law" or exponential process applies) then early participators ought to do many more acts over their lives, more quickly, than late participators. If early acquisition of education (such as a Bachelor's Degree) increases the probability of early participation, and, say, each year earlier of participation increases the life-time amount of participation exponentially, then clearly the timing of education is crucial.[8]

Second, it may be that education gained at different points in the life-span is qualitatively different. The first interpretation above assumes that early attainment of education is important only because political participation itself can be seen to be a source of "civic education" (in the form of civic skills and civic status), and thus a positive feedback

[8]I talk about an exponentially growing process here in order to make vivid the potential gains from early education. Of course, the process as realized within the lives of individuals will vary across people, and, in fact, may not be characterized as a monotonically growing process at all.

process results. One would imagine that any person who is 2 acts into such a process ought to look similar to any other person who is 2 acts into the process, regardless of when in their lives they first began to act. But, perhaps, say, 90% of the "amount" of civic skills acquired early can be "spent" on participation, but only 40% of that same "amount" can be "spent" if acquired later.

In my discussion in this section I have avoided making a strong argument for a single dynamic mechanism by which education drives the process of political participation. That is, I have not specified what it is that each moment of education is providing to the person, be it skills, status, engagement, resources or "roots." This hesitancy occurs because of the lack of previous information about this phenomenon — scholars just don't know very much about how political participation changes and fluctuates over time within the lives of individuals as a univariate series, let alone as the output of a stochastic causal process. However, this chapter will show that the timing of educational attainment is important, at least for the participation trajectories across 33 years of life for the Class of 1965.

3.2 Measuring Political Participation and Education

I have described the concept of "political participation" and its relations in Chapter 1. Appendix A shows the questions that were asked of the respondents in the Political Socialization panel study. I present analyses using all 8 of the items having to do with using time or money on political activity — excluding the item having to do with interpersonal discussions and also excluding voting for reasons I described in earlier chapters.

Figure 3.1 and the preceding discussion suggests a situation in which education may change (i.e. increase by some increment) during the life of an individual. If education is the "heavy hitter" it is supposed to be, then we should see some kind of impact on participation from changes in education. Unfortunately, while the Political Socialization data do allow annual measurements of political participation, they only include measurements of educational status at the 4 panel data collection periods. For example, we only know that an individual received a BA sometime between 1973 and 1982 if they reported having a BA in 1982 but not having one in 1973. One approach to deal with this problem would be to assume something about the probability distribution of the moments within the intervals in which education was acquired across people. However, I don't have good priors about this.

Instead, I propose to do a longitudinal analysis in which people are distinguished by their type of educational trajectory. That is, despite all of the discussion above about participatory responses to increments of education within a person's life, I will use here an across-person analysis — but the differences between people will be on the basis of trajectories of political participation rather than more simple mean differences. For example, I distinguish between those who never received a college degree, those who received a BA by age 26 and those who did so after age 26. That is, inference about the importance of the timing of education in the lives of individuals will be based on comparisons across the trajectories of types of individuals.

Figure 3.2 shows what we ought to expect given the previous literature on the relationship between education and political participation. The four lines represent four trajectories, or ways for political participation to develop over the life of an individual given the moment in her life that she passes a key educational milestone (such as receiving a college degree). These hypotheses assume a single dose of "education" and so the question is about the effect of *when* the dose is received (versus if it is ever received) and ignores questions about the receipt of multiple doses or the time between doses. The bottom line, labeled "Never", represents the effect of education on political participation if the milestone is never passed — i.e. no effect. The next line up, labeled, "Late" represents the expected trajectory for a person who receives a dose relatively late in life. The lines labeled "Middle" and "Early" represent the paths participation ought to take given a mid-life and an early adulthood attainment of a key educational milestone such as a college degree.

The three lines for the individuals who do eventually attain the key quantity of education all end nearly in the same place (I only drew them separated for clarity of exposition). Extant knowledge would suggest that all of these lines should begin at the same level, and all but the "Never" line should end at the same place — all else being equal, of course. That is, education is supposed to have the same effect on participation no matter when in the lifetime it is acquired. At least, no previous literature suggests otherwise, and, in most analyses education is "controlled for" as it if didn't matter when in life it was acquired.

Many people in the sample engaged in non-degree granting educational opportunities. However, in order to address both the literature on civic skills and civic status, I am restricting my focus to the attainment of a bachelors degree. Over the historical period covered by the Political Socialization data, a college degree has been the main and most clear status/skill differentiator among educational experiences. People who have college degrees work in different jobs than people who do not have college degrees. In addition,

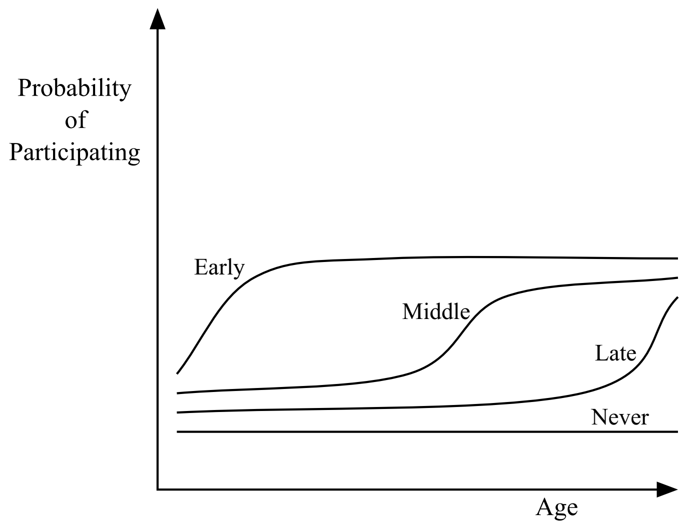


Figure 3.2: Participation Trajectories as a Function of Timing of Educational Attainment

Note: The lines are labeled "early", "middle", "late", and "never" with reference to the timing within a persons life that they pass some key educational milestone (such as receiving a college degree)

those individuals with college degrees tend to do jobs which enable them to learn and hone civic skills. And college degrees in the last 30 years have been a major moment of network placement and status attainment — more so than, say, 4 years of non-degree granting coursework, or even a vocational degree granting coursework. I also choose this focus for analytic simplicity and clarity. While other kinds of educational transitions may well have different (and even greater) effects on participation, the college degree has come to be a major marker of status in our society — kinds of jobs, income, and social status are all roughly divided at the breaking point of a college degree. And so, if any one educational transition ought to matter for political participation, it ought to be this one.

3.2.1 Educational Degrees over the Life Span

Figure 3.3 shows the proportions of respondents in the various education degree categories varies over time within the two generations. No movement across degree categories over time is perceptible from the G1 panel. In fact, there is some movement, but it is very small — from 7.68% having a BA in 1965 to 7.80% in 1982, from 3.34% having a Graduate Degree in 1965 to 3.90% in 1982. Across the years, roughly 36% of the G1 did not have a high school degree, roughly 52% had a high school degree, about 8% had a BA, and about 3 to 4% had a graduate degree. Only about 11 people out of 898 got new college degrees between 1965 and 1982!

The G2 panel shows much more movement. They all began in 1965 with the same educational degree attainment of a high school degree. As of 1973, about 61% had not finished any more degree granting education, but about 29% had received college degrees. Between 1973 and 1997 we see slow and rather steady movement out of high school degrees (from about 61% to about 52%) BAs (from about 29% to about 24%) and into graduate degrees (from about 5% to about 17%). Table 3.1 shows one picture of these transitions focusing on the movements between 1973, when G2 were roughly 26 years old, to 1997, when G2 were roughly 50 years old. For example, of the respondents who had only a high school degree in 1973, roughly 85% of them had not received another degree by 1997. Of those respondents who had received a BA by 1973, about 69% of them did not get other degrees, but about 31% of them went on to get some form of graduate degree by 1997.

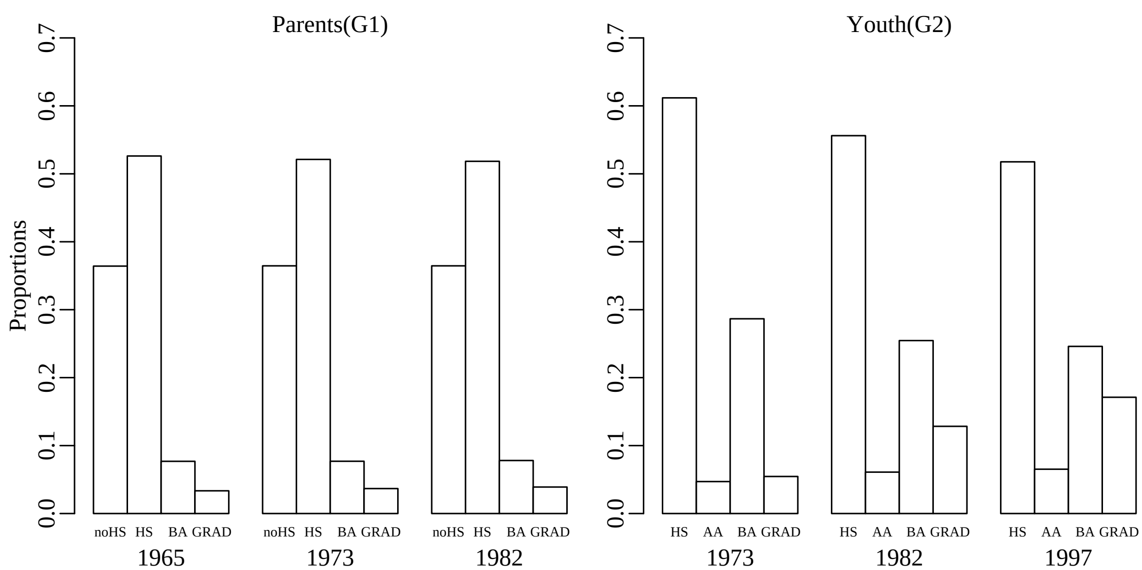


Figure 3.3: Educational Degrees over 3 waves

Across the span of the study, about 42% of G2 received some kind of college degree (including graduate degrees, but excluding associate degrees and college experience without receiving a degree). Only about 11% of G1 received a college degree by the time they were 62 years old.

Table 3.2 focuses specifically on the timing of the acquisition of college degrees

Table 3.1: Proportions of G2 Changing Degrees between 1973 and 1997

		Education Level in 1997			
		HS	AA	BA	GRAD
Education Level in 1973	HS	0.85	0.04	0.07	0.04
	AA		0.86	0.07	0.07
	BA			0.69	0.31

for members of both generations. Of those in G1 who did receive a BA, the overwhelming majority had done so by 1965 (99 out of 103). Among members of the younger generation, most (about 82%) had received their first college degree (BA or graduate degree) by 1973, when they were 26, however a few got their first degree between 1973 and 1982 (10%) or between 1982 and 1997 (8%). Thus, among the older generation it will be impossible to examine the differential trajectories of participation given different timing of college degrees over the life span. However, the younger generation does offer enough variation to make this comparison — although the results based on early BA attainment ought to have less uncertainty associated with them compared to the results based on the smaller numbers of later education attainers.[9]

Table 3.2: Timing of College Degree Acquisition by Generations

	G1		G2	
	%	N	%	N
65	11.02	99		
73	00.33	3	34.12	319
82	00.11	1	04.17	39
97			03.42	32
Never	88.53	795	58.29	545

[9]In the analyses that follow, to avoid having too small a sample of "late" BA respondents, I lump G2 into two groups — the 319 who received their BA by 1973, and the 71 who received their BA after 1973.

3.2.2 Microfoundations

Figure 3.2 suggests that the relationship between education and participation ought to look like a step function of some kind — probably with the "step" being somewhat smooth. Also, the hypothesis specifies that the steps ought to occur early in life among those who receive a dose early, and later among those who receive a dose later. We should not expect to see any such steps in the relationship between education and participation among those who do not receive an additional degree.

Thus, for each type of person, the relationship between political participation at a given moment in time t for an individual i (y_{it}) and Time, ought to be different. Such that,

$$y_{it} = \begin{cases} f_1(\text{Time}_{it}) & \text{if Early Degree} \\ f_2(\text{Time}_{it}) & \text{if Late Degree} \\ f_3(\text{Time}_{it}) & \text{if No Degree,} \end{cases} \tag{3.1}$$

where f_1,f_2, and f_3 represent the different functions relating time to the probability of participation. Specifically, we'd expect each of functions to have two knots closely spaced together — one knot, or kink in the curve, where the participation trajectory begins to increase in the first few years after receiving a dose of education, and another knot where the trajectory flattens out after receiving the dose.

3.3 Comparing Personal Trajectories of Political Participation

Equation 3.1 and Figure 3.2 have shown the differences between individual trajectories that I would expect to find given past literature on the education-participation link. The challenge of the data analysis is to map closely onto the expectations presented above — namely that each type of person ought to have a different sort of trajectory — and that these trajectories ought to involve some more or less smooth "steps".[10] The strategy that I follow here involves representing the f_1,f_2, and f_3 functions of (3.1) as natural cubic b-splines — a way to estimate a flexible functions relating two or more variables that I will describe more below.

Since the education variables are held fixed across the lives of the 935 respondents, but the participation variables vary over up to 33 years of time, some of the variables have

[10] I think about smoothness here as a function of (1) the fact that the actual moment of degree attainment will vary a bit over the years within type of person, (2) the effects of degrees may take some time to occur, and (3) that even if the effects of degrees are sudden, different individuals receive their degrees in different years, thereby "smearing out" the effect of degree attainment across the group of individuals of that type.

935 degrees of freedom while other have 935*33=30,855 degrees of freedom. The way around this problem is to use a model which can take into account the different levels of analysis represented in the model (in this case, years within individuals), and can also deal with unmeasured time-constant heterogeneity across individuals. Equation 3.2 portrays a very simple model as an example, in which the $\beta_{1i}\text{Time}_{it}$ term shows that the trajectory of participation is required to be linear, but for which each individual may deviate from the common trend (β_{1i}) by some amount u_{1i} (shown by (3.4)).[11]

$$\begin{aligned} y_{it} = & \beta_{0i} + \beta_{1i}\text{Time}_{it} + \beta_{2i}X_{it} + e_{it} && (3.2) \\ & \beta_{0i} = \gamma_{00} + \gamma_{01}Z_i + u_{0i} && (3.3) \\ & \beta_{1i} = \gamma_{10} + u_{1i} && (3.4) \\ & \beta_{2i} = \gamma_{20}. && (3.5) \end{aligned}$$

Substituting (3.3), (3.4) and (3.5) into (3.2) yields:

$$\begin{aligned} y_{it} = & \gamma_{00} + \gamma_{01}Z_i + \gamma_{10}\text{Time}_{it} + \gamma_{20}X_{it} + (u_{0i} + \text{Time}_{it}u_{1i} + e_{it}) && (3.6) \\ & \text{with: } (u_{0i}, u_{1i}) \sim N(\mathbf{0}, \mathbf{T}) \text{ and } e_{it} \sim N(0, \mathbf{\Sigma}) && (3.7) \end{aligned}$$

In this model (3.6), X_{it} represents individual level characteristics that change over time, like, say, income, that are not assumed to have significant intra-individual level variation (or at least, not significant variation that matters for the estimation of β_{1i}), and Z_i represents characteristics of individuals that are not expected to change over time, such as sex. The distributional assumptions show that the errors of the random coefficients (i.e. time-constant person level variables) are assumed to be jointly distributed according to a multivariate normal distribution with means 0 and covariance matrix $\mathbf{T}$, that the time-varying within person variables may also be correlated according to the covariance matrix $\mathbf{\Sigma}$. In addition, the within person errors e_{it} are assumed to be uncorrelated with the between person errors u_{0i} and u_{1i}.

The point of showing the model in equation 3.2 thru 3.6 is to demonstrate the

[11] Nearly all available software requires the distribution of the individual specific deviations around the common slope to be normally distributed. Thus, I have assumed normally distributed random coefficients throughout this dissertation. This may not be a realistic or reasonable assumption, but choices for such random effects (or "priors" in the language of Bayesian data analysis) remains an active research project. And the problems of estimation facing non-normal random effects are formidable.

general way in which variables with different degrees of freedom may be integrated into the same framework.[12]

Of course, Equation 3.1 shows a situation in which the trajectories of participation with respect to education (captured by the relationship between participation and Time) are not linear as in (3.2) , but have a specific shape, which varies systematically based on the moments in which individuals acquire their college degree. There are two general ways to estimate the shape of the function relating time to participation within the framework of a random coefficient model. The first, is to specify a particular functional form for $f(\text{Time}_{it})$ — that is, to make the model involve variables that enter in a linear, additive form (for X and Z in equation 3.6) and others that have an inherently non-linear form. The problem with this approach, is that it assumes that the functional form is correct. That is, although my hypothesis displayed in Figure 3.2 seems to involve 3 versions of the same function, with kinks at different locations , in fact, this kind of analysis would not tell us whether or not the actual function has any kinks, or whether is has only 1 or 2 kinks instead of 3.[13] That is, this kind of analysis would *assume* Figure 3.2's accuracy, rather than test for it. Thus, I have a chosen a more flexible approach where $f(\text{Time}_{it})$ is represented using splines. This allows nearly any kind of functional form to emerge from the data, leading to some problems of interpretation and the possibility of using many degrees of freedom, but it forms a very rigorous test. Using the first approach, the hypothesized shapes from Figure 3.2 that are estimated would have to be compared with all possible other shapes, whereas in the nonlinear regression framework, the comparison would just be from the function that best fit the data to the set of functions that characterize Figure 3.2.

The way splines work is that they break the function f into a linear combination of other functions (called "basis functions") that may then be fit as any other variable might be in the framework of a linear model. Thus, to use a spline function to fit a particular function, one needs to specify the number of pieces involved (and the placement of the "knots" at which the function may change). To add extra flexibility, most analysts use cubic-polynomials as the inter-spline pieces, thus, even between knots the function can bend. Equation 3.8 shows how this will work in this case:

[12]This type of model is called by many different names including: multilevel model, hierarchical linear model, and mixed-effects model.

[13]The closest it would come to telling us this information would be if it didn't converge.

$$f(\text{Time}_{it}) = \sum_{k}^{K} \eta_k(\text{Time}_{it})\beta_k. \tag{3.8}$$

where, the $\eta_k(\text{Time}_{it})$ are the natural cubic spline basis functions for each of the k knots, and β_k are the regression coefficients on each basis function such that $f(\text{Time}_{it})$ can be seen as the weighted linear combination of basis functions.[14]

Following this approach, I estimated two sets of models. The first set examined the difference in participation trajectories across individuals distinguished only the highest level of education they had achieved by the end of the study period. This analysis has the benefit of allowing the Parent generation to be analyzed since we have no information on the timing of their degree acquisition. This ignores the issue of timing, but does provide information about the extent to which *ever* receiving certain educational degrees distinguishes among the participation trajectories of individuals. In this model, the functional forms are estimated for each group defined by its members' educational attainment, such that:

$$f(\text{Time}_{it}) = \sum_{d}^{4} \sum_{k}^{K} \text{DegreeType}_{di} \eta_{kd}(\text{Time}_{it})\beta_{kdi}, \tag{3.9}$$

where DegreeType_{di} is a factor representing four dummy variables indicating the presence/absence of a High School degree, an Associates degree, a Bachelors degree, or a Graduate degree for an individual i; $\eta_{kd}(\text{Time}_{it})$ is the natural cubic spline function of Time for the kth knot for the dth type of degree; (β_{kdi} captures the relationship between the given spline function of time and degree with political participation for individual i (i.e. this coefficient is allowed to vary across individuals — that is, this is a random coefficients model)). The point of this model is to provide a sense of whether or not the trajectories for those individuals who ever receive higher educational degrees are dramatically different from those individuals who do not. For example, we wouldn't expect the degrees to begin differentiating individuals until after they began to receive their degrees — around 1973. And all of the individuals should start out in roughly the same location. If education is as powerful a force as it appears from the cross-sectional literature, then we might expect

[14] In this analysis, and in others throughout this dissertation, I use natural b-spline basis functions. The b-spline set of basis functions is particularly useful because it has fewer degrees of freedom, and less inter-correlation among the individual basis functions, than other bases (say, the power polynomial basis). Also the "natural" b-spline is required to be linear outside the bounds of the data, and thus is more likely to make realistic predictions inside the data. For more information about regression splines see Hastie, Tibshirani and Friedman (2001, Chapter 5) and for a rigorous mathematical treatment see de Boor (1978)

to see very different shapes of participation for the different types educational degree (high school, associates, bachelors, graduate). We also ought to see differences in probability of participation at any one moment in time (i.e. changes in level).

In addition to this model, I specified another model to capture the effects of receiving one's degree at different points in one's lifespan — to test the hypothesis displayed in Figure 3.2. This time, individuals were categorized according to when in their lives they received a college degree — never, early (by 1973), or late (after 1973).[15] However, the mechanics of the estimation process remained the same — Time entered into the equation via a different spline function for each different type of person, and the goal was to see if these trajectories are distinguishable from one another, and if they follow the paths laid out in Figure 3.2. The estimation of the different curves for each type of timing of educational attainment (early, late, or never receiving a BA) is via the same kind of spline function as in (3.9), only this time the dummy variables are labeled "DegreeTimingType", such that

$$f(\text{Time}_{it}) = \sum_{d}^{3}\sum_{k}^{K} \text{DegreeTimingType}_{di}\eta_{kd}(\text{Time}_{it})\beta_{kdi}. \tag{3.10}$$

In (3.10), there are 3 spline functions (one for each type of degree timing) with K knots each. In the models that I estimated, I spaced knots 3 years apart across the range of the time variable for G1 and for the electoral activities for G2, but I spaced the knots 4 years apart for the non-electoral activities for G2.[16]. Having between 5 and 7 pieces for each function eats up a lot of degrees of freedom, but it prevents specification error — and provides a more difficult test for the hypothesis in that the function is allowed to twist and turn every three years across it's whole range.

The actual estimation involved substituting (3.10) and (3.9) into (3.6) in place of the linear function of Time. To this equation, I also added controls for presidential year (realizing that the electoral oriented types of political participation are very sensitive to electoral cycles), for interview year (as a very rough way to capture the effect of memory — that people are more apt to remember acts of participation the closer they are to an interview year), for income (interpolated across the panel periods to provide a rough ad-

[15]Only members of G2 were entered into this analysis because we don't have any information about when the members of G1 received their educations.

[16]I first ran all of the models with knots at 3 year intervals. Although these models fit the data most closely in every case, the substantive conclusions are the same between the 3 year knot and 4 year knot models — and the 4 year knot models smooth out what is essentially noise from a substantive point of view, making graphical displays that are easier to interpret

ditional measure of SES and resources), and being female (given what we know about the impact of gender on political participation and education). Other variables that are probably related to participation, but not to education, are left out of this equation (such as residential mobility or parenting). In addition to the variables that enter into the model to prevent misinterpretations of the estimated relationships between education and participation, these models also included a term which estimated the average serial dependence over time within person. Although there are many possible specifications for this, remembering the results of Chapter 2 I chose to use a Continuous AR1 correction for simplicity, such that the diagonal elements of $\mathbf{\Sigma}$ from (3.7) are:

$$\mathrm{cov}(e_{ij}, e_{ik}) = \phi^l,\ j \neq k,\ l \geq 0,\ \phi \geq 0 \tag{3.11}$$

where l represents "position" (which is akin to "lag" in the discrete time, AR1, formulation). Equation 3.11 means that the correlation between adjacent years' observations within a person dies out smoothly (and quickly) as the years become more distant from one another and that these correlations are constant across individuals.[17]

3.4 Results

3.4.1 Maximum Educational Attainment

Figures 3.4 and 3.5 show how the participation trajectories for G2 are hypothesized to differ based on the highest level of formal education attained by the respondents over the study period (where, "study period" refers to ages 18 to 35 for campaign-oriented activities and 18 to 50 for non-electoral activities). The thin, vertical, gray dashed lines mark the locations of the knots for the spline function. The question here is whether educational degrees gained at any point in the life-cycle has any lasting effect on trajectories of political participation.

Figure 3.4 shows that individuals who at some time in their lives received a graduate degree are more participatory across ages 18 to 35 in electoral activities than people who received less education. The education distinction is most clear for "Other" campaign

[17] When modeling binary dependent variables, it is common to use a binomial probability model plus some kind of function that restricts the linear prediction equation to produce values only between 0 and 1. The probability model I assume here specifies that both the responses (y_{it}) and the random effects (u_{0i} and u_{1i}) from (3.6) are normally distributed as shown in (3.7). This probability model may not be ideal for a structural model with a binary dependent variable, but using the spline functions help inhibit the kind of out of bounds predictions that analysts often prevent by using logit or probit link functions within a binomial probability model. Since I don't have strong priors about the shape of the distribution of the random coefficients (u_{0i} and u_{1i}), I have left them as normal in the interests of computational convenience.

related work, where nearly immediately after high school those that eventually receive college degrees and graduate degrees distinguish themselves from the individuals who had not achieved such status. Respondents with graduate degrees also end up having higher probabilities of donating money than their less educated counterparts, but this participatory advantage does not appear until the individuals are in their 20s. It is interesting to note that the additional 2 years or so of education after college of an Associates degree does not do very much to enhance participation above receiving a high school degree. The other clear pattern from the analysis of electoral participation is that those individuals with only a high school education tended to have the lowest probability of participation across their early adult life. What is particularly interesting is that receiving a BA without going on to receive a graduate degree did not always guarantee a participatory advantage over receiving an Associates degree — the trajectories are indistinguishable for wearing buttons or displaying signs, and the results are mixed for attending meetings and rallies (where receiving a BA and an AA are tied for the middle range), and for donating money (where the college graduates do begin to have a higher probability of giving money after about age 26, but then those who receive an AA catch up by age 35).

Figure 3.5 tells a similar story. When it comes to non-electoral activities, it still appears that, in general, those individuals with graduate degrees have a higher probability of participating at any one moment in time (except for writing letters to the editor which nearly no one does and for which, somewhat surprisingly, education does not really differentiate). The spike in predicted probability of protest activity in the late 1960s and early 1970s is only apparent among those who received a college degree — but even more so among those who eventually receive a graduate degree. The probability of protesting is nearly non-existent among individuals other than those who at some time in their lives received a college or graduate degree. In Figure 2.6, I showed that the proportion of G2 who engaged in demonstrations or protests peaked dramatically around the late 1960s and early 1970s. Since the members of G1 did not show such a spike in protest activity at that time, it is tempting to say that protesting was a particularly important activity for the young during that time period — a cohort effect. What Figure 3.5 suggests is that the particular cohort in question it not merely defined by birth year, but by years during which they went to college. And, despite the fact that protesting is often presented as the archetypal act of participation of that period in time, it appears that the mass of Americans did not do it — and that it was mainly restricted to those who were in college at the time. This then

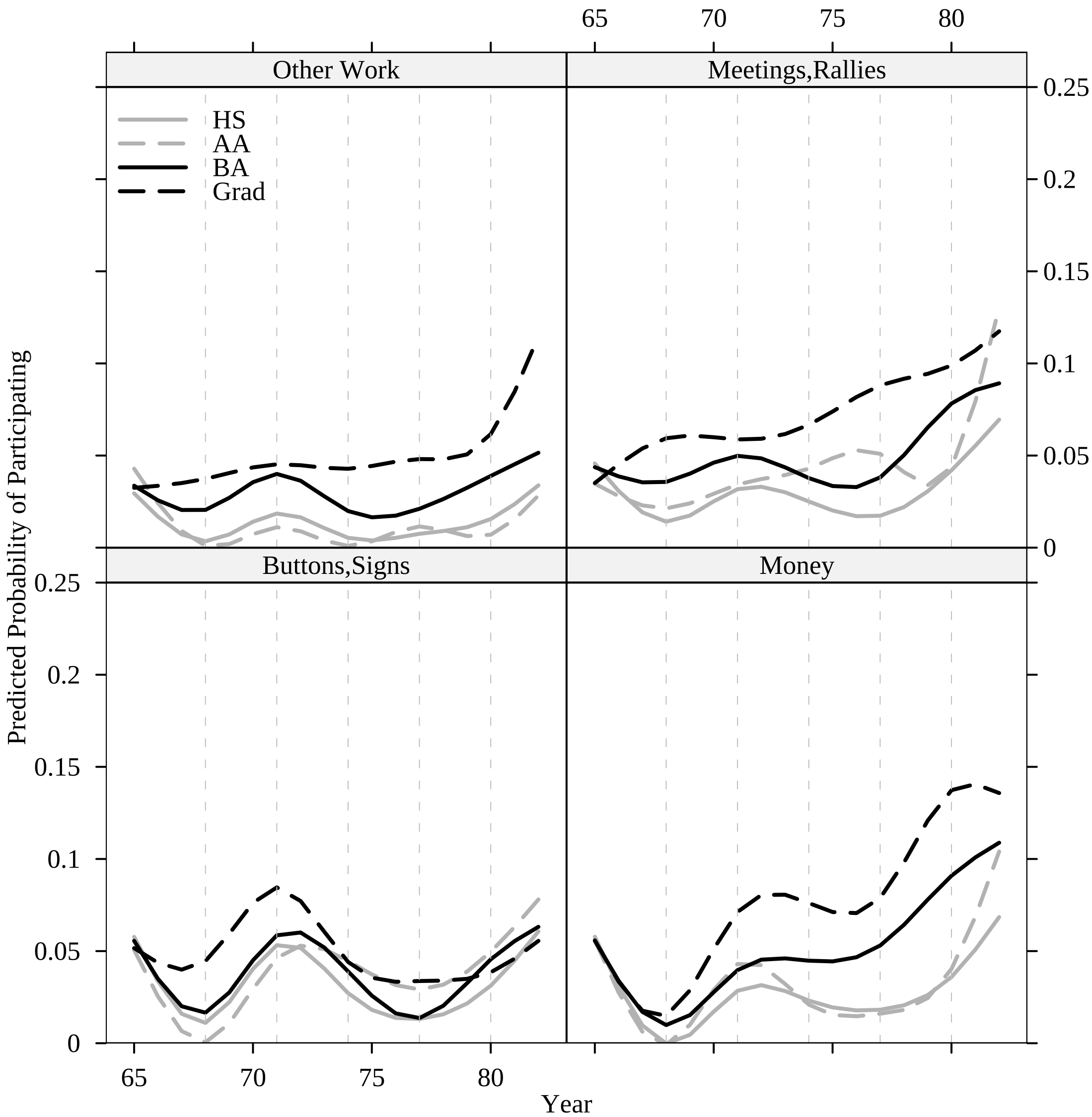


Figure 3.4: Trajectories of Electoral Political Participation by Life-Time Maximum Educational Attainment: G2

suggests whether there is an institutional explanation to the rise in protest activity in the late 1960s — something linked with colleges — rather than something inherent in the youth of most of the participants.

Overall, for G2, those who only received a high school degree are predicted to participate systematically less in *Community Work* and *Contacting Officials*, in general, than those with higher levels of education. And, in the lower panels, reasonably large participatory differences appear between all of the different levels of education.

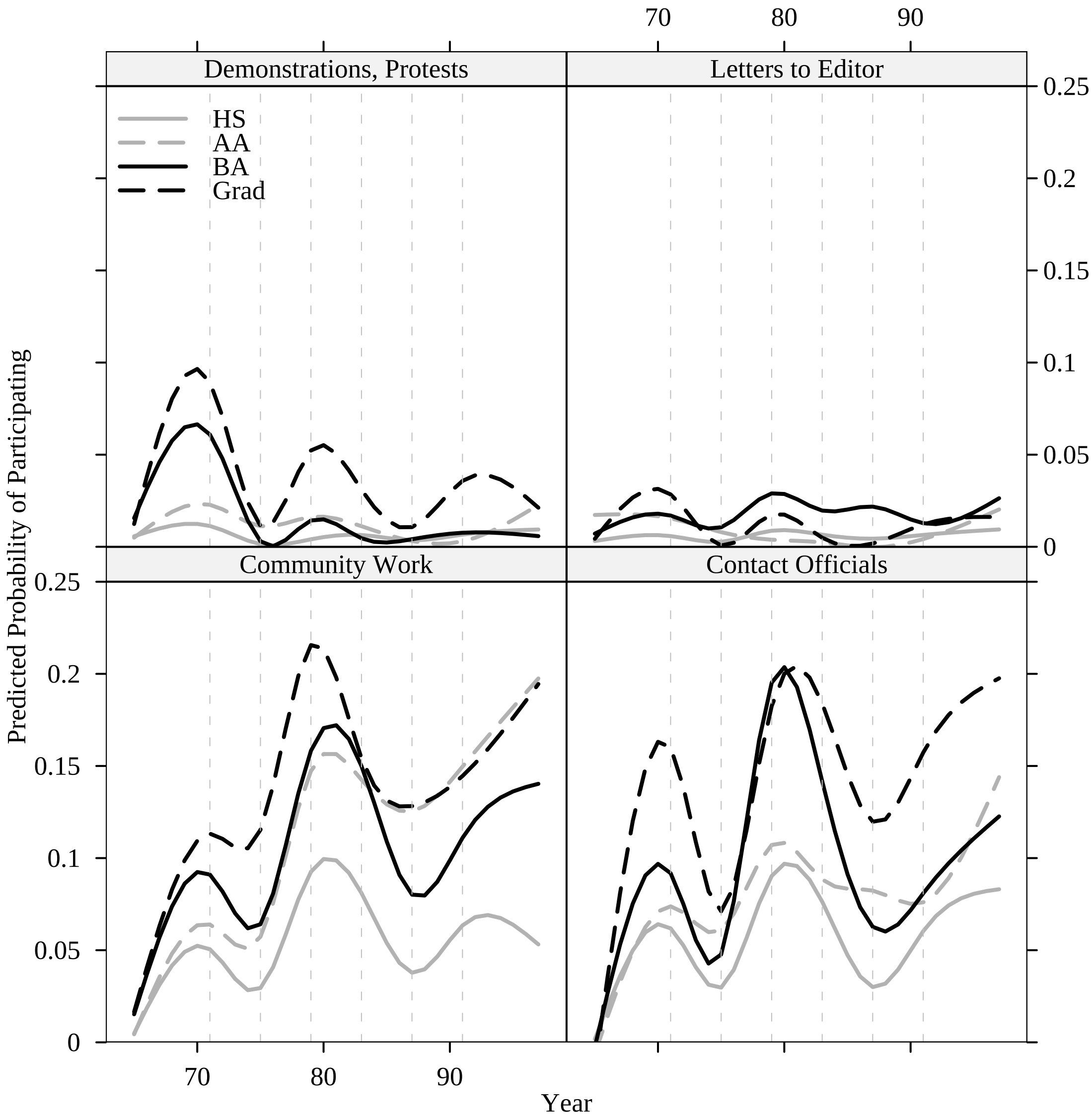


Figure 3.5: Trajectories of Non-Electoral Political Participation by Life-Time Maximum Educational Attainment: G2

Figures 3.6 and 3.7 show similar analyses for the Parent generation, covering the period where these individuals age from roughly 50 to roughly 70. For electoral actions, those (few) individuals with graduate degrees are predicted to be much more likely to participate for other campaign work than those individuals with less education — but college degrees without graduate degrees seemed to provide nearly as many advantages for attending meetings and rallies or donating money. Except for *Buttons/Signs*, graduate degree parents held an edge over all else in the 1968 period. Educational differences are pretty much erased across the board in the 1976 election and are back to some extent by 1980. In general, the "stacking" of the the trajectories representing the probability of participation by education is not as clear for this generation as it was for G2, except in the case of donating money. For electoral participation, those with college or graduate degrees gave money with higher probability across the whole time span, than did those who had high school education or less. However, receiving a college degree without receiving at some point a graduate degree does not seem to systematically advantage individuals in the realm of electoral participation — by this age, perhaps other factors are at work, swamping the enhancements that education ought to provide. If one added uncertainty bounds to these estimates, however, the intervals would surely overlap (especially for the case of graduate degrees).[18]

For non-electoral participation, only those with graduate degrees are predicted to have any appreciable protest activity or letters to the editor across the period. The patterns for *Community Work* and *Contacting Officials* display what one might expect — a stacking of lines where those without a high school degree have the lowest probability trajectory, those with a high school education tend to have a slightly higher probability trajectory, and those with college degrees or higher have the highest probability trajectories. I suspect that the crossing of the graduate and college degree trajectories is due to the relatively small numbers of G1 who had graduate degrees (no more than 4% or about 36 individuals compared with about 8% or about 72 individuals having a BA).

For both generations, participation trajectories are not always cleanly differentiable on the basis of life-time education attainment. However, it is clear that among G2, receiving a graduate degree at some point tends to create higher probability trajectories, while among the G1, graduate degrees and college degrees together seem to improve the probability of participation at any one moment in time. Also, the lowest educational strata (high school only for G2, and no-high school degree for G1) tended to display the lowest probability trajectories. The differences in level are noticeable in the stacking of lines by education for certain activities in both generations (notably electoral activities in the young

[18] I do not present confidence intervals in these charts since there would be too many lines to be interpretable.

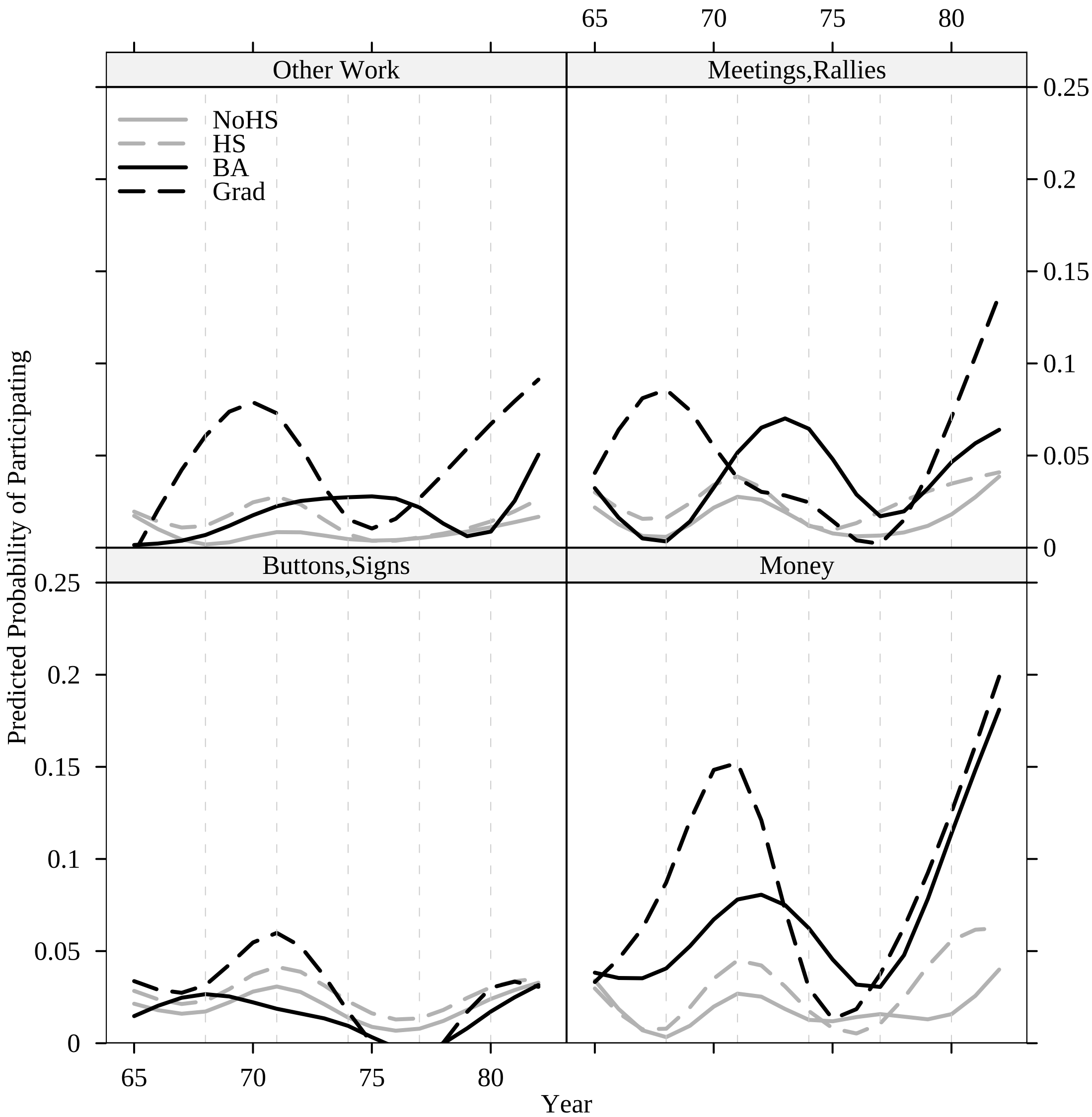


Figure 3.6: Trajectories of Electoral Political Participation by Life-Time Maximum Educational Attainment: G1

adulthood of G2 and *Community Work* and *Contacting Officials* for both G1 and G2). One predominant pattern in these results is that, even where the lines cross, the "peaks" and "valleys" roughly line up (with exceptions of *Other Work* and *Meetings and Rallies* for both generations). This suggests that life-cycle and/or period effects are at least as important in driving the evolution of participation over the life-span as eventual educational attainment — which mainly appears to provide a mean shift in the probability of participation at any

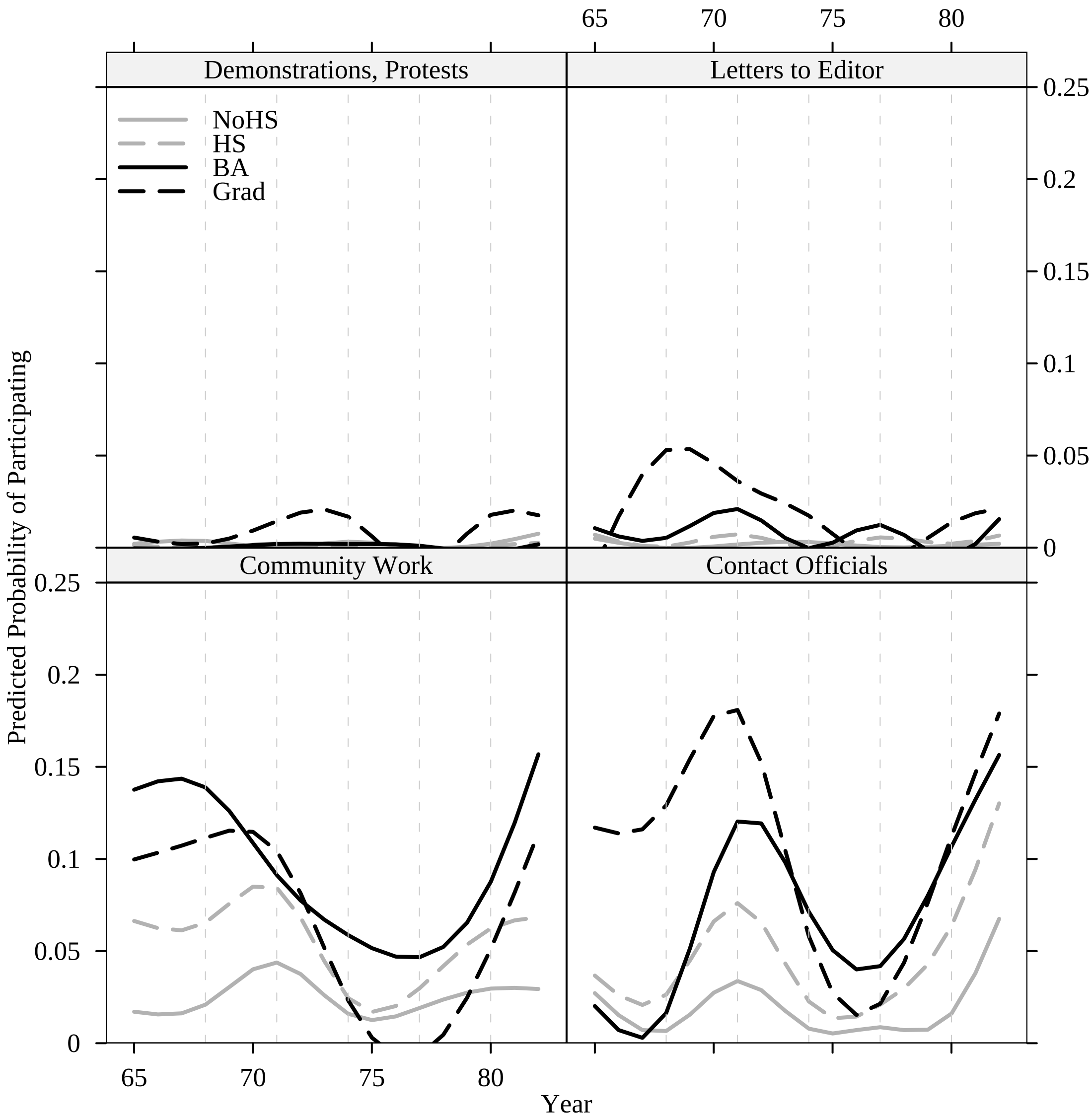


Figure 3.7: Trajectories of Non-Electoral Political Participation by Life-Time Maximum Educational Attainment: G1

one moment in time. This is quite surprising given the power of education as an explanatory force in all previous work on this topic. Such an important variable might have been expected to completely change the *shape* of the trajectories not just the levels. That is, it might have been expected to change the very ways that the life-cycle and period effects interrupt and affect the paths to (or from) participation people take. Instead, it appears from this first analysis, that the total education attained by individuals over their lives tends to shift their trajectories up or down, but not to change the shapes of their trajectories. So, perhaps education is a necessary but not sufficient facilitating factor for participation.

3.4.2 Timing of College Degree

The findings in the previous section roughly support the idea that more education ought to increase the probability of participation at any given moment in time. However, these overall findings ignore the fact that individuals may receive education at different moments in their lives. Do the hypotheses displayed in Figure 3.2 receive any support from the data? If the hypotheses are reasonable, we ought to see *early* bumps upward in the probability of participation for those who receive their BAs early, and *late* bumps upward for those who receive them late.

Figures 3.8 and 3.9 show the results for electoral activities and non-electoral activities respectively, for G2. The solid black lines show the trajectories for those individuals who received a BA before 1973 (labeled "Early"), the dashed black lines show the trajectories predicted for those individuals who received their BAs after 1973 ("Late"), and the gray lines are for those individuals who never received a college degree. This analysis combines those who received a graduate degree with those who received a college degree — it may be the case, for example, that the impact of graduate degrees seen in the previous analyses could actually be caused by the timing of college degree (if those who received graduate degrees tended to get college degrees early in life).

Figure 3.8 shows the expected pattern for *Other Work* and *Meetings and Rallies*, where we see that those who receive a BA early in life have relatively flat trajectories that are mostly higher probability trajectories than others in their cohort who never receive a BA, or even among those who receive a BA late. However, the participation trajectories for both of these types of activities do display an uptick late in the series (close to 1982 or age 35). That is, from these two activities, the suggestion is that late college degrees do not have the same direct causal influence on participation as early — or at least the influence takes more time to manifest itself for this group of people than for those who attained a BA early in life. Somehow, receiving a college degree early ensures a rapid increase immediately after high school, presumably during the college years, whereas receiving the BA late does appear to change the participation trajectories of those who have waited to pursue college education. The late-BA types begin identically to the never-BA types, but as members of that group (about 70 people) begin to receive BAs after 1973, their trajectory gradually leaves behind the never-BA types and catches up to the early-BA types by age 35.[19]

The story is not quite as clear for the trajectories of money donating and button/sign displaying, but it still appears to support the hypotheses. In this case, those who attain BAs before 1973 seem to have a slight early advantage, which disappears quickly, such that all types of people are roughly equal in their button wearing by 1982. The early BA folks do show an early increase (and divergence from the other two groups) in money donating, but the late BA folks catch up by 1982. This leaves the group with only a high school degree on a much lower probability trajectory.

The story for non-electoral participation is less easy to make out. Figure 3.9 suggests that getting a BA early provides a decided advantage for all of the activities except for *Letters to the Editor*. The groups appear to have roughly the same flat trajectories for *Letters to the Editor*. When it comes to *Community Work* or *Contacting Officials*, the early BA folks tended to have higher probability trajectories than the other groups, particularly in the periods from 1965 to 1982. After 1973 the "late" group began to receive their BAs, and they caught up with the early group such that both the late- and early-BA folks look more similar after 1982 than they did before 1973.

The findings about demonstrations suggests that the effect of receiving a college degree on that act was mainly to locate individuals in a situation where they were being mobilized..[20] Both *Community Work* and *Contact Officials* show increases that occur across the periods in which education was acquired. And in both types of participation, receiving a BA late in life does appear to produce late breaking increases that are larger than the ones occurring earlier in life. In essence, the estimated trajectories presented in Figure 3.9 can be seen as roughly follow the paths laid out in Figure 3.2, but forces outside the model are causing dips and bumps to deflect it from the hypothesized contours.

For electoral activities the trajectories for early-BA types either do not change over the period of 1965 to 1973, the period during which they received a BA, or they dip. And the increases in predicted probability of participation over the 1965 to 1973 period for the non-electoral activities among the early-BA types are mirrored (although smaller) among both the late- and the never-BA types (suggesting a life-cycle or period effect rather than a direct effect of education). But of course, the early-BA types increase more. The

[19]Since the folks who received a BA after 1973 could have done so up until 1997, the relatively slow speed of this education effect may be picking up the heterogeneity in times that these people receive their college degrees.

[20]This is congruent with previous findings from this dataset, see for example, Jennings (1987).

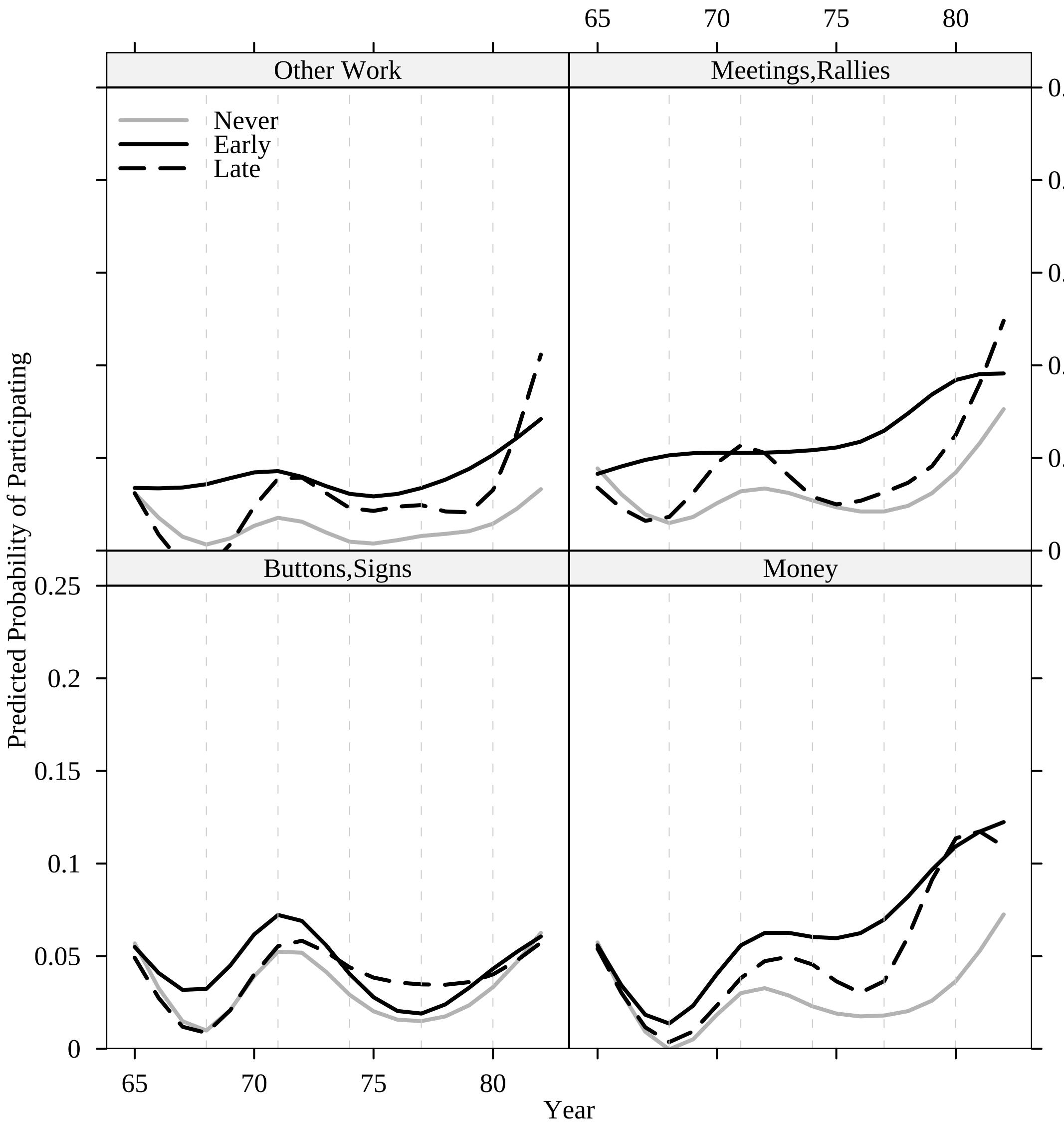


Figure 3.8: Trajectories of Electoral Political Participation by Timing of College Degree: G2

late-BA types do show some increases where expected — toward 1982 for electoral activities, and toward 1982 and 1997 for *Community Work* and *Contacting Officials*. These increases in probability of participation occur in sync with increases among the rest of the sample, just to a lesser degree among the early-BA types (who had already attained a high level of participation) and never-BA types (who have relatively flat trajectories in any case). It is only for the electoral activities that education appears to differentiate the shapes of

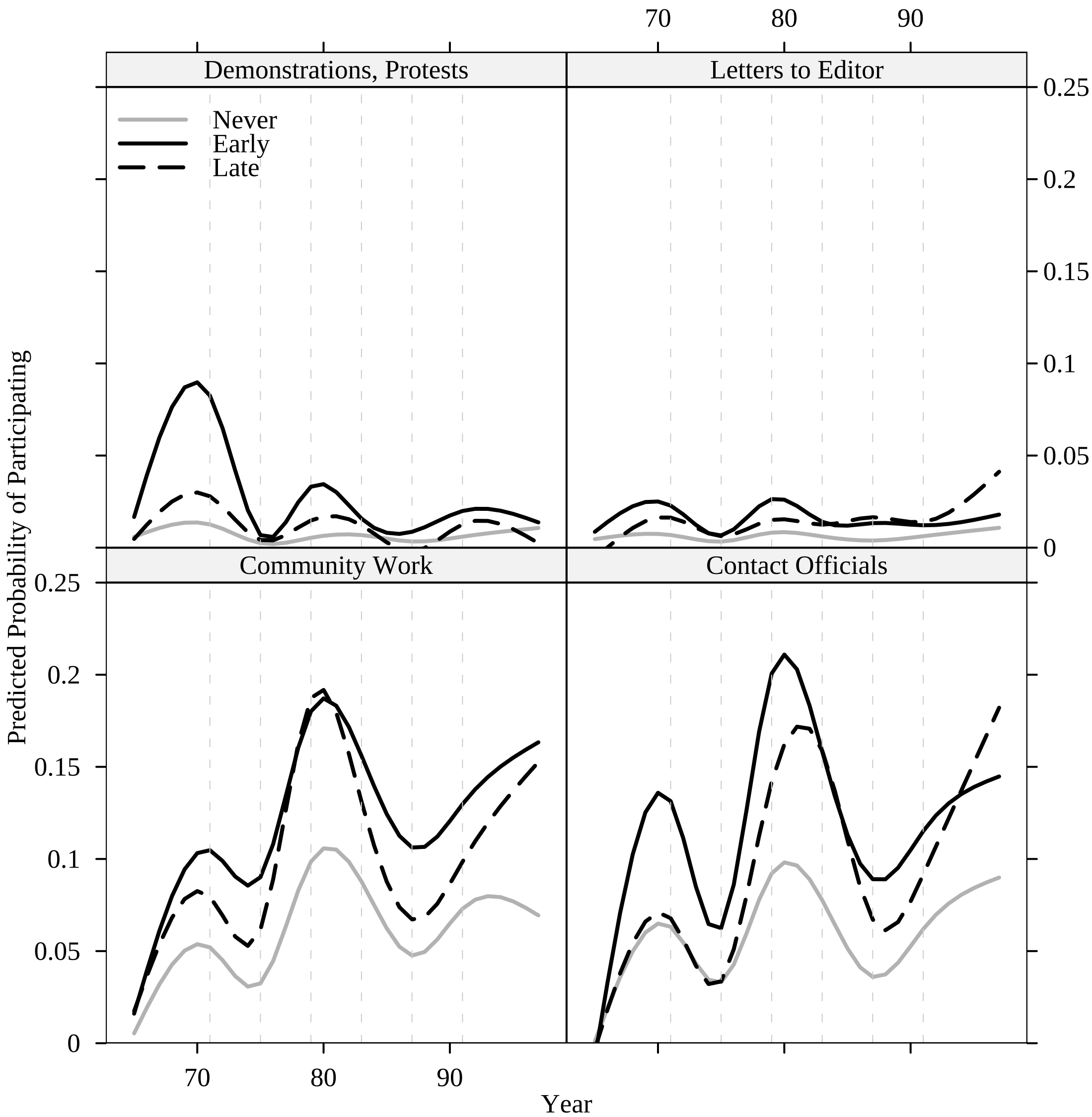


Figure 3.9: Trajectories of Non-Electoral Political Participation by Timing of College Degree: G2

participation trajectories among people. For non-electoral activities, the shapes remain roughly the same.

Thus, education does not appear to be transformative of a person's relationship to politics, but rather it accelerates and intensifies the probability of participation without changing the underlying shape of activity as it responds to the life-cycle and events along the way. The results here largely confirm the hypotheses of Figure 3.2, meaning that the timing of a degree matters, but that the trajectories of participation that begin upon getting a degree look similar whether the degree is attained early or late.

3.5 Conclusion

The analysis presented in this chapter suggests that education does not dramatically alter the shape of political participation as a trajectory of action across the lives of ordinary Americans. This stacking of similar shaped trajectories from low to high in order of educational status is also apparent when the analysis focuses on the timing of attainment of education (represented by the rough period in life when individuals reported receiving a college degree). The analyses do show that early-BA folks will get more involved earlier in their lives than late-BA types, and thus receiving a BA early does seem to spur a longer history of participation across the lives of individuals. However, this chapter also shows that receiving a BA later in life allowed people to catch up with the early-BA types over time. That is, although participation clearly varies over the live of individuals as a dynamic process, the rises and falls don't appear to have too much to do with education. The overall levels of participation (or, at least, the probability of participation at any one moment) does certainly seem to have a strong relationship with education — with the more educated individuals showing a higher probability of participating (and with the earlier educated individuals participating more, sooner, that later educated individuals). This is old news. What is new, however, is that this powerful cross-sectional relationship appears to merely change levels of participation but not the contours of it, not the actual trajectories.

Before writing this chapter, I imagined that education would radically alter the very shape that participation takes across the lives of individuals. After these analysis, I am led to believe that education is an attribute of individuals which enhances (or depresses) their ability to respond to the participatory opportunities or demands of the life-cycle and the historical period. Thus, education is not a good predictor of *when* individuals will get involved. It is just that, in a moment of opportunity or threat or mobilization, it is those who have more education who are more willing (and/or able) to get involved. The fact that the trajectories look so similar suggests something else about participation in general, that perhaps most acts of participation are not cumulative; in other words, little learning may take place over time.

If participation had a positive feedback effect — such that one act tends to lead to more activities — then entering participation early ought to set off a chain reaction of participation throughout the life course. This is the hope of civic education programs which have sprung up across the nation. However, this analysis suggests that even those whose early education enhances early participation do not set off on radically different trajectories of participation than those who never receive education or those who receive it late. That is, there is no "first mover" advantage here, found later in life, or really, not much learning effect translating past participation into future participation.

This speculation is confusing since it goes against what both the civic skills and civic status theories would suggest: in both cases, acts of participation themselves ought to train a person in the skills necessary for participation (thus decreasing costs and increasing future probability of participation) and ought to locate individuals within networks in which individuals are mobilized and know people (another mechanism by which past participation ought to increase future participation). Despite this confusion, it is more important to examine the other important cross-sectional relationships that we know about before launching into new hypotheses. For this reason, in the next chapter, I take up the relationship between residential mobility and political participation.

Chapter 4 Disruption and Recovery?: Residential Mobility and the Dynamics of Participation Within the Lives of Ordinary Americans

America has often been singled out as a "nation of movers" (Lieberson and Waters, 1987, page 802), and it is a common perception among sociologists and social critics that "rootlessness" is both a pervasive feature of American life and a powerful cause of a variety of social and civic ills despite its possible economic benefits.[1] Political scientists have added yet more reason for concern, showing that people who have recently moved to a new place are less likely to vote than people who have lived in that place for a long time.[2] Robert Putnam ties the aggregate and individual level findings together when he suggests, "Just as frequent movers have weaker community ties, so too communities with higher rates of residential turnover are less well integrated...So mobility undermines civic engagement..."(Putnam, 2000, page 205).[3] What is interesting about this link between mobility and civic engagement, however, is that, in the aggregate, residential mobility has not appreciably changed in the last 50 years. And, in fact, any change in mobility that has occurred has been a decrease: around 1950, as many as 1 in 5 Americans were moving, while now that proportion has dropped to about 1 in 6 (Fischer, 2002). Hence, the over-time decline in civic engagement cannot be blamed on residential mobility's effects. Putnam recognizes this fact, asking, "Could rising mobility thus be the central villain of our mystery? The answer is unequivocal: No."(Putnam, 2000, page 205).

[1] Fischer (2002) presents a good summary of what social theorists fear from a rootless nation. Thanks to Paul Martin for suggesting the "nation of joiners"/"nation of movers" link.

[2] This hypothesis has been confirmed with data from the Current Population Survey (CPS) in the 1970s by Wolfinger and Rosenstone (1980), using both CPS and National Election Study (NES) data in 1980 by Squire, Wolfinger and Glass (1987), by Rosenstone and Hansen (1993, pages 157-159) using data from the 1956-1988 NES Cumulative File, and by Highton (2000) using NES data for presidential election years from 1976 to 1996. For other analyses which have shown that residential mobility depresses vote turnout see Verba and Nie (1972, page 139) and Lane (1959, page 267)

[3] "Integrated" in this context refers to the strength or tightness of social ties between individuals within a place. This is not a reference to a place that is racially heterogeneous.

The question I pursue in this chapter concerns the relationship between residential mobility and non-voting political participation within the lives of individuals. This relationship has not been directly studied before. One might assume that the linkage between mobility and non-voting participation has been well established, and, in an important sense it has been. For example, Milbrath (1965) states (with emphasis), "...persons who are well integrated into their community tend to feel close to the center of community decisions and are more likely to participate in politics. One evidence of this is that **the longer a person resides in a given community, the greater the likelihood of his participation in politics**" (page 133, emphasis in the original).[4] We also often use findings which reveal "*who* participates" to infer "*when* people are apt to participate". However, everyone who has analyzed cross-sectional survey data knows that a doubt lurks behind every discussion of such findings. In the end, it is a leap to move from "who participates" to an understanding about how changes in mobility influence changes in participation. To help fill in the gap left by the previous literature, in this chapter I examine the relationship between changes in mobility and changes in participation *within the lives of the same individuals, followed over time.* In a sense this chapter is an attempt to do the "normal science" that needs to be done — to provide evidence confirming or casting doubt upon past findings. The idea is to provide a reasonable base from which to study the idea of political participation as a dynamic process within the lives of individuals (which it surely is), and more seriously, asking, for example, how rates of participation or the persistence of participation react to changes in residence — aspects of participation that have hitherto been very rarely studied, despite the social and political importance attributed to residential stability/mobility by social and political theorists.[5]

Current understanding about the relationship between residential mobility and political activity of individuals can be summarized by the two statements: "Residential mobility decreases civic activity" and "Residential stability increases civic activity." This understanding is based on the finding from cross-sectional studies that more recent arrivals to a place are less likely to participate in politics than established residents, *ceteris paribus.* Unfortunately, findings from cross-sectional studies cannot tell us whether moving (or staying) leads to changes in civic activity since such studies do not observe the same individual's

[4]He supports this statement with nine citations to work ranging from 1937 to 1964.

[5]See Berinsky, Burns and Traugott (2001) and Plutzer (2002) for the only two other recent examinations of the general topic of changes in participation over time, within the lives on individuals, that I know of.

participation before and after moves. In this chapter, I assess this general understanding using longitudinal data that follow two generations of Americans across many years of their lives. The analyses I present do not provide strong evidence to support the common knowledge. This chapter calls into question what we currently know about the link between residential mobility and political participation, and it suggests some directions for further investigation.

4.1 Data: The Political Socialization Panel Study

The analyses in this chapter rely on annual histories of individual level political participation that I derived from the Political Socialization Study (described in previous chapters and in the Appendix A). In addition to the participation questions, the respondents were also asked to report on their past residential mobility, listing the cities/towns and states in which they lived and the number of years they lived in those places. This information allows me to construct series recording the years in which respondents moved, and to examine the relationship between this series and the series recording the years in which respondents got involved in politics.

The raw material for creating this time series of moving came from the following question:

> We'd like to know where you've lived since we interviewed you in [1965, 1973, 1982]. We're not interested in addresses, just in the town or city and state. Can you tell me when and where you've lived since 1965/1973/1982? Let's begin with 1982. [AFTER EACH PLACE OF RESIDENCE: What about the place you lived after that? Where did you live? When did you live there?] [RECORD TOWN OR CITY AND STATE (OR COUNTRY) AND YEAR(s) e.g. 1982-1985 OR LENGTH OF TIME; RECORD UP TO 5 or 11 MENTIONS (for 1973/1982 and 1997 respectively)]

If individuals provided more than 5 (or 11) residences, the places of longest residence were coded. Thus, if a person reported, in 1982, that she had lived in 2 places (other than the interview location) since 1973, and that she lived in the first one for 5 years and the second one for 3 years, and had been in the current location for about 1 year, it would be clear that she moved in 1978 (1973+5 years) and 1981 (1978+3 years). Since this question also collected information on different states of residence, one can tell whether a particular move took a person across states or was just to a new town or city within a state.

Figure 4.1 shows a random sample of 20 individuals, from both generations, who reported engaging in any of the 8 acts of political participation asked about in the Political

Socialization Study. Each of the 20 panels displays the participation series and residential mobility series for a single individual. A vertical line is drawn above the dotted line each time a person reported participating in one or more campaign activities. Similarly, each move is indicated by vertical lines hanging below the dotted line. Person 82, for example, reported moving in each of the years between 1965 and 1970 as well as in 1976, 1982, 1987, and 1992; he also engaged in at least one of the 8 types of non-voting political activity in 1969, 1972, each year from 1979 through 1982, and again in 1996.

Figure 4.1 does not appear to provide much purchase for understanding the dynamic relationship (if any) between residential mobility and political participation. Some individuals, like Person 11043 and Person 374 display patterns in which political participation appears to wait until moving stops. Others, like Person 1638 or 762 seem to participate in politics whether or not they are moving. What this figure does show is that both residential mobility and political participation are stochastic processes occurring across the lives of individuals — and thus must be thought of and modeled as such.

4.2 Current models of political participation

4.2.1 What do these models have to say about mobility?

The effect of residential mobility occurs, according to scholarly consensus, because of disruptions to "social connectedness."[6] New arrivals are supposedly less likely to know their neighbors (or others in their new towns), and this lack of acquaintance with others also prevents mobilizers from knowing the newcomers, and excludes them from mobilization efforts. That is, when people are new to a town, they know fewer people and fewer people know them. For example, Rosenstone and Hansen (1993, page 167) show that people who have lived in the same place for most of their lives are more likely to be contacted by members of a political party than those who are more recent arrivals.

Furthermore, moving imposes costs of time, energy, and money on individuals — and many individuals allocate political participation as a relatively low priority compared to dealing with more quotidian aspects of their lives. The analysis of "who participates" leads scholars to believe that the longer individuals live in one place, the higher the probability

[6]This is one area where the story about voting diverges from stories about other forms of participation. Highton (2000), for example, shows that merely switching houses within the same community is as powerful as changing communities in predicting turnout — implying that the need to register to vote is a much more important "cost" preventing turnout than any "social" costs or benefits affiliated with the length of time a person has lived in a place.

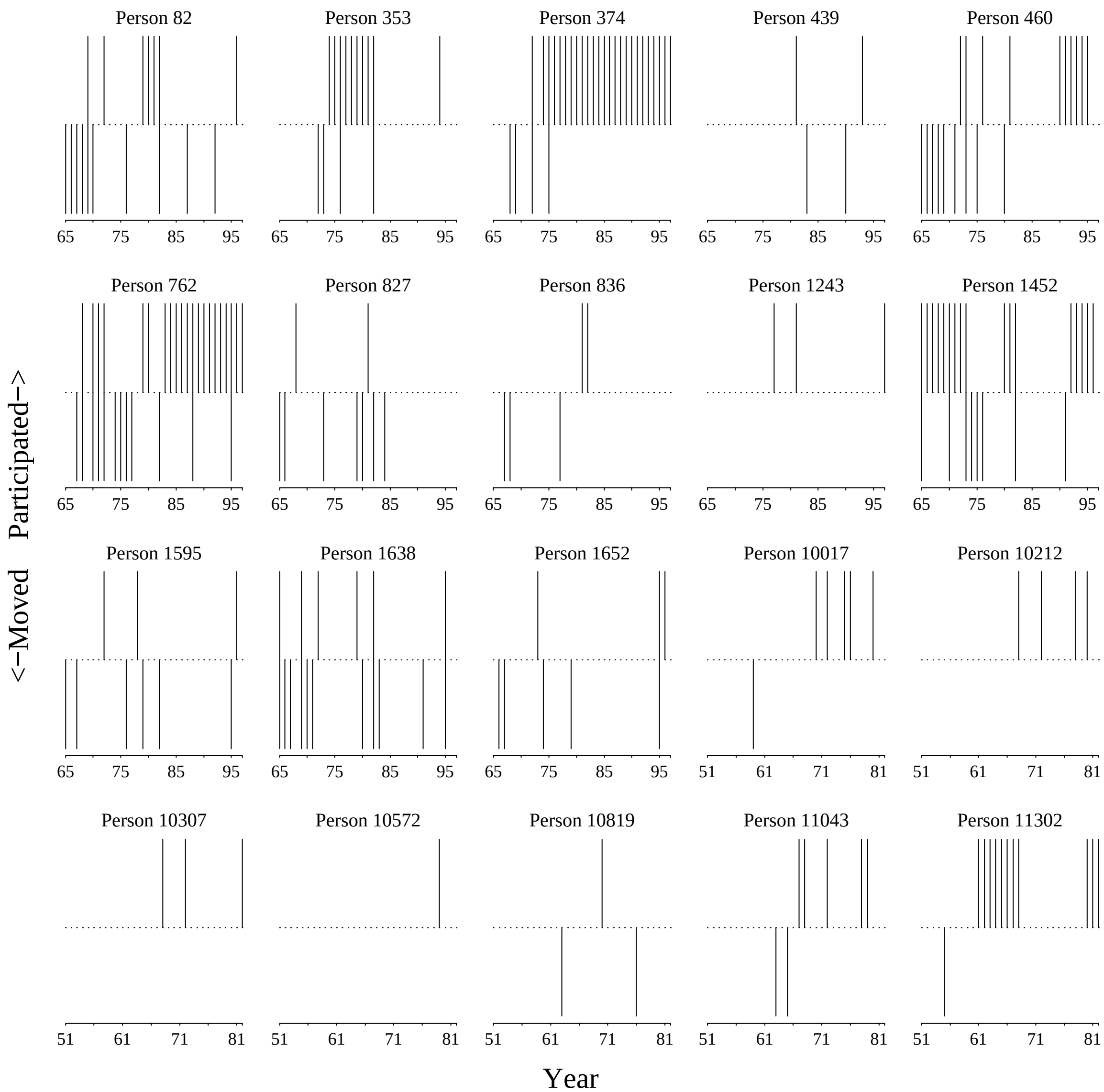


Figure 4.1: Individual Histories of Participation and Residential Mobility

Note: The x-axis runs from 1965 to 1997 for G2 and from 1951 to 1982 for G1 since members of the parent generation in 1965 were able to report both participation and residential mobility for the years preceding 1965 and they were not reinterviewed in 1997.

that they will act at a given moment in time.

Figure 4.2 summarizes these expectations, plotting the expected probability of participation (on the y-axis) against the amount of time a person has lived in a new place after moving, holding constant all of the other attributes of a person or place or time (i.e. life-cycle or period effects) that might otherwise influence the probability of participation. I

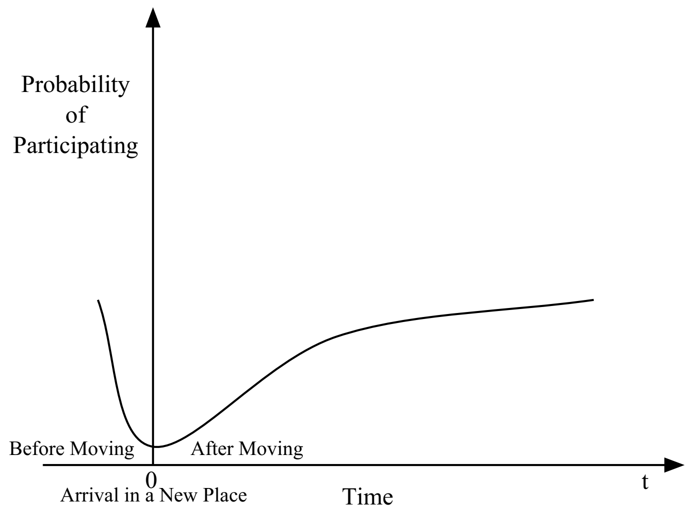


Figure 4.2: A Dynamic Hypothesis from Cross-Sectional Findings

have used "time" instead of years, months, or days for the x-axis here, since previous work does not suggest precisely how fast this process is apt to take. The hypothesis suggests that, immediately prior to moving to a new place, the probability of participation ought to drop as social ties are severed and other costs (in terms of time, money and information) are imposed. I have drawn it to begin before people arrive in a new place, thinking that when individuals prepare to leave their old place (or while they are traveling between the two places), their latent propensity for participation probably decreases as preparations for the move increase. Thus, it is possible to think of time=0 as marking the moment at which individuals arrive in their new home. After a time (months perhaps?), the disruption to the participation trajectory caused by the move ought to end, and a recovery is expected to begin — as the individual becomes known to mobilizers, comes to know more about the intersection of her interests and conditions in the area, and as she has a bit more time freed from the rigors of moving. Current hypotheses do not speak directly about "carry-over" of civic skills or resources from place to place, so the trajectory slowly returns to the initial level, which might be seen as the long-run equilibrium level of participation for a given person over his or her life.[7] And, of course, if a person ends up staying in a new place longer than the old place, one would expect the probability of participation to continue to

[7]This figure is drawn with the x-axis scaled in abstract "time". The actual data analyzed in this dissertation has a temporal resolution of 1 year, no finer. Evidence I present later will suggest that, to the extent residential mobility does follow this patter of disruption and recovery, it may play out much more quickly, and ought to be looked at on the scale of months rather than years.

increase until it is higher than the maximum probability of participation in the old location.

Figure 4.2 is a generalization about participation across the 8 acts I examined here. Of course, not all types of participation ought to react equally to the expected disruption of moving. As I have mentioned before, scholars have very strong priors that voting ought to be much less likely to occur right after a move to a new location. But what about other common types of campaign activity such as volunteering for campaigns, donating money, attending campaign fundraisers and other events, or displaying lawn-signs, buttons, or other symbols of support for a candidate? There are institutional barriers like voter registration preventing participation any of these activities. However, all of them ought to be susceptible to influence by social networks. Certainly showing public support for a candidate or issue via signs and buttons requires some understanding about how other people who see these symbols will react to them; less knowledge about local norms would seem to make public political stands a bit more of a risky business (at least from the point of view of a newcomer considering the creation of a new reputation). Donating one's time to a campaign, via attending events, or more intensively via volunteering as a campaign worker, tends to occur in a person's local area and thus local political knowledge (and being known by local political operatives or at least being within a social network easily reached by such operatives) ought to be necessary conditions for such participation. Huckfeldt (1979) distinguishes between "individually based" participation and "socially based" participation. The idea here is that some activities are more easily (or likely) to be done by individuals by themselves (such as writing a check to a campaign), while other activities are more easily (or likely) done in groups (such as going to a campaign rally). It is thus "socially based" activities which ought to be more influenced by the length of time that a person lives in a place compared to "individually based" activities. Of course, this distinction clearly does not "carve nature at the joints" but it does suggest that one ought to see that the probability of participation in certain activities is more (or less) responsive to the integration of individuals within social networks.

What does it mean to say that social networks affect political participation? At the base of such explanations of political participation is the appreciation that political activities tend to occur in groups, that information about these activities is communicated by word of mouth between people who know one another, and even that a person's understanding of what activities or policies matter to them depends at least in part on social interactions. Knowing others may matter for political participation, but being known may be equally important; having politically active people aware of an individual ought to increase the chances of that individual receiving appeals for help (in time or money) from such activists.

In addition, Huckfeldt (1979), among others, points out that when it comes to political participation, people cannot be thought of as acting in isolation, but that "People also respond to political events, cues, and opportunities which are specific to a given environment" (579). Thus, there are a variety of reasons to expect that moving to a new place ought to affect not just voting but other forms of participation, and that these effects ought to vary depending on the extent to which these acts make it possible for individuals to act alone versus with other people.

Among the acts analyzed in this dissertation, I would expect that donating money and writing letters to the editor would be the two least influenced by social networks in one's local geography — at least both are acts typically done alone. *Community Work* is clearly an act which ought to have the strongest relationship with local community connections, with perhaps the range of other electoral activities (wearing buttons or displaying signs, other campaign work (like volunteering), and attending campaign meetings and/or rallies) also having some such relationship, only less strong. I would expect that contacting officials and demonstrating might be in the middle — both of them are activities that are explicitly mobilized by national level interest groups (thereby diminishing the importance of local social connections), but both also might occur and/or be targeted at the local level (thereby enhancing the important of local social connections). Following this line of reasoning, one should not expect to see as strong effects of residential tenure on donating money and writing letters to the editor as for working with others in the community or the other time-based electoral activities, with contacting and demonstrating in between.

Since the point of this chapter is to look at a relationship between two variables (participation and mobility), Figure 4.2 is drawn as if one were able to ignore many of the other aspects of a person's life that ought to influence their participation trajectory. Also, these pictures are meant to display the "typical" trajectory — it is probable that certain subgroups of the population have different shapes. For example, mothers who are not in the paid labor force may experience much higher participation related costs from a move than men without children — while the men may immediately begin to rebuild civic status, and continue to gain civic skills from their jobs, the women may have just lost civic status and have no way to regain it easily.[8]

[8]The idea that one may both gain and lose civic skills and status over time seems worthy of future

The basic message to take away from the figure is that past theory about the relationship between mobility and political participation should lead us to expect that the probability of participation ought to first decrease prior to a move and then increase as a function of the amount of time a person has lived in one place after moving.

4.2.2 Microfoundations

The hypothesis depicted in Figure 4.2 suggests that the probability of participation for a given individual, i at time, t, should be modeled as

$$\Pr(y_{it} = 1) = f(\text{Tenure}_{it}) + \sum_{p=1}^{P} \beta_p x_{pit} + u_i + e_{it}. \tag{4.1}$$

In this equation, the dependent variable y_{it} can take on values of 0 (meaning no participation at time t) or 1 (meaning some participation at time t); "Tenure" stands for the number of years a person has lived in a new place after moving; the x_{pit} terms represent other variables that might confound the estimates of the relationship between Tenure and Participation; u_i represents the mean level of participation across time for the individual (the long-run equilibrium for that person); and e_{it} represents the extent to which the variables in the model, for person i and time t, do not fully capture the dynamics in the dependent variable.

Figure 4.2 suggests that $f(\text{Tenure}_{it})$ should include at least two "knots" or places where the line can change direction; in other words, there are at least 3 pieces of line — one piece before $t = 0$ (when a person leaves an old place) descending down to the lowest point at $t = 0$ (to represent the disruption presumably caused by the move); another piece climbing back from $t = 0$, the lowest point of the disruption, to the equilibrium level (to represent the recovery of social connections provided by additional years of residential stability); and a third piece that continues relatively flat once equilibrium has been reached. That is, our theoretical model of the function linking Tenure to participation ought to allow for at least 3 distinct relationships depending on the number of years a person has lived in a place. One way to write such a model would be:

exploration.

$$f(\text{Tenure}_{it}) = \begin{cases} \beta_1 \text{Tenure}_{it} & \text{if } K_1 \leq t < 0 \\ \beta_2 \text{Tenure}_{it} & \text{if } 0 \leq t < K_2 \\ \beta_3 \text{Tenure}_{it} & \text{if } K_2 \leq t \end{cases} \tag{4.2}$$

where K_1,0 , and K_2 are the locations where the curve changes direction;and, since the data gathered here are annual, Tenure_{it} stands for the number of years an individual i has lived in a new place (after moving) at calendar year t (since different individuals have different values of Tenure for different calendar years). Tenure obviously is negative for years before arrival in a new place. In theory, β_1 would represent the disruption caused before moving to a new place, β_2 would represent the recovery caused by the increasing social connections acquired from living in a new place; and β_3 would represent the long term level of participation reached by an individual after the recovery period is over. Equation 4.2 has one flaw, which is that it doesn't represent a smooth trajectory — it doesn't make sense to think that individuals will switch from disruption to recovery to equilibrium instantaneously. Figure 4.2 shows these transitions occurring smoothly — also implying that the probability of participation at which one process ends ought to be the same place where the next begins. When I move from this theoretical model to the empirical one, I will be careful to ensure both smoothness and at least three separate slopes.

4.3 A Growth Curve Model of Mobility and Participation

Although some smooth version of Equation 4.2 clearly captures the predominant theory about the individual-level dynamics of the relationship between political participation and residential mobility, how should one approximate this model and assess its fit against the Political Socialization data?

The requirement for a piecewise approach rules out using simple polynomials to represent the function, since polynomials include the entire range of data in their fitting. That is, fitting a collection of squared and cubed terms in the context of a linear model might approximate the hypothesized shape of $f(\text{Tenure})$, but it would not actually produce a piecewise fit — observations from greater than, say, K_2 would have power to change the fit to observations less than 0. This would be inappropriate since past literature has led me to think individuals pass through three distinct stages in response to a move.

If I were modeling only a single individual over time, or a single cross-section, a generalized additive model would work — and would provide the added benefit that I would not have to pre-specify the location of the knots in order to estimate the shape of the function (Hastie and Tibshirani, 1990). The data I am using here involves between 18 and 35 years of observations nested within roughly 800 and 900 individuals from G1 and G2 respectively, and thus, this model would require an extension to allow for (1) correlation of observations across time, within person; (2) unmeasured heterogeneity across people; and (3) the fact that the degrees of freedom are not the same for all variables — specifically that the time-varying variables will have more degrees of freedom than time-constant variables. These requirements point me toward embedding a flexible representation of the function inside a mixed-effects model — also known as a multi-level, or random coefficients, model — which deals appropriately with the nested and dependent structure of the data.[9] A standard formulation of a mixed-effects model for change over time, with residential tenure restricted to a linear effect could be written as follows:

$$\Pr(y_{it} = 1) = \beta_{0i} + \beta_{1i}\text{Tenure}_{it} + \beta_{2i}X_{it} + e_{it} \tag{4.3}$$

$$\beta_{0i} = \gamma_{00} + \gamma_{01}Z_i + u_{0i} \tag{4.4}$$

$$\beta_{1i} = \gamma_{10} + u_{1i} \tag{4.5}$$

$$\beta_{2i} = \gamma_{20}. \tag{4.6}$$

Substituting (4.4), (4.5) and (4.6) into (4.3) yields:

$$y_{it} = \gamma_{00} + \gamma_{01}Z_i + \gamma_{10}\text{Tenure}_{it} + \gamma_{20}X_{it} + (u_{0i} + \text{Tenure}_{it}u_{1i} + e_{it}) \tag{4.7}$$

$$\text{with: } (u_{0i}, u_{1i}) \sim N(\mathbf{0}, \mathbf{T}) \text{ and } e_{it} \sim N(0, \mathbf{\Sigma}) \tag{4.8}$$

Equation 4.3, often known as the level-1 equation because it represents the process at the lowest level of nesting within the data, shows how Tenure, and another time-varying variable, X (such as income), influence y, across time within a person. Equation 4.4 specifies that the mean level of participation within an individual might vary as a function of Z — a characteristic of an individual that is constant over time (such as gender) — plus some random error which allows the intercept of the level-1 equation to vary across individuals — capturing aspects of people which may lead them to be more or less participatory that have not been measured in the model, and thus accounting for the possible heteroskedasticity

[9] See Pinheiro and Bates (2000); Raudenbush and Bryk (2002); Snijders and Bosker (1999); Kreft and Leeuw (1998) for more details about these models, and Diggle, Liang and Zeger (1994) for specific information about longitudinal versions.

induced by the clustering of years within person. Equation 4.5 specifies that the extent to which tenure influences participation may also vary across individuals — this expresses the idea that the variance of participation may change as people spend more time in a place, also that there may be idiosyncratic aspects of individuals that cause them to react differently to residential stability and mobility that are not measured, that can be taken into account here. In addition, allowing both the intercept and slope in the level-1 model to vary allows one to estimate the extent to which those people who tend to have high equilibrium values of participation are also those who tend to recover more (or less) quickly from the disruption of a move.[10]

Equation 4.6 merely notes that the time-varying variable X ought not to have appreciable variance across individuals (or if it does, that variance does not matter for the estimates of the effect of Tenure). That is, the effect of X is seen to be constant across people. The term in parentheses at the end of (4.7) shows that the error variance of the equation consists of three parts: an error associated with the intra-individual variation in mean levels of participation (such as measurement error) (u_{0i}), an error associated with intra-individual variation in mean slope of Tenure (u_{1i}), and an error associated with inter-individual variation in participation not explained by the model e_{it}.[11]

In addition, the measurements within individual are probably correlated from year to year, but each individual does not contribute the same number of measurements at the same points in time (after removing the periods in their lives before they had begun to move from their childhood home, or wherever they had lived for longer than 20 years in the case of G1). To account for this dependence (and to estimate its magnitude) I imposed a continuous AR1 (CAR1) form on the covariances between years, within individual, such that

$$\text{cov}(e_{ij}, e_{ik}) = \phi^l,\ j \neq k,\ l \geq 0,\ \phi \geq 0 \tag{4.9}$$

where l represents “position” (which is akin to “lag” in the discrete time, AR1, formulation), and so the correlation between adjacent observations is considered to be constant across

[10]The random coefficients are assumed to be have been drawn from a multivariate normal probability distribution with some variance-covariance matrix $\mathbf{T}$ as shown in (4.8).

[11]Each of these error terms is required to vary according to a normal distribution, with mean zero, and variance that is estimated as part of the overall estimate process — as shown in (4.8). These assumptions may not be ideal, but alternative probability models proved too difficult to estimate in the context of the kind of functional estimation that these dynamic hypotheses require. The particular estimation strategy and software I use here is described in detail in Pinheiro and Bates (2000).

individuals, but dying out quickly as observations get farther apart.[12]

The mixed-effects model displayed in (4.7) shows Tenure having a simple linear effect on participation. This formulation is not necessary or appropriate given the hypothesized microfoundations explained earlier. In fact, within the framework of a mixed-effects model, I chose to use splines to represent the process of disruption, recovery, and equilibrium suggested by past literature. In particular, I represent the function for a particular person i as

$$f(\text{Tenure}_{it}) = \sum_{k}^{K} \eta_k(\text{Tenure}_{it})\beta_k. \tag{4.10}$$

That is, the influence of Tenure on participation can be broken into K pieces, and for each piece a function, $\eta_k(x)$, is evaluated. These functions are called "basis functions", and are required to join with one another at the "knots" which determine their endpoints. These functions are further required to have continuous second derivatives at each knot — thereby ensuring a smooth transition from one piece to the next. There are several common choices for these basis functions. For the analysis in this chapter I chose to use natural cubic B-spline basis functions — which requires that each $\eta_k(\text{Tenure})$ be linearly independent of the others, and that, beyond the range of the data, the function is linear.[13] The benefits of using this particular form of function include (1) that $f(Tenure)$ is estimated rather than having to be pre-specified other than the placement of the knots; (2) that the pieces of the functions are restricted to join with one another smoothly — another requirement of the theory (and a helpful feature in case the knots are specified a few years off in one direction or another); (3) that the spline bases are linearly independent, thus avoiding the multicollinearity problems that would be posed by simple polynomials, and allowing one to estimate the non-linear relationship between residential tenure and political participation within the framework of a linear model (allowing me to "control for" the linear effects of other possible confounds).

Substituting (4.10) into (4.7) produces the form of model that is used in this

[12]Not all models displayed within-person time-dependence — at least, likelihood ratio tests did not show significant improvements in fit for all of the models. However, for now, I have left all of the models in the CAR1 format since it didn't appear to be a worse fit — and just to make the models comparable. For more details about these auxiliary model statistics, such as the variance components and the time dependency parameters see Appendix B.

[13]For more information about regression splines see (Hastie, Tibshirani and Friedman, 2001, Chapter 5), (de Boor, 1978), Hastie and Tibshirani (1990, page 22-26), (Venables and Ripley, 2002, page 228-230). For an application in the context of a mixed-effects, longitudinal model, see Snijders and Bosker (1999, chapter 12) and Hastie et al. (1999).

chapter. In addition to correctly specifying the time-within-person structure of the data, correcting for temporal dependence, and reasonably approximating the functional form of disruption and recovery suggested by the theory, the model must also include those factors which one might imagine could confound the estimated relationship. To this end, I included several variables in the model (as Xs, in the parlance of model (4.7)). Two of these "controls" capture sources of social network status that may confound the effect of mobility: (1) whether an individual was a homeowner in a given year (since home owners may have stronger social ties to neighbors and have made larger commitments to a particular geographic area), and (2) the extent to which an individual attended church in a given year (since church attendance ought to provide both civic skills and civic status (Verba, Schlozman and Brady, 1995)). In addition, the income level of the individual for that year (normed to 1997 dollars) and the educational attainment of the individual for that year were entered to account for socio-economic status. Since it is well known that men and women participate at different rates, the gender of the respondent (constant over years) was also included. Finally, two other variables seemed important possible sources of confounding effects: (1) a variable indicating whether a presidential election was held that year (since examination of bivariate relationships showed that the electoral participation variables peaked during those years), and (2) a variable indicating whether the year was one during which a panel interview was held (to account for the propensity of individuals to report more participation closer to an interview year). I also restricted the analysis to only consider those moves that forced an individual to cross state lines. This decreased the total number of moves available for study, but it means that the moves should be larger disruptions to the lives of individuals — and therefore, I would expect to see the pattern displayed in Figure 4.2 more clearly. This restriction has the side-effect of excluding all of the people from the analysis who never moved. [14] Whereas I compared people of different educational statuses in the last chapter, the analysis here is within-person, comparing the participation of a person at one time with his or her participation at another time. Thus, no baseline category or control group is necessary.[15]

[14] I did compare the marginal effects of additional years living in a location on participation for people who had stayed in a place their whole lives ("stayers") versus the people who had moved into a place ("movers"). This analysis found that additional years living in a location had no effect on participation for the stayers (all else held constant). The results for the movers will be my concern for the rest of the chapter. In future analyses, I will also look at inter-community moves.

[15] I would like to examine the trajectories of "movers" versus "stayers" in the future, since that might reveal something about the "types" of people who tend to move versus stay. The analyses presented in

Age is one confounding variable that deserves special mention. We know that the young are more likely to move than the old. Census data show that this pattern has been true since at least 1950 (Fischer, 2002). The sample of Americans in the Political Socialization Study are not that different from the rest of the country when it comes to the differences between age groups in moving. Figure 4.3 shows that, among both G1 and G2, moving is a much more frequent occurrence among the young, with the proportion moving in a given year dropping precipitously from around 30% to 40% of the sample in their 20s, to nearly zero after age 40. In fact, the relationship between moving and age for the two different generations looks very similar.[16] In this figure, the heavy black lines show the smoothed proportion while the light grey lines that zig-zag around the black line are the actual sample proportions.

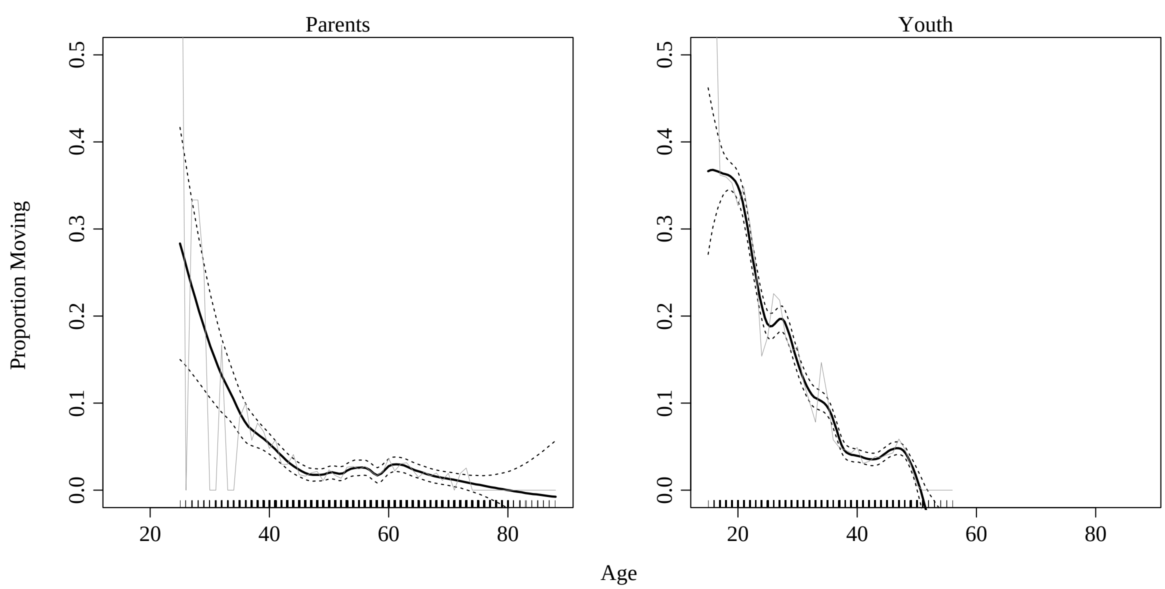


Figure 4.3: Residential Mobility as a Function of Age

Note: Proportion moving in a given year by age for two generations. Actual sample proportions are shown with the light grey line and the smoothed proportion is shown with the heavy black line (and pointwise 95% confidence intervals for the smoothed estimate are shown as dashed lines). The density of the age groups in the sample is shown by the jittered “rug” on the bottom of each plot.

Not only is residential mobility related to age, but age ought to be related to participation itself for number of reasons, including gaining experience in the political world

this Chapter retain the focus on the effect of a move, within the trajectories of individuals, and so cannot compare movers to stayers in this way.

[16] A significance test for the difference between the two smoothed functions will have to wait for the future.

or earning more money. Wolfinger and Rosenstone (1980, page 52) showed that the impact of residential mobility on vote turnout varied greatly by age, affecting the young much less than the old. In the analysis to come, I included age as a control in the model for G1. The model for G2, however, was nearly unidentified if age was allowed to enter in as a continuous explanatory variable. The problem here is that age=time for the G2 cohort. This is because, while G1 is age heterogeneous, G2 is not — nearly all of them are the same age (plus or minus one year). Thus, I ran two separate models for G2: one for the moments during which the individuals were less than 27 years old, and another for the moments during which they were older than 27. I chose 27 to enable more than a few of them to have both moved and lived in their new locations for more than 6 years.[17]

Finally, I estimated this model as a linear probability model. Although it is fashionable, and often correct, to constrain linear predictors' effects on binary dependent variables to run from 0 to 1 via some non-linear link function (such as the logit, probit, or clog-log link functions), the spline basis functions tend to constrain the fitted values from this model in a similar manner. In addition, the various corrections imposed by the mixed-effects modeling framework provide adjustments to the standard errors of the linear model akin to the "Goldberger" correction suggested by Aldrich and Nelson (1984, page 14-19). Another reason I made this choice is because political participation is a relatively rare event — and because King and Zeng (2001) show that binomial models tend to underestimate the frequency of events in such situations.[18] Also, the technology for estimating mixed-effects models for binary dependent variables is still very much a research topic in statistics, and the approximation methods currently available display much less numerical stability (and therefore trustworthiness) than the techniques for estimating linear mixed-effects models. In theory, Markov-Chain Monte-Carlo techniques do allow the analyst to characterize completely the posterior distribution of the set of parameters (in contrast to other methods which often approximate the posterior with normal distributions). In practice, however, they are not a panacea because (1) it is difficult to be completely certain when they have done an adequate job of sampling from the posterior, and (2) they are sensitive to choice of priors and starting values just as other methods are.[19]

[17] In future analyses I might be able to pool the G1 and G2 samples in order to get more leverage on the relationship with age.

[18] It is possible that I can address this problem in the future by thinking of individuals who never participate at all as censored, and thus use a mixture of a binomial and a mass-point as the distribution of the responses.

[19] I did replicate the linear mixed-effect model analysis using a "generalized linear mixed effects model"

4.4 Results

Although the analyses presented here pursue the hypothesis presented in Figure 4.2, it became cumbersome to estimate the function for years before a given move — since so many people moved so often. Thus, for this chapter, Tenure begins at $t = 0$. Thus, the expectation from that figure, is that as Tenure increases from $t = 0$, so should the probability of participation.

Figure 4.4 and 4.5 summarize the results of this analysis from the model I described above for G2, and Figures 4.6 and 4.7 summarize the results for G1.[20] The thick, bold line, in each panel displays the population average predicted probability of participating as it changes across the years that a person stays in a location after a move.[21] The predicted probabilities were created holding all of the other variables at their mean (or most common value for dichotomous variables such as sex) — meaning, in this case the predicted values represent the values that would be predicted for a woman, with average income, with average educational attainment, who is not a home owner, does not attend church, during years that are neither presidential election years nor interview years. Of course, any changes in these variables would not affect the shape of the curve, but would only shift it up or down. I do not plot confidence intervals for these curves to avoid cluttering the graphs — I will discuss the uncertainty associated with the estimation of these curves below. Given the hypothesis at stake here, it is not the actual levels of the predicted values that matter, but the shape of the trajectory. The vertical dashed lines in each panel show the locations of the knots that I specified for the different pieces of the spline. I chose the locations at 3 and 6 years to cover the early years of residential tenure, during which the disruption and recovery process

(aka GLMM) using a binomial family and a variety of link functions and found that the shape of the functional form for tenure was replicated, but the fitted values were restricted within a very narrow range (0 to .02) due to the problems the binomial model has with rare events. In the end, the linear probability model I present produces values that are more in line with the actual means and bivariate results (using "plain", single-level, glm and gam models).

[20] Detailed results may be found in Appendix B. I do not present coefficients here, since the coefficients of interest, on the spline function, are not interpretable as mapping directly on the individual intra-knot pieces, but rather only in combination. Thus, the graphs do a much clearer job of displaying the results of this analysis than the coefficients.

[21] The models for G2 restricted the number of years of tenure to be ≤ 15 for the over 27 age analysis and ≤ 9 for the under 27 age analysis since age and tenure become ever more closely related as tenure increases, such that only 35 year olds have a tenure of 17. This is another reason for the piecewise fitting procedure — such that the lack of data for the piece from 12 to 15 years of tenure doesn't have much impact on the estimates for the other pieces. The models for G1 restricted tenure to run from 0 to 20, assuming that after 20 years in one place, no additional change would be observed (and because, predicted probabilities from models I ran allowing 60 years of residential stability among G1 confirmed this assumption and also hid any changes in the earlier years, when the disruption/recovery process is supposed to occur).

ought to play out.[22]

Across Figures 4.4 to 4.7, the only apparent demonstration of a disruption-recovery process is for monetary donations, among G2, when they were less than 27 years old.[23] The disruption in Figure 4.2 is BEFORE the move, which is not portrayed in Figures 4.4 to 4.7. For the parent generation, both *Community Work* and *Donate Money* show very slight disruption-recovery patterns. F-tests for the hypothesis that the entire spline has a slope of zero only allowed for rejection of this hypothesis at any conventional significant level for G2, and then only for the over-27 analysis, for *Community Work* (p=.01), *Contacting Officials* (p=.01), and *Donate Money* (p=.02). These three cases are typified *not* by any initial dip followed by a recovery, but rather by relatively steady trend of increasing participation with additional years of tenure. These results suggest that the process of disruption and recovery may be occurring as hypothesized in Figure 4.2, with the disruption beginning before $t = 0$.[24]

While previous literature indicates that for some acts of participation, integration into a community is likely to be more important (e.g. attending meetings, working in the community, or contacting officials), the results here are mixed. *Community Work*

[22] The estimation of knot locations for spline, or other piecewise or changepoint models, is a difficult non-linear least squares problem that increases in difficulty with the number of knots one is trying to discover. When I ran a non-linear mixed effects model (and non-linear least squares models) I discovered great numerical instability in knot estimates (when any were supplied), which was a hint to me that perhaps there were no changepoints for many of these variables (that is, it makes little sense to say, "estimate a knot location for me" when the function is truly flat!). As a model check, I did run a grid-search procedure for a one and two-knot linear piecewise B-splines from linear mixed effects model fits: for the single knot search, log-likelihood was maximized by knots at 7,2,11,11 years for buttons, money, other work, and rallies respectively; the double knot search yielded the following pairs (4,7),(2,12),(6,11),(6,9). As the results of the two-knot spline models show, for many of these variables, there are no changepoints to estimate — but if there were, the grid-search suggested that knots at 3 and 6 years of tenure were at least in the ball park.

[23] I'm not paying attention to the steep drop-offs predicted for nearly all of the acts for G2 among the under-27 group since the data from 6 to 9 years of Tenure for those people are very sparse. I take the drop-offs to reflect the general tendency for individuals not to get involved in politics — and thus, as the sample size decreases, the non-participators become a greater proportion due to the asymmetry of the binomial distribution.

[24] Unfortunately, the joint hypothesis tests do not provide information about the intra-knot slopes separately. I did assess the individual slopes for each segment separately. To do this, I re-ran all of the models, substituting a piecewise linear specification for the natural cubic B-spline. I still required the slopes to share intercepts — such that where one segment ended would have to be were another segment began — but this time, the linear mixed-effects model coefficients could be directly interpretable (and testable) as referring to specific segments. Using these models, I then used Wald tests for the specific linear combinations of coefficients governing the relationship between residential mobility and participation for each segment (0 to 3 years, and 4 to 6 years). I did not do the tests for the 7+ years segments since any changes after 7 years cannot be seen as part of the disruption that residential mobility is supposed to cause in the civic engagement of individuals. I do not present the tests here, because, in the end, I decided that it was misleading to show uncertainty estimates for one set of estimates as if they ought to apply directly to another set of estimates.

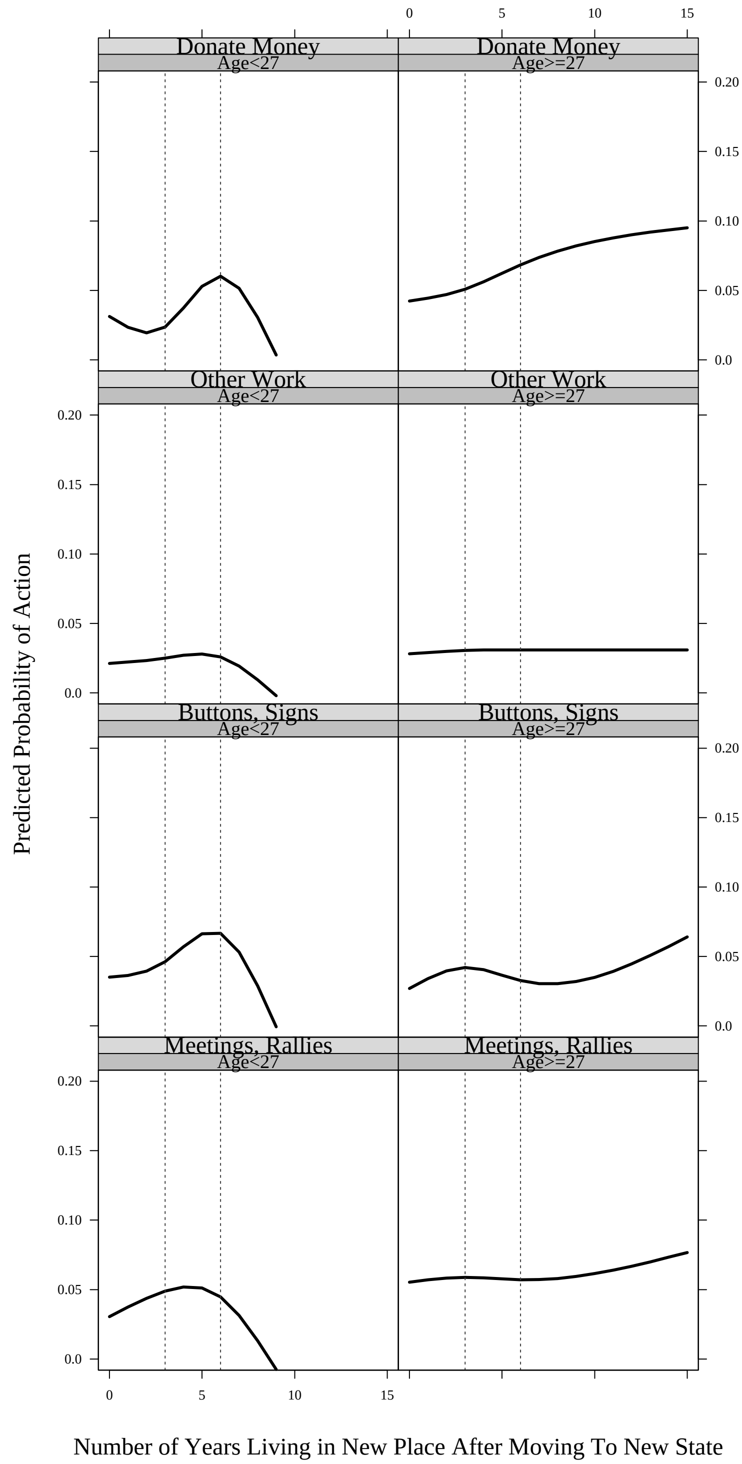


Figure 4.4: Estimated Participation Trajectories After Moves: Electoral Activities for G2

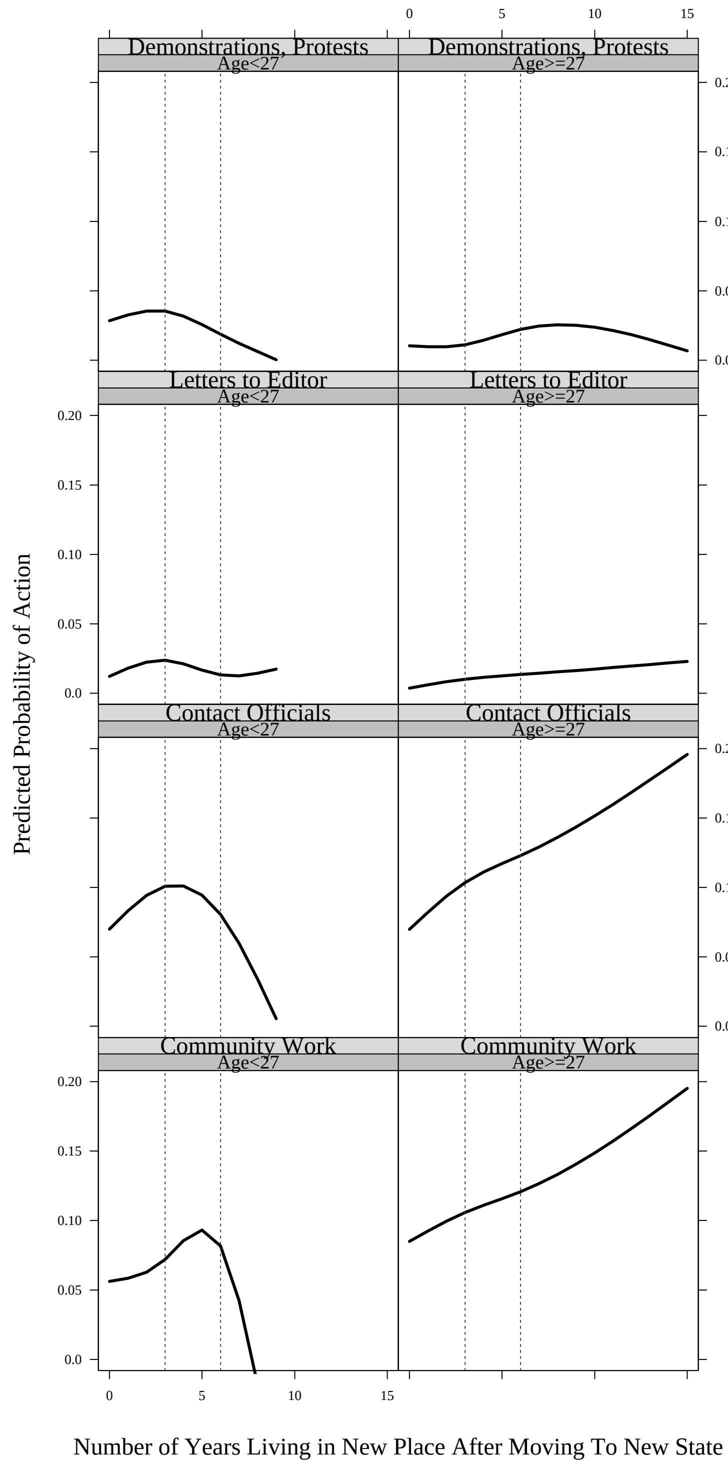


Figure 4.5: Estimated Participation Trajectories After Moves: Non-Electoral Activities for G2

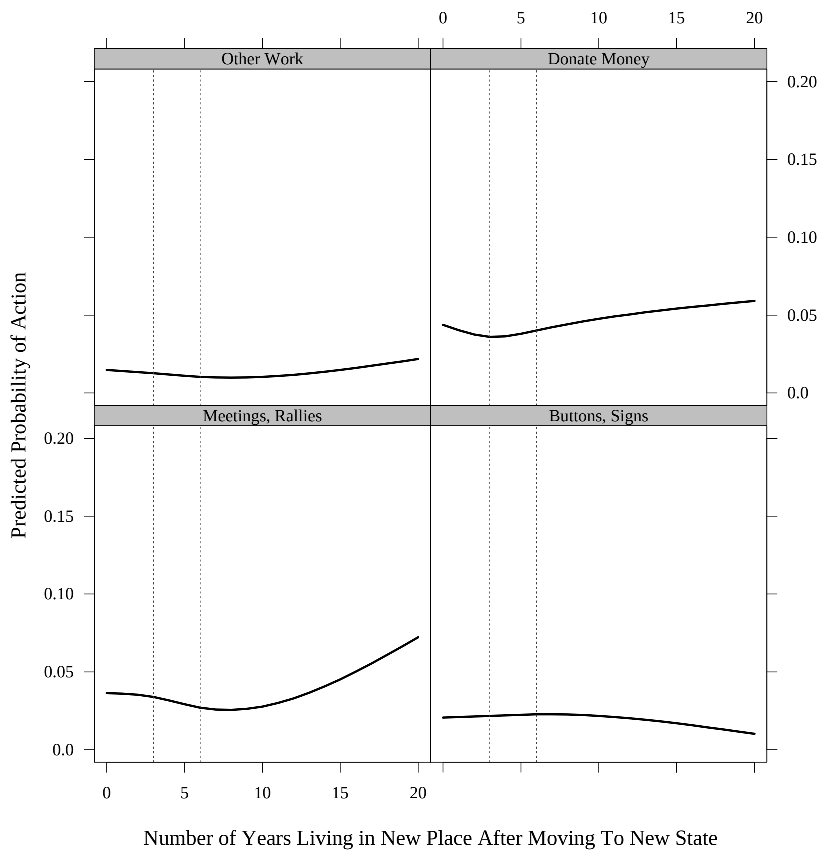


Figure 4.6: Estimated Participation Trajectories After Moves: Electoral Activities for G1

shows some effect of residential mobility and tenure for both generations, but so, too, does donating money. Nevertheless, perhaps, for both generations community recognition (e.g. having one's name on local mailing lists for solicitation of funds, along with recruitment to attend meetings and work for the community) is sufficient, and community integration is not necessary.

The predominant pattern in the data analysis is that of slightly increasing relationships between residential mobility and political participation in the first few years after a move to a new place (and a new state). For acts that *a priori* might be assumed to be especially sensitive to the disruption caused by moving (such as *Community Work* and *Contacting Officials*), the analysis show either early increases which appear to taper off over time or steady upward trends.

It is possible, of course, that these finding are idiosyncratic to the dataset containing the members of a special cohort — the High School Class of 1965 and their parents.

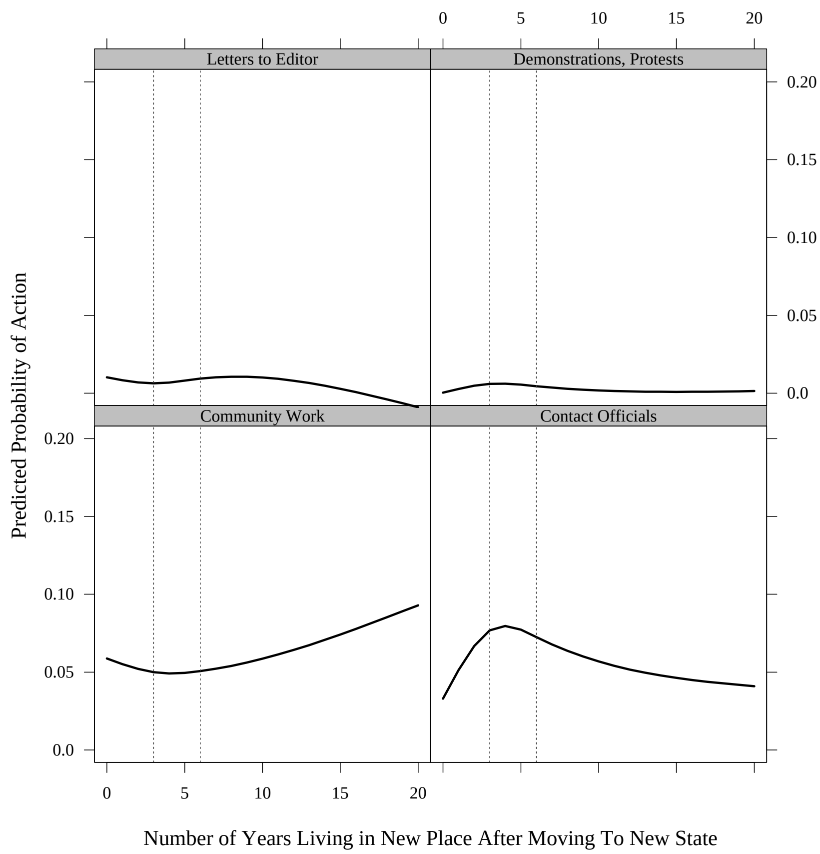


Figure 4.7: Estimated Participation Trajectories After Moves: Non-Electoral Activities for G1

In order to check these results, I replicated the analyses using the Cumulative File of the National Election Studies for the years 1970 to 2000. The strength of the analysis of the NES data is that both year and age can be controlled for — since each NES includes a wide diversity of people of different ages, and the same questions have been asked over time. Of course, the weakness, compared to the Political Socialization study, is that all of the inferences must be across people rather than across time, within individuals. Given the wealth of data in the NES, in this analysis I also allowed for 4 knots, at 3,6,9, and 12 years.[25]

Figure 4.8 is the cross-sectional version of the previous figures using the same kind of spline, but this time estimating the model using a "plain-old" binomial generalized linear model (glm) with a logit link function (aka a "logit" model or "logistic regression").[26]

[25]As control variables, I included nearly the same set as for the Political Socialization analysis; namely home ownership, church attendance, year, age, gender, income, education, and a dummy indicating whether it was a presidential election year.

[26]The NES analyses were restricted to those individuals who had lived in a place for 15 years or less —

Again, monetary donations is the only variable displaying an initial dip in Figure 4.8.[27] The other variables display a rather flat relationship across individuals. Unfortunately, the National Election Studies Cumulative file does not contain variables measuring non-electoral participation over time.

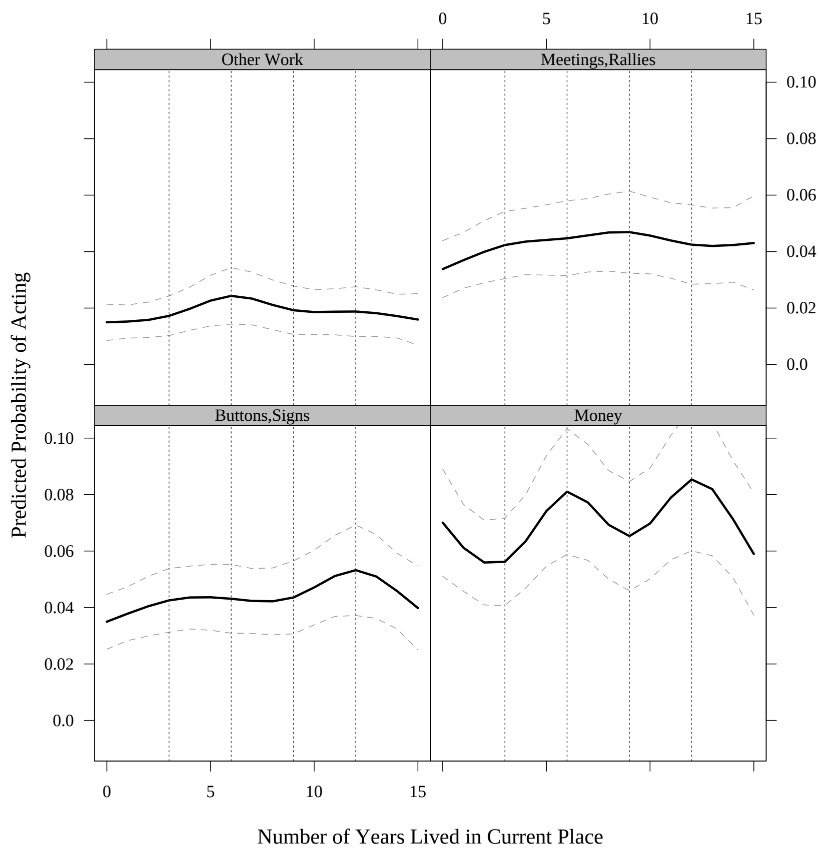


Figure 4.8: Predicted Participation Trajectories after Moves (NES 1970-2000 Data)
Note: The dotted lines represent pointwise 95% confidence intervals for the predicted values from the logistic regressions.

So, it is clear that the understanding of residential mobility derived from the cross-sectional research is not holding up strongly when examined using data from both

thereby capturing more "movers" (i.e. people who had moved into the current location from somewhere else) and fewer (if any) "stayers" (i.e. people who had lived in that location all of their lives). The NES question also only asks about length of residence in the current "city/town/township/county", and so, many of the respondents in my analysis are folks who have not left their state. In addition, the NES analyses only included those people who were interviewed face-to-face, to enhance comparability of responses across years.

[27]Tables and details of this analysis may be found in Appendix B.2.

the NES and the Political Socialization study. The dominant hypothesis, however, is still intuitively plausible (that moving, especially long distance moves) ought to disrupt one's social connections — social connections which really ought to have important consequences for the political participation of individuals. What could be going on?

Besides needing to take into account time before a move, these relatively flat trajectories — or at least curves that do not display much in the way of disruption due to moves crossing state lines — may not be the best way to test the hypothesis displayed in Figure 4.2. In fact, when past researchers have said that "tenure affects participation", what they mean is that "social connectedness affects participation" and "social connectedness is disrupted after moves, and grows with tenure." The idea would be that the influence of tenure on participation differs among groups of more or less socially connected people. Take home ownership and church participation as examples of activities that may speed up or slow down the integration of an individual into local social networks (which place them "in the path" of mobilizers, provide them with information and interests to protect, etc., all of which either decrease the costs of participation, or increase the costs of non-participation). The idea would be that the influence of tenure on participation differs among groups of more or less socially connected people. Therefore, individuals who move to a new place and buy a house may not experience the disruption hypothesized above — or their disruption may occur less severely, have a shorter duration, and their recovery more quick — than those individuals who move to a new place without owning a home. A parallel argument could be made for church goers. Frequent church attendance may allow new arrivals to a place to quickly build the social networks that have been disrupted by a move. Figure 4.9 suggests how this process might occur. Individuals with many, or strong, social connections might experience a smaller disruption which ends more quickly, than individuals with fewer, weaker social connections.

If this is the case, then perhaps the quick recovery times of the highly-socially connected individuals are causing the early parts of the curves estimated for the whole sample to look flat. Unfortunately, initial analysis allowing for different trajectories for home owners and church goers do not, as yet, support this hunch.[28]

Another interpretation is that the disruption-recovery process occurs much faster than annual data can detect. That is, whether or not an individual buys a house or joins

[28]The results of this preliminary analysis are presented in Appendix B.3.

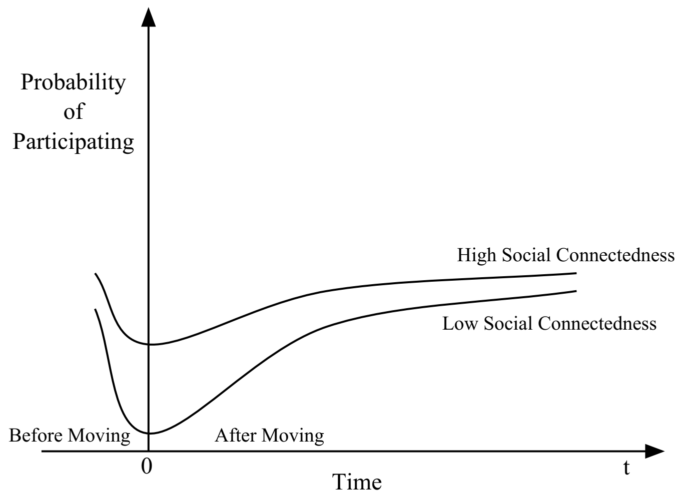


Figure 4.9: A Hypothesis About Social Connectedness, Residential Tenure, and Political Participation

a church, people may tend to suffer a downturn in their participation and then begin to experience a recovery all within the first year, or two, of their new residency.[29] If this is so, then all of the upward sloping lines for the 0 to 3 year segment might in fact be capturing the recovery phase for a disruption phase hidden within the first year or so of residence.

This second interpretation also makes sense in light of findings by previous scholars studying vote turnout. The findings presented by Wolfinger and Rosenstone (1980, page 52) among others, do suggest that the impact of mobility may occur over shorter periods of time than years — such that the disruption/recovery process could play out within a single year or a few years.[30] If this is true, then, the findings presented here can be interpreted as "smoothing over" the actual dynamics expected by the hypotheses. We will have to wait to analyze data with a more fine grained temporal resolution to understand if the hypothesis does hold, but just at a level beyond our current ability to perceive (the social scientific equivalent of "microscopic").

[29] I just discovered that the Citizen Participation Study does ask about the number of months lived in a town and neighborhood among other measures of residency. In the future, I will look at that data to get a better sense for the time-scale on which these dynamics are really occurring — or at least rule out the hunch that my lack of results is really just a function of the coarseness of the data.

[30] Wolfinger and Rosenstone (1980, page 52), for example, predict a 19 percentage point negative difference in probability of voting for individuals who had lived in the same place for 1-2 years compared to those who had lived in the same place for 10 or more years, and that individuals who had lived in their current location for 4 months or less were predicted to be 23 percentage points less likely to turnout than people who had lived in the same location for 10 years or more, compared to 9 percentage points among those who have lived in their current location for 3-5 years.

One possible explanation for this combination of findings is that participation in electoral activities may not be a truly local-oriented activity. If people (from 1965 to 1982) mainly gave money, volunteered, attended rallies, etc. via mobilizers who found these individuals via national level data sources, then, whether or not local level social networks were disrupted by moving should not matter for participation. Although this is a possibility, I think it is not plausible, given the fact that at least some of the relationships I've displayed are not zero. Rather, the relationships are not monotonic (which is what we'd expect given the hypothesis presented earlier), but the form of their non-monotonicity is not what would be expected, nor is the form consistent or varying in easily recognizable ways across acts and specifications.

4.5 Conclusion

This chapter can be seen as providing evidence that the effect of residential mobility on non-voting participation within the lives of individuals is different from the effect of residential mobility previously discovered to exist between individuals, mainly, on the act of voting. Even though I do not analyze voting here, I do not doubt that residential mobility has a large effect on vote turnout (largely via the need to re-register to vote). But, the analyses I present here suggest that it is not a major influence on non-voting participation — at least, that it doesn't appreciably disrupt all kinds of political participation when it occurs.

This might mean that social networks do not in fact have much to do with political participation (if, indeed, moving across state lines does disrupt social networks). It might also mean that moving across states lines does not disrupt social networks to the extent we previously might have imagined — and that the link between political participation and social networks is quite tight.

Of course, one might ask whether it would make sense to expect to see effects for all of the 8 activities analyzed here. Over the historical period from 1965 to 1997, many kinds of political participation were mobilized from a national level — such that being on a mailing list for a political party or interest group could be a more important factor in determining monetary donations, say, than close connections with neighbors or others in the same geographic area. That is, perhaps geography matters less for certain of these activities than others — or perhaps geographically constrained social connections matter less, in any case (compared, to say, social connections over long-distance phone lines, the Internet, and periodic visits).

The expectations about the differential impact of mobility across participatory acts that I laid out in the introduction to this chapter are not completely born out by the analysis for either generation or of the cross-sectional analysis of the NES. Where there are any non-null findings they exist for *Community Work*. However, the other activities that are typically done around campaign time at the local level do not tend to display strong relationships with tenure. And, *Contacting Officials* and *Donating Money* tend to be the only other activities that display any appreciable relationship with the number of years a person has lived in a new place after crossing state lines. It is possible that these null findings are a product of flaws in the data analysis (a possibility I detail below). It is also possible that the lack of relationship for campaign oriented activities merely reflects the decline in local level mobilization by parties over this period.

This finding about the lack of relationship with campaign oriented activities may have the simple explanation I just suggested. However, such a finding does not coincide with the well established relationship between mobility and vote turnout. I would expect that anyone who participates in a campaign activity to vote: to be so involved in the arena of electoral politics as to spend time or money on it, perhaps even on “get out the vote” drives, would seem to cause great psychological distress if one did not vote. In a way, I would see voting and campaign participation as pieces of an additive scale, in the way that simple arithmetic and calculus are: if you can do calculus, we know you can add and subtract; if you can donate money or time to a campaign, we know you can vote for that candidate. If this is sensible, then why don’t we see the strong relationship with residential mobility that we otherwise see with voting? If moving is disruptive, it ought to be even more disruptive of higher cost activities than voting (even considering that voting has relatively higher costs than we might otherwise think given the need to register).

These findings suggest that we really don’t understand the phenomenon of non-voting participation over time as well as we thought we did. For example, in this chapter I did not take into account the characteristics of the places people moved between. Does moving from an urban to a rural location mean something different for changes in participation than moving from a rural place to an urban one?

They also suggest that the data available here were gathered at possibly too coarse a level of resolution — that, to the extent that any disruption-recovery curve is visible, mostly it is only the recovery portion of the curve that is visible after time $t = 0$.[31] An-

[31] In the future, I will change the analyses presented here to account for the time before a move. Perhaps,

other possibility is that participation is basically random responses to events — and that knowledge and caring about these events is not correlated with amount of time living in a place

The empirical findings of students of vote turnout suggest that staying in one place enhances turnout and the theories of studies of social capital suggest that staying in one place is a civic good (and perhaps an individual good, a part of the "good life"). At the same time, economic theory would suggest that freedom to move (and actual moving) allows individuals and firms to take advantage of new economic opportunities (or to move away from economic problems). And, the class-mobility story of the American Dream (which some claim has lent great stability to our polity, for good or ill) is not at all about staying in the same neighborhood and participating in the governance of the community and nation, but about buying newer and bigger houses outside of town, in new towns, exploring new places and ways of life. The size of our country and its political and economic structures have enabled many people to pursue this vision of the good life (although, as I noted earlier, this number has not increased since the 1950s). My father's parents, children of homesteaders in Montana, have told me that they tend not to ask about the background of newcomers, that it doesn't really matter, and that "the deeds make the man." And, even my mother's parents, children of Italian and Sicilian immigrants living in ethnic ghettos in New Jersey, have stories about freeing themselves from the past — specifically from the web of social connections that so constrained their choices in the Old World. Up until now, the predominant social scientific findings have suggested that you can't have both the freedom celebrated by both the homesteaders and the immigrants and the civic life perceived to be at the heart of a healthy democracy. What this chapter suggests is that moving may not have the severe import that one might have thought. We have long known that homesteaders and immigrants are remarkably resilient people, so perhaps it should not be surprising in the end, to find that in the few cases where moving appears to have some effect on participation, disruption is rapidly followed by recovery.

even with annual data the disruption phase will yet be uncovered.

Chapter 5 Changing Contexts by Changing Social Status: Bearing and Rearing Children

Parenthood is an excellent example of a case in which the stakes for participation and resources of time and money change dramatically in response to an easily definable event in the lives of individuals. The association at the individual level within a cross-section was observed by Jennings in his article on parenthood and political participation (Jennings, 1979), and the influence of parenthood on aggregate civic participation was recognized by Rosenstone and Hansen:

> As the baby boom generation graduated from high school in the late 1970s, parents lost many of the concerns and social involvements that had motivated their participation in meetings at the local level. The removal of education issues from citizens' personal agendas caused participation in local politics to decline. (Rosenstone and Hansen, 1993, page 126)

Becoming a parent is a life-changing event in many ways.[1] The most immediately obvious changes involve the physical struggles of women to deal with the impact of bearing children on their bodies, followed closely by the physical challenges of caring for a baby whose activity schedule seems calculated to induce sleep-deprivation psychosis on even the most well-balanced of psyches. However, child rearing has other far-reaching impacts on a person's life, including new and larger economic needs, changes in relationships between partners and within families and communities. Bearing and rearing children, also, leads to changes in some of the factors that have been previously found to influence political participation. For example, men with children are more likely to be employed than men

[1]The sociological literature on childbearing is quite large. However, a few examples suggest that the commonsense about parenthood producing major changes in a person's life is true. For example, Taniguchi (1999) shows how early child bearing is associated with lower life-time wages and "the timing of childbearing significantly influences the extent to which this event shapes women's life chances."(1008). Also, McLanahan and Adams (1987) summarize many studies which show that the transition to parenthood is marked by unhappiness and stress caused by economic, time, social, and psychological demands on the parent (and especially on the mother).

who do not have children, and women with children are more like to be unemployed than women who do not have children (Burns, Schlozman and Verba, 2001, pages 310-311).

Therefore, this chapter addresses the question of what impact children have on the political participation of their parents across the parents' lives. If having and raising children affect factors known to be related to participation — time, interest in issues outside one's family, and socioeconomic status — does the fact of having children (and its timing) affect the trajectories of people's political participation, *ceteris paribus*.

5.1 Data

This chapter, like the others in this dissertation, relies on the data from the four waves of the Political Socialization study. The analysis here will only focus on the members of the Class of 1965. The reason for this choice is that it is only for this group that we can observe them during the first few years after their children are born — their parents already had at least one 18 year old child when they were first interviewed, so their immediate reactions to child-bearing and rearing are not observed.

From age 18 to 50 (1965 to 1997), over 83% of the Class of 1965 cohort had at least one child. About 83% of women became mothers in this period, and about 80% of men became fathers. Table 5.1 shows the distribution of family size across the respondents. By age 50, very few respondents had large families; no more than about 9% (or 84 people) had more than 3 children. Nearly as many people had 3 children as had 1 child, and a plurality (about 39%) had 2 children. The fact that such a large part of the sample were parents provides more confidence in the results of the within-person, across-time, analysis that I will display later in the chapter. It also allows me to look at the effects of having multiple children compared to just having one.

Table 5.1: Parenting by the Class of 1965 (1965-1997)

	Number of Children							
	0	1	2	3	4	5	6	7
Percent of Sample	16.68	17.22	38.50	18.40	6.42	2.14	.21	.43
Number of Respondents	156	161	360	172	60	20	2	4

Note: Up to 10 children were able to be recorded in the 1997 wave of data collection, while up to 8 children were recorded in the 1982 wave. Nobody in the combined 4 wave dataset reported more than 7 children at any time over the 3 decades of the study.

Figure 5.1 shows how this distribution of child bearing changes over time. The dark black portion of each bar represents the proportion of respondents who had their first child in a given year. The lighter gray sections of each bar represent the proportions having a 2nd, 3rd, etc. child in a given year. For example, no one reported having a 2nd+ child (let alone 3rd child) in 1965; and about 2% of the sample (or about 15 people) reported having their first child in 1965. The height of each bar shows the proportion of the sample reporting the birth of any child during that year. For example, a little more than 15% of the sample reported the birth of a child for 1971: for about 9% of the sample, this was the birth of their first child; about 5% had their second child that year; and about 1% had their third child that year (2 people had a fourth child that year, and 1 person had a fifth child that year).[2]

Figure 5.1 shows that at least some members of the Class of 1965 made the transition into parenthood (by having their first child) every year from 1965 until 1996. The bulk of births of first children occurred roughly before the individuals turned 35 (in 1982) and it peaked around 1971 and 1972. This means that an analysis of the transition to parenthood will involve individuals at a variety of ages — covering nearly the whole range in the dataset, with most between 18 and 35 years old. As individuals aged, more and more of them had 2nd and 3rd children. And child bearing mostly ended after about age 40.

The child-bearing behavior of the people in this cohort of the Political Socialization Study roughly mirrors the behavior of their counterparts in the general population. Table 5.2 compares the percent of women who do not have children in this cohort (born between 1947 and 1951 for the Current Population Survey of the Census and mostly during 1947 and 1948 for the Political Socialization study). The women in the Political Socialization sample appear to be slightly more likely to be childless at age 29 (in 1976) than the

[2]There is some error in my measurement of birth years of children. I created this variable using reports of children's ages at the interview years of 1973, 1982, and 1997. In between these interview moments many changes happened to the families in the panel. Thus, for some of these individuals, they were not present when one of their children was born because that child is a step-child (but step-children are not distinguished from biological children in this dataset). Other individuals may experience the first few years of a child's life before a divorce effectively ends their relationship with the child — or certainly significantly changes that relationship. Unfortunately the data do not provide easy and accurate ways to discern all of the moments that a child was living with an adult in between the panel years (for example, we do not know about boarding school attendance or foreign exchange programs). So, I have assumed that any child reported in a given panel year is exerting some influence on the parent for the years since the previous interview — at the ages attained by that child during that time. In the future, I may need to combine the analysis of children with an analysis of changes in marital status in order to deal more effectively with these measurement difficulties. For now, I felt it more important to provide a basic analysis of the relationship between parenting and political participation as a baseline for future work.

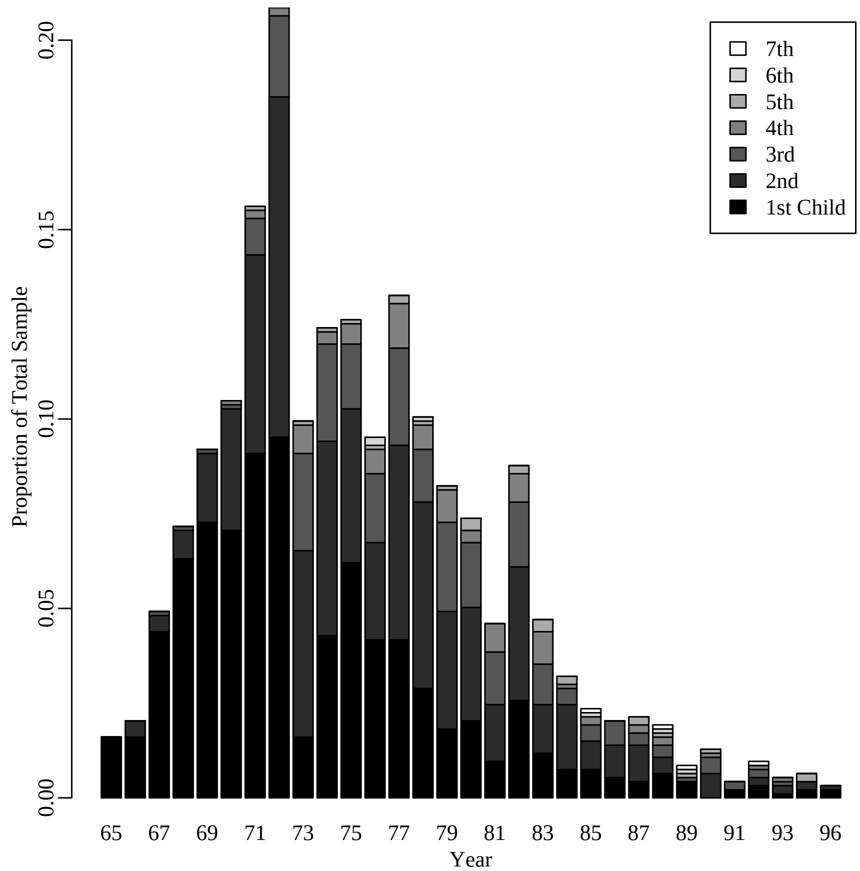


Figure 5.1: Child Bearing by the Class of 1965

comparable cohort (25-29 year olds) in the population. However, the two groups end up at nearly the same place by 1990, with 17.6% of the Socialization sample women having had no children by age 43, and 16% of the Current Population Survey sample of women having no children by age 40-44. Table 5.3 shows that the size of families eventually attained by the women in the Socialization sample (by age 43, in 1990) is quite similiar to the sizes attained by the women in the Census sample (by age 40-44, in 1990). Overall, the women in the Socialization sample are slightly more likely not to have children than the women in the Census samples, and slightly more likely to have smaller families (by around age 40). However, these differences are not very large, such that findings based on the child bearing and rearing of the Socialization sample ought not to be extremely different from the results one would find if one had followed the entire cohort in the population over time. Furthermore, this discrepancy makes sense because individuals with higher education tend to have kids later than those with low education, and the Political Socialization sample is skewed upward in terms of education.

Table 5.2: Percent of Women Without Children by Age/Year

	Year				
	1976	1980	1982	1984	1990
Socialization Sample	36.1	25.4	23.3	19.5	17.6
Census Cohort	30.8	19.8	14.4	15.4	16.0

Note: The Census data in this table come from Current Population Surveys. The "Census Cohort" was born between 1947 and 1951. Nearly 96% of the Socialization Sample was born in 1947 or 1948.

Table 5.3: Percent of Women by Number of Children Born by Age 40-44.

	Number of Children				
	0	1	2	3	4
Socialization Sample (age 43)	17.2	18.0	38.8	15.9	6.7
Census Sample (age 40-44)	16.0	16.9	35.0	19.4	8.0

Note: The Census data in this table come from the 1990 Current Population Survey.

5.2 How do children enhance or impede participation?

There seem to be at least four main ways to conceive of parenthood as influencing political participation. First, the most intuitively obvious effect that children have on the lives of their parents is to decrease the amount of time and money available for non-child-oriented use. Thus, to the extent that individuals spend time on politics pre-children, one would expect them to spend less time on politics during-children (and then more time on politics again after children are grown and out of the house). And, since women tend to spend more time on child rearing than men, one ought to expect to see this inhibiting effect to be more severe for women than for men. Evidence for the importance of free time on political participation, however, has been surprisingly weak. For example, neither Verba, Schlozman and Brady (1995) nor Burns, Schlozman and Verba (2001) found evidence suggesting that having children dramatically decreased the amount of time individuals had available for politics — either for men or for women. Such findings are quite surprising given the strong sense that time is a resource that is essential for participation and that since it is distributed unequally across the population, it ought to help explain the inequalities in participation that are so apparent within democratic nations.

Second, the mere change of state from non-parent to parent occasioned by the birth (or adoption) of a child may be enough to change the stakes of political participation perceived by an individual. That is, having a child ought to raise a new set of reasons to care about politics — new interests, a new (and additional) stake in politics. Having more to care about in the political realm ought to increase political participation. This idea, that one must care about politics in order to participate has received ample evidence, most recently by Verba, Schlozman and Brady (1995), but also in the specific case of parents who begin to participate more in education related activities (compared to activities targeted more at national level audiences) by Jennings (1979). Using cross-sectional data from the 1968 National Election Study (NES) Survey, Jennings found that individuals with school-aged children were more likely to participate in school-oriented political activities than individuals without school aged children. He explained his findings with the notion that people get involved in politics when they perceive a clear reason to do so:

> This article proceeds on the assumption that, given adequate resources, individuals respond to participation opportunities when they perceive a clearly defined stake (1979, page 756).

Third, we know that mothers are less likely to work outside of the home than women without children, and that fathers are more likely to work outside the home than men without children. Thus, having children appears to change the ways that people interact with institutions outside the home. In fact, Burns, Schlozman and Verba (2001) find that it is the fact that mothers are less likely to work full time that creates the biggest impact on their political participation (decreasing it), and the fact that men are more likely to work full time also leads indirectly to an increase in their political participation.[3] Thus, it may not be time and money which decreases the political participation of women while they raise children, but instead it may be the indirect effect of removing women from the workforce (where they acquire skills and integrate into politically relevant social networks) that depresses their political participation.

Fourth, growing children tend to create new connections between parents and local communities, thus increasing the odds of exposure to mobilization (a process which

[3] The fact that full time work enhances participation via the provision of skills and network position (and money) rather than inhibits participation via decreases in leisure time is supported both by Burns, Schlozman and Verba (2001) and Verba, Schlozman and Brady (1995).

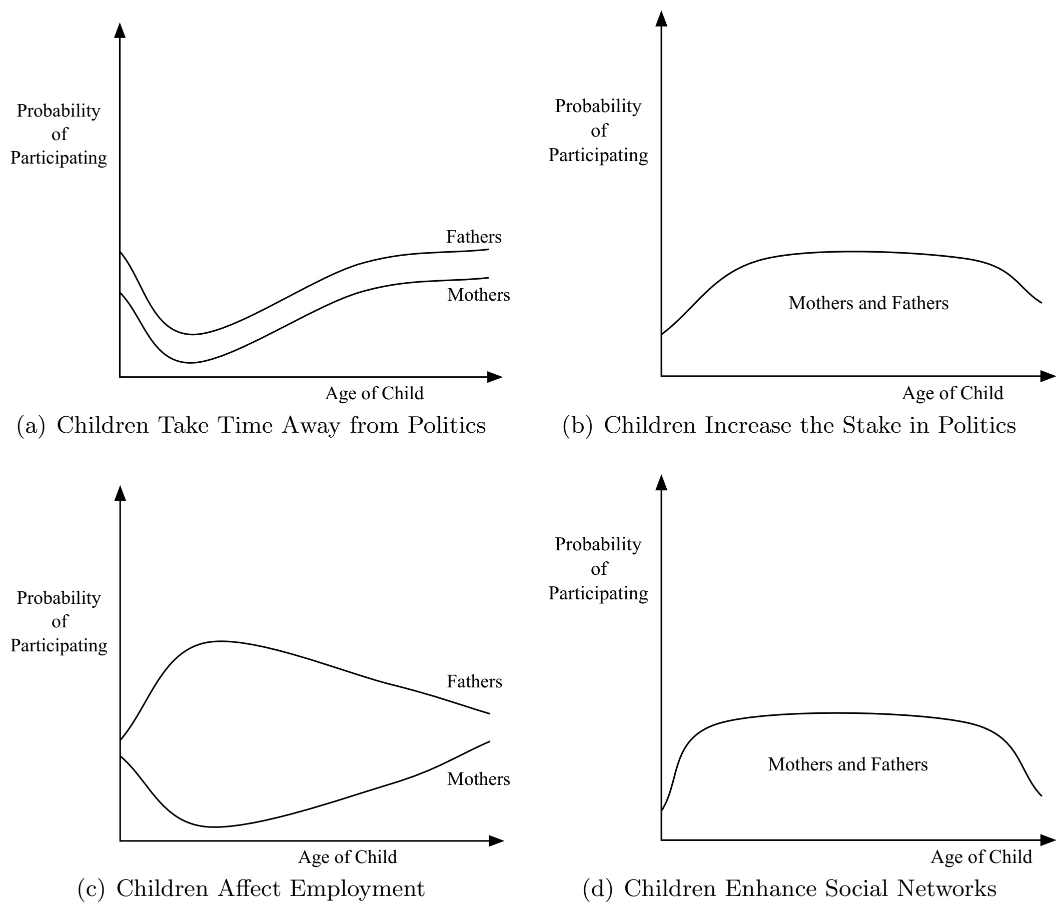
presumably stops when children leave home). One argument for this positive effect suggests that by meeting other parents and joining organizations for one's children (from informal play groups to a variety of school and other youth related activities), individuals are pushed into the path of mobilizers. In addition, interacting with others who are structurally similar (i.e. parents), may help individuals elucidate their interests (another role that more explicit mobilizers are apt to play as well.) The fact that mobilization is an important determinant of political participation has been shown by many authors, most recently by Rosenstone and Hansen (1993).

5.2.1 Dynamic Hypotheses

The four stories posed in the previous section about the ways that parenting may influence the political participation of individuals conflict with one another in terms of general direction of predicted effects. Two suggest parenting ought to increase participation, one suggests a decrease, and another suggests both a decrease for women and an increase for men. All of them have some support based on findings comparing parents to non-parents (although the findings about time availability appear to be the weakest ones so far). The actual question, however, is about how becoming a parent ought to create changes within the life of a person. The comparison that most directly speaks to the theories and questions is within a person (comparing the situation before having children to the situation afterwards). This is the type of comparison I will be making in this chapter. In order to do so, however, I must translate the general findings described above into the form of hypotheses about comparisons within a person's life.

Previous chapters have shown that such dynamic hypotheses tend to have non-linear characteristics. In this case, the "treatment" variable is a child as it ages. Figure 5.2 displays my attempt at representing the dynamic implications of these different hypotheses. It is apparent that, in each case, the hypothesis becomes more complex when the question is about the impact of parenting (as a process) on political participation as it changes over time. Each sub-figure displays the shape participation ought to take if that particular story were the sole determinant, in a world in which individuals only had a single child, and in which nothing else is assumed to affect participation. The x-axis is assumed to run from the birth of the child to well into adulthood (say, from age 0 to 30).

Figure 5.2(a) represents what I think ought to happen if the "Children Take Time Away from Politics" story is correct. Both parents ought to be affected, mothers more than fathers, but the shapes of their trajectories ought not to differ. Immediately upon the birth of a child, participation ought to decrease due to the severe time demands of dealing with an infant. However, as the children become more autonomous as they age and eventually leave home, I expect that participation ought to rebound.



(a) Children Take Time Away from Politics

(b) Children Increase the Stake in Politics

(c) Children Affect Employment

(d) Children Enhance Social Networks

Figure 5.2: Hypotheses about Political Participation as Function of Age of Child

Figure 5.2(b) suggests a completely different pattern. Both mothers and fathers have more reason to participate in politics, especially as the child enters school, and this increase in participation due to an increased stake in politics (since a child provides the parents more to care about, i.e. additional interests to advance and protect), tails off a bit after the child leaves home. One dynamic implication of the statement by Rosenstone and Hansen (cited above) is that the main influence of children on the political activity of parents is to get parents involved in education-related activities in the local area.

Figure 5.2(c) shows a different story again. This time, the operation of children on participation is completely different for men and women (on average). Immediately upon having a baby, fathers tend to remain in or re-enter the full time work force and mothers tend to leave it. Since working full time provides civic skills and civic status, the probability of participation for fathers ought to rise during the period in which the main financial burden for a family rests on their shoulders (since Burns, Schlozman and Verba (2001) show that men without children are less likely to be employed full time than men with children). Thus, their trajectory rises, in this picture, early in the life of the child, but begins to decrease back to their starting level as their mates begin to be able to enter the workforce again. Since, in this story, motherhood pushes women out of the workforce, they lose the skills and status benefits it provides, and thus their participation probabilities ought to decrease, with a gradual increase (on average) as they (re)enter the workforce as the child ages.[4]

Figure 5.2(d) depicts the idea that children quickly connect their parents to social networks — enhancing and increasing the amount of contact that adults have with other adults in their local area. The only difference between this hypothesis and that shown in Figure 5.2(b) is that it ought to take a bit longer for the parents to become aware and make sense of their new political interests and concerns (Figure 5.2(b)) than for parents to be wrapped up in networks of other parents, friends, and neighbors upon the birth of a baby (Figure 5.2(d)).

Figure 5.2 shows a situation for "Political Participation" in general. But, it is possible that children have no effect on overall level of participation, but instead cause parents to redistribute their amount of participation differently. For example, parents may become more involved in education and youth oriented activities than in, say, partisan politics or national level politics. Although this is a compelling alternate hypothesis to the models depicted here, I will not pursue it in this chapter — opting instead to analyze the relationship with overall participation before delving more deeply into the particular issues at stake for any given act.

These stories are all plausible in different ways, but they also leave out two aspects

[4]Future analyses of these data will take into account both the marital and workforce status of fathers and mothers. For this chapter, I have chosen to keep the analyses as simple as possible and so do not differentiate among respondents on these grounds.

of parenting: the timing in a person's life when they have a child and having multiple children.

I suspect that the models in Figure 5.2 would maintain their general shapes for multiple children. However, I expect that the magnitude of the effects would be lessened. That is, I expect that each additional child does not provide an entire additional "unit" of stake in politics, or decrease in leisure time, or need to work full-time. And that, in general, the effect of additional children will lead to decreasing marginal effects — whichever general story is supported by the effects. It is also possible that additional children increase the time in the life-span during which the effects occur — such that, to the extent that the different models suggest some tailing off of an effect as the single child leaves home, the effects would not tail off until all of the children have left home. Thus, while I do not expect multiple children to significantly lower or raise the probability of participation at any given moment in time, I do expect that they will help the effect of children to endure over more years.

One of the most pervasive stereotypes of poverty in current decades has been the teenage mother. Although part of the political rhetoric sees early child bearing (which tends to be out of wedlock) as morally wrong, another, perhaps larger, group of commentators have been concerned about the extent to which early child bearing affects the future income and career possibilities of young women. Although having children early is highly related to socio-economic status, it will be important to see if early parenthood also acts as a powerful initial condition establishing the future path of political involvement for both young men and women, married or not. More generally, it is possible that bearing a child in one's 20s may exert a different effect on a participation trajectory than child-bearing (and rearing) in one's 30s, 40s or 50s and beyond.[5] For example, although Burns, Schlozman and Verba (2001) don't find much evidence for significantly lower probabilities of participation among women with children compared to women who don't have children (in a multivariate model), they do find that one particular group of women does have a particularly low participation rate — single, unemployed mothers. To the extent that being a single, unemployed mother also tends to occur during youth (where individuals are in high school rather than employed, for example), this may show up as an effect of timing of motherhood here.[6]

[5]For example, Plutzer and Wiefek (2001) shows early parenthood depresses future income and vote-turnout among a cohort of African-American women.

[6]I have decided not to examine the effects of marital status and parenting jointly here in order to keep this chapter to a manageable length.

5.2.2 Microfoundations

The hypothesis depicted in Figure 5.2 suggests that the probability of participation for a given individual, i at time, t, should be modeled much as the hypotheses of the previous two chapters, where the probability of participation y for a given person i in time t is some function of the age of a child.

$$\Pr(y_{it} = 1) = f(\text{Age of Child}_{it}) + \sum_{p=1}^{P} \beta_p x_{pit} + u_i + e_{it}. \tag{5.1}$$

As before, the x_{pit} terms represent other variables that might confound the estimates of the relationship between Age of Child and Participation; u_i represents the time-constant aspects of a person that have not been measured but which might otherwise confound the estimates of f(Age of Child); and e_{it} represents the extent to which the variables in the model do not fully capture the dynamics in the dependent variable.

The different hypotheses in Figure 5.2 suggest that $f(\text{Tenure}_{it})$ include functions that have between two and four places where the curve can change direction. In all cases, there is a piece representing the short-term influence of children (from, say, 0 to 3 years old), and then at least two more pieces representing the response of the parents to the aging of their children. Thus, the theoretical model of the function linking Age of Child to participation ought to allow for at least four distinct relationships, such that:

$$f(\text{Age of Child}_{it}) = \begin{cases} \beta_1 \text{Age of Child}_{it} & \text{if } -\varepsilon \leq t < K_1 \\ \beta_2 \text{Age of Child}_{it} & \text{if } K_1 \leq t < K_2 \\ \beta_3 \text{Age of Child}_{it} & \text{if } K_2 \leq t < K_3 \\ \beta_4 \text{Age of Child}_{it} & \text{if } K_3 \leq t \end{cases} \tag{5.2}$$

where K_1, K_2, and K_3 are the locations where the curve changes direction;and Age of Child$_{it}$ is the Age of a Child for person i at calendar year t (since different individuals have different values of Age of Child for different calendar years). In (5.2) ε represents the period just before the birth of a child. An additional assumption is not shown here, but is made in the previous two chapters, which is that the pieces of line represented here must join smoothly, so as to allow the underlying probability of participation of a person to change in a continuous fashion. Also, the theoretical model in (5.2) shows the situation for a single child, but in the empirical models I will estimate the effects of up to four children.

5.3 A Growth Curve Model of Parenting and Participation

At this point in the dissertation, the data analysis strategy for the models presented in Figure 5.2 should be obvious. As in the other chapters, the questions about model estimation and testing boil down to questions about the shape of the function relating some explanatory variable (the age of children in this case) to the probability that an individual will participate in politics in a given year. The strategy that I have used so far, and that I will use here, is to use spline functions to estimate the shape of this function. As in the other chapters, concerns about the clustering of observations within an individual, and heterogeneity across individuals, will lead me to embed the function-estimation within the framework of a random coefficients model, such that:

$$\Pr(y_{it} = 1) = \beta_{0i} + \sum_{k=1}^{K} \eta_k(\text{Age of Child}_{it})\beta_{ki} + \beta_{K+1i}X_{it} + e_{it} \tag{5.3}$$

$$\beta_{0i} = \gamma_{0,0} + \gamma_{0,1}Z_i + u_{0,i} \tag{5.4}$$

$$\beta_{1i} = \gamma_{1,0} + u_{1,i} \tag{5.5}$$

$$\vdots$$

$$\beta_{Ki} = \gamma_{K,0} + u_{K,i} \tag{5.6}$$

$$\beta_{K+1i} = \gamma_{K+1,0}. \tag{5.7}$$

$$\text{with: } (u_{0,i}, \ldots, u_{K,i}) \sim N(\mathbf{0}, \mathbf{T}) \text{ and } e_{it} \sim N(0, \mathbf{\Sigma}) \tag{5.8}$$

where, I have used commas in (5.4) to (5.7) to keep the parameter identifiers separate. In this model, I have left the number of knots K unspecified, since I will use different numbers of knots for the 18 year long electoral activity series (4 knots) versus the 33 years long non-electoral activity series (6 knots). However, the basic form of the model is identical to those used before. Again, the estimation of the function shape is broken down into the estimation of the coefficients for the basis functions of a natural cubic spline function $\eta(\text{Age of Child}_{it})$. And those coefficients are allowed to randomly vary across individuals, and to covary within individuals, according to a normal distribution as shown in (5.8).[7] Possibly confounding variables enter in as linear, additive fixed terms (that do not vary across individuals), denoted here by the placeholder X_{it}.

This time, I will estimate separate functions for mothers and fathers, since it is probable that men and women will react differently to becoming parents. And I will also

[7] The error structure within person is assumed to be continuously auto-correlated at lag 1 (CAR1).

estimate separate functions for first children (which represent the transition from non-parent to parent), and second, third, and fourth children.

The other variables that I expect might matter both for political participation and for the moment of child bearing (which I include in my analysis as controls) are: education and income for that year (since SES is well known as a driver of political participation and since timing and number of children born is also strongly related to SES, it makes sense to include these variables to guard against possible confounds), whether an interview or presidential election occurred that year (since both kinds of years show more participation than other years), and the age at which the person first became a parent (since one might expect people who are different "types" might have correlated and different patterns of participation and reproduction).

Although I examine a single birth cohort of parents, their transitions into parenthood did not happen all at once (as shown by Figure 5.1). One might easily imagine that the moment in one's life when a child is born ought to affect the impact that parenthood has on subsequent political participation. As I said above, I include the age that a person became a parent in the analysis as a time-constant variable. I also ran a second set of analyses separately for those individuals who became parents before age 25, and for those who became parents after 25.

5.4 Results: Overall Model

Figures 5.3 and 5.4 show the predicted values relating the age of children to their parent's political participation. For each act there are two panels, one containing the results for mothers, and the other showing the findings for fathers. Each panel has 4 lines which display the relationship predicted between political participation and the ages of the first, second, third, and fourth children in a family. The gray vertical lines show the locations of the knots that I specified for the spline functions — the thickest of these lines marks the year that a child is born (i.e. age=0). The x-axis runs from -10 to 17 (for electoral activities) and to 32 (for non-electoral activities) although the actual estimation allowed children's ages to run from -30 to 32 (at the maximum). The inclusion of negative ages here allows me to assess the changes in political participation in the years just before the birth of a child — to more accurately see a decline in participation or an increase starting at the birth of a child it is useful to be aware of the pattern that preceded that moment. The trajectories below zero for the first child, show the predicted probability of participation during the 10 years before the individuals become parents. The trajectories for ages below zero for the 2nd, 3rd, and 4th children show the years before the births of those children, during which the previous children are aging.

Figure 5.3 presents the predicted probabilities of participation for electoral activities. It is immediately clear that each kind of activity tends to have a different pattern of relationship, and also that the curves estimated for mothers are different from the curves estimated for fathers. The differences between children, however, are not pronounced, and are not substantively distinguishable in most of the cases. For *Meetings and Rallies*, fathers do not seem to experience any immediate effect of the birth of a child (other than perhaps a very mild increase for the first child, and mild decreases upon the births of subsequent children.). One can also see that perhaps the effects of children occur at the same time — for example, the curve for the first child peaks around 12 years old, while that for the 2nd and 3rd children peak around 9 years old, and that for the 4th child around that same time, if not a bit earlier. This suggests that, for individuals with multiple children, something which has to do with *all* of the children moves them into politics, briefly, rather than having a consistent peak at age 6 for each child (such that each child would have an effect at a different year). There is also a definite rise and fall evident in the father's pattern for *Meetings and Rallies*, such that by the time the children are 17 year old the predicted level of involvement is nearly back to the levels at which it began when the children were born. It is also possible that this is evidence of a parent age effect, however, rather than a child effect.

The effects of children on women's participation in meeting and rallies are a bit different. Women are predicted to have a slight decrease in participation during the first 3 years of their first child's life (coming after an increase as they get older before becoming a mother). Then their participation probabilities increase more or less steadily from age 3, with a slight plateau/slowing of growth in probability after age 12 (for all children except for the 4th one — the instability of the curve for which reminds us of the paucity of data in this analysis — only about 84 individuals compared with the roughly 320 individuals in the 2nd child analysis). Thus, parenthood clearly enhances participation in meetings and rallies, after an initial dip, among mothers — and this enhancement lasts at least through the high school years, whereas for men, it is a more fleeting effect.

The impact of parenthood on *Other Work* having to do with political campaigns is basically flat — the peak in probability attributable to the 4th child when it is 6 years old,

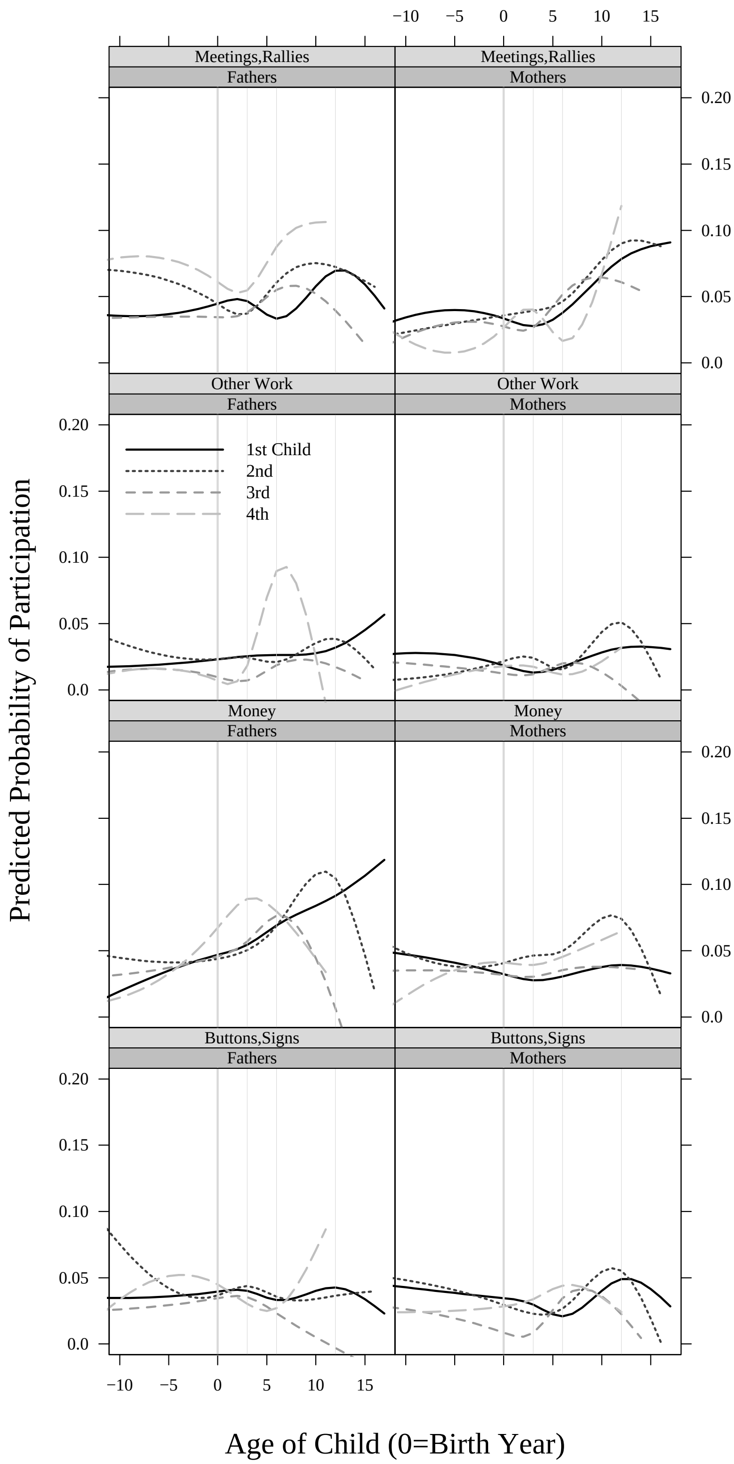


Figure 5.3: The Influence of Parenthood on Political Participation: Electoral Activity

for Fathers, can only be an anomaly. Men show absolutely no change in their probability of engaging in this kind of activity upon the births of their children — with only a very small shift in level of the probability of participation predicted between the first two and second two children. The other campaign work of women does not seem to be affected by the births of multiple children — showing only the slightest of decreases when the first child is born, and not much change around the births of other children.

Displaying buttons and signs during and around campaigns, seems similarly unassociated with individual transitions into parenthood for either men or women. The functions estimated that different children for men display a bit more diversity than those estimated for women. However, the dramatic increases or decreases apparent for men do not show easily interpretable patterns (i.e. it is a bit difficult to interpret a dramatic increase after age 6 for 4th children, but a dramatic decrease after age 6 for 3rd children, and a decrease in the years between the birth of the first and second child.).

The only campaign-oriented action for which there appears to be some influence of the age of children is *Donating Money*. For first children, among men, there is a strong positive relationship — but it doesn't appear to have any changepoints relative to the age of the child. In fact, the upward slope of this relationship appears at least 10 years before the birth of the first child, on average. However, as 2nd, 3rd, and 4th children age, the estimated functions show a sharp drop off in the probability of donating money — and this drop off occurs at earlier ages depending on the order of the child, such that by the 4th child, the drop off begins around age 3, whereas for the second child it begins around age 12, and for the third child around age 6. This suggests that a single child doesn't tend to disrupt the pre-established donation behavior of men, but multiple children do (perhaps one child is enough to enhance the network effects to enhance giving, but multiple children create more demands for time/money which prevent such participation). Women, however, are predicted to have a much weaker relationship between the ages of their children and giving money — their probabilities begin roughly about the same place as the men, but never increase, nor do they show large effects attributable to children (except for the increase and then drop off for the 2nd children). These findings suggest that the story of women dropping out of the labor force, losing economic power and network status, may be operating since the action that one would most expect to see effects on, donating money, is the one for which the effects are seen — however the analysis as currently configured is not setup to capture the interactions between the employment behavior of individuals, their parenting behavior, and the participation.

Thus, for *Meetings and Rallies*, either the "social network" story or the "increased stake" story seems to hold best — since the increases don't appear to begin until school age (where parents are apt to be drawn in to participation via other parents). And for *Donating Money*, the economic division of labor that has tended to be maintained across households appears to have political ramifications. If decreased time were a powerful inhibitor of participation, I would have expected to see declines in *Meetings and Rallies* and *Other Work*. However, the increase in *Meetings* and flat lines for *Other Work* suggest that time devoted to politics doesn't decrease spectacularly, but that the networks that might recruit individuals into certain kinds of activities do increase their pull.

Figure 5.4 shows a similar pattern for non-electoral activities: two activities show little or no relationship with child bearing and rearing for either gender, and two show effects that are different for mothers and fathers. This time, however, the range of age of child is much greater — up until age 32 at the oldest with "knots" at 0,3,6,12,18, and 25 years old. The function for *Letters to the Editor* is flat: few individuals ever write letters to the editor, and those individuals apparently do not let their child rearing duties influence their propensity to write (at least in any substantively meaningful way). For *Demonstrations and Protests* the only clear pattern (other than the overall rarity of this kind of activity) is that the propensity to protest decreases before the birth of the first child. This is probably a period effect — after the early 1970s the protest activity of most of these individuals died away (as has been shown in other chapters), and coincidentally (or not?) this is about the same time in these individuals' lives when they began to reproduce.

Both *Contacting Officials* and *Community Work* display the general pattern found for *Donating Money*: a generally positive relationship with the age of the first child exists for fathers, a curve that doesn't seem to be interrupted by the birth of the first child, and women have more or less flat trajectories with perhaps more sensitivity to the particular life stages of the children. For *Contacting Officials*, men are more homogeneous with respect to multiple children than women — each additional child for men appears to increase their odds of contacting elected officials, with dropoffs of probability after the youngest children turn 18 (thus, the fact that the curve representing the effect of the first child doesn't show a strong drop off is because many of those with first children are still participating due to their second and third (and fourth) children even when the first child is older than 25 years old. The birth of children, for men, does not seem to interrupt or inhibit this relationship.

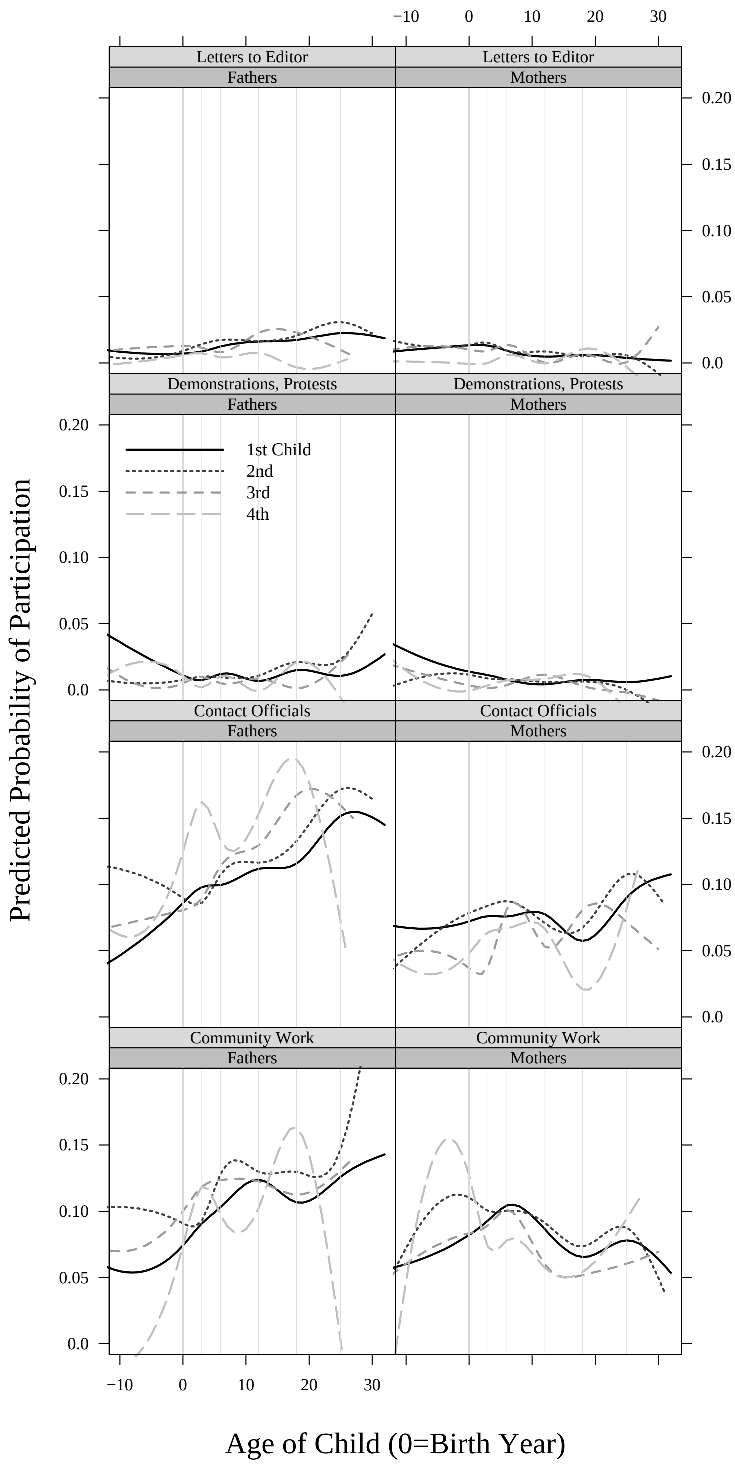


Figure 5.4: The Influence of Parenthood on Political Participation: Non-electoral Activity

Women show more different effects for different children — the birth of third children (and possibly fourth children) appears to be associated with a decline in contacting, but first and second children do not appear to interrupt this kind of activity.

For *Community Work*, men display immediate increases which seem to accelerate upon the birth of children, but which plateau after about when children are 12 (with the dramatic drop off for fourth children after age 18 perhaps indicating that much of community work for men of this cohort at least, lasts as long as the youngest child is still in the house). Women also show modest increases in predicted probabilities of community work immediately upon the birth of their first and third children (and a sharp decrease upon the birth of their 4th children). However, the pattern thereafter is a slow decline punctuated by moments of activity around age 6 for all children. The increase in activity predicted after 18 for the youngest children (4th children) suggests that women are freed from child care duties and may re-enter political life.

This analysis has shown two main results. Rather than clearly pointing to one of the 4 possible hypotheses that I laid out in Figure 5.2, the results so far are a mixture of no effects and some modified versions of the others. Certain types of activity appear immune from interruption or influence by children. Some of this occurs due to the sheer paucity of data. Very few people write letters to the editor, and thus those who do show very little relationship with age of children, or the transitions of birth of new children. Certain other activities show differences in the ways that parental status influences participation by gender. The most significant cases of this involve donating money, contacting officials, and community work, all of which show increasing probabilities of participation by fathers, but decreasing or relatively flat probabilities of participation by mothers — the predicted probabilities from the models look like bumpy versions of Figure 5.2(c). It is possible that there is some division of political labor in a household — such that donating money, contacting officials, and community work could tend to be done by men, but with the urging or "in the name" of the family rather than as solo actors. Other activities, such as wearing buttons or displaying signs and engaging in "other" campaign work, show smaller effects for both mothers and fathers, with no easily discernible trend — perhaps a weak version of "children take time away from politics" for the mothers, and no effects for the fathers.

The different activities for which there is any effect at all also suggest that multiple processes may be operating. In general, women appear disadvantaged by children (except perhaps for attendance at campaign rallies and meetings), and men advantaged. The actual birth of children does not appear to have immediate and dramatic effects (except possibly for community work by fathers). But, instead, the effects of children appear more tied to way that they push their parents into contact with other parents as they age (or they pull their mothers away from mobilizing structures when they are born).

5.5 Results: By Age of Reproduction

Although the age at which individuals became parents was entered into these equations as a time-constant control variable it is possible that these flat effects are really conditional on class or context (despite the income and education controls, and the within person nature of the analysis). To check for this, I reanalyzed all of the data, focusing specifically on the moment at which individuals transition to parenthood (i.e. the birth of their first children), run separately by whether this child was born before or after the individual was 25 years old. These analyses were run just like the others — controlling for education, income, presidential election year, interview year.[8]

Figure 5.5 and 5.6 show the results of this analysis. The black solid lines represent the predicted probability of participation at each year of age for the first child born to an individual for a person who made the transition to parenthood *after* age 25. The dashed gray lines show the predicted probabilities of participation for persons who became parents *before* age 25. Both of these figures tend to show that those individuals who become parents after age 25 tend to have high probabilities of participation at any given moment.

Figure 5.5 shows that becoming a parent seems to have a small increased effect on attendance at *Meetings and Rallies* among men who pass through this transition later in life, while the men who become parents early in life are predicted to not only participate with lower probability before becoming parents, but also to experience a dip in their probability of participation that lasts until the children turn 6 years old. The dramatic crash in the probability of participation among men who have their first children late in life (which is apparent among all of the acts) is due to a paucity of data — and the fact that, overall, electoral activities are rare.[9] Women who bear their first children after age 25 are predicted to have much higher participation before they become parents than women who bear their first children before age 25. This is sensible since we know the young are less apt to participate than all other adult groups. What is particularly interesting about the

[8]And allowing the intercepts and slopes of the spline functions vary across individuals according to a normal distribution, and with continuously annual autocorrelated errors within person.

[9]I'm not sure why this crash does not appear for the women.

patterns for women's participation in campaign oriented *Meetings and Rallies* is that the late-parenting women and early-parenting women converge in their predicted probabilities of participation at the moment they become mothers. After this convergence (which occurs mainly due to the late-parenting mothers having a decrease in their predicted probability of participation), the women who have their first child after age 25 are predicted to rebound immediately; by the time their child is 3 years old they are predicted to rise in probability of action. The women who have their first child before age 25 also rebound, but they take longer; not until their first child is 6 is their predicted probability of participation expected to rise, and it does not rise as high as that for the late-child bearing women.

A similar pattern holds for both genders when it comes to other campaign oriented work. For late-parenting men, a fairly rapid increase is predicted upon the birth of a child (at least until age 6), while for early-parenting men, making the transition to parenthood seems to lead to a slight decrease if any effect at all. The late-parenting women again precede parenting with a higher predicted probability of participation than the early-parenting women, and the two converge (again largely due to the movement downwards in the curve representing the over-25 parenting women), followed by a fairly rapid rebound among the late-child-bearing women (and a slight increase or no effect among those women who have children earlier in life).

Donating Money still seems to respond more positively to child bearing among men than among women — both groups of men show an increasing propensity to donate as their first children are born and age (but the over-25 fathers increase more rapidly). Among women, the late-parenting group again begins high, decreases to join the early-parenting group at the moment of transition to parenthood, and rebounds. The overall picture among men appears to be upward trending while that among women, for this act, is relatively flat (the quick rebound of the late-child bearing women is followed rapidly by a sharp decrease).

The pattern for *Buttons, Signs* is a bit different, and perhaps reflects that becoming a parent has little to do with this activity. In this case, the early-child bearing men are predicted to have an immediate (yet slight) increase in their probability of wearing buttons or displaying signs, while the late-child bearing men show an equally slight decrease. The women again converge at the moment of transition, both due to a decrease among the late-parenting women, and a slight increase among the early-parenting women. However, over all, there appears to be little effect of becoming a parent on displaying buttons and signs.

The patterns for non-electoral behavior (shown in Figure 5.6) also describe a situ-

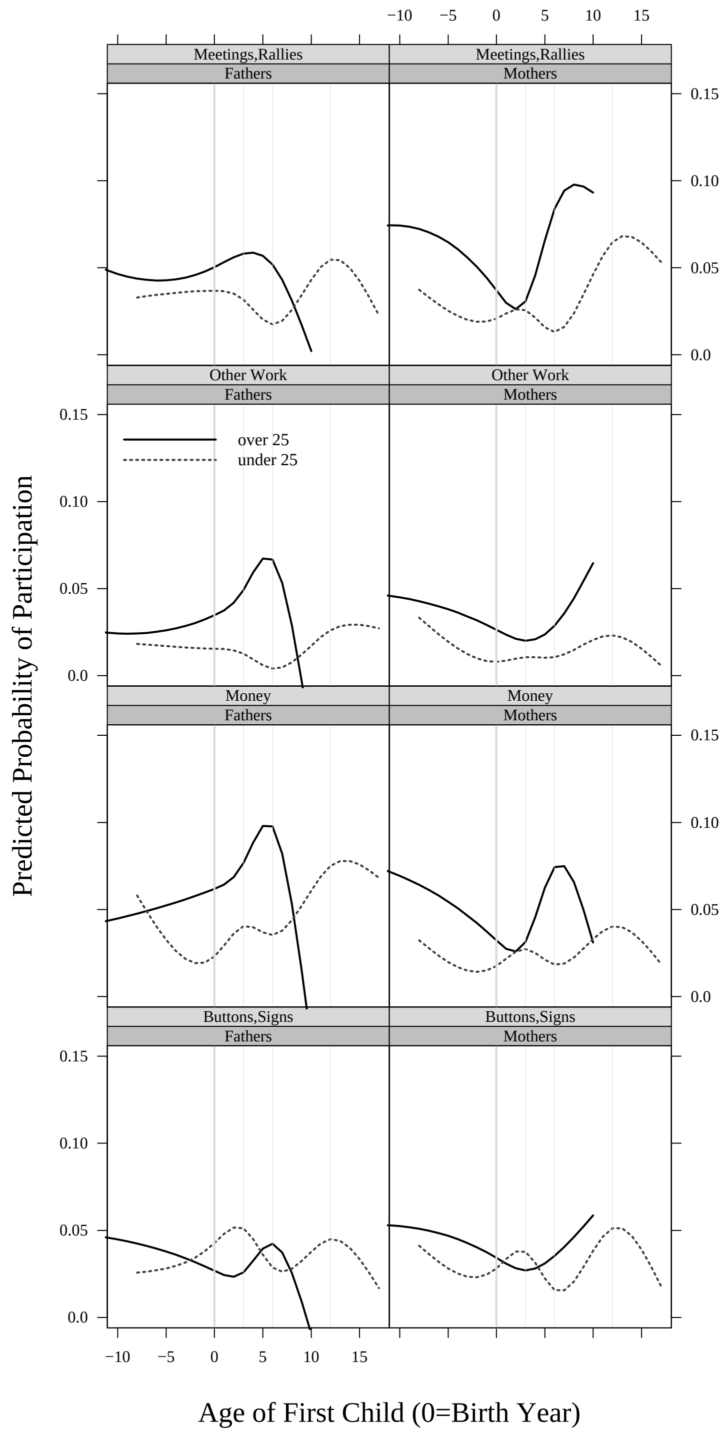


Figure 5.5: The Influence of Transitions to Parenthood on Political Participation:Age of First Child and Electoral Activity

ation where the older parents have a participatory advantage over younger ones. However, writing letters to the editor remains largely untouched by parenthood. The effect of the transition to parenthood on Demonstrations is also probably largely explained by the period effect of the Protest Generation — having children later is related to decreasing protest behavior since the era of protests was over by then.

The results for *Contacting Officials* and *Community Work* are a bit different. Among men, becoming a parent is associated with increased Contacting among those who have children after age 25, while it is associated with decreased *Contacting* (at least until the child is 6 years old) among those who become parents early in life. The two types of fathers converge around the time the children are 12, and then diverge again (with the early-parents decreasing and the late-parents increasing) after the child is 12 years old. Among women, the two groups converge as well, also at age 12. The convergence among the mothers, is largely driven by a decrease in the probability of contacting officials among late-mothers from age 6 to age 12 of their first child. This could be the moment when their second child is born. These convergences/divergences, however, are probably displacements of period effects. Regardless, these late-mothers rebound from their decrease dramatically and immediately.

The two groups nearly mirror one another among both men and women for *Community Work*. In each case, the late-parents appear to increase in their probability of participation more quickly and to participate at higher levels than the early-parents. This pattern is weakly present among the men and more strongly so among the women. Across the two genders, however, the patterns are quite different. The men show increasing probability of community work from about age 0 to about age 6 of the first child followed by a relative plateau (among the late-fathers and a peak at age 12 among the early-fathers). Both groups of mothers show early increases in probability of participation upon the birth of their first child (until about age 3 for late-mothers, and about age 12 for early mothers), followed by an unsteady, stepwise-seeming drop in participation from then on. At each point (except the point of convergence at age 12 and possibly age 25), the late-mothers are predicted to have a higher probability of participation than the early-mothers.

The moment in a person's life that their first child is born matters for the future shape of their political participation. If the child is born early in life (in this case, before age 25), participation is affected immediately and is predicted to be lower than the participation of late-parents for the next 20 years. The question raised by this second set of

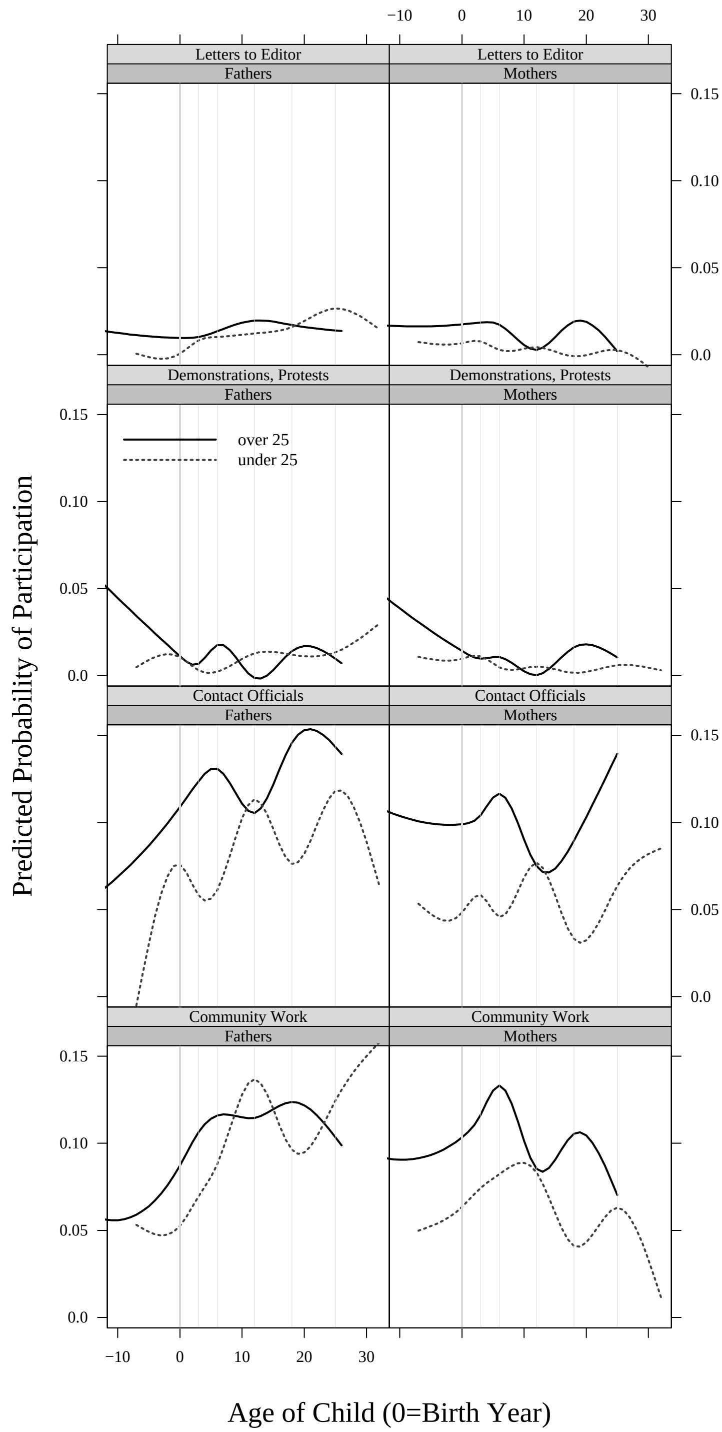


Figure 5.6: The Influence of Transitions to Parenthood on Political Participation:Age of First Child and Non-electoral Activity

analyses is what this timing of parenthood in the life of a person stands for. Is it capturing something about social status that has lingered despite the presence of education, income, and person specific random effects in the model? Is it something causal in itself — something about waiting to have children allowing individuals the time to develop an interest in politics, learn skills, or acquire resources? The sociological literature finds that early child bearing has dramatic (and negative) consequences for the future earnings of women. Some suggest that this finding results from the fact that late-mothers have time to gain education, build networks, and otherwise prepare themselves for the wage-earning world, while early-mothers never get around to gaining more education or preparation for work. Perhaps the same dynamic is in effect when it comes to politics. People have to be ready to participate at the moment when participation is demanded or opportune. Those who have built up the networks and skills to some critical point are *ready to act*, while others are not. Bearing children may stop the process of "getting ready" to participate for young women. In contrast, older women, may pass up particular opportunities during the infancy of the child, but somehow they are ready to act fairly quickly after bearing a child.

5.6 Conclusion

The previous literature on parenthood and participation has suggested that the dominant intuition (that children take time and money away from politics) is wrong. As I noted before, neither Verba, Schlozman and Brady (1995) nor Burns, Schlozman and Verba (2001) found evidence linking being a parent to the time that either men or women had available for political activities. And, Jennings (1979) showed that particular kinds of activities (those that have to do with schools and youth) are more likely to be done by parents than non-parents. These findings produce the sensible suggestion that children may give parents more to care about when it comes to politics. And caring more about what is going on in politics (even if it is just pertains to the local school), seems a natural precursor to doing more about politics.

My findings here suggest that this is true for some kinds of activity, for some kinds of people: notably, donating money, contacting officials, and community work increase among men. Compared to men, women appear not to react to motherhood with strong long term increases or decreases in political participation. For example, while men increase their community work as their children age, women tend to slightly increase their involvement, until their children are around 6 years old and then to decrease their community activity.

Men increase their donations of money over time (until their second, third, and fourth children are in high school), while women tend not to show much relationship between donating money during campaigns and the ages of their children.

Some of the results for men do not show large changes at the moment when children are born, and it is tempting to interpret these finding as due to increasing money and social networks as people age. However, at each point of "age of child" the respondents in the Class of 1965 are many different ages themselves since they bore children at many different times in their lives. So, trends evident for men are not simply evidence that the men are getting older — they are aging along with their children, but they are beginning the process of parenthood at many different points in the life-span.

I've also found that all children do not have the same effect on the political participation trajectories of either their fathers or mothers. I did not spend much time interpreting these findings in detail in this chapter, but it is clear that second, third, and fourth children do not just produce neatly stacked lines (with the subsequent children producing the same general shape, but a muted effect). Rather, since most individuals had more than one child, it is possible that bumps in the trajectory for the first child refer to the moment of birth or school transitions for the second child. Alternatively, it could be that it takes something that matters for all of a person's children, and not just one, for them to be moved to act.

I've also shown that the timing of the transition to parenthood matters. Early parenting is associated with lower probabilities of participation across the board — controlling for time-varying and time-constant individual characteristics. This finding suggests that children may actually have something to do with resources of time or money (contrary to past findings), or perhaps that early-parents do not feel the same child-oriented cares about politics that are supposed to stimulate the activity of all parents. However, it is true that, even among early-mothers (and late-mothers, and both kinds of fathers), the first few years after a first child is born is associated with increasing probabilities of work with others in the community — a finding that doesn't square with the "children take time and money away from politics" point of view.

I began this chapter by suggesting that events and experiences which transform a person's life in many ways are also apt to change their propensity to participate in politics. This seems quite obvious, since political participation tends to be a relatively small-scale and rare event within the lives of individuals as seen in Chapter 2. What I've found by examining what happens to political participation during the process of parenthood is that this particular life-changing event doesn't matter much for a number of types of political participation (probably because they are so rare, that it takes a truly determined person to accomplish them), and that it doesn't uniformly inhibit or catalyze participation across the board either. The fact that the strongest effects are positive ones, for contacting officials, community work, and donating money, particularly for men, suggests that children do play some role in enhancing or creating social networks for men, and that perhaps there is a division of "political work" in households as well.[10] It is possible that children also increase the amount that parents care about politics, but it is also clear that the increase in caring is not enough to dramatically increase the number of buttons or yard signs displayed, campaigns volunteered for, protests joined, letters to the editor written, or campaign meetings and rallies attended. The amount of threat or opportunity posed during the moments when most people decide about these kinds of activity cannot be that large compared to the costs even parents face to get involved. However, it is possible that increased caring may have combined with networks (inhabited mainly by fathers) to produce the increased participation in presumably lower cost activities of community work and contacting officials.[11] It is hard to believe that men would care more than women about the impact of public policies on their children, and it is also hard to believe that having children increases the financial well-being of men more than women (since, the push into full-time work for men, and out of it for women, presumably has to do with the need to pay for the expenses of children — assuming, of course, that the men and women are married). Thus, I am led to the conclusion advocated by Burns, Schlozman and Verba (2001), that parenthood pushes men into social networks where they are more apt to be mobilized to act than are women. This is surprising given my intuition that mothers would be more likely to expand their circle of acquaintances as they act as the primary caregiver. However, it is possible that mothers' child-oriented networks do not push individuals into the path of mobilization the way that fathers' work-oriented ones do. This chapter suggests that some kinds of networks may be better for political participation than others, and that gender divisions of labor in the economic realm may manifest themselves in the political realm. Furthermore, the effects of this gender division may be even more powerful for fathers and

[10] Although Burns, Schlozman and Verba (2001) show that having children urges men to work full-time, it is not clear that the men have more disposable income due to this.

[11] To more directly understand and examine shifts in the stake or interest individuals have in politics after bearing children, one would have to examine how the issue focus of their activities change in response to parenthood. This is possible to do using the Political Socialization dataset, and will be a topic for future research.

mothers.

Chapter 6 Conclusions and Future Directions

Those who are concerned about democracy have always worried about political participation. In the past fifty years, this concern has resulted in much scholarly work focused on identifying the sources of participatory inequality within countries. Verba, Schlozman and Brady (1995) express the core reasons for this focus when they write, "Since democracy implies not only governmental responsiveness to citizen interests but also equal consideration of the interests of each citizen, democratic participation must also be equal" (page 1). The problem is, as they see it, that the reality is far from this ideal. The few people who participate at any given time in a democracy are quite different from those who do not, and so, "...the voice of the people as expressed through participation comes from a limited and unrepresentative set of citizens" (page 2). This kind of information about differences between activists and an inert mass of ordinary citizens, suggests an obvious speculation — that policy outcomes within a democracy in which participators do not represent non-participators will tend to disadvantage those types of people who are not active in politics. Since decades of research have shown that those who do not participate also tend to be those who are disadvantaged in many other ways, participatory inequality has the potential to exacerbate and perpetuate many other kinds of inequalities — and certainly does not involve "equal consideration of the interests of each citizen."[1]

The research program that aims to understand the reasons behind this inequality is vigorous and continuing, but this dissertation is not directly part of that debate. Instead, I have tried to suggest that focusing on inequality between citizens is only part of the picture of political participation. Citizen participation is not only a dividing line across a society at a given moment in time. It is also a dynamic process that occurs sporadically across the lives of individuals. In order to understand exactly who the participators and non-participators are, one must know something about how the participation of individuals

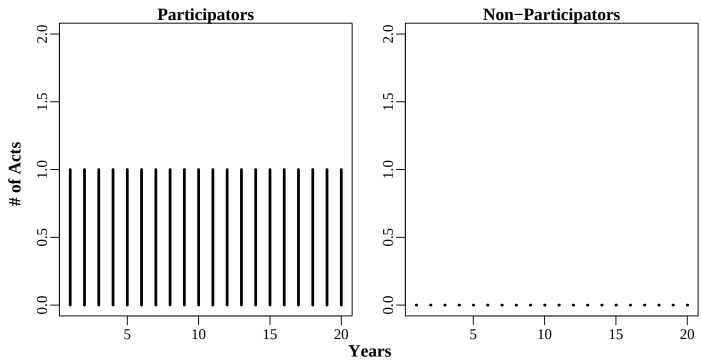
[1] And, as Fiorina (1999) notes, this inequality may also form a vicious cycle in which decidedly less representative activists drive ordinary citizens ever further away from politics.

plays out over time. After all, a citizen who stays at home one year but attends a rally the next year is different from one who continually stays at home, removed from politics. Without data over time within the lives of individuals, cross-sectional results may lead to incorrect interpretations about the state of political participation in any given country.

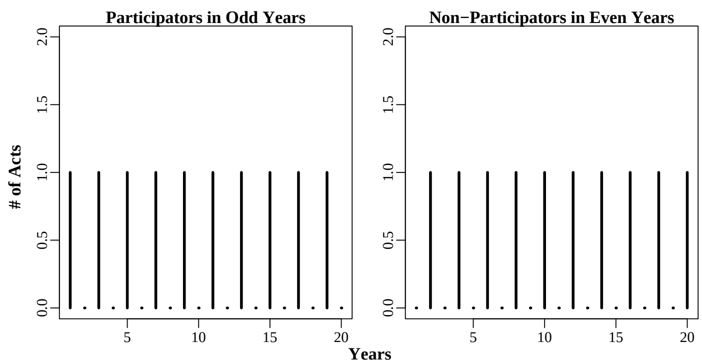
Figure 6.1 illustrates how a single cross-sectional result can hide very different dynamic patterns — each of which has a very different substantive and normative interpretation. Figure 6.1(a) shows a hypothetical democracy consisting of two groups of individuals over 20 years of their lives. At any one point in time, the percentage of the population that is active is only 50% — and a cross-sectional survey done at any one point in time will confirm this. However, in this democracy, the same group always participates. To the extent that the participators do not share the interests and values of the non-participators and are not otherwise altruistic, and to the extent that citizen participation influences government outcomes, one would expect to see severe inequality in the society. Figure 6.1(b) represents a democracy that is similarly divided in terms of the participation trajectories observed over the 20 year spans of life of its citizens. This time, half of its citizens participates during odd years, and half participates during even years. Just as in Figure 6.1(a), at any one point in time, the percentage of the population that is active is only 50%. Since participation is traded off from year to year across the population, neither group can control the government, and so we'd expect less inequality in this society.

By presenting this figure, I do not mean to suggest that the participation of Americans can be thought of as either divided into participators versus non-participators, or that participation appears to be handed off between groups in society. Instead, this discussion illustrates how dangerous it can be to make normative claims and policy prescriptions based on snapshots of different random samples of the electorate. If the cross-sectional surveys tend to say that around 10% of Americans get involved in politics, there might be some cause for concern (depending on one's normative predisposition) if this number means that 10% of the population participate all of the time and 90% participate none of the time. But should proponents of participatory democracy still be concerned if 100% of the population participate 10% of the time?

This line of reasoning raises the question about how many different kinds of participation rates and distributions of these rates over populations are compatible with any given cross-sectional finding. To think this through, it is useful to use a very simple case, where the population is divided into two groups, and the individuals in each group participate at a



(a) Half the population participates all the time



(b) All the population participates half of the time

Figure 6.1: Two Democracies from A Single Cross-Sectional Finding

constant rate over time. This is the kind of case I've just discussed (e.g. where members of Group 1, consisting of 10% of the population, participates all the time (i.e. at a rate of 1.0), and Group 2, consisting of 90% of the population, never participates (i.e. at a rate of 0.0)). Table 6.1 illustrates the fact that for this very simple case of 2 groups with constant participation, there are many possible dynamic patterns underlying any single cross-sectional finding. Every row in this table represents a population that would lead to an observed cross-sectional participation proportion of 10%. The first and second columns show possible distributions of the population between two groups — one where the population is divided 90/10, and other distribution where the population is evenly divided into the two groups. The third and fourth columns contain the rates of participation over time within the two groups that are compatible with observing the given cross-sectional proportion. For example, a cross-sectional observation of 10% participation could result from any of the following: a situation where 10% of the population participates and 90% does not; where

10% never participate and 90% participate about 11% of the time; where 10% participate 33% of the time, and 90% participate about 7% of the time; where 50% participate 20% of the time and 50% do nothing; where everyone participates 10% of the time; and so on.

Table 6.1: Some Participation Rates Compatible with Observing 10% Participation in a Cross-Section

Proportions		Rates	
Group 1	Group 2	Group 1	Group 2
0.1	0.9	1.000	0.000
0.1	0.9	0.000	0.111
0.1	0.9	0.330	0.074
0.5	0.5	0.200	0.000
0.5	0.5	0.100	0.100
0.5	0.5	0.150	0.050

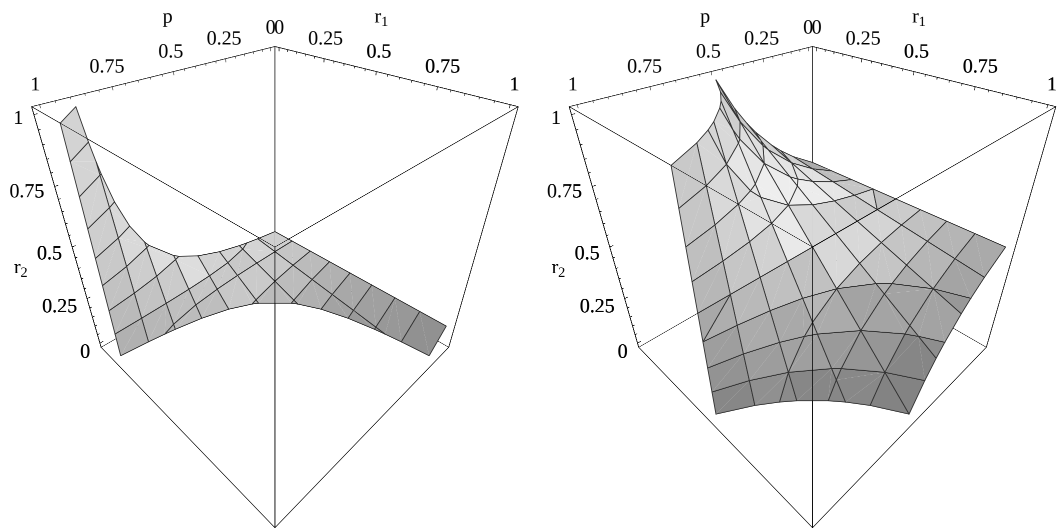
The fact that any given aggregate cross-sectional result can occur via multiple dynamic mechanisms is an example of a more general problem about aggregation in general. For example, in Table 6.1, Group 1 could be blacks (which make up 10% of the population of some hypothetical country) and Group 2 could be whites (which make up 90% of the population of that country), and the aggregate number of 10% could represent a situation where all blacks participated in a single election but none of the whites did, or a situation where none of the blacks participated but 11.1% of the whites did. This relationship between aggregates and their components can be written in a general form like so:

$$pr_1 + (1-p)r_2 = o \tag{6.1}$$

where p is the proportion of the population in group one and $(1-p)$ is the proportion of the population in group 2; r_1 and r_2 are the constant rates of participation over time for members of group one and two, respectively; and o is the observed proportion of participators in the population at a single moment in time.[2] In fact, for each distribution of the population across groups and each observed cross-sectional proportion, the rates (r_1 and r_2) that satisfy the relationship in (6.1) consist of a line: thus, there are infinitely many combinations.

[2]This setup assumes only two groups, but this can easily be generalized to account for any number J of groups such that: $\sum_{i=1}^{J} p_i r_i = o$ as long as $\sum p_i = 1$.

For a full view of the complexity that even this simple story can produce for any given cross-sectional measurement, I produced Figure 6.2. Each of the two panels shows all of the combinations of p, r_1, and r_2 which are compatible with a single proportion of participation observed in a cross-section. Figure 6.2(a) shows the range of possibilities for the hypothetical finding that 10% of the population participate at a given moment. Figure 6.2(b) shows the range for the hypothetical finding that 50% of the population are participators in a single survey. Instead of a line with infinitely many points as solutions, when p is allowed to vary, the space of possible solutions becomes a surface. Within each surface there are also infinite points, although the surfaces themselves do provide some general bounds on possible states of the world.



(a) 10% Participation in Cross-Section (b) 50% Participation in Cross-Section

Figure 6.2: Possible proportions (p) and participation rates over time (r_1 & r_2) for a population with two groups and constant participation rates given two different cross-sectional findings

The point here is that it would be wrong to interpret any single cross-sectional finding as if it meant anything in particular about the rates at which people participate over their lives. I suggest that scholars and policy makers ought to judge the world based on such rates rather than on snapshots. I hasten to add here that the complex picture I have painted up until now is in fact much too simple — remember that I have shown several times in this dissertation that rates of participation change across the lives of individuals (although, for the sake of simplicity, the table and figures just presented assume constant rates of participation over time). If the uncertainty in interpreting cross-sectional findings appears large so far, one can only imagine the indeterminacy that would result from applying a single cross-sectional finding to a world of many and shifting groups, within which individuals change their rate of participation over time. Thus, the great concerns over inequalities in participation or general levels of participation may be overstating the extent of the problem — or these concerns may be right on target — but without longitudinal data we just can't know. And, any policy based on assumptions about longitudinal behavior based only on cross-sectional information would have to be called shots in the dark because cross-sectional group differences in participation are consistent with many dynamic stories.

This dissertation has not taken direct issue with students of aggregate participation rates. But, instead has sought to translate the findings of past decades based on differences across individuals to guide analyses of political participation as it changes over time within peoples' lives — looking at differences across time, within person. However, the past individual level findings which show that those with more education are more likely to participate, or those who have recently moved are less likely to participate, have caused great concern about participatory inequality precisely because of the aggregate interpretations that have easily arisen. Such that the causal theory linking education to participation (which has received ample support from individual-difference analyses) has fed fears that our nation consists of a very few well-educated active people who effectively rule, and a mass of less educated, comparatively inactive people, who effectively are ruled. Recent work has directly addressed these concerns by showing, for example, that church attendance and union membership can overcome the disadvantage of low-education (Verba, Schlozman and Brady, 1995). The individual level findings that I have endeavored to replicate and explore have always had most motivating power (when it comes to public policy) based on translations to aggregate consequences. And, although astute scholars have recognized the problems of compositional indeterminacy before, I hope that my work makes more scholars and policy makers more aware of the little we truly know about how political participation works.

Of course, previous scholars have not ignored the fact that timing of participation and its dynamics are crucial components for diagnoses of the "health" of a democracy or the stability of a regime.[3] However, until recently no data on participation over long time

[3]One group of scholars has been concerned about the timing and consequences of participation on the

periods within people's lives have existed. In fact, there is only one such dataset, and it exists only because of work done over 30 years by M. Kent Jennings and his collaborators. It is rare for science to have to wait 30 years for a dataset, but once such a source becomes available, it is immensely valuable.

6.1 Summary of Results

Using this extraordinary dataset, I've discovered that Americans don't appear to be divided into two groups of those who are active and those who are not. Instead, in Chapter 2, and in Figures 1.1 and 4.1, I've shown that participation occurs sporadically across the lives of many individuals, and that spells of participation tend to last only one year (which is the minimum temporal resolution of this dataset). Compared to cross-sectional findings that between 5% and 10% engage in participation in any one year, this dissertation has shown that up to 68% of the Youth generation (and up to 38% of the Parent generation) engage in at least one kind of non-voting participation over 30 years of time.[4] I did not assess the representativeness of the shifting groups of people who reported participating in each type of activity, so I cannot speak especially rigorously about the dynamics of inequality within these patterns. However, in a world in which 65% of a cohort contacts elected officials at least once in their lives, participation cannot be said to be strictly the province of a small group of elite participators.

I've also shown large differences between the participation patterns of two generations of Americans. The older generation has been recently lauded as especially "great" from the point of view of their civic engagement (Brokaw, 1998), and I have found that these individuals are likely to vote in every presidential election over the study period.[5] However, across all other measures, they are *less* participatory than their children — they

stability of regimes. Some celebrate the power of the people to force transitions to democracy on totalitarian and authoritarian regimes (See, for example Havel, 1990). Others are concerned about how democratic regimes can themselves be brought low when the mob moves into the streets (See, for example Huntington, 1968). To my knowledge, neither set of scholars has focused on predicting the behavior of individuals, and instead, they have all tended to use "group", "regime", "government", or "social movement" as units of analysis. Their concern with the dynamics has focused on how precise timing of mass uprisings (or their absence) have had consequences for regime stability. Since the units of analysis in this dissertation have been individuals, I have not focused on or extensively reviewed this other literature.

[4] For the Youth generation, 68% worked with others in their community and 65% contacted an electoral official at least once over the 30 year study period. The Parent generation was less participatory in general, with 38% and 36% ever reporting community work or contacting officials over their study period, respectively. For more details, see Chapter 2.

[5] The Parent cohort here encompasses some members of this group (i.e. those individuals who were old enough to have served in World War II), but it also contains individuals who were older, so it cannot be seen as containing *only* members of the so called "greatest" generation.

engage in fewer activities and spend fewer years per spell. For example, although fewer of the Youth generation voted in absolutely all of the presidential elections for which they were eligible than the Parent generation, fewer of the Youth also remained non-voters over time than the Parent generation. Of course, members of the Parent generation are also much less education than members of the Youth generation, and so this participatory difference could be traced to the operation of education, rather than anything inherent about either birth cohort.

I've also shown some evidence that the participation rates of individuals are not constant over time. And, despite the fact that some people persist in multi-year spells of action, participation is largely a one-shot type of activity that occurs at irregular intervals (and the intervals between moments of action are quite diverse across individuals).

In addition to this evidence about participation itself, this dissertation has provided a new perspective about how participation relates to education, residential mobility and parenting. In each case, the hypotheses drawn from previous literature were not completely supported. In fact, each set of analyses seemed to raise new questions and reveal new complexities, rather than to simply bolster or undermine the hypotheses that I distilled from previous literature. However, a few results do stand out.

Despite the power of education to distinguish those who are apt to participate from those who are not, education does not appear to radically change the shape of participation trajectories within people's lives. In other words, the dynamics of political participation do not appear to be driven by education, even if the level is. The findings about education that I have presented here imply that participatory inequality by education is apparent and persists across the lives of individuals. I explored the idea that the timing of attainment of a college degree within the lives of individuals might itself have dynamic effects, and found evidence that those individuals who receive college degrees earlier in life are more likely to participate at any one moment in time, over their lives, than those individuals who receive a college education later in life. However, the shapes of the functions relating time to participation did not change appreciably. Thus, although education has long been seen as one of the most powerful factors influencing participation, I have found that it appears not to have major consequences for dynamics — modeled here as the shapes of participation trajectories over time.

My findings about education suggest that participation is not well characterized by positive feedback or increasing returns. If it were, the gap in the level of participation between those who received college degrees early in life and others ought to increase over time (since early attainment of higher levels of participation would lead to yet more participation, which itself would lead to more). Nor does the attainment of a college degree seem associated with big changes in slope or direction in the trajectories. Those who receive college degrees relatively early are *always* predicted to have a higher probability of participation than others — even before these individuals have finished college.

It is unclear how these results relate to the skills vs. status debate. The lack of dramatic effects perhaps supports the idea that a college degrees sets a person on the *path* to civic skills without directly providing a package of such skills. If a college degree were to change a person's civic status, one would expect this effect to occur rather quickly and immediately. However, it is also possible that individuals only come to realize the benefits of educationally based changes in civic status and civic skills later in life when they are confronted by moments where participation is either possible or necessary. My general interpretation of these results is that education functions to prepare a person to take advantage of participatory opportunities via both skills and status, but that the frequency that such opportunities arise in the life of a person is driven by other factors.

Residential mobility has also been a key explanation for who participates. I have found that moving may not have as large an impact on all types of non-voting political participation within the lives of individuals as one might have expected given the cross-sectional findings. Of course, one might not expect strong relationships between mobility and activities that are less tied to local networks of personal relationships. Since much of politics in the U.S. has become centrally organized by direct mailings and television broadcasts orchestrated by professional activists, one might expect that what matters most for participation is having a television and a postal address. In fact, the strongest findings from this chapter are the positive relationships between residential tenure, community work, and contacting elected officials, precisely the kinds of activities that ought to be most strongly stimulated by personal ties. These relationships are relatively steady (and appear fairly linear) for the Youth generation, but for the Parents, the relationships are weaker and nonlinear, perhaps due to the paucity of data on moving among the relatively immobile older generation. For campaign activities or higher cost non-electoral activities (such as letter writing or protesting), there appears to be no relationship with moving. This lack of relationship, along with the moderate positive relationships with community work and contacting officials, suggests that moves may not disrupt the participation of individuals as severely as may have previously been thought based on evidence about voting. And so perhaps people rather quickly re-integrate themselves into the political life of campaigns and communities after they move — perhaps even faster than the one year time resolution captured here.

The relationship between parenting and political participation has not been well studied, despite the fact that parenthood is as common as moving or attaining some level of formal education, and that it has as far-reaching consequences in terms of social status, economic resources, and psychological well-being. Those that have investigated the relationship began with the strong prior that parenthood ought to decrease participation, but the cross-sectional studies to date have not found evidence to support this idea. Neither men nor women appeared to have less time or money for use in participation, nor were parents especially less likely to participate than people without children. The sole exception to this pattern were single, unemployed women who, as mothers, were much less likely to participate than similar women without children (Burns, Schlozman and Verba, 2001).

This dissertation has shown that men and women react differently to the transition to parenthood and to the aging of their children. Men appear to participate more as their children age (and this effect cannot be due solely to the aging of the men themselves, since at any given age of a child, fathers have diverse ages), while women do not appear to react to motherhood with long term changes in their participation trajectories. For example, while men increase their community work as their children age, women tend to slightly increase their involvement until their children are around 6 years old, and then they tend to decrease their community activity. Men increase their donations of money over time, while for women, there tends not to be much of a relationship between donating money during campaigns and the ages of their children.

This chapter on parenting also shows that the timing of the transition to parenthood has long term consequences for the participation trajectories of individuals, both men and women. In general, people who have children early are less likely to get involved than people who have children later at any moment in time from that point on (controlling for education and income as they change over time and for unmeasured time-constant aspects of individuals). This result suggests that resources may still play a role in the link between children and participation: since one could probably assume that the two sets of parents (early vs. late) have similar interests in their children's welfare. These analyses also show that women are likely to increase their involvement in community work in the first few years after their first child is born — a finding that runs against the resources argument, but which perhaps indicates the “pull” into new social networks (where political mobilization can occur) exerted by children. In the end, I agree with Burns, Schlozman and Verba (2001) when they argue that the major impact of parenthood has to do with the kinds of social networks men and women fall into (or out of) as a result of becoming parents. The particular social networks for fathers appear to have more mobilizing potential than those for women. The precise testing of this intuition will have to occur when better measures of the mobilizational characteristics of social networks become available.

6.2 More Questions

Given the discussion in the introduction, it is clear that the analyses presented in this dissertation only scratch the surface of what is possible using these data. One major way that I hope to push this work forward in the future is by adding voting as an outcome to be explained. It will have to be analyzed using a different concept of time than the other activities (instead of year, the unit of time will have to be presidential/congressional election). Nevertheless, it will make the link to cross-sectional findings clearer given the preponderance of research on voting in the literature.

I would also like to extend the analysis of transitions, sequences, and durations beyond the mainly descriptive explorations that were restricted to Chapter 2. Information about the persistence of participation, or the influence of participation in one type of event on activity in another, is substantively important and still unknown. We would like to know if individuals engage in ten different types of activity over time, of it they repeat a single type of activity ten times in that same period. It is clear that it would be important to discover what qualities of an activity tend to increase the length of spells of action, change the intensity with which it is pursued in a given moment, and the extent to which participation in certain types of activities tends to complement participation in other types. For example, if we learn that participation in one kind of activity is the stepping stone to both future acts of that types as well as to future acts of other types, then mobilization attempts can focus on the “gateway” activities rather than types that are more rare or harder to convince people to do.

Most current discussions about participation refer to the idea of bringing “new voices” into politics or to the exclusion of such voices — where “voices” represent different people with diverse interests and values. What I have suggested so far is that we should not only ask how to encourage non-participators to become involved, but also ask (1) at what moment in time (their lives, within a campaign, or in the history of a regime) are they to be involved and (2) once we've succeeded in stimulating one moment of participation from someone who has never before acted, how can we help this person continue (if we think that continuing is in the person's and society's best interests)? This dataset has the potential to offer some preliminary answers to these questions as logical elaborations of the work done in the dissertation.

6.3 Next Steps

Despite its unique strengths, the dataset at the core of this dissertation has a number of limitations. Obviously, a study to answer the specific questions posed here could be designed better.

The sample design of such a study would necessarily involve multiple interviews with the same people over decades of their lives. However, it would be ideal to include multiple birth cohorts in the study from the beginning and to select more as time went on. To assess and adjust for non-random panel attrition, fresh random cross-sections ought to be included at each interview point as well.[6]

In order to attain reports about the timing of political behavior with high resolution and accuracy, the study could involve annual 15 minute telephone interviews interspersed between 1 hour in-person interviews occurring every 5 years or so. Such a study would also include some way to verify some of the reports of participation — tracing the receipt of some random set of letters reportedly sent; checking the public record for receipt of campaign contributions; examining minutes of organizations for attendance at meetings; or even asking respondents themselves for "harder" evidence of their activities.

The questionnaire for such a study would involve detailed questions about organizational membership, mobilization attempts by other people and groups, civic skills and civic status, and SES as it changes over the lives of the individuals. Also, it would be important to measure the political, economic, and social contexts in which individuals live and work. That is, I would want to know about the events occurring around the individual and the perceptions of, interpretations of, and reactions to these events.[7] One hint that has

[6] In some ways, this design would be like that used by the National Election Studies since the early 1990s — panel studies with fresh cross-sections — only it would follow people over longer spans of time at more regular intervals.

[7] The measurement of contexts, or "ecometrics" (Raudenbush and Sampson, 1997), could occur via the observation of interviewers, by automated coding of newpapers for events, or via use of historical and

emerged from this dissertation is that participation may look sporadic because it mainly occurs as a *reaction* rather than a habit. This questionnaire must also be designed to encourage accurate recall of past participation information — what individuals did, how they explain their behavior, and most importantly, when they did it. Other scholars have made fruitful use of calendars for such recall, and I envision using the same methods (See Belli, Shay and Stafford, 2001, for a comparison of such methods).

Eventually, it might also be important to expand such a study to countries other than the U.S.A. In some ways, limiting the study to participation in this country is equivalent to restricting the values of both the outcome and the explanatory variables. The results so far have hinted that particular repertoires of political activity may be determined largely by historical period (e.g. the lack of protest activity among the older generation in the Political Socialization dataset). Including other countries in a long-term study of political activity might reveal more about the diversity of what "political participation" means in different places and situations. Do certain types of political systems or social, political, or economic circumstances appear to lead individuals to choose certain types of activities over others? What are the consequences of these kinds of choices for the stability of regimes and the "health" of democracies?

6.4 Final Thoughts

In many ways, this dissertation has been more about re-defining the questions we ask about political participation than about uncovering new causal variables. Looking over the analyses that have been included here (and those that have been excluded, as well), it is hard to escape the feeling that what I have been analyzing are *reactions* to events. The chapters suggest that there is a distinction between events which are the time-varying stimuli for particular moments of participation and relatively time-constant attributes of people which enable individuals to react effectively to such stimuli. However, this distinction between an event and an attribute or status is not strict. For example, one might think of education or parenthood as being attributes, but the transition from uneducated to educated, or non-parent to parent, can be seen as an event in and or itself. And moving from one place to another is easily thought of as an event, but it may be that the status of "newcomer" or "old-timer" is more important from the point of view of participation than the event of

contemporary data from the federal or local governments on conditions and characteristics of places and elections.

moving itself. My hunch is that events tend to be local (such as a neighbor requesting a signature on a petition or a controversial school board or city council vote).

In the end, I think that past literature has taught us much about which attributes are associated with participation, but that the relationship between events and political action has barely begun to be explored.

The most general question that I think proponents of participatory democracy ought to be asking about political participation is about how to enable people to react effectively when the opportunity to participate arises. Ensuring effective reactions involves making sure that people can perceive and interpret events and situations that pose threats to, or provide opportunities for, their interests and concerns. Once individuals have accurately understood what is at stake in a particular moment, then individuals must be able to take action. Such a normative stance also seems to hew more closely to what we currently know about participation than a position that aims to have everybody always participating regardless of what is at stake (or a position where only elites participate and they do so full-time). Such a concern is also congruent with worries about inequality. That is, proponents of participatory democracy should not stop attempting to bring new voices into politics, but they should start asking when such voices ought to be brought in, how long anyone should stay, and how often individuals should go through this cycle of entrance into and exit from politics. In some ways, this calls for an integration between the literature that I have mostly relied upon in this study, which analyzes the behavior of individuals, and the literature that I have mostly left out of this study, which analyzes the collective action problems of groups.

My findings have shown that more individuals engage in political participation than previously thought when seen from the perspective of years of life. This does not mean that concerns about inequality are misplaced. Instead, I think that the evidence that scholars have been using about inequality is ambiguous due to limitations inherent in the data. I think that the most substantively important information about political participation in a democracy consists of the rates, timing, duration, and sequence of activity over time within peoples' lives. And all who judge and assess our democracy, or who desire to change the ways that ordinary individuals relate to politics, ought to focus their energy on understanding and gathering data about the dynamics of political participation.

Appendix A Measures of Political Participation

The Study of Political Socialization includes a wide array of measures of political participation, based on closed- and open-ended questions.

Electoral Participation Questions about the occurrence, timing, and content of acts of this type were asked of the class of 1965 in 1973 and 1982. In 1997 detailed timing information was not asked for these items. The focus of the actions were collected as open-ended responses to the "what was it about" questions. These open-ended responses were then aggregated into very detailed numeric codes. I constructed the variables indicating school oriented participation using these codes. The questions were:

- **Campaign Influence** First, did you talk to any people and try to show them why they should vote one way or the other? When was that? What issue/candidate was it about?
- **Campaign Rallies** Have you gone to any political meetings, rallies, dinners, or other things like that since (1965/1973/1982)? When was that? What issue/candidate was it about?
- **Campaign Work** Have you done any other work for a party, candidate or issue since (1965/1973/1982)? When was that? What issue/candidate was it about?
- **Campaign Button** Have you worn a campaign button or put a campaign sticker on your car since (1965/1973/1982)? When was that? What issue/candidate was it about?
- **Campaign Donation** Have you given money or bought any tickets to help a particular party, candidate, or group pay campaign expenses since (1965/1973/1982)? When was that? What issue/candidate was it about?

Non-electoral Participation Much political activity occurs outside the periodicity marking elections. These include contacting public officials, writing letters to the media, taking part in demonstrations, and working on local issues. The timing as well as the nature of these efforts are available.

The following questions were asked about such activities in the 1973, 1982, and 1997 waves of the Study of Political Socialization for the panel of respondents who were 18 years old in 1965:

"Aside from activities during election campaigns, there are other ways people can become involved in politics."

Contacting For example, since (1965/1973/1982) have you written a letter, sent a fax or e-mail message, or talked to any public officials, giving them your opinion about something? (IF YES) When was that and what was it about?

Letter to Editor Since (1965/1973/1982) , have you written a letter to the editor of a newspaper or magazine giving any political opinions? (IF YES) When was that and what was it about?

Demonstration Since (1965/1973/1982), have you taken part in a demonstration, protest march, or sit-in? (IF YES) When was that and what was it about?

Community Work Since (1965/1973/1982), have you worked with others to try to solve some community problems? (IF YES) When was that and what was it about?

Appendix B Details of the Analysis for Chapter 4

B.1 Political Socialization Results

The analysis of the Political Socialization Data was conducted using only years during which those individuals had lived in a new state for less than 30 years (for G2) and 40 years (for G1). This means that the analysis excluded individuals who never left their home state. Data from the Parent cohort interview was used to ascertain the length of residence for the Youth as of their senior year in high school. The need to match members of the two cohorts meant that the total available sample size for the Youth dropped from roughly 900 to roughly 600 individuals. Since roughly 200 of them never moved, the models used about 400 respondents as the base for analysis for G2. Since fewer member of G1 moved, the base for that analysis was even smaller — 122 individuals.

The variables were coded as follows: Home(1=Home Owner, 0=Not); Church (1=Attend At Least Once Per Week to 0=Never Attend); Education (Mean=0,SD=1), Income(1997 dollars)(Mean=0,SD=1), Pres. Election Year and Interview Year were each coded 1 for those years, and 0 otherwise. Tenure ran from 0 to 15 years living in a place after a move. For G1, Age was added as a control variable as well, centered at 50 years.

The Political Socialization dataset is the result of a 4 wave panel study. To construct the 35 years of annual information about participation and mobility, retrospective reports of the respondents were used. Home ownership, church attendance, education, and income were collected at each interview point, and then their values were linearly interpolated for the intervening (non-interview) years.

B.1.1 Analyzing the Probability of Events

The analysis was described in detail in the text. The coefficients on the spline basis functions of residential tenure η_k(Tenure) are not interpretable directly as corresponding to the intra-knot pieces. For example, the coefficient associated with η_1(Tenure) is *not* the slope of the line over the first segment (0 to 3 years) of residential tenure. Rather, the piecewise polynomial emerges from a combination of these coefficients. Although z-tests of the individual coefficients of the spline function are presented here, the only substantively relevant hypothesis tests involve joint hypothesis about all of the spline coefficients (the p-values for some of which I presented in the text).

Tables B1 to B4 contain the coefficients from a linear probability mixed-effects, or multi-level, model for G2 and tables B5 to B8 contain the results for G1. All models were estimated using Restricted Maximum-Likelihood (REML), which provides better estimates of the variance components (i.e. the random effects). The variation of the random coefficients is labeled sd(u_0) for standard deviation of the Intercept, sd(u_1) for the standard deviation of the first component of the spline function (η_1), sd(u_2) for the second spline component, etc.. For this analysis, the random coefficients were restricted to be uncorrelated — partially because of the construction of the spline components, and partially because there was no reason to believe there to be any correlation among them. For the under-27 analysis, the range of the tenure variable ran from 0 to 9 years, for the over-27 analysis the tenure variable ran from 0 to 15 years, and for G1 the variable ran from 0 to 20. The parameter of the CAR1 error process (ϕ) is also shown. For the G2 models, I present here all of the models with the ϕ estimated, even though, for many of the models, it's inclusion did not add appreciably to the fit of the model (as measured by likelihood ratio tests and inspection of the AICs). The row labeled "N" refers to the number of individuals in the model, and the row labeled "Time" refers to the total number of years analyzed.

Table B1: Non-Electoral Participation as a Function of Residential Tenure [Part 1]: G2

	Community Work				Contact Officials			
	Age<27		Age≥27		Age<27		Age≥27	
	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
(Intercept)	0.032	0.087	0.021	0.438	0.068	0.001	0.024	0.363
η_1(Tenure)	0.072	0.059	0.034	0.286	0.010	0.806	0.048	0.099
η_2(Tenure)	−0.068	0.241	0.100	0.005	−0.001	0.982	0.131	0.000
η_3(Tenure)	−0.149	0.079	0.099	0.007	−0.088	0.332	0.108	0.000
Home	−0.025	0.271	0.070	0.003	−0.037	0.125	0.039	0.084
Church	−0.025	0.246	0.006	0.846	−0.021	0.381	−0.051	0.071
Education	0.075	0.003	0.117	0.000	0.092	0.001	0.121	0.000
Income	0.099	0.027	0.005	0.919	0.016	0.742	0.006	0.899
Female	−0.003	0.841	−0.042	0.051	−0.012	0.420	−0.012	0.585
Pres.Elect.Year	0.032	0.002	0.008	0.390	0.036	0.005	0.016	0.097
Interview Year	−0.004	0.751	0.027	0.058	−0.032	0.045	0.046	0.002
sd(u_0)	0.071		0.163		0.082		0.177	
sd(u_1)	0.124		0.290		0.053		0.273	
sd(u_2)	0.000		0.010		0.004		0.003	
sd(u_3)	0.380		0.244		0.282		0.081	
ϕ	0.155		0.277		0.061		0.114	
LogLik	189.518		−380.854		−243.406		−484.639	
N	369		416		369		416	
Time	2345		3357		2345		3357	

Table B2: Non-Electoral Participation as a Function of Residential Tenure [Part 2]: G2

	Letters to the Editor				Demonstrations, Protests			
	Age<27		Age≥27		Age<27		Age≥27	
	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
(Intercept)	0.029	0.009	−0.003	0.705	0.048	0.001	0.000	0.977
η_1(Tenure)	−0.010	0.563	0.009	0.390	−0.015	0.581	0.026	0.022
η_2(Tenure)	0.018	0.547	0.022	0.066	−0.011	0.762	0.007	0.637
η_3(Tenure)	−0.005	0.904	0.016	0.114	−0.035	0.477	−0.003	0.836
Home	−0.011	0.373	0.000	0.967	−0.015	0.364	−0.009	0.270
Church	−0.025	0.051	0.006	0.449	−0.030	0.066	−0.006	0.544
Education	−0.005	0.732	0.011	0.115	0.033	0.077	0.034	0.000
Income	−0.027	0.282	0.004	0.775	−0.020	0.542	0.003	0.884
Female	0.001	0.904	−0.002	0.669	0.002	0.869	0.004	0.534
Pres.Elect.Year	0.012	0.008	−0.002	0.651	−0.004	0.588	0.002	0.710
Interview Year	0.005	0.371	0.003	0.532	−0.023	0.016	0.017	0.008
sd(u_0)	0.076		0.000		0.065		0.050	
sd(u_1)	0.055		0.113		0.000		0.070	
sd(u_2)	0.001		0.116		0.001		0.000	
sd(u_3)	0.213		0.053		0.002		0.000	
ϕ	0.013		0.000		0.177		0.158	
LogLik	1908.243		2974.548		892.838		2394.113	
N	369		416		369		416	
Time	2345		3357		2345		3357	

Table B3: Electoral Participation as a Function of Residential Tenure [Part 1]: G2

	Money				Other Work			
	Age<27		Age≥27		Age<27		Age≥27	
	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
(Intercept)	0.011	0.321	−0.041	0.032	0.020	0.055	−0.031	0.031
η_1(Tenure)	0.064	0.055	0.040	0.041	0.011	0.640	0.002	0.901
η_2(Tenure)	−0.022	0.569	0.053	0.032	−0.009	0.791	0.005	0.765
η_3(Tenure)	−0.013	0.801	0.050	0.016	−0.025	0.618	0.001	0.934
Home	−0.011	0.402	0.026	0.107	−0.006	0.639	0.038	0.001
Church	−0.037	0.002	−0.021	0.304	−0.031	0.013	0.003	0.821
Education	0.041	0.005	0.103	0.000	0.024	0.093	0.076	0.000
Income	0.105	0.000	0.129	0.000	0.008	0.749	0.007	0.786
Female	0.007	0.322	−0.022	0.170	0.012	0.141	0.008	0.516
Pres.Elect.Year	0.071	0.000	0.062	0.000	0.046	0.000	0.020	0.000
Interview Year	−0.032	0.001	−0.001	0.895	−0.021	0.016	0.015	0.055
sd(u_0)	0.016		0.128		0.040		0.102	
sd(u_1)	0.279		0.178		0.097		0.037	
sd(u_2)	0.002		0.169		0.001		0.044	
sd(u_3)	0.156		0.020		0.169		0.086	
ϕ	0.000		0.009		0.000		0.046	
LogLik	888.553		606.950		1142.564		1655.164	
N	369		416		369		416	
Time	2345		3357		2345		3357	

Table B4: Electoral Participation as a Function of Residential Tenure [Part 2]: G2

	Rallies, Dinners				Buttons, Signs			
	Age<27		Age≥27		Age<27		Age≥27	
	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
(Intercept)	0.023	0.058	−0.002	0.935	−0.007	0.582	−0.014	0.353
η_1(Tenure)	0.020	0.564	−0.005	0.801	0.055	0.111	−0.019	0.194
η_2(Tenure)	0.001	0.981	0.015	0.546	−0.003	0.945	0.035	0.081
η_3(Tenure)	−0.050	0.419	0.019	0.410	−0.037	0.505	0.026	0.171
Home	−0.033	0.029	0.028	0.081	−0.003	0.833	0.021	0.096
Church	−0.023	0.108	0.013	0.540	−0.002	0.870	0.013	0.420
Education	0.023	0.185	0.098	0.000	0.024	0.164	0.020	0.174
Income	0.039	0.205	0.001	0.970	0.075	0.019	0.000	0.990
Female	0.005	0.578	−0.020	0.189	0.002	0.807	0.002	0.850
Pres.Elect.Year	0.066	0.000	0.052	0.000	0.136	0.000	0.067	0.000
Interview Year	−0.020	0.079	0.009	0.383	−0.029	0.015	0.011	0.209
sd(u_0)	0.000		0.124		0.000		0.098	
sd(u_1)	0.208		0.094		0.219		0.001	
sd(u_2)	0.116		0.158		0.159		0.000	
sd(u_3)	0.168		0.132		0.002		0.122	
ϕ	0.000		0.001		0.000		0.000	
LogLik	546.119		565.329		486.393		1169.289	
N	369		416		369		416	
Time	2345		3357		2345		3357	

Table B5: Non-Electoral Participation as a Function of Residential Tenure [Part 1]: G1

	Community Work		Contact Officials	
	$\hat{\beta}$	p	$\hat{\beta}$	p
(Intercept)	−0.032	0.355	0.048	0.081
η_1(Tenure)	−0.001	0.967	−0.005	0.885
η_2(Tenure)	0.007	0.851	0.060	0.100
η_3(Tenure)	0.039	0.201	−0.019	0.492
Home	0.046	0.045	−0.036	0.047
Church	−0.026	0.351	−0.018	0.373
Age	−0.001	0.891	−0.001	0.910
Eductation	0.093	0.015	0.105	0.000
Income	0.104	0.019	−0.016	0.642
Female	0.007	0.749	−0.027	0.084
Pres.Elect Year	−0.005	0.624	0.041	0.001
Interview Year	0.004	0.776	−0.008	0.560
sd(u_0)	0.077		0.000	
sd(u_1)	0.005		0.057	
sd(u_2)	0.000		0.073	
sd(u_3)	0.077		0.003	
ϕ	0.406		0.248	
LogLik	244.533		119.768	
N	122		122	
Time	1829		1829	

Table B6: Non-Electoral Participation as a Function of Residential Tenure [Part 2]: G1

	Letters to Editor		Demonstrations	
	$\hat{\beta}$	p	$\hat{\beta}$	p
(Intercept)	−0.019	0.030	0.000	0.972
η_1(Tenure)	0.009	0.384	−0.003	0.670
η_2(Tenure)	−0.012	0.324	0.007	0.421
η_3(Tenure)	−0.016	0.053	−0.003	0.618
Home	0.013	0.012	0.003	0.536
Church	0.004	0.486	−0.003	0.532
Age	−0.001	0.495	−0.002	0.302
Education	0.039	0.000	0.010	0.140
Income	−0.005	0.602	−0.003	0.696
Female	0.004	0.390	−0.003	0.525
Pres.Elect Year	0.013	0.012	0.002	0.510
Interview Year	0.013	0.023	−0.001	0.623
sd(u_0)	0.000		0.000	
sd(u_1)	0.046		0.056	
sd(u_2)	0.000		0.057	
sd(u_3)	0.000		0.000	
ϕ				
LogLik	1843.401		2851.487	
N	122		122	
Time	1829		1829	

Table B7: Electoral Participation as a Function of Residential Tenure [Part 2]: G1

	Money		Other Work	
	$\hat{\beta}$	p	$\hat{\beta}$	p
(Intercept)	−0.025	0.328	−0.019	0.112
η_1(Tenure)	0.010	0.765	−0.008	0.592
η_2(Tenure)	0.002	0.955	−0.002	0.925
η_3(Tenure)	0.020	0.397	0.008	0.517
Home	0.056	0.001	0.021	0.004
Church	−0.044	0.027	−0.003	0.681
Age	0.016	0.019	−0.004	0.223
Education	0.069	0.009	0.041	0.000
Income	0.040	0.201	0.002	0.907
Female	0.002	0.882	0.001	0.910
Pres.Elect.Year	0.068	0.000	0.029	0.000
Interview Year	−0.001	0.958	0.005	0.456
sd(u_0)	0.048		0.007	
sd(u_1)	0.216		0.058	
sd(u_2)	0.001		0.001	
sd(u_3)	0.089		0.042	
ϕ	0.188			
LogLik	431.792		1322.543	
N	122		122	
Time	1829		1829	

Table B8: Electoral Participation as a Function of Residential Tenure [Part 2]: G1

	Rallies, Dinners		Buttons, Signs	
	$\hat{\beta}$	p	$\hat{\beta}$	p
(Intercept)	−0.021	0.305	0.003	0.821
η_1(Tenure)	−0.023	0.276	0.004	0.828
η_2(Tenure)	0.008	0.749	−0.003	0.875
η_3(Tenure)	0.036	0.139	−0.011	0.418
Home	0.044	0.001	0.015	0.103
Church	−0.024	0.130	−0.011	0.319
Age	0.010	0.064	−0.011	0.007
Education	0.001	0.980	−0.003	0.849
Income	0.076	0.003	0.012	0.502
Female	−0.008	0.511	−0.013	0.128
Pres.Elect.Year	0.055	0.000	0.072	0.000
Interview Year	0.013	0.256	0.010	0.282
sd(u_0)	0.039		0.019	
sd(u_1)	0.051		0.095	
sd(u_2)	0.001		0.046	
sd(u_3)	0.152		0.002	
ϕ			−0.069	
LogLik	539.760		953.105	
N	122		122	
Time	1829		1829	

B.2 NES Results

The analysis of the NES data (1970-2000) was conducted using only those respondents who (1) were interviewed face-to-face and (2) who had lived in their current place (meaning, a town, city, township, or county) for less than 15 years. All of variables were coded to have meaningful values at zero; multi category variables such as year(aka time), age and income were coded to have mean=0 and variance=1. Education was coded such that -1=grade school, 0=high school,1=some college, and 2=BA+.

The analysis was conducted using a generalized linear model, binomial family, logit link (aka a logistic or logit regression). The coefficients in the tables are odds ratios. Looking at the first column, of Table B9, one can see that women are estimated to be 1.44 times more likely to engage in "Other Work" for campaigns than men (holding all else constant), but the odds of participating in other campaign work are decreased by more than half (.42) for home owners compared to those who do not own homes (ceteris paribus). The coefficients on the spline basis functions of residential tenure η_k(Tenure) are *not* interpretable directly as corresponding to the intra-knot pieces. See the description of the Political Socialization analysis for more details.

Table B9: Modeling Electoral Participation as Function of Residential Tenure (NES Data 1970-2000) [Part 1]

	Other Work		Meetings, Rallies	
	$\exp(\hat{\beta})$	p	$\exp(\hat{\beta})$	p
(Intercept)	0.01	0.00	0.04	0.00
η_1(Tenure)	2.01	0.01	1.31	0.17
η_2(Tenure)	1.14	0.68	1.49	0.09
η_3(Tenure)	1.33	0.37	1.12	0.61
η_4(Tenure)	1.19	0.64	1.60	0.07
η_5(Tenure)	1.04	0.87	1.10	0.60
Home (1=Owner,0=Not)	0.42	0.00	0.60	0.00
Church (1=Freq to 0=Never)	1.20	0.01	1.11	0.04
Time(Mean=0,SD=1)	1.35	0.01	0.92	0.26
Age (Mean=0,SD=1)	1.24	0.10	0.92	0.38
Female(Female=1,Male=0)	1.44	0.05	1.63	0.00
Income (Mean=0,SD=1)	1.12	0.09	1.24	0.00
Education (0=HS)	1.77	0.00	1.86	0.00
Pres. Election Year	1.06	0.56	1.11	0.17
Null Deviance	3296.39		5480.01	
Deviance	3086.91		5123.36	
N	10097		10109	

Table B10: Modeling Electoral Participation as Function of Residential Tenure (NES Data 1970-2000) [Part 2]

	Buttons, Signs		Money	
	$\exp(\hat{\beta})$	p	$\exp(\hat{\beta})$	p
(Intercept)	0.04	0.00	0.07	0.00
η_1(Tenure)	1.26	0.21	1.54	0.02
η_2(Tenure)	1.15	0.53	0.71	0.13
η_3(Tenure)	1.68	0.01	1.79	0.01
η_4(Tenure)	1.65	0.04	0.70	0.15
η_5(Tenure)	0.99	0.98	1.08	0.66
Home(1=Owner,0=Not)	0.67	0.00	0.42	0.00
Church (1=Freq to 0=Never)	0.83	0.00	1.58	0.00
Time(Mean=0,SD=1)	0.99	0.85	0.95	0.47
Age (Mean=0,SD=1)	0.98	0.82	1.05	0.62
Female(Female=1,Male=0)	1.77	0.00	1.09	0.50
Income (Mean=0,SD=1)	1.05	0.25	1.83	0.00
Education (Mean=0,SD=1)	1.36	0.00	1.85	0.00
Pres. Election Year	1.67	0.00	1.04	0.61
Null Deviance	6000.42		6037.30	
Deviance	5795.65		5233.44	
N	10104		9446	

B.3 Preliminary Results on Social Connectedness, Residential Tenure, and Political Participation

The test for the hypothesis presented in Figure 4.9 is a simple extension of the model I used to examine the hypothesis in Figure 4.2. This time, however, the intercept and slope of the spline function is allowed to be different for different values of home ownership and church attendance using interaction terms, changing the level-1 equation (the growth, or time-level, equation) to look like

$$\begin{aligned} y_{it} =& \beta_{0i} + \sum_{k=1}^{5} \beta_{ki}\eta_k(\text{Tenure}_{it}) + \beta_{6i}\text{Home}_{it} + \beta_{6i}\text{Church}_{it} + \\ & \left(\sum_{k=1}^{5} \beta_{k+6,i}\eta_k(\text{Tenure}_{it}) \cdot \text{Home}_{it}\right) + \\ & \left(\sum_{k=1}^{5} \beta_{k+11,i}\eta_k(\text{Tenure}_{it}) \cdot \text{Church}_{it}\right) + \beta_{17i}X_{it} + e_{it}, \end{aligned} \tag{B1}$$

where X is a placeholder for the variety of time-varying variables that enter into the equation as controls for level-shifts in the participation/tenure trajectory.[1]

Figure B1 presents the predicted values resulting from this model — which were generated using the same procedure as Figure 4.4 and 4.5, only this time the values were generated using Home Ownership and Church Attendance set to their minimum and maximum values ("No Home" and "Home", meaning not a home owner versus a home owner) and ("No Church" and "Max Church" representing those who never attend church and those who attend church at least once per week). Each panel of Figure B1 displays the value predicted for a given combination of values for home ownership and church attendance. Each row contains the values for one of the four acts of campaign participation.[2] For example, the top left hand panel shows the predicted probabilities of participation in campaign meetings and rallies, during years in which a respondent never attended church (marked by the strip header labeled "No Church"), and during years in which the respondent did not own a home (marked by the strip header labeled "No Home"). The spline knot locations are marked in each panel by thin, dotted, vertical lines. The comparison between people who own homes,

[1] In this model, the number of random effects (i.e. the u's) is still 2, one for the intercept, and another for the linear effect of Tenure. I did attempt to estimate models allowing for a random effect for each spline basis, but this was just clearly asking too much of the data and the models never converged.

[2] For this preliminary analyses, I restricted the focus to campaign activity of G2 only [so that I could examine 16 estimated functions rather than 128].

and those who do not (among those people who never attend church) is between the first two columns, and among those people who attend church at least once per week, between the third and fourth columns.

If the hypothesis displayed in Figure 4.9 is correct, then the panels in the second column (home owners who do not attend church) ought to display less disruption, and faster recovery than the panels in the first column. Similarly, the panels in the third column (frequent church attenders who do not own homes) ought display less disruption, and quicker recovery, than the panels in the first column (people who never attend church who do not own homes). And, if both home ownership and church attendance do not somehow cancel one another out, then the fourth column should display less disruption, and quicker recovery than all the rest.

The patterns clearly vary by act, but some regularities are apparent.

During moments in which individuals both own homes, and attend church frequently, moving seems to cause little to no disruption, followed by a pattern of participation *above* the level they began at, in their first year of residence. For each type of act, the probability of participation is "bumpy" across the years that an church-going/home-owner lives in a new place — with more rapid increases after about 9 years in a new place, and a lull (not below starting levels) between 5-9 years of residence.[3]

The tradeoff between church-going and home-owning, however, is not so clear. For attending campaign meetings and rallies, each of the four panels in the top row shows a similar starting place (at 0 years living in a place), with very little early disruption. Frequent church goers who do not own homes are predicted to participate at a constant rate, regardless of the number of years they've lived in a new place. Home-owners, who do not attend church, are predicted to have much wider swings in their probabilities of attending meeting and rallies — with an increase after the first three years, followed by a sharp decrease after 6 years, and an increase after 9 years. The people representing the "low social connectedness" category look the same as the other categories during the first three years of residence, only to experience a long term decline afterwards. Since these models have controls for educational attainment and income, it is hard to see this pattern as only reflecting SES differences in type of person. But it is puzzling to see the decline occur *after*

[3]This "bumpy" behavior, controlling for presidential election year, interview year, age, etc. suggests that individuals might have an activation/exhaustion cycle to their participation. Note that the knots do not require such bumpiness — see, for example other smooth curves in other panels of this figure.

the first three years of residence.

Attending church alone is not enough to cause tenure to increase (or really decrease) the probability of participating in "Other Campaign Work." Rather, home owners, in this case, are predicted to start out at a (relatively) high rate, and then experience a steady decline (until 12 years, after which the data are too sparse to be worth interpreting seriously). In this case, home-owners who are church-goers are predicted to be able to sustain a relatively high level of participation regardless of the number of years they live in a place.

Donating money seems to respond most strongly to owning a home — home owners who do not attend church experience no disruption at all, and instead just are predicted to have a steadily increasing probability of donating money across their years living in a new place. This trajectory is much steeper and higher in level than that predicted for the maximum socially connected individuals (home-owning, church-goers) — who may be diverting more of their income stream to church and away from political campaigns explicitly. In fact, both of the panels for church-attending show lower probabilities for donating money to political parties or campaigns than either of the "secular" panels. The lowest social connection group, who neither own homes nor attend church, does experience a small, short, disruption, followed by a recovery that surpasses their initial values.

The predominant pattern for wearing buttons and displaying signs does not involve any early disruption at all. Rather, in all cases, individuals are predicted to increase (slightly for the non-home owning, non-church goers, and more strongly for the other groups), followed by a decrease after about 6 years, followed by an up-tick after about 9 years.

In sum, although the highly socially connected individuals are predicted to engage in participation with higher probability, without early disruption, it is not as though other groups are predicted to have deep, declines in participation associated with their first few years living in a new place. After living in a place for a few years, many of the types of individuals depicted in Figure B1 are predicted to have declines in participation (after 3 or 6 years), followed by an upturn after about 9-12 years. Again, the sequence of disruption followed by recovery is not apparent — instead, the pattern is (as summarized in Figure 4.4 and 4.5) is one of no early impact of moving, a late decline, and an even longer term increase.

These results clearly do not square with hypotheses gleaned from the current, common knowledge about residential mobility. Instead they suggest that there is some aspect of residential mobility associated with the loss of social networks, and that home

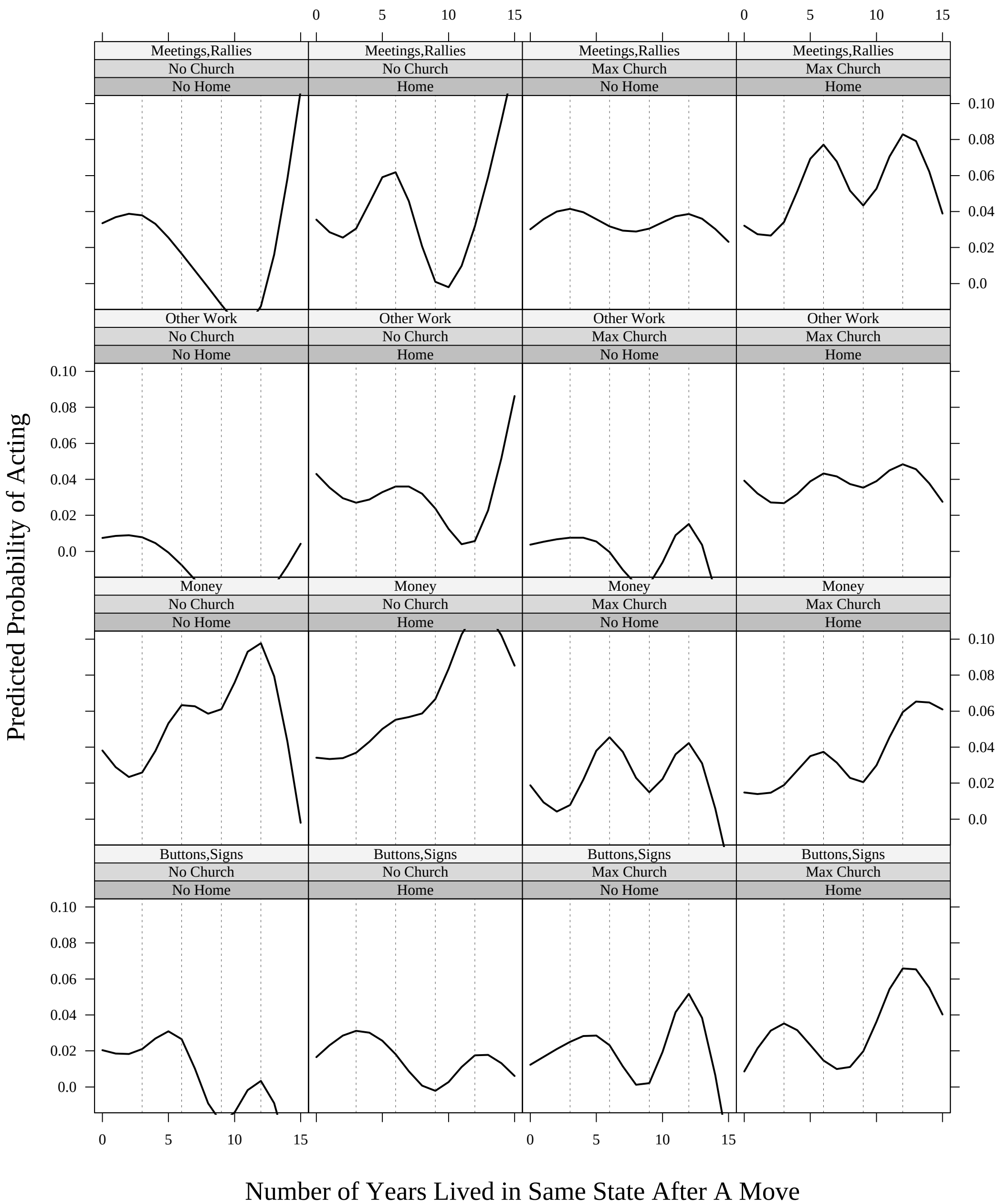


Figure B1: Estimated Campaign Participation Trajectories After Moves By Home Ownership and Church Attendance: G2 Only

ownership and church attendance together seem to help individuals maintain high levels of participation despite moves across state lines. However, different acts seem to respond differently to the two ways that individuals can gain social connections: that the route by which individuals gain their social connections might be as consequential for their participation opportunities and choices as the fact that social connections exist at all — connecting to a community by owning a home clearly places individuals directly in the path of those mobilizing to raise funds, while connections built through a church does not have the same effect. However, church goers seem to attend meetings and do other work (when they own homes) at a kind of steady rate.

Figure B2 uses the NES data and roughly matches the general findings from Figure B1: no clear pattern of disruption and recovery emerges, even when "social connectedness" is represented more directly using home ownership and church attendance instead of "years living in a place."[4] As before, it appears that individuals with both attributes (church-going, home-owners) have a slight edge, in that, among that group, individuals who have lived in a place for a longer amount of time appear to have a slightly higher probability of participation than new arrivals. However, this pattern is not strong or consistent.

Overall, the results from the NES confirm my (null) findings using the Political Socialization study — a pattern of short term disruption followed by longer term recovery is not the norm. In Figure 4.8, individuals who have lived in a place longer do not appear to be appreciably different from new-comers. If anything, the patterns in this figure are muted version of those shown in Figure 4.4 and 4.5 — with monetary donations showing more "up-and-down" activity with respect to tenure — an activity that is not easily explained by the theory about the loss and rebuilding of social connectedness.

[4]The functions for the NES analyses here were fit using a natural cubic b-spline function within generalized linear model (binomial family, logit link). The analysis was restricted to individuals who were born between 1945 and 1949 — i.e. an age cohort that is roughly the same as the G2 cohort in the Political Socialization study.

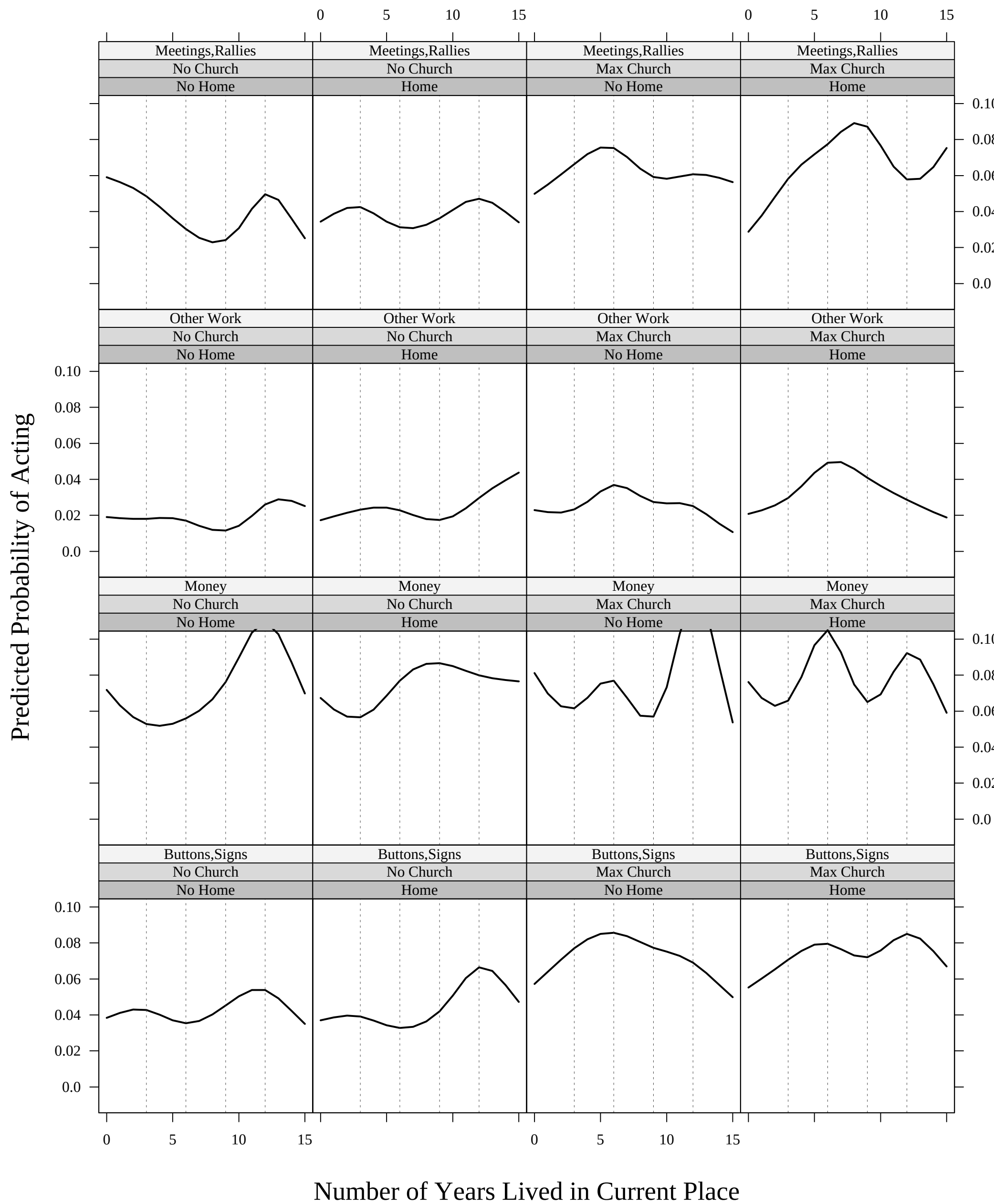


Figure B2: Predicted Participation Trajectories After Moves By Home Ownership and Church Attendance (NES 1970-200 Data)

Appendix C Measuring Skills and Status

To what extent are the variables that are available in the Political Socialization dataset reasonable indicators of "civic skills" or "civic status"? How might we find out? There are two main steps to answering such questions about measurement. First, one must choose those variables that exist in the Political Socialization dataset whose domain of content might be agreed upon by the relevant community of scientists to include "civic skills" and "civic status". Next, one must ask whether the selected variables are related to other variables that are agreed to be good measures of the concepts I care about.

The first task involves specifying what "civic skills" and "civic status" mean (i.e. what job these concepts are expected to do for social scientists). Then describing what criteria a measure of these concepts must fulfill in order to be plausible. I have done this in the chapter on this topic.

The second task requires good measures of the concepts that I'm aiming to measure. In the ideal case, one could take the group of variables chosen in the first step, and then examine the extent to which each of them is related to previously constructed, and validated, measures of "civic skills" and "civic status". Of course, the fact that I'm engaging in this measurement exercise at all is proof that I don't have access to excellent measures of these concepts in the Political Socialization dataset. However another Citizen Participation dataset Verba et al. (1995) does have excellent indicators of civic skills/status as well as versions of the variables that exist in the Political Socialization data set. Given this fortuitous situation, I can then ask whether the variables that exist in the Political Socialization data are good predictors of either civic skills or civic status as operationalized in the Citizen Participation dataset.

C.1 Specifying a Measurement Model

Table C1 shows the percentages of respondents reporting engagement in the activities that Verba, Schlozman and Brady (1995) consider part of the "civic skills" scale, and that Nie, Junn and Stehlik-Barry (1996) consider measures of "civic status". The items are sorted to show the relative "difficulty" of the various items — for example, many more people report feeling comfortable speaking at in public than report ever writing a letter as a part of their participation in an organization. Similarly, while 37% of the respondents reported having an acquaintance with a member of a local council, only 5% reported having an acquaintance with a member of the national media. In order to figure out how the types of variables available in the Political Socialization dataset relate to "civic skills" and "civic status", I need to create scales out of the items shown in Table C1.

Ideally, any scales that measure civic skills and status using the items in Table C1 will be an interval level scale — meaning that, for an example, a score of 4 on such a scale could be seen as twice as big as a score of 2. Interval level scales will allow me to estimate relationships, such as correlations, between the two scales and will also facilitate the search for relationships between other variables (ones available in the Political Socialization dataset) and the scale themselves. Also, ideally, the scale would be created taking into account for random variation in who each person answered these questions.

C.1.1 Creating the Scales: Unconditional Models

The procedure that I use here embeds an item-response model within the framework of a mixed-effects, or multilevel, model.[1]

This method creates an interval level scale for each group of items, estimates the "difficulty" of each item, and allows estimation of the "abilities" of each of the individuals in the dataset on the particular scales (i.e. allows the assignment of interval level scores for "civic skills" and "civic status"). It also takes into account the fact that there are two sources of randomness in the model: random variation across items within individuals due to measurement error and random variation across individuals due to differential abilities.

The structure of the model is as a logistic multi-level model, otherwise known as a generalized linear mixed model (glmm). The response variable, Y_{ij}, consists of 1s for

[1] I have hewn closely to the examples of this kind of model provided by Raudenbush and Bryk (2002, Chapter 11) and Cheong and Raudenbush (2000). This kind of model is also known as a Rasch model with random effects, or a multilevel model with latent variables.

Table C1: Item Responses for Civic Skills and Civic Status

	Percent	
	No	Yes
Civic Skills		
Speak well enough to make statement in public	31	69
Job: gone to a meeting	50	50
Job: written a letter	58	42
Job: given a presentation	70	30
Job: planned a meeting	71	29
Church: gone to a meeting	78	22
Org: gone to a meeting	83	17
Could write a convincing letter to govt	84	16
Church: given a presentation	86	14
Church: planned a meeting	88	12
Org: given a speech	91	9
Org: planned a meeting	91	9
Church: written a letter	92	8
Org: written a letter	92	8
Civic Status		
Member of local council	63	37
Some other local board	67	33
Someone workng for local media	69	31
Member state legislature	80	20
Member of congress	88	12
Someone workng for nat'l media	95	5

Data: American Citizen Participation Study, 1990

each item that the respondent answered in the affirmative, and 0s for each item that the respondent answered in the negative. We assume that there is some underlying probability of an affirmative answer for each item i within each individual j denoted μ_{ij}. Since Y can only take on values of 0 or 1, $E(Y_{ij}) = \mu_{ij}$. It is common in binary response models using explanatory variables to model the *log-odds* of a correct response rather than the probability of a correct response because probabilities are bounded between 0 and 1 and the fitted effect of a linear predictor may fall outside this interval. That is, the log-odds of a correct response (represented by the logit function) links the probabilities of a correct response to the explanatory variables. This is the standard setup for a Generalized Linear Model (glm). That is, while the probabilities of a correct response are μ_{ij} the log-odds of a correct response can be written:

$$g(E(Y_{ij})) = g(\mu_{ij}) = \eta_{ij} = \ln\left(\frac{\mu_{ij}}{1-\mu_{ij}}\right) \tag{C1}$$

Given this link function, we can specify that the log-odds of answering in the affirmative on item i depends on 1) whether the item measures skills or status and 2) which particular item in involved. Within each person j the model of item response is

$$\begin{aligned} \eta_{ij} =& D_{SKILLS_{ij}}\left(\pi_{SKILLS_{ij}} + \sum_{m=1}^{13} \alpha_{mij} S_{mij}\right) \\ &+ D_{STATUS_{ij}}\left(\pi_{STATUS_{ij}} + \sum_{m=1}^{5} \delta_{mij} T_{mij}\right), \end{aligned} \tag{C2}$$

where,the dichotomous variables $D_{SKILLS_{ij}}$ and $D_{STATUS_{ij}}$ indicate with 1s whether or not the ith response is to an item that measures Skills or Status (and where, $D_{SKILLS_{ij}} = 1 - D_{STATUS_{ij}}$) and 0s otherwise, and S_{mij} and T_{mij} indicate that the ith item is item m in the particular scales. The skills and status "true scores" (i.e. log-odds of an affirmative response) across the sample are represented by $\pi_{SKILLS_{ij}}$ and $\pi_{STATUS_{ij}}$, adjusted for the different contributions of the items, and the α_{mij} and δ_{mij} parameters capture the difficulty of item m within the skills and status scales respectively. Although there are 14 items in the Skills scale, and 6 in the Status scale, only 13 and 5 are included here — with one item in each scale having a difficult set to 0 in order to identify the equation.[2]

In order to account for person-level variation in skills and status, this model allows the π coefficients to vary randomly across respondents around a mean value β with random effects u which are assumed distributed bivariate normal with means 0 and a variance-covariance matrix of $\boldsymbol{\tau}$:

[2]If there were missing data on these items, the m subscript would allow different items to measure skills and status for different people, and the π coefficients would be adjusted for this. In my case, there was no missing data.

$$\begin{aligned} \pi_{SKILLS_{ij}} &= \beta_{SKILLS} + u_{SKILLS_j} \\ \pi_{STATUS_{ij}} &= \beta_{STATUS} + u_{STATUS_j} \end{aligned} \tag{C3}$$

The combined model can be written as:

$$\begin{aligned} \eta_{ij} =& D_{SKILLS_{ij}} \left(\beta_{SKILLS} + \sum_{m=1}^{13} \alpha_m S_{mij} + u_{SKILLS_j} \right) \\ &+ D_{STATUS_{ij}} \left(\beta_{STATUS} + \sum_{m=1}^{5} \delta_m T_{mij} + u_{STATUS_j} \right), \end{aligned} \tag{C4}$$

with the added constraint that the item difficulties be held constant across respondents (an assumption from the Rasch model literature, and not that important for this task).

Figure C1 displays the α and δ parameters estimated from this model.[3] The left column of the figure shows the item difficulties for the Civic Skills scale. The right column shows the item difficulties for the Civic Status scale.

More important than the item difficulties are the estimates for the latent traits of Civic Skills and Civic Status. A model based on equation C4 using mean-deviated versions of the S and T variables provides estimates for *pi* which can be interpreted as the latent levels of Civic Skills and Civic Status for the "typical respondent", adjusted for the two levels of error (item-level and person-level) in the model. Since *pi* are allowed to vary randomly across individuals, we can estimate the amount of variation in the estimates of skills and status between respondents as well as the correlation between skills and status. In this case, the typical Civic Skills score on the logit scale is -8.0455 [exp(-8.0455)=0.0003205366, 1/(1+exp(-8.0455))=0.9996796] and the typical Civic Status score is -2.5929 [exp(-2.5929)=0.07480523, 1/(1+exp(-2.5929))=0.9304011] (both estimated to be different from zero with p¡.0001). The standard deviation for Civic Skills is 1.28 (95% CI: 1.23 to 1.33). The standard deviation for Civic Status is 2.06 (95% CI: 1.97 to 2.14). The amount of variation in the Civic Status scale is much more than that in the Civic Skills scale — ought to lead to more efficient estimates, making it easier to detect smaller effect sizes than for the Civic Skills scale. The correlation between Civic Skills and Civic Status is estimated to be .48 (95% CI: .43 to .52). While this correlation is clearly not zero, it is

[3]Estimation of GLMM models continues to be an area of active research. The models in this section were estimated using the penalized quasi-likelihood approach.

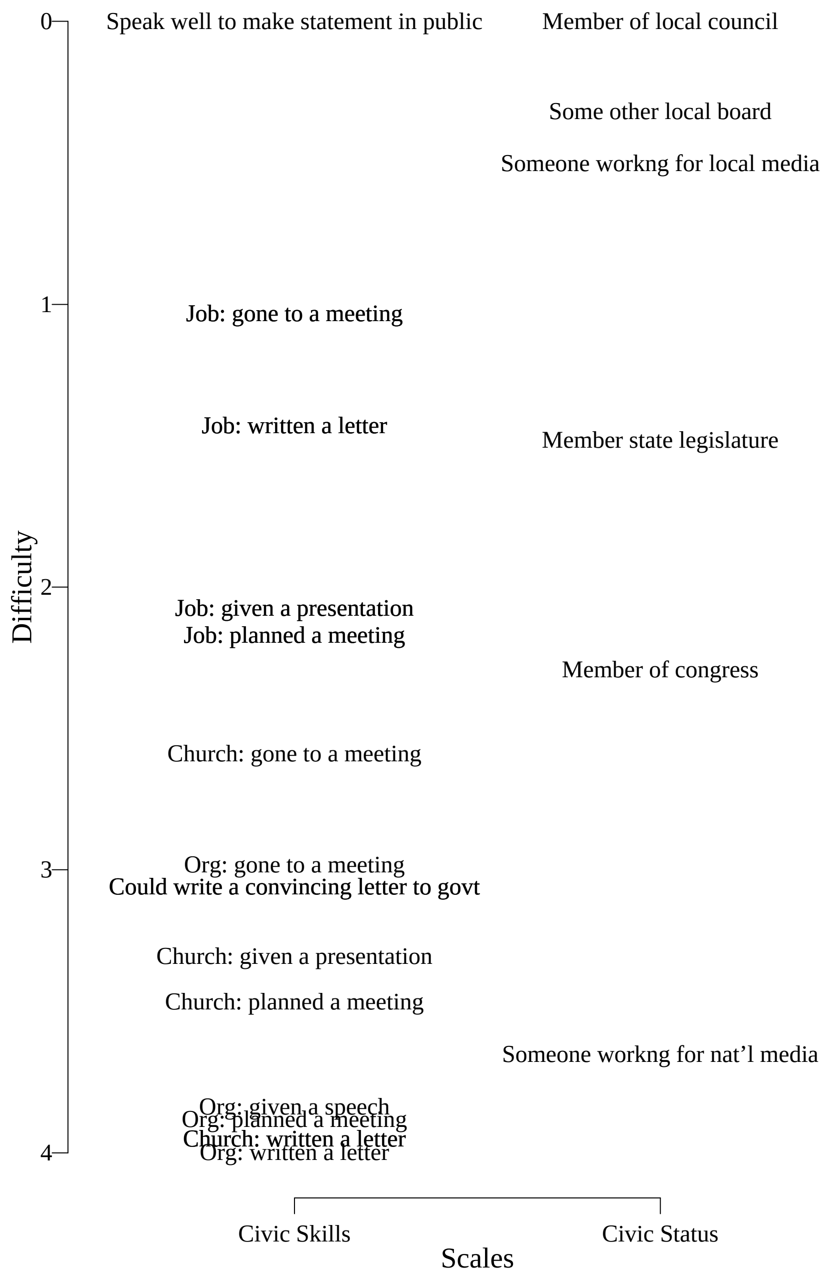


Figure C1: Item Difficulties for Civic Skills and Civic Status

not high enough to make me think that these two scales ought to be combined into a single scale.

C.1.2 Relating Other Items to the Scales: Conditional Models

The unconditional model suggested that Civic Skills and and Civic Status are two separate constructs, yet that they definitely are related to one another. The question motivating this work, here, is whether the variables available on the political socialization study are reasonable indicators of these latent traits. The variables that I selected as candidates for the role of indicators of Civic Skills and/or Civic Status are: Whether the respondent has a college degree (BA), Whether at least one of the respondent's parents have a college degree, The self-reported high school academic performance, Whether the respondent is a protestant, Whether the respondent is catholic, Whether the respondent attends religious services, The number of non-political organizations with which the respondent is affiliated, whether a family member is a member of a union, the family income, whether the respondent was employed in the last week, the total number of correct responses to a vocabulary test, and the number of years the respondent had lived in the same town. Each of these variables has been used as a proxy for "civic skills" and "civic status" in the past literature, and a version of each of them is available in the Political Socialization dataset.

By modifying the model presented in equation C4, I was able to estimate the influence of each of these variables on the latent variables of Civic Status and Civic Skills. The modification to the model occurred simply by adding a series of Z terms for the various variables I listed above:

$$\begin{aligned}\eta_{ij} =& D_{SKILLS_{ij}} \left(\beta_{SKILLS_0} + \sum_{q=1}^{12} \beta_{SKILLS_q} Z_{qj} + \sum_{m=1}^{13} \alpha_m S_{mij} + u_{SKILLS_j} \right) \\ &+ D_{STATUS_{ij}} \left(\beta_{STATUS_0} + \sum_{q=1}^{12} \beta_{STATUS_q} Z_{qj} + \sum_{m=1}^{5} \delta_m T_{mij} + u_{STATUS_j} \right),\end{aligned} \tag{C5}$$

where now the β_{SKILLS_q} terms capture the relationship between the Political Socialization variables (labeled Z) and the Civic Skills latent variable, and the β_{STATUS_q} captures their relationship with Civic Status. One interesting side-effect of this model is that the correlation between the two latent variables is reduced (accounted for by the addition of characteristics of individuals omitted from the "unconditional" model of equation C4) — the correlation now is estimated to be 0.2949859(95% CI: 0.2319310 to 0.3555722). Also, the variance of the traits across people has decreased such that for Civic Skills it is 0.8869212(95% CI: 0.8438043 0.9322413), and for Civic Status it is 1.6663045 (95% CI: 1.5890042 1.7473652).

Figure C2 plots the β_{STATUS_q} against the β_{SKILLS_q}. Although all of the variables were re-coded to only take on values between 0 and 1 and to be dichotomous, the substantive meaning of a shift from 0 to 1 (which is what is captured by the coefficients) differs. For example, a shift from 0 to 1 in the variable representing the whether or not a respondent was a member of any non-political organizations does not have the same substantive meaning as a shift from 0 to 1 in the Union membership variable which indicates whether or not a member of the respondents' family is a member of a union.[4]

This figure presents the extent to which the various "candidate" variables relate to the latent variables of Civic Skills and Civic Status, controlling for measurement error.[5] The solid black lines represent coefficients of 0, meaning no influence or relationship between the item and the latent trait. For example, being employed (versus, unemployed in some manner or another) has no influence on whether an individual reports knowing different political actors, but instead has a strong influence ($p < .001$) on the Civic Skills latent trait (of course, 4 out of the 14 items defining that trait are about activities occurring at one's workplace). Conversely, reporting a Protestant religious affiliation has no effect on the Skills trait, but a strong ($p = .001$) effect on the Status trait (while, it is interesting to see that being Catholic seems to be related to Skills (negatively) than to Status).

Items on, or close to, the 45° dashed line are those that appear to measure both Skills and Status equally. Table C2 displays the information from Figure including the p-values for Wald tests for two different hypothesis tests. The p values under the headings of Skills and Status present the probabilities associated with Wald tests for the null hypothesis that each individual coefficient is equal to zero. For example, the p-value for whether a respondent has lived in the same town for less than 5 years is $p = .294$ for the coefficient of $\beta = -.061$ on the Skills scale, and $p < .001$ for the coefficient of $\beta = -.699$ on the

[4]These results also hold when the explanatory variables are not dichotomous. I chose to dichotomize them, however, for purposes of presentation since the coefficients varied quite widely, making the figure nearly unreadable, and I think misleading.

[5]This analysis, akin to a confirmatory factor analysis, makes certain assumptions, such as that each of the items in question is related to the underlying, latent traits in a linear and additive fashion. I also allowed all of the items to enter into one equation — thus, these coefficients reflect the influence of each item which is linearly independent of the influence of the other items. These different assumptions should certainly be explored further in future analysis of the measurement properties of these scales. Since, my own data is more rough-and-ready, I felt it reasonable to proceed with the simplest possible assumptions in order to get a general sense for how these items relate to the latent traits that I wise to represent in the Political Socialization dataset

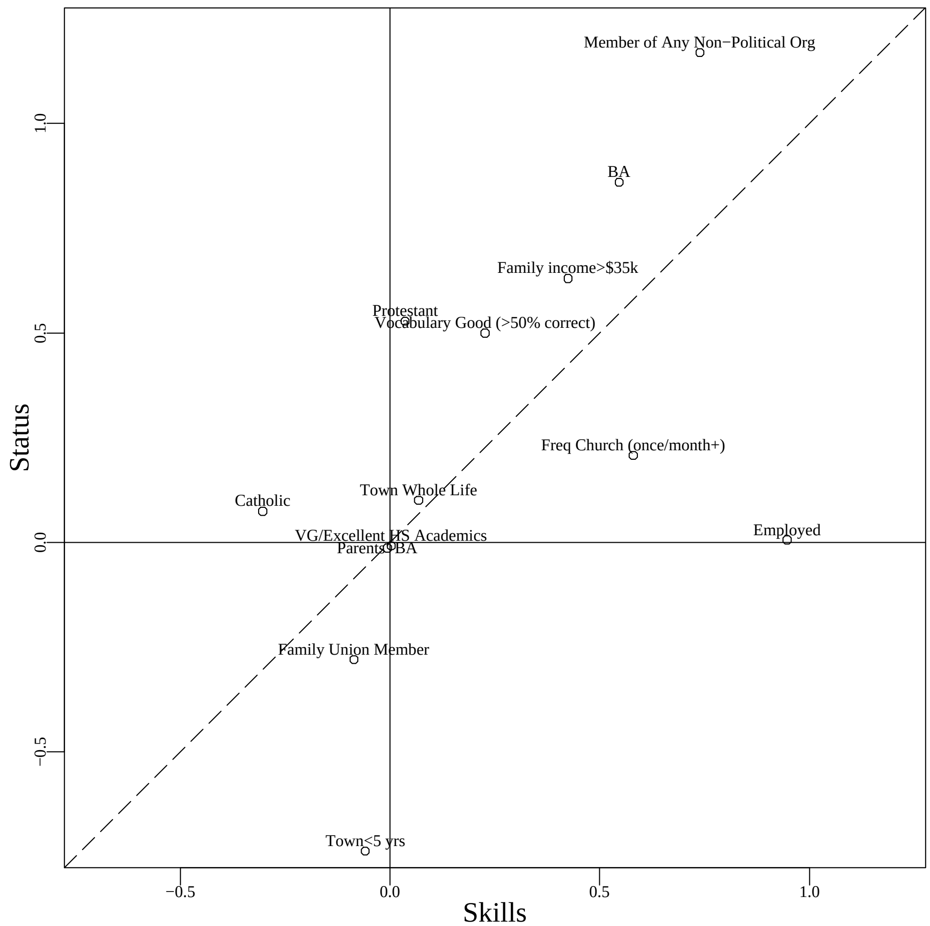


Figure C2: Loadings for Candidate Variables on Civic Skills and Civic Status

Status scale. This suggests what is also obvious from Figure C2 — that living in the same town for less than 5 years drastically decreases the probability that a person will know relevant local political actors (compared to people who have lived in the same town for more than 5 years, but not their whole lives), but it is not related to the skill level of a person. The difference between the coefficients for Skills and Status for this item is -.639, and this difference is significantly different from zero with $p < .001$. That is, living in the same town for a short time is a strong indicator of Status, but not of Skills. Although having a family member in a union significantly decreases the odds that the respondent will know political actors (thereby decreasing Status), the difference in coefficients between the insignificant ($p = .157$) effect on Skills and the significant effect on Status is not strongly significant at conventional levels ($p = .103$). Thus, union membership is not a very useful indicator in that it does not distinguish between the two scales.

Table C2: Performance of "Candidate" Items as Measures of Civic Skills and Civic Status

	Skills		Status		Skills-Status	
	β	p	β	p	$\beta - \beta$	p
BA	0.531	0.000	0.808	0.000	0.277	0.010
Parents' BA	-0.006	0.922	-0.013	0.904	-0.007	0.949
VG/Excellent HS Academics	0.004	0.937	-0.011	0.901	-0.015	0.872
Protestant	0.035	0.657	0.498	0.001	0.463	0.002
Catholic	-0.301	0.001	0.072	0.655	0.373	0.026
Freq Church (once/month+)	0.574	0.000	0.193	0.040	-0.381	0.000
Member of Any Non-Political Org	0.731	0.000	1.115	0.000	0.384	0.002
Family Union Member	-0.084	0.157	-0.267	0.013	-0.183	0.103
Family income>$35k	0.418	0.000	0.597	0.000	0.179	0.071
Employed	0.924	0.000	0.006	0.948	-0.918	0.000
Vocabulary Good (>50% correct)	0.226	0.000	0.471	0.000	0.245	0.020
Town Whole Life	0.066	0.390	0.091	0.509	0.024	0.865
Town<5 yrs	-0.061	0.294	-0.699	0.000	-0.639	0.000

Note: Coefficients are on the logit scale. The test for differences between coefficients is a Wald test. Total N= 2291.

In the end, this analysis suggests several items that I can use to represent skills and status in the Political Socialization dataset.[6]

To measure Civic Status, the variables indicated as best by this analysis are: Recency of Tenure in the same Town, and being a Protestant. Other items which influenced Status significantly more than Skills (but also were significantly different from zero as loading on Skills) were Having a College Degree, being a Member of Nonpolitical Organization. Although scoring high in a vocabulary test is a stronger measure of Status than Skills, the difference is relatively small compared to those for the other variables ($p = .02$).

[6]Of course, 1) these measurement relationships may vary by period and so, in fact, the idea that I would use a single item represent the same concept over 30 years may be an error; 2) these measurement relationships may be different for different generations; 3) differences in the specific question wording and mode of interview between the Citizen Participation study and the Political Socialization study may make these relationships different between the two datasets.

To measure Civic Skills, the variables indicated as best by this analysis are: Frequency of Church Attendance, Being Employed, and Being Catholic.

Family income (over the median for the sample) does significantly predict both skills and status, but is not significantly different. And reports of Parents' college degrees or Academic Achievement in High school had no effects on either skills or status once all of the other variables were entered in.

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